5) Make sure that the B-nut [463] is fully seated.

6) If the B-nut [463] is not fully seated, do the above steps again to tighten the B-nut [463].

7) Secure the B-nut [463] with safety cable, G50065 [CP8006] (TASK 70-41-00-400-802-G00).

(p) Tighten the B-nut [465] to the LPT ACC fuel manifold [469] as follows:

1) Tighten the B-nut [465] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut [465] until you can turn them with your fingers.

3) Tighten the B-nut [465] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the B-nut [465] for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nut [465] is fully seated.

6) If the B-nut [465] is not fully seated, do the above steps again to tighten the B-nut [465].

7) Secure the B-nut [465] with safety cable, G50065 [CP8006] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-05-420-321-G00

(3) Install the 7th-stage air manifold [452] as follows (View D, E, F, G, H, and J, Figure 414):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [450], four bolts [453], two bolts [455], and four bolts [457].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [624], bolts [453], two bolts [626], and four bolts [457].

PTF ALL

(c) Apply a layer of stick wax, G51435 [CP2635], to the new seal [454] and new seal [458] to hold them during installation.

(d) Put the 7th-stage air manifold [452] in position on the engine.

(e) Install the new seal [454] on the 7th-stage air manifold [452] or on the compressor case [65].

(f) Install the new seal [458] on the 7th-stage air manifold [452] or the HPT port [459].

(g) Install the four bolts [453] that attach the 7th-stage air manifold [452] and seal [454] to the compressor case [65].

(h) Install the four bolts [457] that attach the 77th-stage air manifold [452] and seal [458] to the HPT port [459].

(i) Tighten the bolts [453] and bolts [457] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(j) Connect the air tube [460] and B-nut [461] to the 7th-stage air manifold [452].

(k) Tighten the B-nut [461] to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(l) Install the two bolts [450] that attach the 7th-stage air manifold [452] to the support bracket [451].

(m) Tighten the bolts [450] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007 (Continued)

(n) Install the two bolts [455] that attach the 7th-stage air manifold [452] to the support bracket [456].

(o) Tighten the bolts [455] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(p) Install the two bolts [624] and strap clamp [625] that attach the 7th-stage air manifold [452] to the support bracket [451].

(q) Tighten the bolts [624] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(r) Install the two bolts [626]and strap clamp [627] that attach the 7th-stage air manifold [452] to the support bracket [456].

(s) Tighten the bolts [626] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

SUBTASK 73-11-05-420-301-G00

(4) Install the wye duct [28A] as follows (Figure 420):

(a) Put the wye duct [28A] onto the core module to interface with the High Pressure Turbine Active Clearance Control (HPTACC) inlet duct at the 9:30 o'clock position.

1) If the wye duct [28A] already has silicone rubber tape, G51669 [CP2975], in the clamping area, proceed to step (b).

2) If the wye duct [28A] does not have silicone rubber tape, G51669 [CP2975], in the clamping area, apply as follows:

a) Apply the two adjacent rows of silicone rubber tape, G51669 [CP2975].

NOTE: Each row will be three plies thick of silicone rubber tape, G51669 [CP2975], 0.02 in. (0.51 mm) x 1.0 in. (25.4 mm) width to the wye duct directly aft of the forward edge bead of the wye duct.

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [30A].

(c) Attach the wye duct [28A] to the HPTACC inlet duct with the bolts [30A] and saddle clamp [29A].

(d) Tighten the bolts [30A] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

(e) Measure the gap between the saddle clamp [29A] with the feeler gage (View D, Figure 420).

1) Measure each gap in a location that is between the centerline of the bolt and the bend radii for the relief holes.

NOTE: The gaps must be measured at the forward and aft sides of both bolts.

2) Acceptable gap between the saddle clamps [29A] adjacent to the wye duct [28A] is to 0.045 in. (1.143 mm) - 0.075 in. (1.905 mm).

3) If measured gap exceeds 0.075 in. (1.905 mm), remove the bolts [30A] from the saddle clamps [29A].

a) Remove one ply of silicone rubber tape, G51669 [CP2975], from wye duct [28A].

b) Repeat steps (c), (d), and (e).

4) If measured gap is less than 0.045 in. (1.143 mm), remove the bolts [30A] from the saddle clamps [29A].

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a) Add one ply of silicone rubber tape, G51669 [CP2975], to wye duct [28A].

b) Repeat steps (c), (d), and (e).

SUBTASK 73-11-05-420-302-G00

(5) Install the Low Pressure Turbine Active Clearance Control (LPTACC) valve (TASK 75-24-01-400-801-G00).

SUBTASK 73-11-05-420-322-G00

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Install the engine upper CDP duct assembly (TASK 36-11-01-400-801).

W. Fuel Nozzle Installation (Position 15)

SUBTASK 73-11-05-420-303-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 15) as follows (View C, Figure 416):

(a) Install the fuel nozzle [1] (position 15) with the new seal [524] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 15) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [524] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 15) aft to engage the fuel nozzle [1] (position 15) tip in the combustor radial mixer (cup).

(d) Align the seal [524] and fuel nozzle [1] (position 15) bolt holes to the combustion case [5].

(e) Install the support bracket [468] in position on the fuel nozzle [1] (position 15).

(f) Install the four bolts [520] that attach the fuel nozzle [1] (position 15) and support bracket [468] to the combustion case [5].

(g) Do not tighten the fuel nozzle bolts at this time.

(h) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(i) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 15) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(j) Tighten the bolts [520] as follows (Figure 416):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [520].

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2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(k) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(l) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

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c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-304-G00

(2) Attach the support bracket [526] to the support bracket [527] as follows (View E, Figure 416):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [525].

(b) Install the bolt [525] that attaches the support bracket [526] to the support bracket [527].

(c) Tighten the bolt [525] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-305-G00

(3) Install the LPTACC valve [526B] as follows (View E, Figure 416):

(a) Apply oil, D00599 [CP2442], on the threads of two bolts [525A].

(b) Hold the LPTACC valve [526B] and install the two bolts [525A] that attach the LPTACC valve [526B] to the support bracket [526] and support bracket [526A].

(c) Tighten the bolts [525A] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-306-G00

(4) Install the HPTACC valve (TASK 75-24-04-400-801-G00).

(a) If it is necessary, cut off the hose clamp tab more than 1 in. (25.4 mm) from the hose clamp screw housing.

1) Remove all burrs from the hose clamp tabs.

SUBTASK 73-11-05-420-307-G00

(5) Install the wye duct [28A] as follows (Figure 420):

(a) Put the wye duct [28A] onto the core module to interface with the HPTACC inlet duct at the 9:30 o'clock position.

1) If the wye duct [28A] already has silicone rubber tape, G51669 [CP2975], in the clamping area, proceed to step (b).

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2) If the wye duct [28A] does not have silicone rubber tape, G51669 [CP2975], in the clamping area, apply as follows:

a) Apply the two adjacent rows of silicone rubber tape, G51669 [CP2975].

NOTE: Each row will be three plies thick of silicone rubber tape, G51669 [CP2975], 0.02 in. (0.51 mm) x 1.0 in. (25.4 mm) width to the wye duct directly aft of the forward edge bead of the wye duct.

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [30A].

(c) Attach the wye duct [28A] to the HPTACC inlet duct with the bolts [30A] and saddle clamp [29A].

(d) Tighten the bolts [30A] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

(e) Measure the gap between the saddle clamp [29A] with the feeler gage (View D, Figure 420).

1) Measure each gap in a location that is between the centerline of the bolt and the bend radii for the relief holes.

NOTE: The gaps must be measured at the forward and aft sides of both bolts.

2) Acceptable gap between the saddle clamps [29A] adjacent to the wye duct [28A] is to 0.045 in. (1.143 mm) - 0.075 in. (1.905 mm).

3) If measured gap exceeds 0.075 in. (1.905 mm), remove the bolts [30A] from the saddle clamps [29A].

a) Remove one ply of silicone rubber tape, G51669 [CP2975], from wye duct [28A].

b) Repeat steps (c), (d), and (e).

4) If measured gap is less than 0.045 in. (1.143 mm), remove the bolts [30A] from the saddle clamps [29A].

a) Add one ply of silicone rubber tape, G51669 [CP2975], to wye duct [28A].

b) Repeat steps (c), (d), and (e).

X. Fuel Nozzle Installation (Position 16)

SUBTASK 73-11-05-420-308-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 16) as follows (View C and J, Figure 417):

(a) Install the fuel nozzle [1] (position 16) with the new seal [547] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 16) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [547] and combustion case [5] pad.

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(c) Move the fuel nozzle [1] (position 16) aft to engage the fuel nozzle [1] (position 16) tip in the combustor radial mixer (cup).

(d) Align the seal [547] and fuel nozzle [1] (position 16) bolt holes to the combustion case [5].

(e) Install the support bracket [541] in position on the fuel nozzle [1] (position 16).

(f) Install the four bolts [540] that attach the fuel nozzle [1] (position 16) and support bracket [541] to the combustion case [5].

(g) Do not tighten the fuel nozzle bolts at this time.

(h) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(i) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 16) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(j) Tighten the bolts [540] as follows (Figure 417):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [540].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(k) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(l) Apply oil, D00599 [CP2442], on the threads of the two bolts [596].

(m) Install the two bolts [596] that attach the support bracket [541] to the support bracket [584].

(n) Tighten the bolts [596] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(o) Apply oil, D00599 [CP2442], on the threads of the two bolts [543], bolt [545], and bolt [545A].

(p) Install the two bolts [543] that attach the two loop clamps [544] and EGT harness [542] to the support bracket [541].

(q) Tighten the bolts [543] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(r) Install the bolt [545] that attaches the loop clamp [546] and T3 sensor [595] to the support bracket [541].

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(s) Tighten the bolt [545] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

(t) Install the bolt [545A] that attaches the loop clamp [546A] and TCF accelerometer cable [595A] to the support bracket [541].

(u) Tighten the bolt [545A] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(v) Apply oil, D00599 [CP2442], on the threads of the two bolts [609] (View C and XC, Figure 417).

(w) Install the two bolts [609] that attach the loop clamps [610], W200 electrical harness [611], and W300 electrical harness [612] to the support bracket [541].

(x) Tighten the bolts [609] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(y) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

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f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-309-G00

(2) Connect the turbine center frame accelerometer at the connector and attach the turbine center frame accelerometer to the mounting bracket as follows:

(a) Do the applicable steps in this task: Turbine Center Frame Accelerometer Installation, TASK 77-31-06-400-801-G00.

SUBTASK 73-11-05-420-310-G00

(3) Connect the EGT sensor harness 1 and 2 as follows (TASK 77-21-04-400-801-G00 and TASK 77-21-05-400-801-G00):

(a) Connect the EGT sensor harness 1 and 2 at the connectors.

(b) Attach the EGT sensor harnesses to the mounting bracket.

SUBTASK 73-11-05-420-311-G00

(4) Install the T3 sensor (TASK 73-21-64-400-801-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 73-11-05-420-312-G00

(5) Connect the fuel hydraulic manifold [447] as follows (View D, E, F, G, and H, Figure 417):

(a) Apply oil, D00599 [CP2442], to the threads of the bolt [530], bolt [538], four bolts [448], three bolts [478], bolts [534], bolt [536], bolt [549], bolt [552], bolt [555], bolt [557], bolt [559], bolt [589], bolt [590], bolt [592], and two bolts [598].

(b) Install the gasket seal [476] between the fuel hydraulic manifold [447] and SCU [151].

(c) Install the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

1) Tighten the bolts [448] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(d) Connect thefuel manifold [588] B-nut to the fuel fuel hydraulic manifold [447].

1) Tighten the fuel manifold [588] B-nut as follows:

a) Tighten the fuel manifold [588] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) Tightly attach the fuel manifold [588] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(e) Connect the fuel hydraulic manifold [447] B-nut to the fuel manifold [588].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

g) Tightly attach the fuel hydraulic manifold [447] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(f) Connect the LPT ACC fuel manifold [469] B-nut to the fuel hydraulic manifold [447].

1) Tighten the LPT ACC fuel manifold [469] B-nut as follows:

a) Tighten the LPT ACC fuel manifold [469] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

g) Tightly attach the LPT ACC fuel manifold [469] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(g) Connect the hydraulic fuel tube [528] B-nut to the fuel hydraulic manifold [447].

1) Tighten the hydraulic fuel tube [528] B-nut as follows:

a) Tighten the hydraulic fuel tube [528] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

g) Tightly attach the hydraulic fuel tube [528] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(h) Connect the fuel hydraulic manifold [447] B-nut to the hydraulic fuel tube [529].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

g) Tightly attach the fuel hydraulic manifold [447] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(i) Connect the fuel hydraulic manifold [447] B-nut to the ACC fuel tube [446].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

g) Tightly attach the fuel hydraulic manifold [447] B-nut with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(j) Install the bolt [589] that attaches the fuel hydraulic manifold [447] to the compressor case [65].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 AND PRE LEAP-1B SERVICE BULLETIN 75-0004

(k) Install the bolt [589] that attaches the loop clamp [597A] and fuel hydraulic manifold [447] to the support bracket [599].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0004

(l) Install the bolt [589] that attaches the loop clamp [597A] and fuel hydraulic manifold [447] to the support bracket [599A].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

(m) Tighten the bolt [589] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(n) Install the two bolts [598] that attach two loop clamp [597] and fuel hydraulic manifold [447] to the support bracket [599].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(o) Install the bolt [590] that attaches the loop clamp [591] and fuel hydraulic manifold [447] to the support bracket [594].

(p) Install the bolt [592] that attaches the loop clamp [593] and fuel hydraulic manifold [447] to the support bracket [594].

(q) Install the bolt [559] that attaches the loop clamp [558] and fuel hydraulic manifold [447] to the support bracket [587].

(r) Install the bolt [557] that attaches the loop clamp [556] and fuel hydraulic manifold [447] to the support bracket [587].

(s) Install the bolt [555] that attaches the loop clamp [554] and fuel hydraulic manifold [447] to the support bracket [553].

(t) Install the bolt [552] that attaches the loop clamp [551] and fuel hydraulic manifold [447] to the support bracket [553].

(u) Install the bolt [534] that attaches the loop clamp [535] and fuel hydraulic manifold [447] to the support bracket [533].

(v) Install the bolt [536] that attaches the loop clamp [537] and fuel hydraulic manifold [447] to the support bracket [533].

(w) Install the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

(x) Install the bolt [549] that attaches the loop clamp [548] and fuel hydraulic manifold [447] to the support bracket [550].

(y) Install the bolt [530] that attaches the loop clamp [531] and fuel hydraulic manifold [447] to the support bracket [532].

(z) Install the three bolts [478] that attach the three loop clamps [477] and fuel hydraulic manifold [447] to the support bracket [445].

(aa) Tighten the bolt [590], bolt [592], bolt [559], bolt [557], bolt [555], bolt [552], bolt [534], bolt [536], bolt [538], bolt [549], bolt [530], two bolts [598], and three bolts [478] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 73-11-05-420-313-G00

(6) Connect the fuel hydraulic manifold [447] as follows (View D, E, F, and G, Figure 417):

(a) Apply oil, D00599 [CP2442], to the threads of the four bolts [448], two bolts [478], two bolts [534], and bolt [538].

(b) Install the gasket seal [476] between the fuel hydraulic manifold [447] and SCU [151].

(c) Install the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

1) Tighten the bolts [448] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(d) Connect the fuel hydraulic manifold [447] B-nut to the hydraulic fuel tube [528].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

(e) Connect the fuel hydraulic manifold [447] B-nuts to the hydraulic fuel tube [529] at two locations.

1) Tighten the fuel hydraulic manifold [447] B-nuts as follows:

a) Tighten the fuel hydraulic manifold [447] B-nuts to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nuts until you can turn it with your fingers.

c) Tighten the B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nuts for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nuts are fully seated.

f) If the B-nuts is not fully seated, do the above steps again.

(f) Connect the fuel hydraulic manifold [447] B-nut to the ACC fuel tube [446].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

(g) Install the two bolts [534] that attach the two loop clamps [535] and fuel hydraulic manifold [447] to the support bracket [533] at two locations.

(h) Install the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

(i) Install the two bolts [478] that attach the fuel hydraulic support block [477], fuel hydraulic manifold [447], and ACC fuel tube [446] to the support bracket [445].

(j) Tighten the bolts [478], bolts [534], and bolt [538] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-329-G00

(7) Connect the fuel hydraulic manifold [447] as follows (View D, E, F, G, XA, H, and XB, Figure 417):

(a) Apply oil, D00599 [CP2442], to the threads of the four bolts [448], two bolts [478], bolt [538], two bolts [534], bolt [530], two bolts [552], bolt [601], bolt [557], bolt [604], bolt [592], bolt [606], and bolts [598].

(b) Install the gasket seal [476] between the fuel hydraulic manifold [447] and SCU [151].

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(c) Install the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

1) Tighten the bolts [448] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(d) Connect the fuel hydraulic manifold [447] B-nut to the hydraulic fuel tube [528A].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated

f) If the B-nut is not fully seated, do the above steps again.

(e) Connect the fuel hydraulic manifold [447] B-nuts to the hydraulic fuel tube [529A] at two locations.

1) Tighten the fuel hydraulic manifold [447] B-nuts as follows:

a) Tighten the fuel hydraulic manifold [447] B-nuts to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nuts until you can turn it with your fingers.

c) Tighten the B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nuts for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nuts are fully seated.

f) If the B-nuts are not fully seated, do the above steps again.

(f) Connect the fuel hydraulic manifold [447] B-nut to the ACC fuel tube [446].

1) Tighten the fuel hydraulic manifold [447] B-nut as follows:

a) Tighten the fuel hydraulic manifold [447] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the B-nut until you can turn it with your fingers.

c) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make sure that the B-nut is fully seated.

f) If the B-nut is not fully seated, do the above steps again.

(g) Secure the fuel hydraulic manifold [447] B-nuts with a safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(h) Install the two bolts [534] that attach the two loop clamps [535] and fuel hydraulic manifold [447] to the support bracket [533].

(i) Install the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(j) Install the bolt [549] that attaches the loop clamp [548] and fuel hydraulic manifold [447] to the support bracket [550].

(k) Install the two bolts [478] that attach the fuel hydraulic support block [477], fuel hydraulic manifold [447], and ACC fuel tube [446] to the support bracket [445].

(l) Install the bolt [530] that attaches the loop clamp [531] and fuel hydraulic manifold [447] to the support bracket [532].

(m) Install the two bolts [552] that attach the loop clamps [551] and fuel hydraulic manifold [447] to the support bracket [553].

(n) Install the bolt [601] that attaches the loop clamp [600] and fuel hydraulic manifold [447] to the support bracket [602].

(o) Install the bolt [557] that attaches the loop clamp [556] and fuel hydraulic manifold [447] to the support bracket [587].

(p) Install the bolt [604] that attaches the loop clamp [605] and fuel hydraulic manifold [447] to the support bracket [603].

(q) Connect the fuel hydraulic manifold [447] B-nuts to the hydraulic fuel tube [529] and hydraulic fuel tube [528A] at two locations.

1) Tighten the fuel hydraulic manifold [447] B-nuts to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
2) Loosen the B-nuts until you can turn it with your fingers.
3) Tighten the B-nuts again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
4) Tighten the B-nuts for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
5) Make sure that the B-nuts are fully seated.
6) If the B-nuts are not fully seated, do the above steps again.

(r) Secure the fuel hydraulic manifold [447] B-nuts with a safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

(s) Install the bolt [592] that attaches the loop clamp [593] and fuel hydraulic manifold [447] to the support bracket [594].

(t) Install the bolt [604] that attaches the loop clamp [605] and fuel hydraulic manifold [447] to the support bracket [606].

(u) Install the two bolts [598] that attach the two loop clamps [597] and fuel hydraulic manifold [447] to the support bracket [599].

(v) Connect the fuel manifold [588] B-nut to the fuel hydraulic manifold [447].

1) Tighten the fuel manifold [588] B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
2) Loosen the B-nut until you can turn it with your fingers.
3) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
4) Tighten the B-nut for a third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).
5) Make sure that the B-nut is fully seated.
6) If the B-nut is not fully seated, do the above steps again.

(w) Secure the fuel manifold [588] B-nut with a safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(x) Tighten the bolts [448], bolt [478], bolt [538], bolts [534], bolt [530], bolts [552], bolt [601], bolt [557], bolt [604], bolt [592], bolt [606], and bolts [598] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL

SUBTASK 73-11-05-420-314-G00

(8) Install the HPTACC valve (TASK 75-24-04-400-801-G00).

(a) If it is necessary, cut off the hose clamp tab more than 1 in. (25.4 mm) from the hose clamp screw housing.

1) Remove all burrs from the hose clamp tabs.

Y. Fuel Nozzle Installation (Position 17)

SUBTASK 73-11-05-420-315-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 17) as follows (View C and P, Figure 418):

(a) Install the fuel nozzle [1] (position 17) with the new seal [586] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 17) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [586] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 17) aft to engage the fuel nozzle [1] (position 17) tip in the combustor radial mixer (cup).

(d) Align the seal [586] and fuel nozzle [1] (position 17) bolt holes to the combustion case [5].

(e) Install the support bracket [584] in position on the fuel nozzle [1] (position 16) and fuel nozzle [1] (position 17).

(f) Install the bolt [585] that attach the support bracket [584] to the fuel nozzle [1] (position 17).

(g) Install the support bracket [581A] in position on the fuel nozzle [1] (position 17).

(h) Install the remaining three bolts [585] that attach the fuel nozzle [1] (position 17) and support bracket [581A] to the combustion case [5].

(i) Do not tighten the fuel nozzle bolts at this time.

(j) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(k) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 17) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

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(l) Tighten the bolts [585] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [585].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(m) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(n) Apply oil, D00599 [CP2442], on the threads of the two bolts [596].

(o) Install the two bolts [596] that attach the support bracket [541] to the support bracket [584].

(p) Tighten the bolts [596] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the two bolts [582].

(r) Install the two bolts [582] that attach the two loop clamps [583] to the support bracket [581A].

(s) Tighten the bolts [582] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(t) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] PPM and PS fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-316-G00

(2) Install the 7th-stage air tube [574] as follows (View D, K, L, M, and N, Figure 418):

(a) Apply a layer of stick wax, G51435 [CP2635], to the new seal [576], and new seal [580] to hold them during installation.

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(b) Install the new seal [576] on the7th-stage air tube [574] or on the HPC extension case [101].

(c) Install the new seal [580] on the 7th-stage air tube [574] or on the HPT port [581].

(d) Apply oil, D00599 [CP2442], on the threads of the four bolts [575].

(e) Install the four bolts [575] that attach the 7th-stage air tube [574] to the HPC extension case [101].

(f) Tighten the bolts [575] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(g) Apply oil, D00599 [CP2442], on the threads of the four bolts [579].

(h) Install the four bolts [579] that attach the support bracket [578] and 7th-stage air tube [574] to the HPT port [581].

(i) Tighten the bolts [579] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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(j) Apply oil, D00599 [CP2442], on the threads of the two bolts [577].

(k) Install the two bolts [577] that attach the 7th-stage air tube [574] to the support bracket [565].

(l) Tighten the bolts [577] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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(m) Apply oil, D00599 [CP2442], on the threads of the two bolts [623].

(n) Install the two bolts [623] and strap clamp [622] that attach the 7th-stage air tube [574] to the support bracket [565].

(o) Tighten the bolts [623] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

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(3) Install the 4th-stage cooling air manifold [560] as follows (View D, E, F, G, H, and J, Figure 418):

(a) Install the seal [563] on the 4th-stage cooling air manifold [560] or on the compressor case [65].

(b) Install the seal [572] on the 4th-stage cooling air manifold [560] or on the adapter [573].

(c) Install the coupling clamp [571] to the 4th-stage cooling air manifold [560] and adapter [573].

(d) Install the 4th-stage air tube [570] and B-nut [569] to the 4th-stage cooling air manifold [560].

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(e) Apply oil, D00599 [CP2442], on the threads of the two bolts [70].

(f) Install the two bolts [70] that attach the 4th-stage cooling air manifold [560] to the support bracket [79] (View FA, Figure 418).

(g) Place the harness support bracket [78] and install the two bolts [69] that attach the support bracket [76], support bracket [77], and harness support bracket [78] (View F, Figure 418).

1) Do not tighten the bolts [69] at this time.

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(h) Install the HJ8A harness [72], W200 harness [74], and W300 harness [75] to the push clamps [71] on the harness support bracket [78] (View F, Figure 418).

NOTE: The support bracket [562] is two brackets assembled together that provides attachment for the booster anti ice tube [67], HJ8A harness [72], and HJ8B harness [73].

(i) Install the HJ8A harness [72] and HJ8B harness [73] to the push clamps [71] on the support bracket [562] (View F, Figure 418).

(j) Remove the two bolts [68] and strap clamp [66] that attach the booster anti ice tube [67] to the support bracket [562] (View F, Figure 418).

(k) Tighten the bolts [70], bolts [69], and bolts [68] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(l) Install the the reverse bleed valve assembly (TASK 75-34-01-400-801-G00).

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(m) Apply oil, D00599 [CP2442], on the threads of the four bolts [561].

(n) Install the support bracket [562] in position on the 4th-stage cooling air manifold [560].

(o) Install the four bolts [561] that attach the 4th-stage cooling air manifold [560] and support bracket [562] to the compressor case [65].

(p) Tighten the bolts [561] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

(q) Apply oil, D00599 [CP2442], on the threads of the bolt [566].

(r) Install the bolt [566] that attaches the loop clamp [567] to the support bracket [568].

(s) Tighten the bolt [566] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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(t) Apply oil, D00599 [CP2442], on the threads of the two bolts [564].

(u) Install the two bolts [564] that attach the 4th-stage cooling air manifold [560] to the support bracket [565].

(v) Tighten the bolts [564] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

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(w) Apply oil, D00599 [CP2442], on the threads of the two bolts [620].

(x) Install the two bolts [620] and strap clamp [621] that attach the 4th-stage cooling air manifold [560] to the support bracket [565].

(y) Tighten the bolts [620] to 106 in-lb (11.98 N·m) - 124 in-lb (14.01 N·m).

PTF ALL

(z) Tighten the coupling clamp [571] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(aa) Tighten the B-nut [569] to 79 ft-lb (107.1 N·m) - 91 ft-lb (123.4 N·m).

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Z. Fuel Nozzle Installation (Position 18)

SUBTASK 73-11-05-420-318-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzle [1] (position 18) as follows (View C, Figure 419):

(a) Install the fuel nozzle [1] (position 18) with the new seal [604] aligned into the combustion case [5] pad with the fuel nozzle [1] tip forward.

(b) Turn the fuel nozzle [1] (position 18) 180 degrees counter-clockwise and put the fuel nozzle [1] in position on the seal [604] and combustion case [5] pad.

(c) Move the fuel nozzle [1] (position 18) aft to engage the fuel nozzle [1] (position 18) tip in the combustor radial mixer (cup).

(d) Align the seal [604] and fuel nozzle [1] (position 18) bolt holes to the combustion case [5].

(e) Install the support bracket [616] and bracket [601] in position on the fuel nozzle [1] (position 18).

(f) Install the four bolts [600] that attach the fuel nozzle [1] (position 18) and bracket [601] to the combustion case [5].

(g) Apply oil, D00599 [CP2442], on the threads of the two bolts [614] and two bolts [602].

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(h) Install the two bolts [614] that attach the support bracket [615] to the support bracket [616] (View E, Figure 419).

(i) Tighten the bolts [614] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(j) Install the two bolts [602] that attach the fuel manifolds, and two loop clamps [603] to the bracket [601].

(k) Do not tighten the fuel nozzle bolts at this time.

(l) Do not apply lubricant on the fuel nozzle [1] or B-nut [4] threads.

(m) Hand tight the fuel manifold B-nuts [4] on their fuel nozzle [1] (position 18) fittings, before you tighten the fuel nozzle bolts.

1) Do not torque the B-nuts [4] at this time.

NOTE: This helps the fuel manifold correctly center onto the fuel nozzle fittings.

(n) Tighten the bolts [600] as follows (Figure 422):

1) Make sure that Tiodize T8E-H grease, D50355 [CP2961], is on the thread of the fuel nozzle bolts [600].

2) Tighten the fuel nozzle bolts and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m).

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3) Do the sequence to tighten the fuel nozzle bolts again and start at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129 in-lb (14.57 N·m) - 151 in-lb (17.06 N·m).

NOTE: One of the four fuel nozzle bolt holes is elongated.

(o) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

1) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 423.

2) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

3) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

(p) Tighten the bolts [602] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(q) Tighten the fuel manifold B-nuts [4] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings as follows:

NOTE: False tightening is a possibility, mainly in hard to reach locations. A wrench/attachment/extension can hit other hardware when you tighten them and give a false torque.

NOTE: It is necessary to triple tighten the fuel nozzle B-nuts in this sequence: TIGHTEN – LOOSEN – TIGHTEN– TIGHTEN.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (TASK 70-51-00-800-801-G00).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple tighten the fuel nozzle B-nuts [4] as follows:

a) Tighten the two B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

b) Loosen the two B-nuts [4] until you can turn them with your fingers.

c) Tighten the two B-nuts [4] again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

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d) Tighten the two B-nuts [4] for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

e) Make a record of the completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 401).

f) Paint a stripe line across the fuel nozzle B-nuts [4] and fuel nozzle fitting with temporary marker, G50530 [CP5061].

3) Do the fuel nozzle B-nuts [4] tightening inspection as follows:

a) Tighten the fuel nozzle B-nuts [4] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

NOTE: On each fuel nozzle tighten the PS B-nut first followed by the PPM B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector must be a different person from the crew that does the steps to tighten fuel nozzle B-nuts.

b) Record completion of inspection and torque value at each fuel nozzle B-nut [4] fitting (Table 401).

c) Tightly attach each fuel nozzle B-nuts [4] with safety cable, G50065 [CP8006], or with safety wire, G02495 [CP8002] (PAGEBLOCK 70-41-00/201).

NOTE: If lockwire is used, stop the lockwire at the fuel nozzle end.

SUBTASK 73-11-05-420-319-G00

(2) If disconnected, install the bleed air precooler (TASK 36-12-01-400-803).

AA. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-05-420-324-G00

(1) Install the thrust reverser compression rod No. 2 (Figure 424).

SUBTASK 73-11-05-860-025-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-860-026-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-05-860-027-G00

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 73-11-05-860-028-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

AB. Fuel Nozzle Installation Test

SUBTASK 73-11-05-860-030-G00

(1) Do the tests as follows:

(a) Do an idle check (TASK 71-00-00-790-804-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

a) Electronic Engine Control (EEC) can find fuel system faults after fuel system maintenance because of residual air in fuel lines, do these steps:

<1> Do an engine idle run (TASK 71-00-00-790-804-G00).

<a> Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table, note 7.a.

b) Check the Onboard Maintenance Function (OMF) for any new fuel system faults.

<1> If the faults are found, clear the faults (TASK 71-00-00-700-811-G00 and TASK 71-00-00-700-812-G00).

<a> Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table, note 7.b.

(b) Do a pneumatic leak test (TASK 71-00-00-790-801-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

NOTE: This is necessary for HPSOV, Environmental Control System (ECS) ducts, PRSOV, and TBV duct installation.

a) After the tests are done, follow the applicable inspections:

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<1> Do a visual inspection of the engine after idle leak check and do as follows:

<a> Do a visual inspection of the fuel lines and connections and check for fuel leaks.

<b> Do a visual inspection of the air ducts, tubes, and seals to check for hot air leakage indication and/or damage.

(c) Do an idle test menu mode (TASK 71-00-00-700-808-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

SUBTASK 73-11-05-790-004-G00

(2) If you opened the EEC air signal manifolds, do this step:

(a) Do the test for the air ducts and air tubes (TASK 71-00-00-800-804-G00).

SUBTASK 73-11-05-700-003-G00

(3) If it is necessary, do a wet motor leak test (TASK 71-00-00-790-803-G00).

AC. Triple Tightening and Torque Inspection of B-nuts

SUBTASK 73-11-05-420-320-G00

(1) Triple tightening and torque inspection of PS and PPM B-nuts.

Table 401/73-11-05-993-803-G00 Triple tightening and torque inspection of Pilot Secondary (PS) and Pilot Primary Main (PPM) B-nuts

FN Position	B-nut Position	Confirm First Torque (Yes/ No)	Confirm Loosen (Yes/No)	Confirm Second Torque (Yes/ No)	Record Third Torque at Each B-nut	Completion Date and Mechanic Signature	Inspection: Record Torque at Each B-nut	Inspection: Completion Date and Signature
1	PS							
	PPM							
2	PS							
	PPM							
3	PS							
	PPM							
4	PS							
	PPM							
5	PS							
	PPM							
6	PS							
	PPM							
7	PS							
	PPM							
8	PS							
	PPM							
9	PS							

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Table 401/73-11-05-993-803-G00 Triple tightening and torque inspection of Pilot Secondary (PS) and Pilot Primary Main (PPM) B-nuts (Continued)

FN Position	B-nut Position	Confirm First Torque (Yes/ No)	Confirm Loosen (Yes/No)	Confirm Second Torque (Yes/ No)	Record Third Torque at Each B-nut	Completion Date and Mechanic Signature	Inspection: Record Torque at Each B-nut	Inspection: Completion Date and Signature
	PPM							
10	PS							
	PPM							
11	PS							
	PPM							
12	PS							
	PPM							
13	PS							
	PPM							
14	PS							
	PPM							
15	PS							
	PPM							
16	PS							
	PPM							
17	PS							
	PPM							
18	PS							
	PPM							

——— END OF TASK ———

TASK 73-11-05-000-802-G00

4. Fuel Nozzle Set Removal

(Figure 424, Figure 425, Figure 426, Figure 427, Figure 429, Figure 430, Figure 431, Figure 432, Figure 433, Figure 434, Figure 435, Figure 436, Figure 437, Figure 438, Figure 439, Figure 440, Figure 441, Figure 442, Figure 443, Figure 444, Figure 445, Figure 446, Figure 447, Figure 448, Figure 449, Figure 450, Figure 451, Figure 452, Figure 453, Figure 454, Figure 455, Figure 456, and Figure 457)

A. General

(1) This task is to show the removal and installation procedures to do a full set replacement of the combustion chamber fuel nozzle.

(2) There are eighteen fuel nozzles installed on the combustion case.

(3) The fuel nozzle position numbers are in a Clockwise (CW) sequence when viewed from the rear of the engine facing forward or Aft Looking Forward (ALF).

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(4) It is recommended to dry motor the engine up to 5 minutes to help cool down the engine (TASK 71-00-00-910-808-G00).

(a) Make sure that the Exhaust Gas Temperature (EGT) is below 150°F (66°C).

NOTE: The dry motoring will help to cool down the engine to start the work of the fuel nozzle [1] replacement.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
36-11-01-000-801	Engine Pneumatic Duct Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-07-000-801	HPSOV Removal (P/B 401)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-64-000-801-G00	T3 Sensor Removal (P/B 401)
73-22-01-000-808-G00	Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal (P/B 401)
75-24-01-000-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal (P/B 401)
75-24-04-000-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal (P/B 401)
75-34-01-000-801-G00	Reverse Bleed Check Valve Assembly Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25114	Device - Ultrasonic Part #: 9447M49P01 Supplier: 58828
SPL-25115	Kit - Adapters, Ultrasonic Part #: 956A1742P01 Supplier: 58828
STD-200	Container - Fuel Resistant, 10-gallon (38-Liter)
STD-562	Lens - Magnifying, 10x
STD-858	Tag - DO NOT OPERATE
STD-1242	Hammer - Standard
STD-1285	Container - Fuel Resistant, 10 Gallon (38 Liter)
STD-3842	Punch - Drift
STD-6847	Ratchet

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G00834	Cloth - Lint-free Cotton	

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(Continued)

Reference	Description	Specification
G50530 [CP5061]	Marker - Temporary	
G51403 [CP2166]	Tape - Polyester-Backed, Adhesive	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-11-05-040-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-05-010-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-040-004-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 73-11-05-000-023-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-11-05-040-005-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

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(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 73-11-05-000-024-G00

(6) Read and obey the recommendation, while you do this procedure:

(a) Use good lighting and a 10x magnifying lens, STD-562, to help you see the left and right engine core areas.

(b) Catch the fuel that will drain from the combustor area in a 10 gallon (38 liter) fuel resistant container, STD-1285.

(c) Use lint-free cloth, G00834, or rags around the combustion case.

SUBTASK 73-11-05-000-026-G00

(7) To prevent broken bolts and damaged inserts during the removal process, do the following steps:

NOTE: The steps below apply to the 72 fuel nozzle pad bolts, and any bolts that can be removed from the engine cases to get access to the fuel nozle pads (i.e., ECS pad, TBV pad, and/or 7th stage air manifolds). This is done to keep the possible risk of broken bolts and damaged inserts to a minimum.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(a) Do an initial soak of all the 72 fuel nozzle bolts [130]bolts [135], bolts [136] , bolts [138], bolts [2], bolts [140], bolts [141], bolts [142], bolts [146], bolts [150], bolts [153], bolts [154], bolts [155], bolts [158], bolts [160], bolts [175], bolts [187], bolts [195], bolts [197], bolts [203], bolts [209], bolts [210], bolts [212], and bolts [222] with penetrating fluid, D50186 [CP2691] at the very beginning of the job.

(b) Do an initial soak of all the case pad bolts listed below at the very beginning of the job.

1) The four bolts that attach the upper CDP duct assembly [25] to the case (VIew B, Figure 428).

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2) The four bolts [32] that attach the transient bleed air tube [31] to the case (View C, Figure 430).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

NOTE: The transient bleed air tube is referenced as the reverse bleed check valve assembly in Post LEAP-1B Service Bulletin 72-0378.

PTF ALL

3) The four bolts [56] and four bolts [58] that attach the 7th-stage air tube [55] to the case (View C and View D, Figure 431).

4) The four bolts [92] that attach the start bleed tube [91] to the case (View B, Figure 433).

5) The eight bolts [102] that attach the 7th stage air tube [101] to the case (View C and View F, Figure 434).

6) The eight bolts [117] that attach the 7th-stage air tube [115] to the engine cases (View C and View E, Figure 436).

7) The two bolts [173] that attach the start bleed bracket [172] to the case (View C, Figure 445).

SUBTASK 73-11-05-000-028-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING MAKE SURE THAT ALL PLUGS OR CAP-COVERS ON OPEN HOLES ON TUBES, DUCTS, MANIFOLDS THAT YOU REMOVE. IF YOU DO NOT FOLLOW THIS WARNING, YOU CAN CAUSE INJURY.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(8) Remove the ECS/CDP ducts (Figure 428).

(a) Remove the Pressure Regulating and Shutoff Valve (PRSOV) [21] (TASK 36-11-04-000-801).

(b) Remove the mix manifold [22] (TASK 36-11-01-000-801).

(c) Remove the High Pressure Shutoff Valve (HPSOV) [24] (TASK 36-11-07-000-801).

(d) Remove the lower CDP duct assembly (TASK 36-11-01-000-801).

(e) Remove the upper CDP duct assembly (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-029-G00

(9) Remove the HPTACC valve [26] (TASK 75-24-04-000-801-G00).

SUBTASK 73-11-05-020-170-G00

(10) Remove the LPTACC valve [27] (TASK 75-24-01-000-801-G00).

SUBTASK 73-11-05-020-171-G00

(11) Remove the wye duct [28], as follows (Figure 429):

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(a) Remove the two bolts [30] and two saddle clamps [29] that attach the wye duct [28] to the High Pressure Turbine (HPT) inlet duct at the 9:30 o'clock position.

(b) Remove the wye duct [28].

SUBTASK 73-11-05-020-172-G00

(12) Remove the transient bleed air tube [31], as follows (View B, Figure 430):

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

NOTE: The transient bleed air tube is referenced as the reverse bleed check valve assembly in Post LEAP-1B Service Bulletin 72-0378.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the two bolts [34] that attach the transient bleed air tube [31] to the support bracket.

(b) Remove the coupling clamp [35] and seal [20] from the transient bleed valve [36].

1) Discard the seal [20].

(c) Remove the four bolts [32] that attach the transient bleed air tube [31] to the combustion case.

(d) Remove the transient bleed air tube [31].

(e) Discard the metal seal ring [33] from the transient bleed air tube [31].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(f) Remove the reverse bleed check valve assembly (TASK 75-34-01-000-801-G00).

PTF ALL

SUBTASK 73-11-05-020-173-G00

(13) Remove the booster anti-ice tube [37], as follows (View E, Figure 430):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the four bolts [42] and two strap clamps [43] that attach the booster anti-ice tube [37] to the support brackets.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [42] and strap clamp [43] that attach the booster anti-ice tube [37] to the support bracket.

PTF ALL

(c) Remove the coupling clamp [46] and seal [45] from the booster anti-ice tube [38].

(d) Remove the coupling clamp [46] and seal [45] from the booster anti-ice tube [39].

(e) Discard the seals [45].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 OR PRE LEAP-1B SERVICE BULLETIN 72-0144

(f) Remove the bolt [47] that attaches the loop clamp [48] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(g) Remove the two bolts [47] that attach the strap clamp [43] to the support bracket.

PTF ALL

(h) Remove the coupling clamp [40] and seal [41] from the booster anti-ice tube [44].

1) Discard the seal [41].

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(i) Remove the booster anti-ice tube [37].

(14) Remove the booster anti-ice tube [44], as follows (View G, Figure 430):

(a) Remove the two bolts [50] and clamp [49] that attach the booster anti-ice tube [44] to the support bracket.

(b) Remove the two bolts [53] and clamp [49] that attach the booster anti-ice tube [44] to the support bracket.

(c) Remove the coupling clamp [51] and seal [52] that attach the booster anti-ice tube [44] to the SBV valve.

1) Discard the seal [52].

(d) Remove the booster anti-ice tube [44] from the engine.

SUBTASK 73-11-05-000-030-G00

(15) Remove the 7th-stage air tube [55], as follows (View B, Figure 431):

(a) Remove the four bolts [56] that attach the 7th-stage air tube [55] to the HPC extension case.

1) Remove and discard the seal ring [57].

(b) Remove the four bolts [58] that attach the bracket [59] and 7th-stage air tube [55] to the HPT case.

1) Remove and discard the seal [60].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0005

(c) Remove the two bolts [62] that attach the 7th-stage air tube [55] to the 7th-stage air tube bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(d) Remove the two bolts [62] that attach the strap clamp [61] and 7th-stage air tube [55] to the 7th-stage air tube bracket.

PTF ALL

SUBTASK 73-11-05-020-174-G00

(16) Disconnect the HJ8A harness [65] (TASK 73-22-01-000-808-G00) (View B, Figure 432).

SUBTASK 73-11-05-020-175-G00

(17) Remove the bracket [73], as follows (View D, Figure 432):

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(a) Loosen the two bolts [71] and two loop clamps [72] that attach the SBV TBV manifold [70] to the bracket [73].

1) Keep the loop clamps [72] installed on the bracket [73] and do not remove them from the engine.

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0071 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217

(b) Remove the bolt [76] and loop clamp [77] that attach the drain tube [75] to the bracket [73].

(c) Remove the two bolts [78] that attach the bracket [73] to the support bracket.

(d) Remove the bolt [85] that attaches the CCC air manifold [74] to the bracket [73].

PTF ALL

(e) Remove the bracket [73] from the engine.

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SUBTASK 73-11-05-020-176-G00

(18) Detach the SBV TBV manifolds [70], as follows (View E, Figure 432):

PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0071 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217

(a) Remove the four bolts [80] and four loop clamps [81] that attach the SBV TBV manifolds [70] to the bracket [82].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(b) Remove the two bolts [83] and four loop clamps [84] that attach the SBV TBV manifolds [70] to the bracket [82].

PTF ALL

(c) Detach the SBV TBV manifolds [70].

SUBTASK 73-11-05-020-177-G00

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(19) Remove the CCC air manifold [74] (TASK 73-11-05-000-801-G00) (View F, Figure 432).

PTF ALL

SUBTASK 73-11-05-020-178-G00

(20) Disconnect the PS3 tube [87] from only one tube end to make it flexible, as follows (View G, Figure 432):

NOTE: It is not necessary to remove any components from the engine.

(a) Loosen the two bolts [89] that attach the two loop clamps [88] and PS3 tube [87] to the support brackets.

(b) Disconnect the PS3 tube [87] from the fitting.

SUBTASK 73-11-05-020-179-G00

(21) Remove the start bleed tube [91], as follows (View B, Figure 433):

(a) Remove the four bolts [92] that attach the start bleed tube [91] to the HPC extension case.

1) Remove and discard the seal [93].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(b) Remove the two bolts [94] that attach the start bleed tube [91] to the start bleed bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(c) Remove the two bolts [98] that attach the start bleed clamp [99] to the start bleed bracket.

PTF ALL

(d) Remove the coupling clamp [96] that attach the start bleed tube [91] to the SBV valve [90].

1) Remove and discard the seal [97].

SUBTASK 73-11-05-020-181-G00

(22) Remove the 7th stage air tube [101], as follows (View B, Figure 434):

(a) Disconnect the air tube [107] from the 7th stage air tube [101].

(b) Remove the four bolts [102] and seal [103] that attach the 7th stage air tube [101] to the compressor case.

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(c) Remove the four bolts [102] and seal [103] that attach the 7th stage air tube [101] to the HPT port.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(d) Remove the four bolts [105] that attach the 7th stage air tube [101] to the support brackets.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(e) Remove the four bolts [106] and two strap clamps [104] that attach the 7th stage air tube [101] to the support brackets.

PTF ALL

(f) Discard the seals [103] from the 7th stage air tube [101].

SUBTASK 73-11-05-020-180-G00

(23) Remove the LPT ACC valve drain tube [108], as follows (View H, Figure 434):

(a) Put a fuel resistant container (10 gal), STD-200, below the ACC valve drain tube that you will remove.

(b) Disconnect the B-nut from the ACC valve drain tube and let it drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [109] that attaches the loop clamp [110] to the drain tube bracket.

SUBTASK 73-11-05-020-182-G00

(24) Disconnect the T3 sensor [111], as follows (TASK 73-21-64-000-801-G00) (Figure 435):

(a) Loosen all the bolts that attach the loop clamps of the T3 sensor [111].

NOTE: There is no need to remove the bolts and loop clamps.

(b) Loosen the B-nut connected to the combustor case threaded insert adapter.

SUBTASK 73-11-05-020-183-G00

(25) Remove the 7th-stage air tube [115], as follows (View B, Figure 436):

(a) Remove the four bolts [117] that attach the 7th-stage air tube [115] to the compressor case.

1) Remove and discard the seal [116].

(b) Remove the four bolts [117] that attach the 7th-stage air tube [115] to the HPT port.

1) Remove and discard the seal [116].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0005

(c) Remove the four bolts [118] that attach the 7th-stage air tube [115] and support bracket [119] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(d) Remove the two bolts [124] that attach the block clamp [125] to the bracket [120].

(e) Remove the two bolts [121] that attach the bracket [120] to the bracket.

(f) Remove the two bolts [122] and strap clamp [123] that attach the 7th-stage air tube [115] to the bracket.

PTF ALL

(g) Remove the 7th-stage air tube [115].

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-184-G00

(26) Remove the HPT ACC valve drain tube [126], as follows (View F, Figure 436):

(a) Put a 10 gallon (38 liter) fuel resistant container, STD-1285, below the HPT ACC valve drain tube [126].

(b) Disconnect the HPT ACC valve drain tube [126] from the drain tube (View K, Figure 436).

1) Let the fuel drain into the 10 gallon (38 liter) fuel resistant container, STD-1285.

(c) Remove the bolt [127] that attaches the loop clamp [128] to the fuel manifold bracket.

(d) Remove the bolt [127] that attaches the loop clamp [128] to the tube bracket.

(e) Disconnect the HPT ACC valve drain tube [126] from the LPT ACC valve tube [129].

(f) Remove the bolt [127] that attaches the loop clamp [128] to the support bracket.

(g) Remove the bolt [127] that attaches the loop clamp [128] to the ACC drain bracket.

(h) Disconnect the HPT ACC valve drain tube [126] from to the ACC valve.

(i) Remove the HPT ACC valve drain tube [126].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-221-G00

(27) Remove the fuel hydraulic drain tube [185] as follows (View F, G , H and I, Figure 436):

(a) Put a 10 gallon (38 liter) fuel resistant container, STD-1285, below the fuel hydraulic drain tube [185].

(b) Disconnect the fuel hydraulic drain tube [185] and B-nut from the drain tube (View I, Figure 436).

1) Let the fuel drain into the 10 gallon (38 liter) fuel resistant container, STD-1285.

(c) Remove the bolt [127] that attaches the loop clamp [128] to the fuel manifold bracket.

(d) Remove the bolt [127] that attaches the loop clamp [128] to the tube bracket [198].

(e) Remove the bolt [127] that attaches the loop clamp [128] to the ACC drain bracket.

(f) Disconnect the B-nut [214] that attaches the HPT ACC valve drain tube [126] to the fuel hydraulic drain tube [185].

NOTE: It is not necessary to remove the HPT ACC valve drain tube from the engine.

PTF ALL

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SUBTASK 73-11-05-000-032-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(28) Loosen the fuel manifolds B-nuts as follows and do these steps to drain the three fuel manifold sections as follows:

NOTE: There are no fuel manifold drain plugs on the engine. The three fuel manifold sections are drained by decreasing the fuel pressure and draining each fuel manifold tube. Each fuel manifold tube is independent and must be drained at its lowest connection point on each manifold section. Each manifold section has two separate manifold tubes.

NOTE: Both fuel manifold section tubes that supply the fuel nozzle to be removed must be drained before the fuel nozzle removal. The fuel manifolds can have a small amount of pressure remaining after the fuel manifold tube is drained.

(a) Remove the thrust reverser compression rod No. 2 (Figure 424).

(b) Put fuel resistant container (10 gal), STD-200, below the fuel manifold drain point of the fuel nozzle [1] you will remove.

(c) Remove all the safety wire from the B-nuts [3] (at 18 fuel nozzles positions).

(d) Loosen the fuel manifold B-nuts [3] at the lower 6 o'clock position.

1) Fully remove the B-nuts [3] from the fuel nozzle [1].

(e) Loosen the fuel manifold B-nuts [3] at the upper 12 o'clock position.

1) Fully remove the B-nuts [3] from the fuel nozzle [1].

(f) Drain all the fuel into fuel resistant container (10 gal), STD-200.

(g) Loosen the remaining B-nuts [3] from the engine.

SUBTASK 73-11-05-840-001-G00

(29) Make sure that you cover all ports and openings.

(a) If you do not obey, Foreign Object Debris (FOD) can go into the engine and damage can occur.

G. Fuel Nozzle Removal

SUBTASK 73-11-05-020-185-G00

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(1) Remove the fuel nozzle bolts as follows:

(a) Carefully shock the fuel nozzle bolts heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

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(b) Break torque on the fuel nozzle bolts.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION

DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

(c) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts again.

1) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts.

(d) Start the removal of the fuel nozzle bolts.

1) Use a ratchet, STD-6847, to turn the bolts slowly.

(e) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

1) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts slowly.

NOTE: If you turn the fuel nozzle bolts too fast, they can get hot and seize.

2) Turn the machine bolts counterclockwise.

3) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt hole.

4) Loosen the bolts again.

5) If the bolts cannot be removed, repeat the steps above.

(f) Remove all 72 fuel nozzles bolts.

(g) Discard the fuel nozzle bolts.

SUBTASK 73-11-05-720-001-G00

(2) If it is necessary to remove any support bracket that is attach to the fuel nozzle, use temporary marker, G50530 [CP5061], to do a temporarily position mark to the associated bracket to the associated fuel nozzle [1] position, before you start removing brackets and fuel nozzles.

SUBTASK 73-11-05-020-186-G00

(3) Remove the fuel nozzles bolts that attach the fuel nozzle [1] to the combustion case.

(a) Make sure that the fuel nozzles bolts were removed before the fuel nozzles and support brackets.

NOTE: It is recommended to keep the brackets assembled on top of the fuel nozzles until the fuel nozzles are removed from the engine and to mark each fuel nozzle position/bracket, to make the installation easy.

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SUBTASK 73-11-05-020-187-G00

(4) Loosen the two bolts [11] and two loop clamps [12] that attach the fuel manifold [13] to the support bracket [14] at the fuel nozzle [1] (positions 17, 18, 3, 6, 7, 10, 11, and 14) (Figure 426).

(a) Do not remove the hardware from the engine.

SUBTASK 73-11-05-000-031-G00

(5) Remove or disconnect the support brackets attached to the fuel nozzle [1] and combustion case on QUADRANT 1, as follows (Figure 425):

(a) If necessary, loosen the two bolts [132] that attach the support bracket [131] to the support bracket [133] at the fuel nozzle [1] (position 17) (View B, Figure 437).

(b) Move the support bracket [133] away (position 17, and 16).

NOTE: The support bracket [133] and support bracket [131] are two brackets assembled together that span across fuel nozzle [1] (position 17) and fuel nozzle [1] (position 16).

1) Move the support bracket [134] away at fuel nozzle [1] (position 17).

(c) Move the support bracket [137] away at the fuel nozzle [1] (position 18) (View B, Figure 438).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [217] that attach the support bracket [218] to the support bracket [219] (View C, Figure 438).

(e) Move the support bracket [219] away from the engine at the fuel nozzle [1] (position 18) (View C, Figure 438).

PTF ALL

(f) Remove the bracket assembly [139] from the engine at the fuel nozzle [1] (position 2 and 3) (View B, Figure 440, View B, Figure 441).

NOTE: The bracket assembly [139] is bracket that span across the fuel nozzle [1] (position 2) and fuel nozzle [1] (position 3).

(g) Move the support bracket [143] away at the fuel nozzle [1] (position 3) (View B, Figure 441).

SUBTASK 73-11-05-020-188-G00

(6) Remove or disconnect the support brackets attached to fuel nozzle [1] and combustion case on QUADRANT, 2 as follows (Figure 425):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(a) Remove the bracket [147], bracket [145] and bracket [144] from the engine core at the fuel nozzle [1] (position 5 and 6) (View B, Figure 443).

NOTE: The support the bracket [147] and bracket [145] are two brackets assembled together that span across the fuel nozzle [1] (position 5) and fuel nozzle [1] (position 6).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(b) Remove the bracket [149], and bracket [151] from the engine core at the fuel nozzle [1] (position 5 and 6).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217 (Continued)

1) Keep the bracket [149] and bracket [151] assembled with the bolts [148] (View B, Figure 443).

NOTE: The bracket [149] and bracket [151] are two brackets assembled together that span across the fuel nozzle [1] (position 5) and fuel nozzle [1] (position 6).

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0135 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217

(c) Remove the bracket [152] from the engine core at the fuel nozzle [1] (View B, Figure 444).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(d) Remove the support bracket [156] from the engine core at the fuel nozzle [1] (View B, Figure 444).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(e) Loosen the bolt [162] that attaches the support bracket [161] to the bracket at the combustion case (View B, Figure 445).

1) Move the support bracket [161] away at the fuel nozzle [1] (position 7).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(f) Loosen the two bolts [168] and two bolts [169] that attach the SBV and TBV manifold [170] to the bracket [164] with the four p-clamps [171] at the fuel nozzle [1] (position 7) (View D, Figure 445).

1) Do not remove the hardware from the engine core.

(g) Loosen the bolt [166] that attaches the PS3 air tube [167] to the bracket [164] with the loop clamps [165].

1) Do not remove the hardware from the engine core.

(h) Loosen the two bolts [163] that attach the bracket [164] to the support bracket [159].

1) Do not remove the hardware from the engine core.

PTF ALL

(i) Remove the support bracket [159] from the engine core at the fuel nozzle [1] (position 7) (View B, Figure 445).

(j) Remove the two bolts [173] and start bleed bracket [172] attached to the combustor case at the fuel nozzle [1] (position 7) (View C, Figure 445).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(k) Loosen the bolt [177], bolt [180], bolt [182], and bolt [183] that attach the oil tubes to the bracket [176] with the p-clamp [178], p-clamp [179], p-clamp [181], and p-clamp [184] at the fuel nozzle [1] (position 9) (View C, Figure 447).

1) Do not remove the hardware from the engine.

2) Move the bracket [176] away at the fuel nozzle [1] (position 9).

PTF ALL

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SUBTASK 73-11-05-020-189-G00

(7) Remove or disconnect the support brackets attached to the fuel nozzle [1] and combustion case on QUADRANT 3, as follows (Figure 425):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(a) Loosen the two bolts [193] with the two loop clamps [191] that attach the SBV TBV manifolds [192] to the support bracket [186] at the fuel nozzle [1] (position 10) (View C, Figure 448).

1) Do not remove the hardware from the engine core.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(b) Loosen the bolt [193] that attaches the SBV TBV manifolds [192] to the support bracket [186] with loop clamp [191] at the fuel nozzle [1] (position 10) (View C, Figure 448).

1) Do not remove the hardware from the engine core.

PTF ALL

(c) Loosen the bolt [189] that attaches the tube to the support bracket [186] with the loop clamp [190] at the fuel nozzle [1] (position 10) (View B, Figure 448).

1) Move the bundle of support bracket [186] away for clearance.

2) Move the support bracket [188] away at the fuel nozzle [1] (position 10).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(d) Remove the four bolts [206] with the four hinged loop clamps [205] that attach the SBV TBV manifolds [207] to the tube bracket [198] at the fuel nozzle [1] (position 11) (View C, Figure 449).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the two bolts [206] with the fuel hydraulic support block [194] that attach the SBV TBV manifolds [207] to the tube bracket [198] at the fuel nozzle [1] (position 11) (View C, Figure 449).

PTF ALL

(f) Remove the bolt [127] with the loop clamp [128] that attaches the HPT ACC valve drain tube [126] to the tube bracket [198] at the fuel nozzle [1] (position 11) (View D, Figure 449).

(g) Remove the tube bracket [198] from the engine core.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

1) Keep the tube bracket [198] assembled to the tube bracket [202] with the bolt [204].

PTF ALL

(h) Move the support bracket [196] away at the fuel nozzle [1] (position 11) (View B, Figure 449).

(i) Move the drain tube bracket [208] away at the fuel nozzle [1] (position 12) (View B, Figure 450).

NOTE: The drain tube bracket [208] spans across the fuel nozzle [1] (position 12) and fuel nozzle [1] (position 13).

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SUBTASK 73-11-05-020-190-G00

(8) Remove or disconnect the support brackets attached to the fuel nozzle [1] and combustion case on QUADRANT 4, as follows (Figure 425):

(a) Move the support bracket [211] away at the fuel nozzle [1] (position 14) (View B, Figure 452).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(b) Loosen the two bolts [216] with the two loop clamps [215] that attach the fuel manifold to the support bracket [213] at the fuel nozzle [1] (position 14) (View C, Figure 452).

1) Move the support bracket [213] away at the fuel nozzle [1] (position 14).

PTF ALL

(c) Remove the two bolts [224] and two loop clamps [225] that attach the LPT ACC fuel manifold to the support bracket [221] and support bracket [223] at the fuel nozzle [1] (position 15) (Figure 453).

(d) Remove the bundle of the support bracket [221], support bracket [223], and support bracket [220] from the engine core at the fuel nozzle [1] (position 15) (View B, Figure 453).

(e) Remove the three bolts [231], bolts [232], three loop clamps [230], and loop clamp [233] that attach the TCF accelerometer cable, EGT harnesses, and T3 sensor at the fuel nozzle [1] (position 16) (View C, Figure 454).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(f) Remove the two bolts [226] that attach the loop clamp [227], W200 harness [228], and W300 harness [229] to the support bracket [131] at the fuel nozzle [1] (position 16) (View B and C, Figure 454).

PTF ALL

(g) Remove the four bolts [130] that attach the support bracket [131] and fuel nozzle [1] (position 16) to the case (View B, Figure 454).

1) Move the support bracket [131] away at the fuel nozzle [1] (position 16).

SUBTASK 73-11-05-000-033-G00

(9) Remove the fuel nozzle [1] (positions 1 through 18) as follows:

(a) To remove the multiple fuel nozzles or the full set of fuel nozzles, do these steps:

NOTE: If you will remove multiple fuel nozzles or the full set of fuel nozzles at the same time, it is preferential to remove the nozzles in sets (Figure 425).

NOTE: The fuel nozzle removal sequence is for reference only and may be changed as necessary.

1) If it is necessary, loosen and partially remove an adjacent fuel nozzle in order to provide access to rotate the fuel nozzle out of the combustion case during fuel nozzle removal.

a) If you loosen three or more fuel nozzle bolts, replace the fuel nozzle seal assembly.

2) Support the fuel nozzles [1] (positions 1 through 18) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle from the combustor radial mixer (cup).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Lift the fuel nozzles [1] (positions 1 through 18) and move the mounting flange aft and down.

4) Turn the fuel nozzles [1] (positions 1 through 18) to remove the fuel nozzles [1] from the combustion case as follows:

a) Fuel Nozzle (FN) at positions 1, 2, 3, 18, and 17 (removed at the 1:30 o'clock position)

NOTE: FN 2 and FN 3 need to be turned right to get FN 1 removed.

<1> The FN 1 turn to the RIGHT and out of the case.

<2> The FN 2 turn to the LEFT and out of the case.

<3> The FN 3 turn to the LEFT and out of the case.

<4> The FN 18 turn to the RIGHT and out of the case.

<5> The FN 17 turn to the RIGHT and out of the case.

b) FN 4, 5, 6, 7, 8, and 9 (removed at the 3 to 5 o'clock position)

NOTE: FN 5 and FN 6 need to be turned right to get FN 4 removed.

<1> The FN 4 turn DOWN and out of the case.

<2> The FN 5 turn UP and out of the case.

<3> The FN 6 turn UP and out of the case.

<4> The FN 7 turn UP and out of the case.

<5> The FN 8 turn to the RIGHT and out of the case.

<6> The FN 9 turn to the RIGHT and out of the case.

c) The FN 11, 10, and 12 (removed at the 7 to 8 o'clock position)

NOTE: FN 12 needs to be turned up to get FN 11 removed.

<1> The FN 11 turn UP and out of the case.

<2> The FN 10 turn LEFT and out of the case.

<3> The FN 12 turn DOWN and out of the case.

d) FN 16, 15, 14, and 13 (removed at the 9 o'clock position).

NOTE: FN 15 and FN 14 need to be turned down to get FN 16 removed.

<1> The FN 16 turn DOWN and out of the case.

<2> The FN 15 turn UP and out of the case.

<3> The FN 13 turn UP and out of the case.

(b) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(c) Remove and discard the seals [4] from the fuel nozzles [1] (positions 1 through 18) or the combustion case.

(d) Install protective covers on the fuel nozzle, the fuel nozzle ports, and fuel manifold connections.

(e) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case.

(f) Attach DO NOT OPERATE tags, STD-858, to the fuel nozzles [1].

(g) Put caps on the fuel nozzle inlet fittings and aft heat shield.

(h) Discard all 18 removed fuel nozzle seals [4].

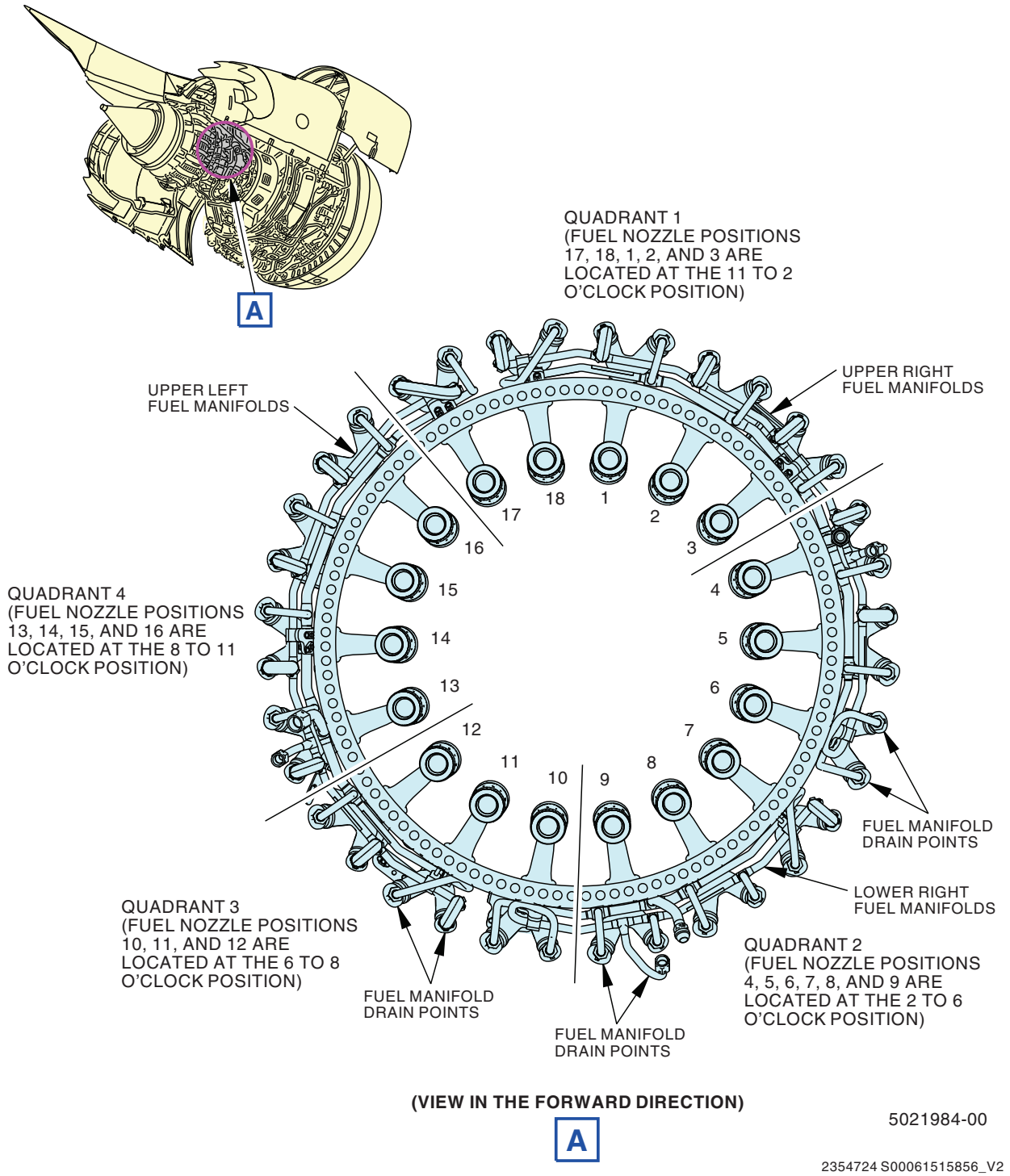
(i) Record serial number and respective cup position of the removed fuel nozzles [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Record engine TSN and CSN.

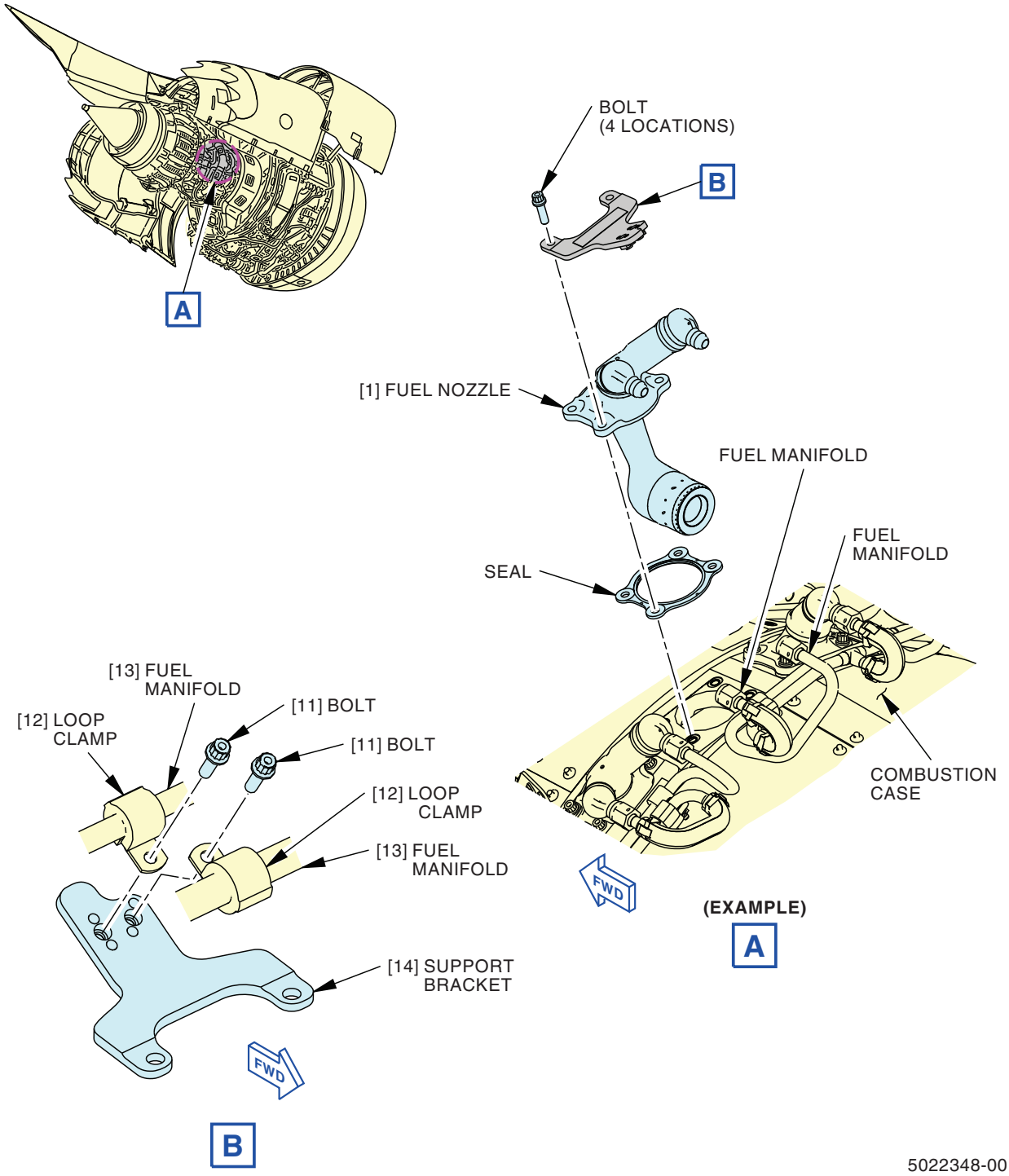
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



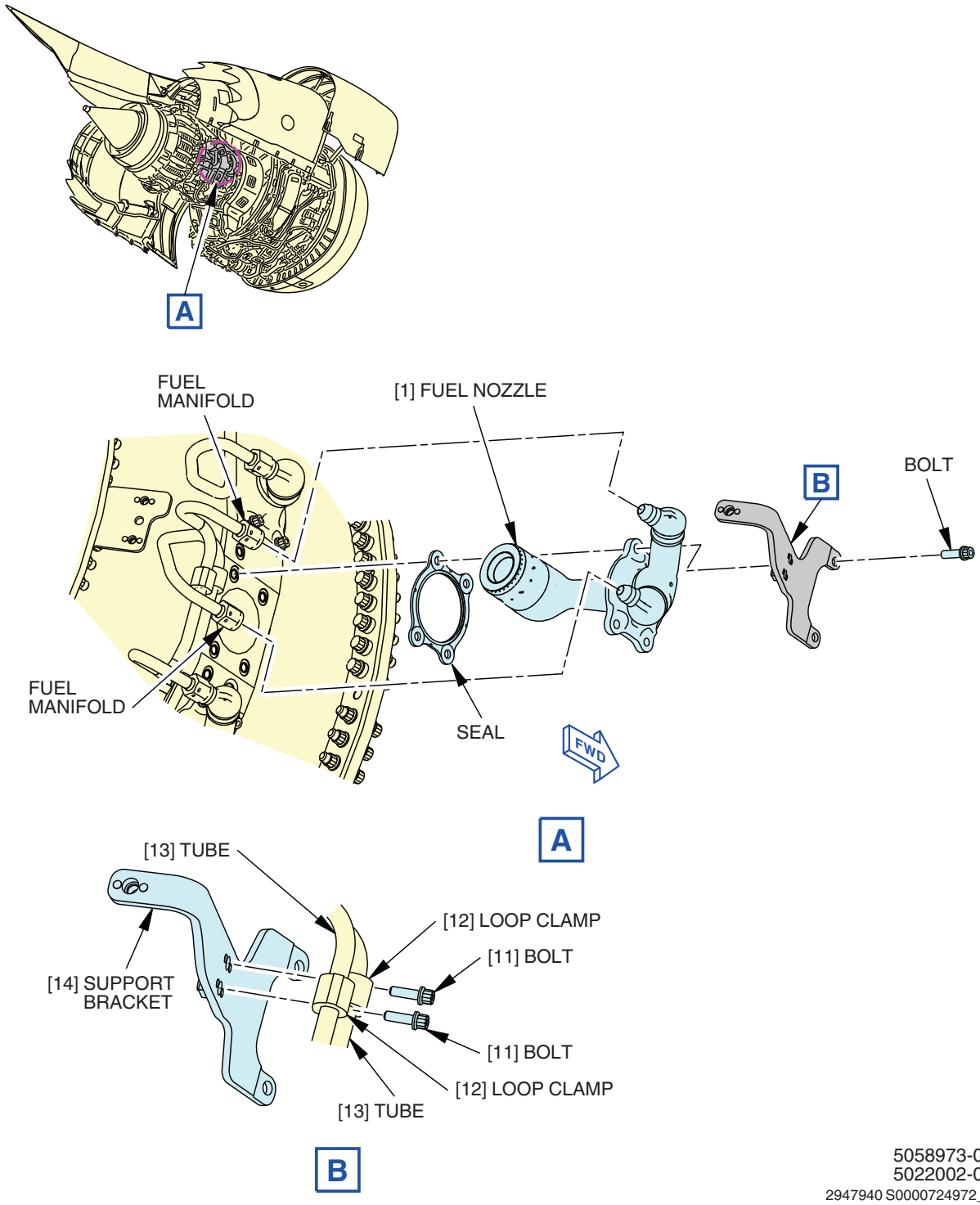
Fuel Nozzle Positions and Fuel Manifold Drain Points
Figure 425/73-11-05-990-855-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loosen p-clamps on fuel nozzles (positions 17, 18, 3, 6, 10, and 14) Installation
Figure 426/73-11-05-990-856-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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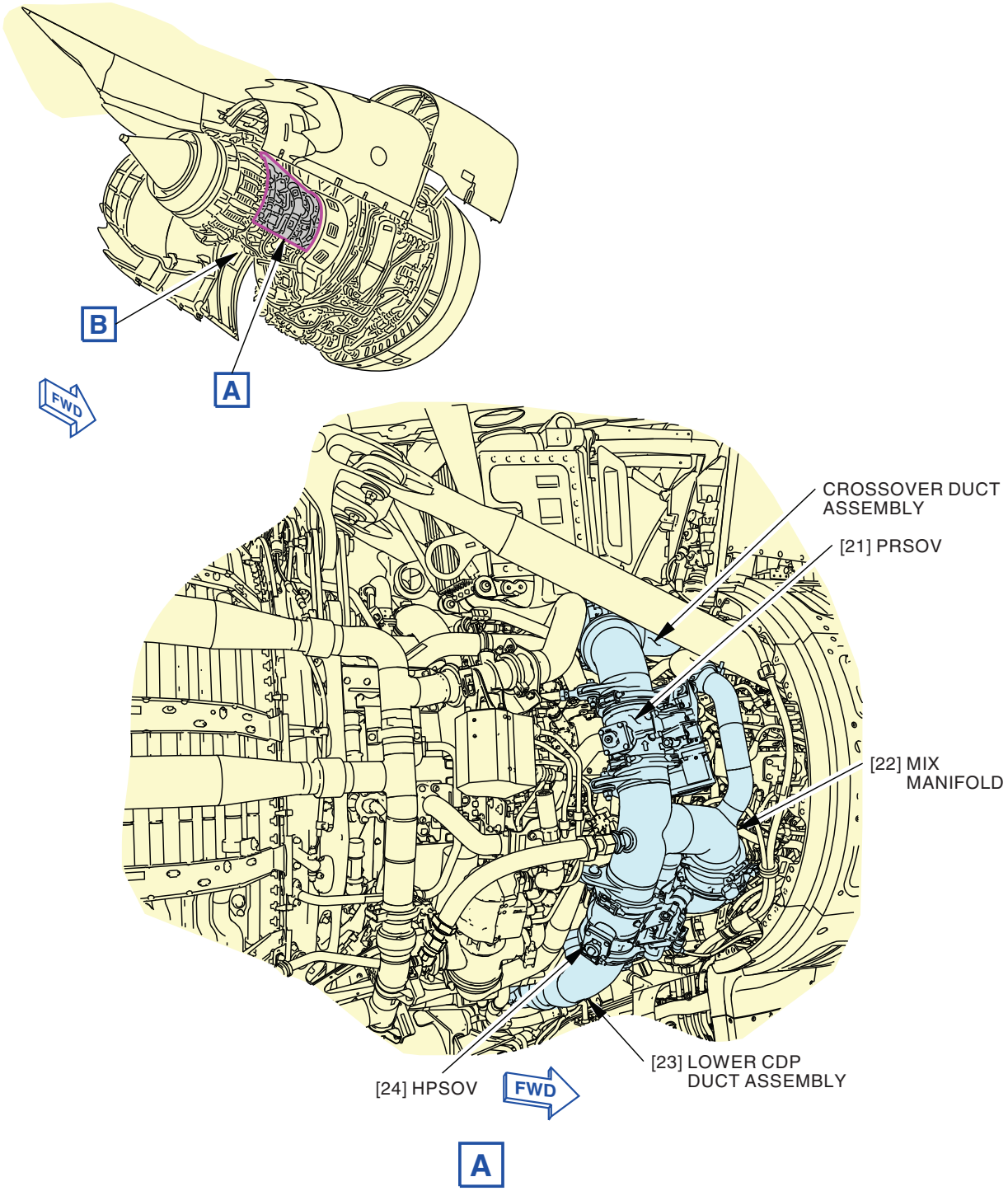
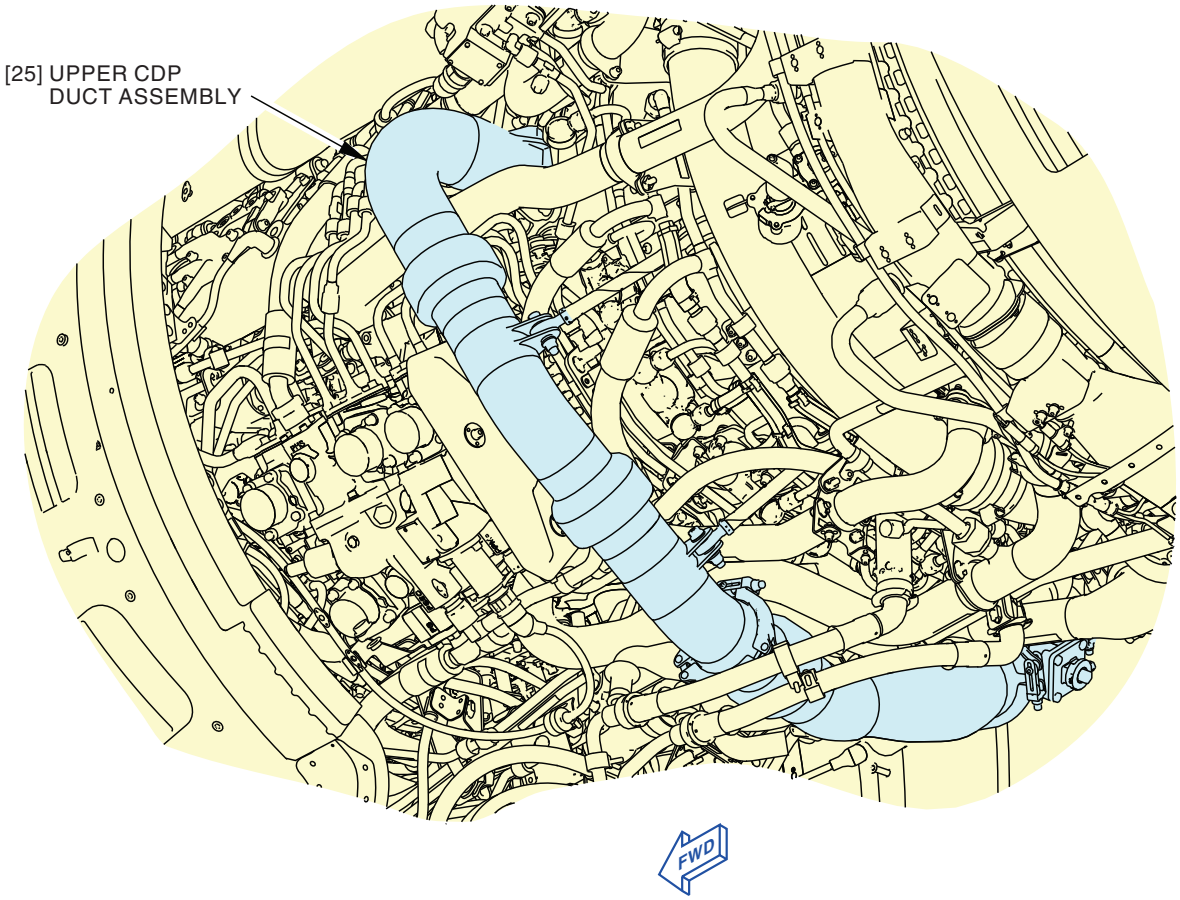
Loosen p-clamps on fuel nozzles (positions 7, and 11) Installation
Figure 427/73-11-05-990-857-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Build Up (EBU) Hardware Installation
Figure 428/73-11-05-990-858-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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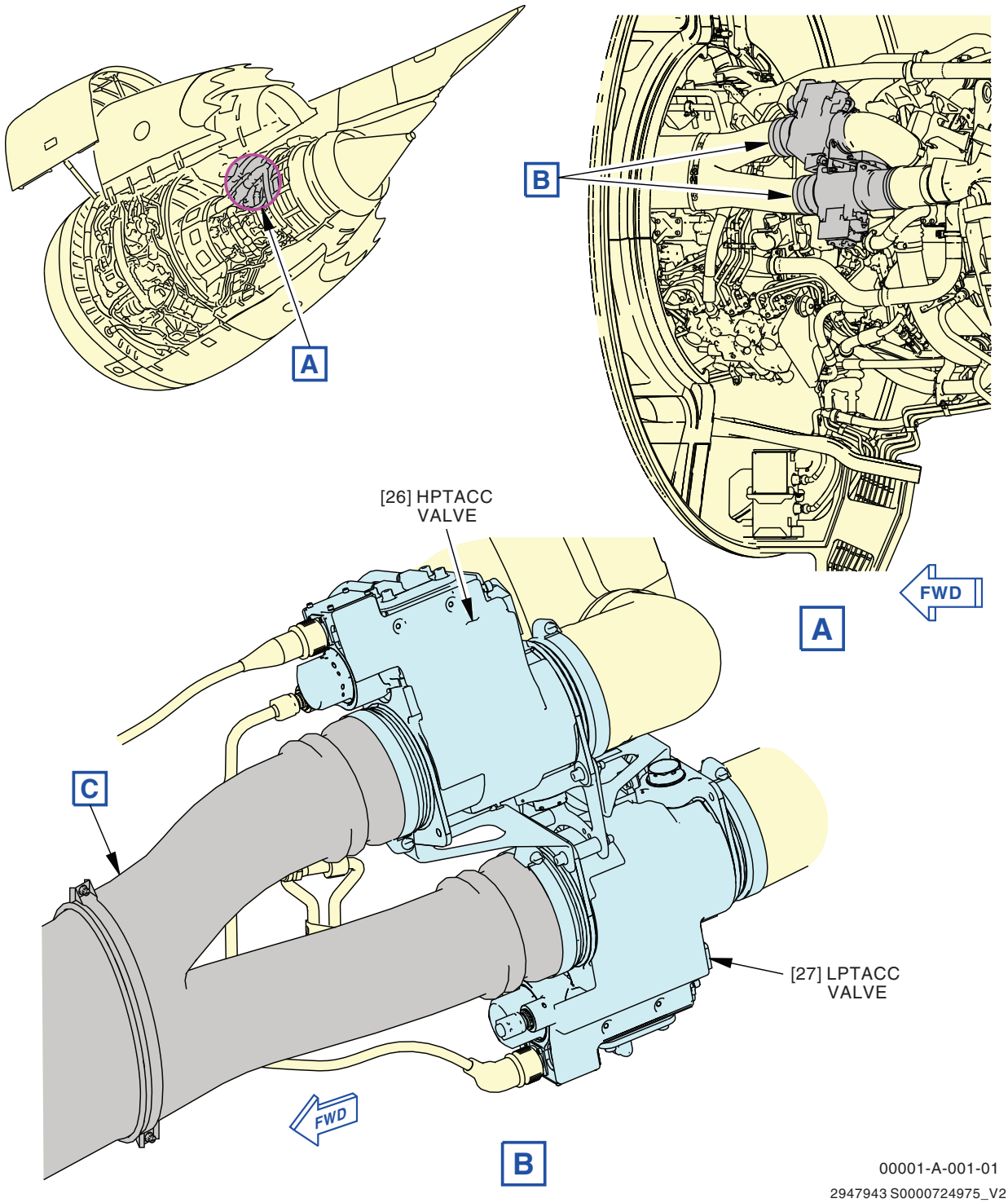
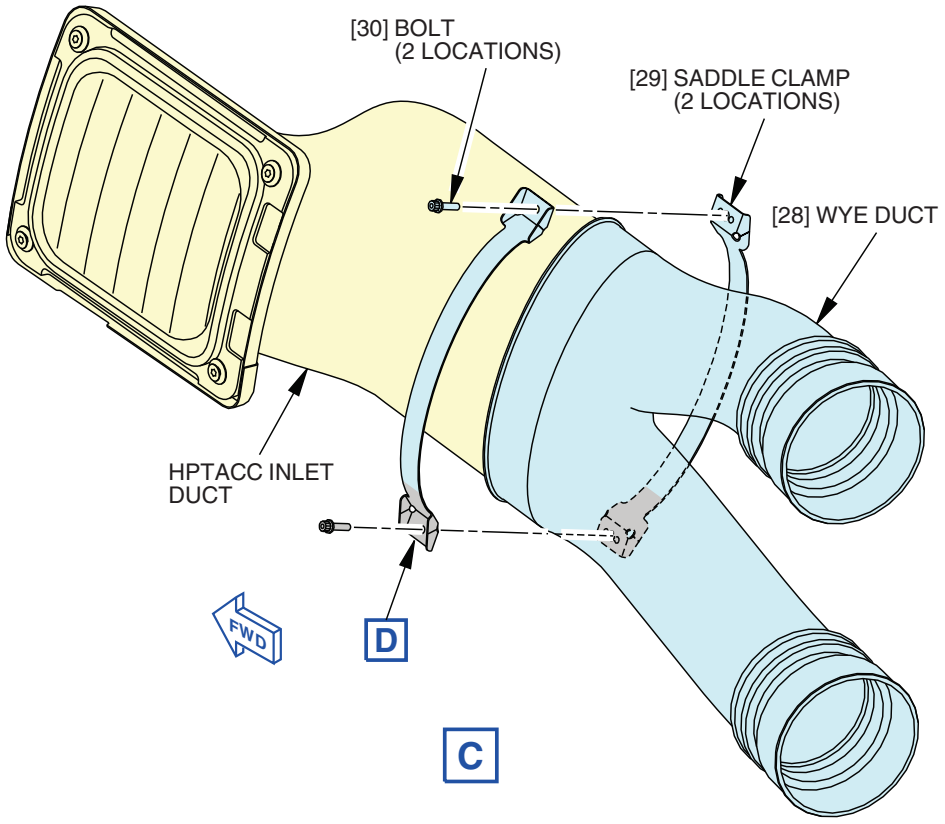
Engine Build Up (EBU) Hardware Installation
Figure 428/73-11-05-990-858-G00 (Sheet 2 of 2)

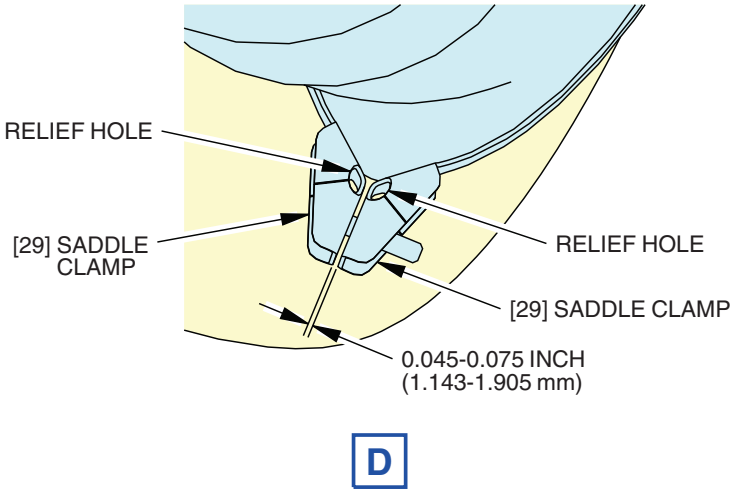
ECCN 9E991 BOEING PROPRIETARY - See title page for details



HPT and LPT Valve Installation
Figure 429/73-11-05-990-859-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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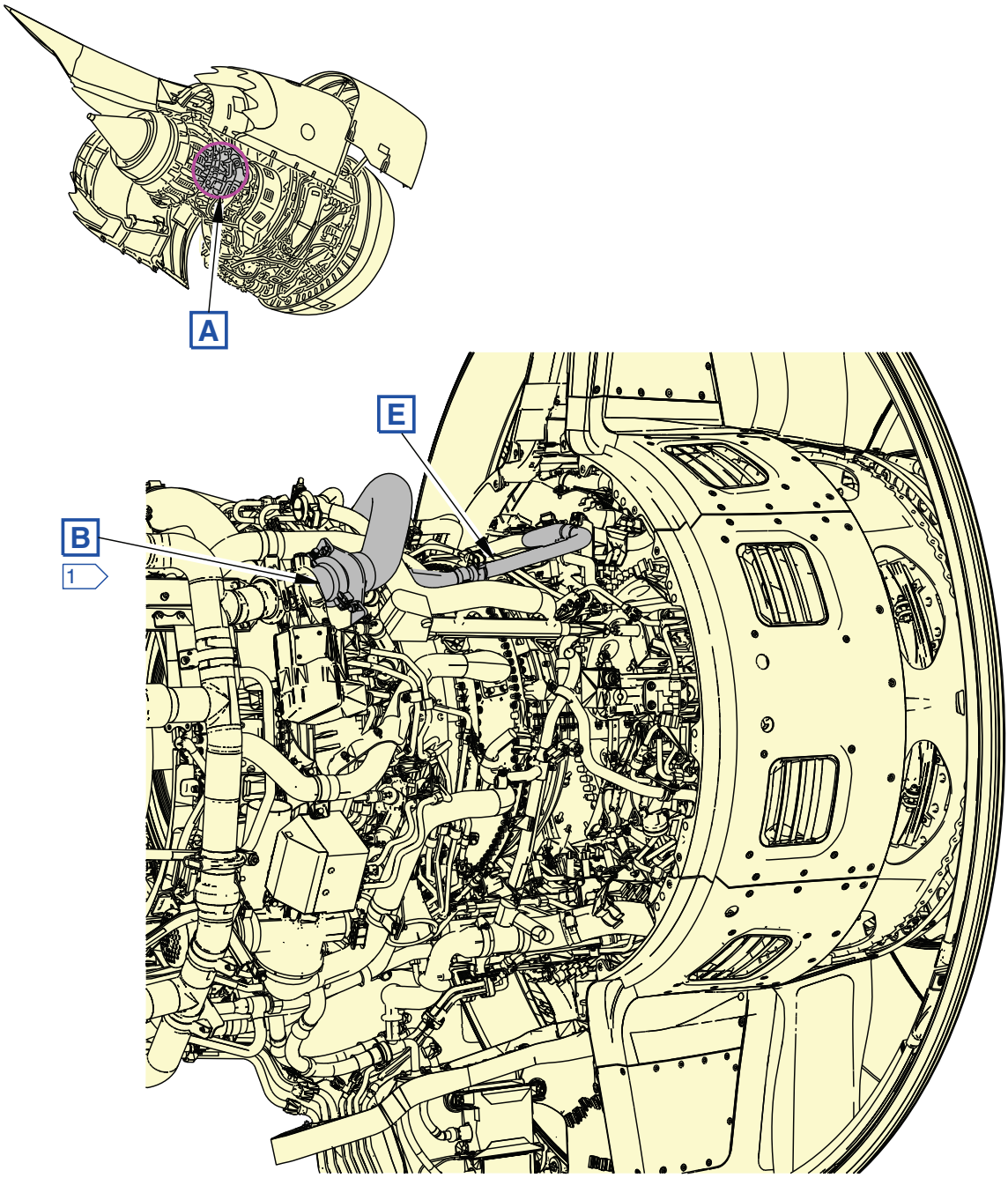
HPT and LPT Valve Installation
Figure 429/73-11-05-990-859-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PRE LEAP-1B SERVICE BULLETIN 72-0378

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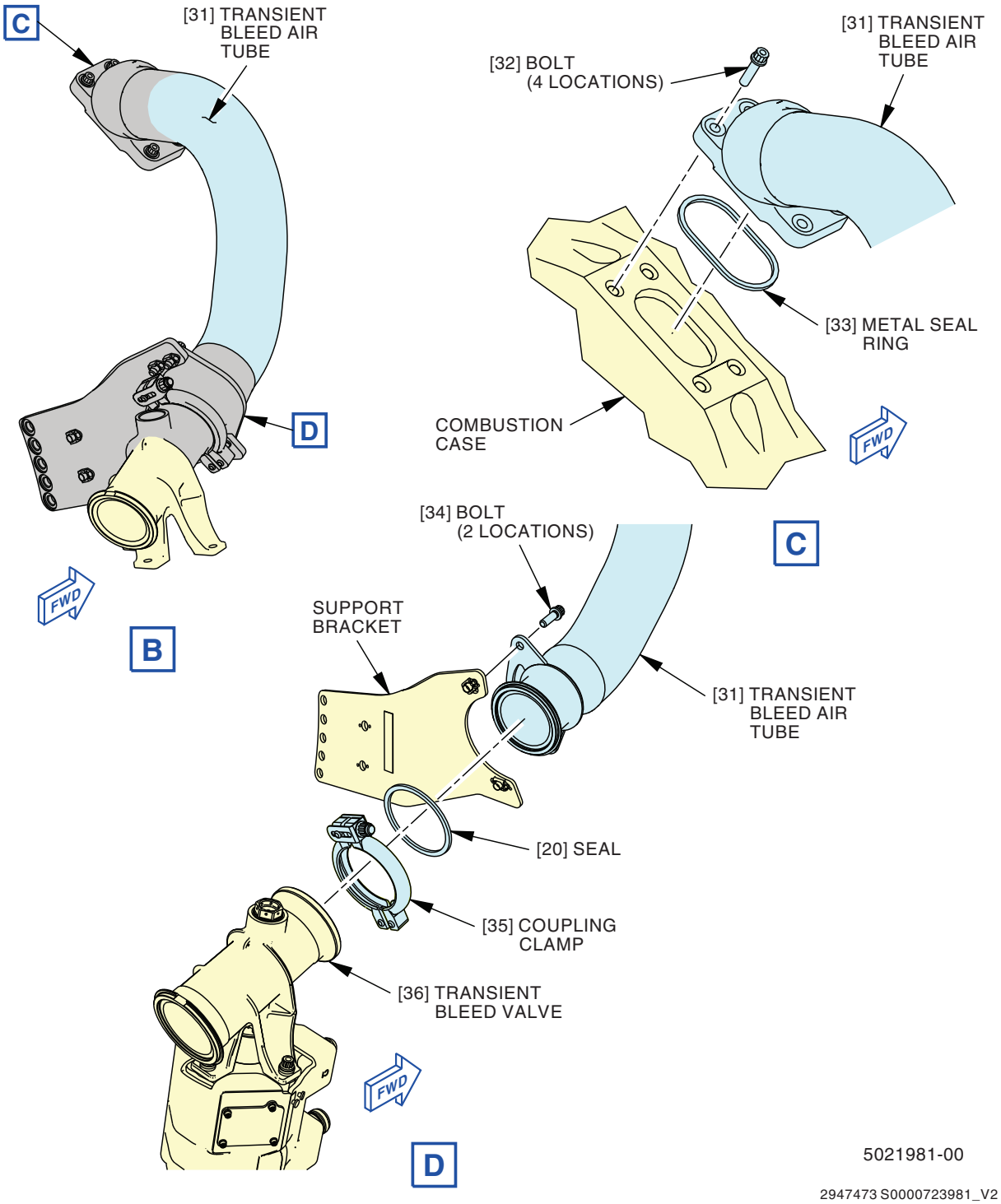
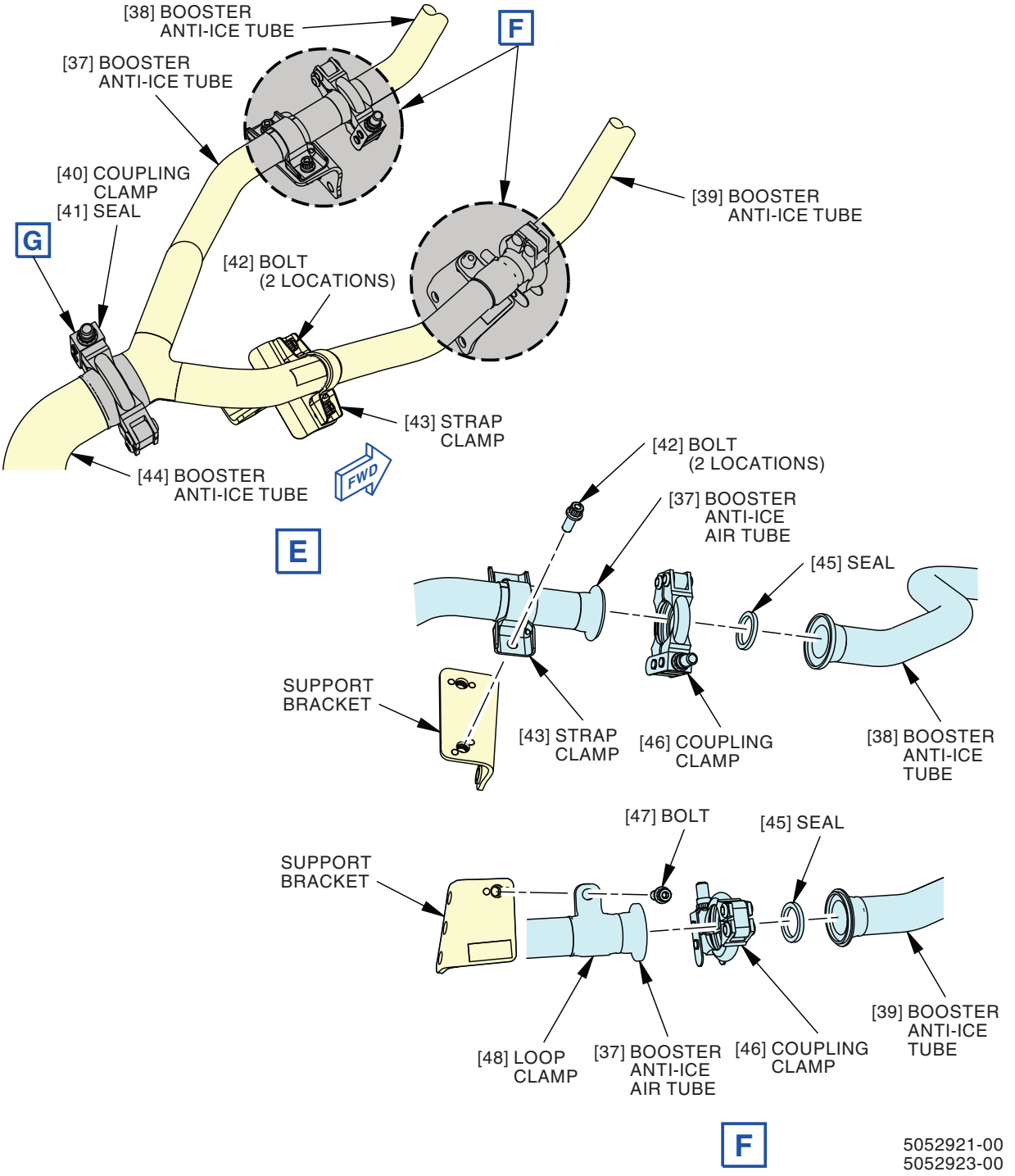
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



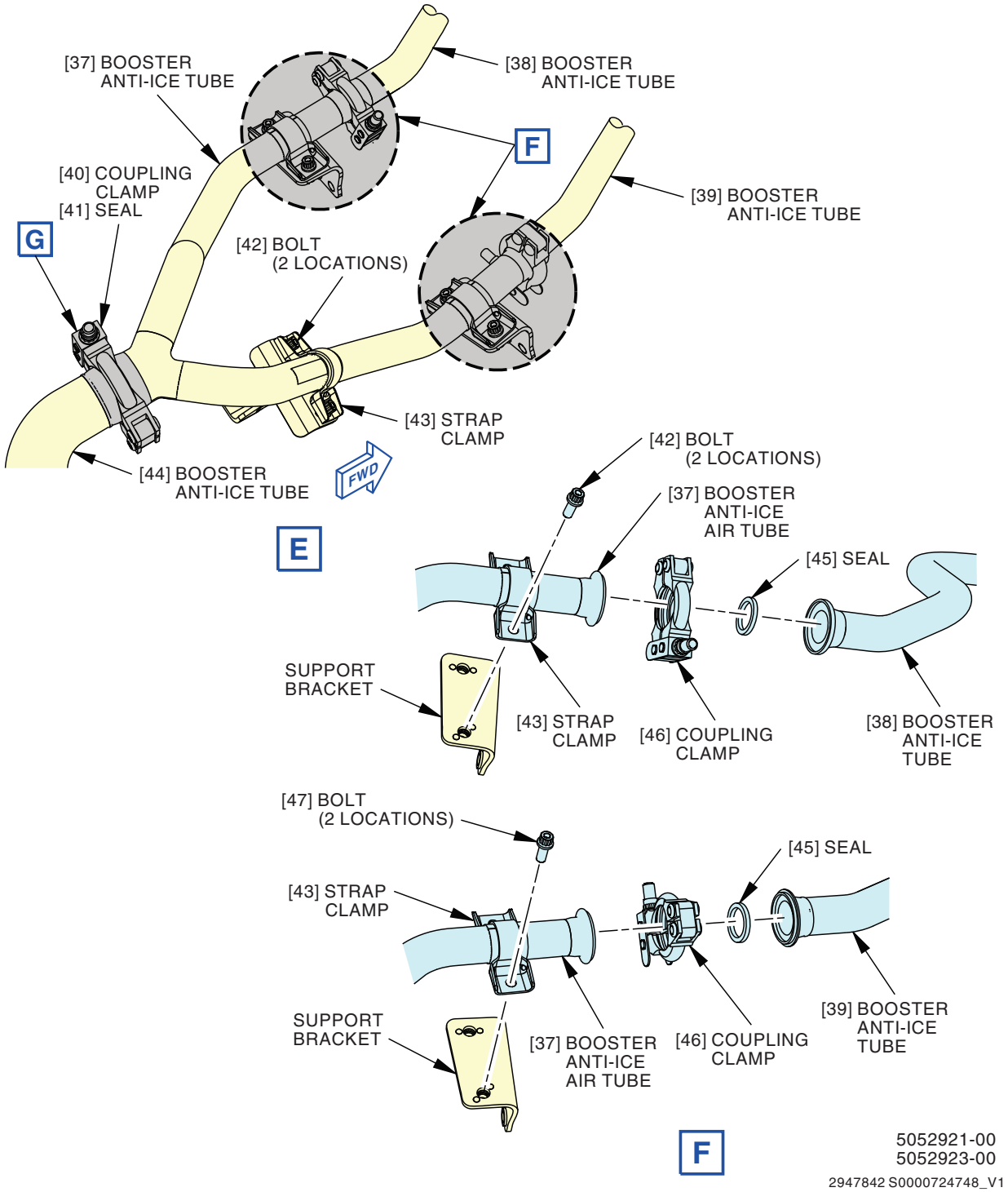
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



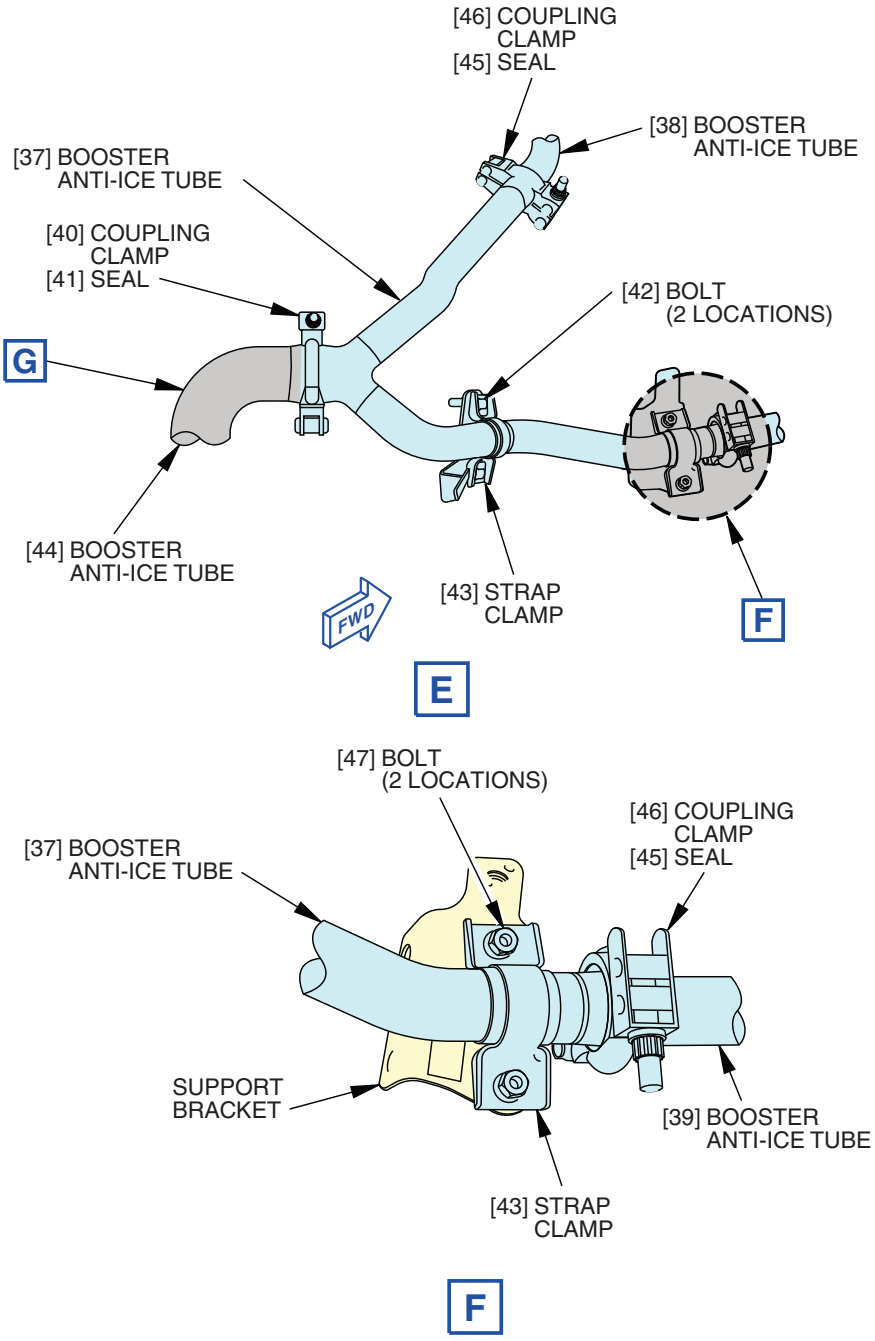
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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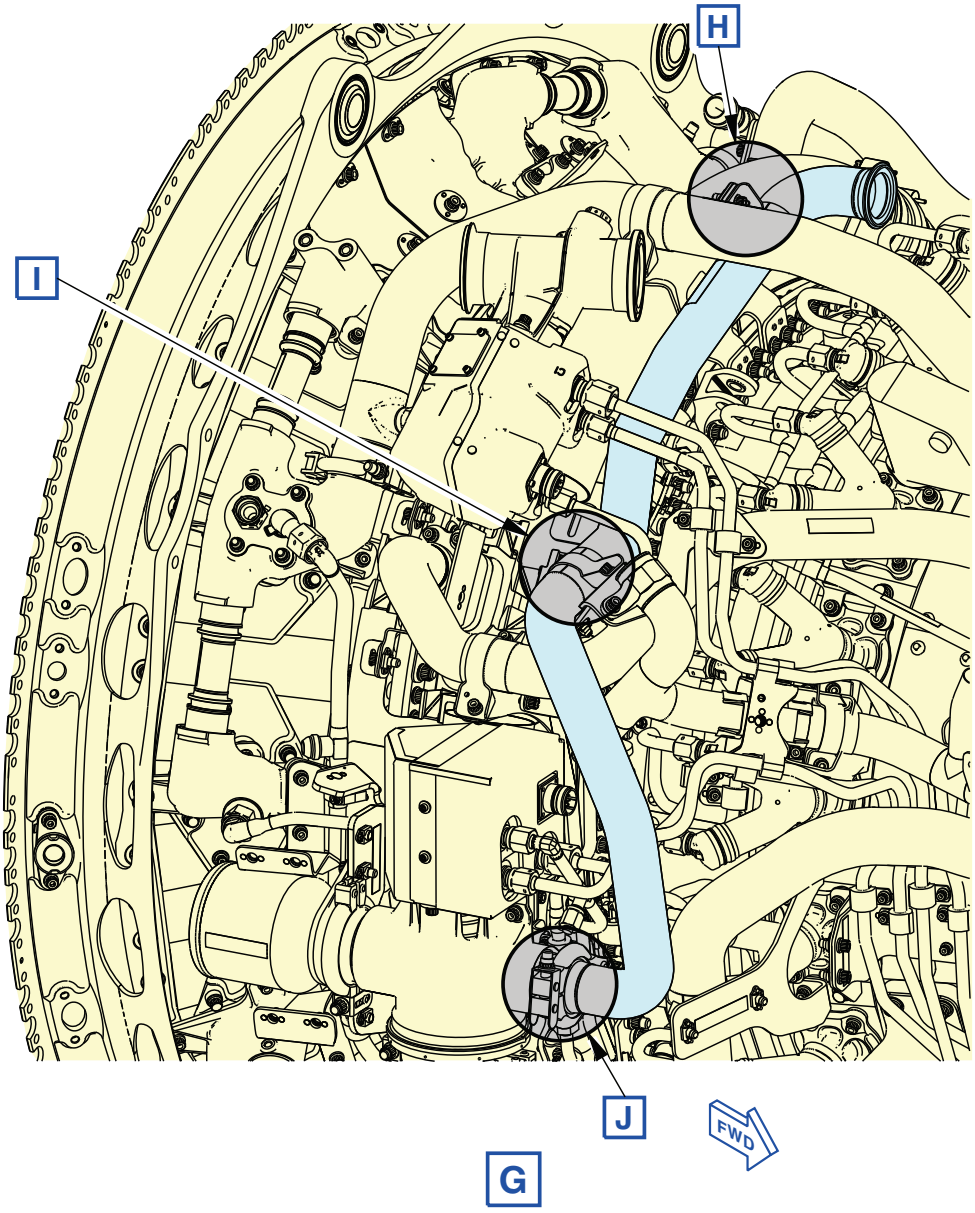
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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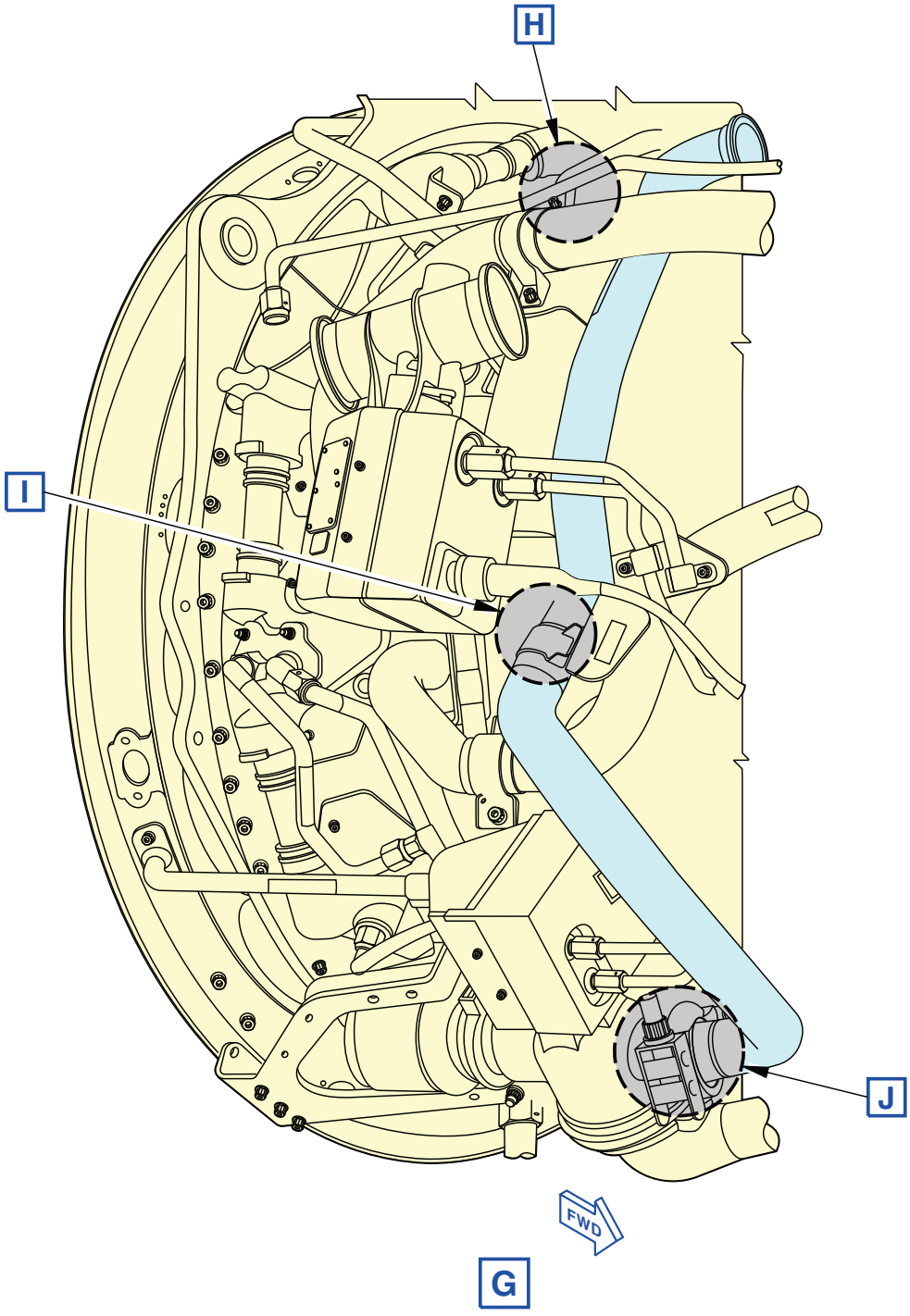
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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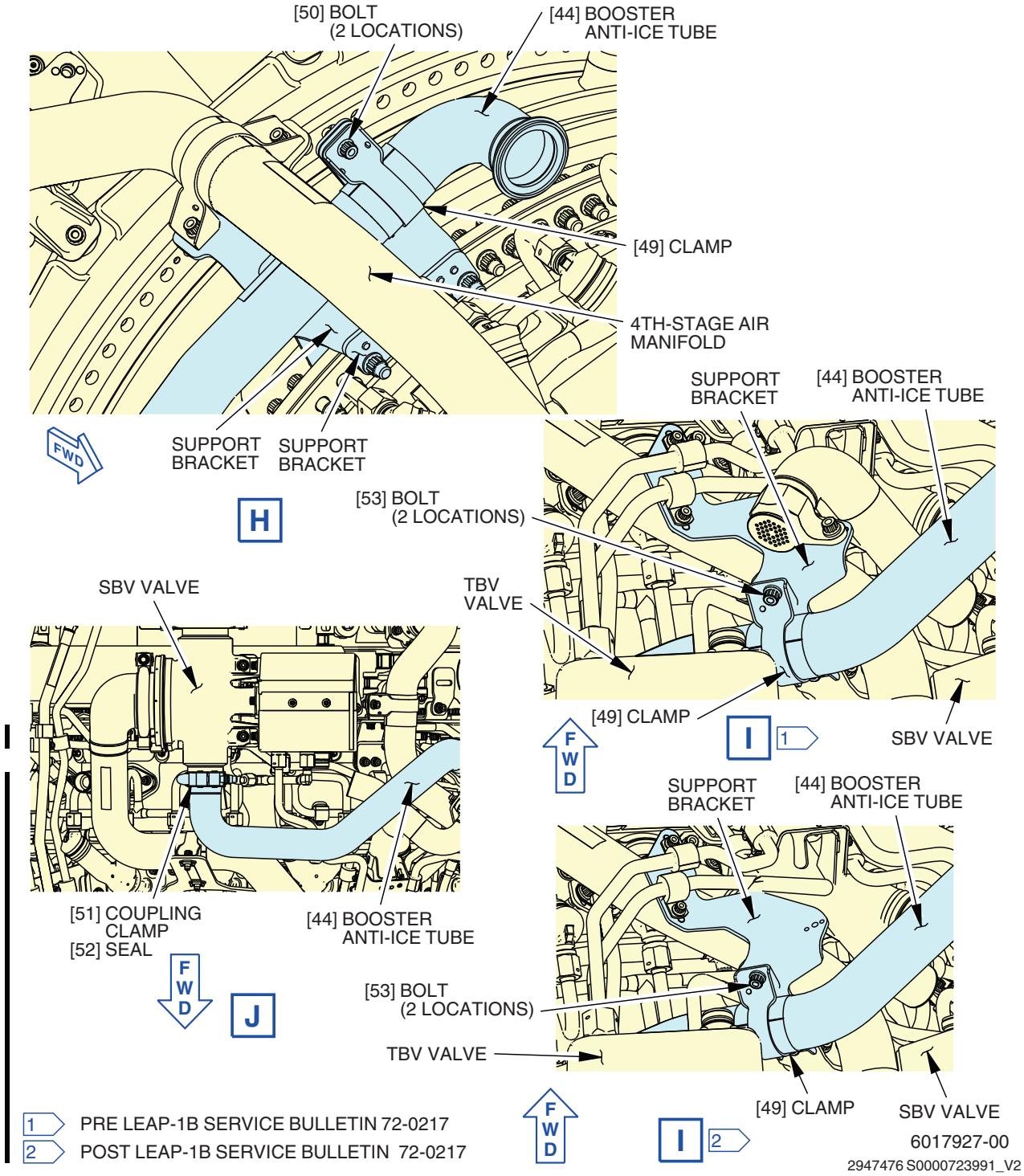
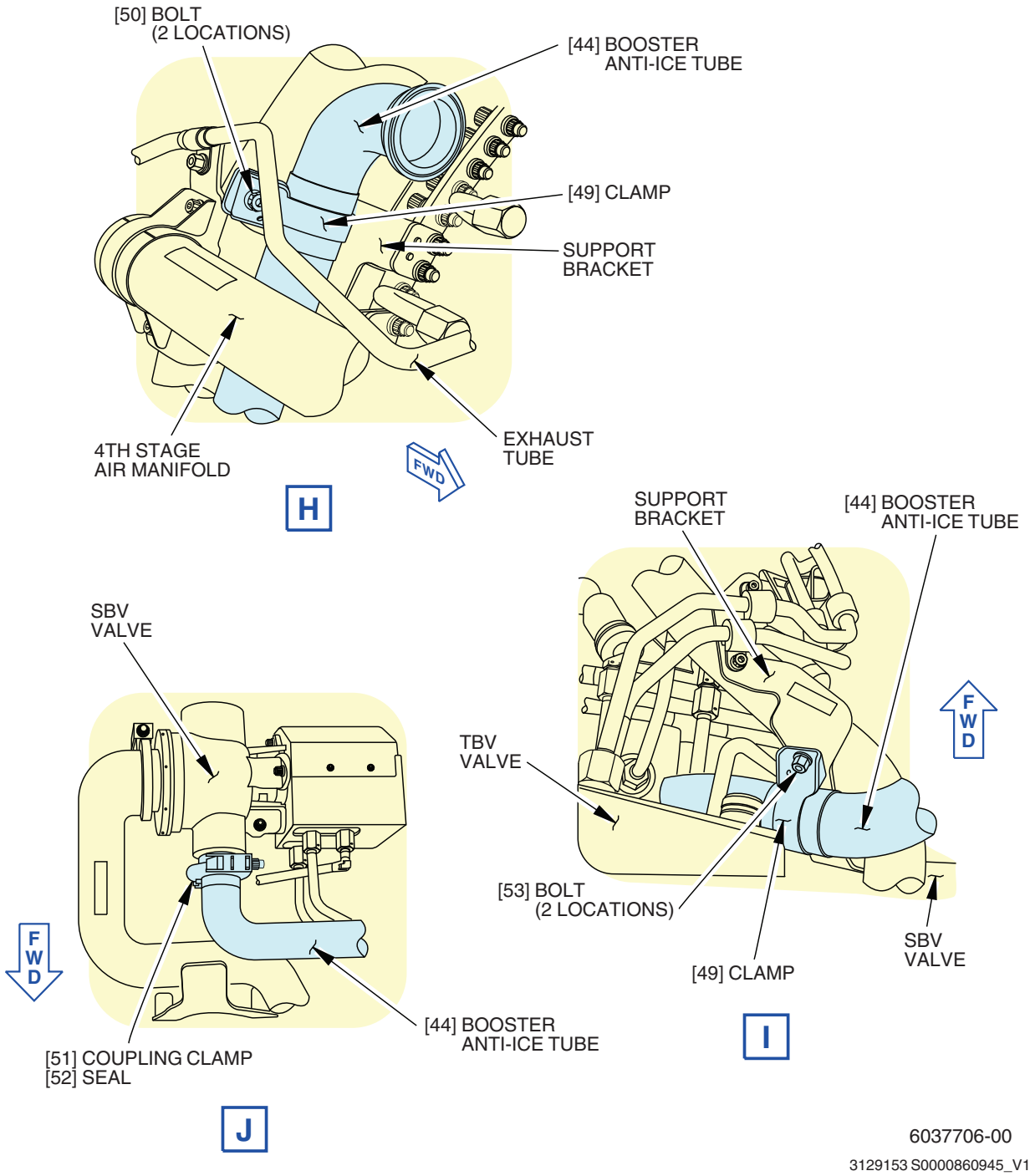
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



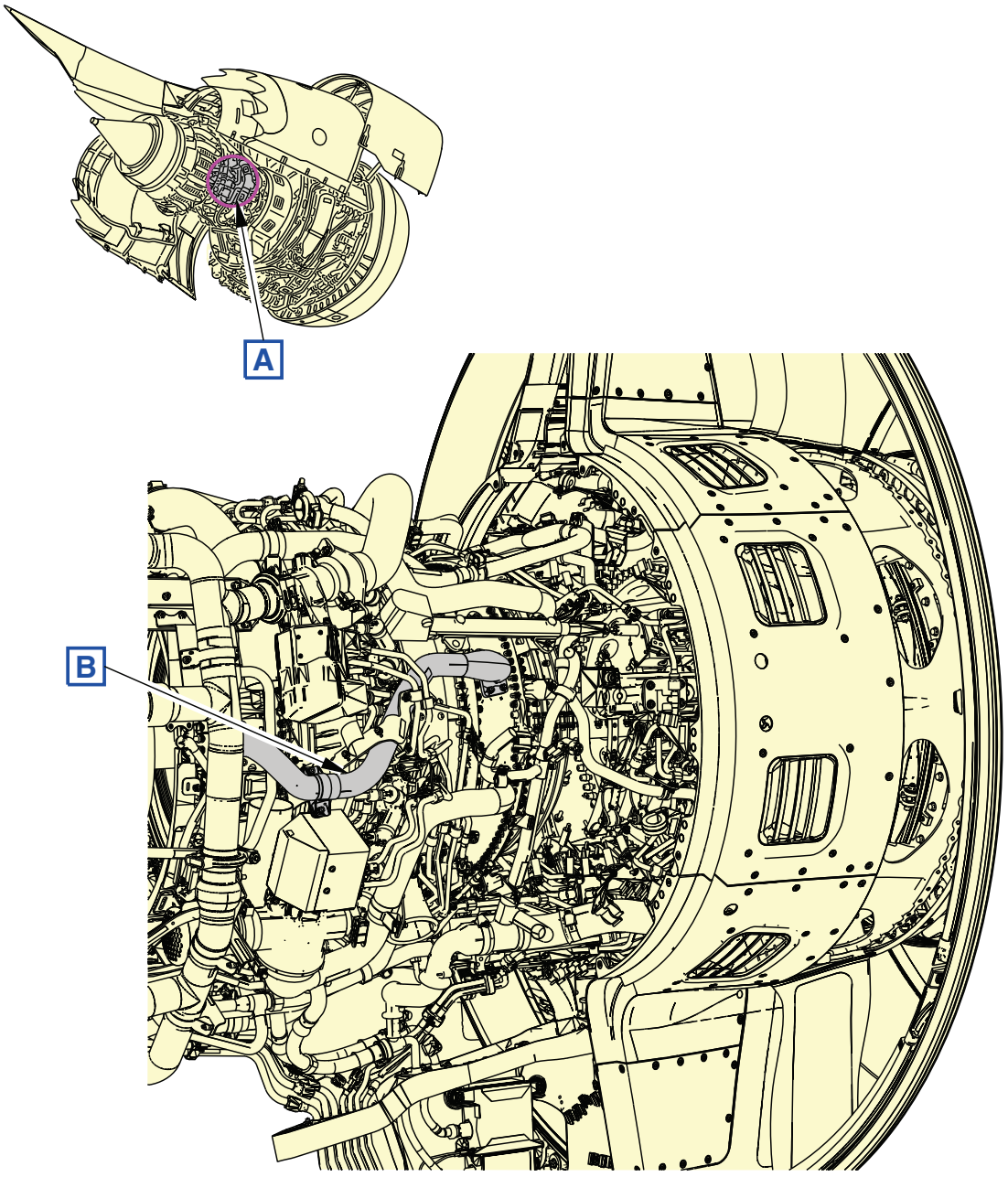
Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Transient Bleed Air Tube and Booster Anti-ice Air Tube Installation
Figure 430/73-11-05-990-860-G00 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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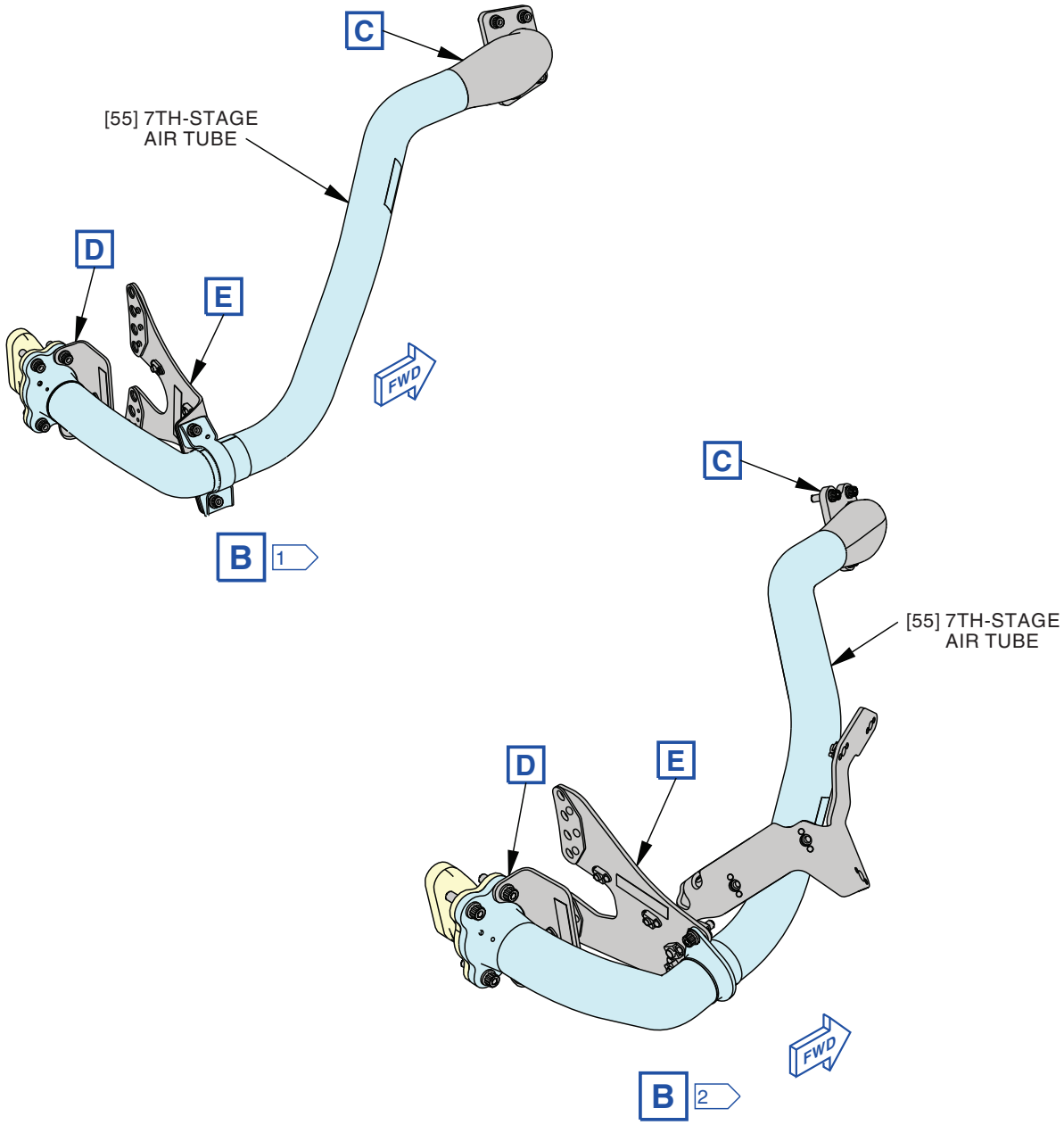
7th Stage Air Tube Installation
Figure 431/73-11-05-990-861-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 POST LEAP 1-B SERVICE BULLETIN 75-0005
2 PRE LEAP 1-B SERVICE BULLETIN 75-0005

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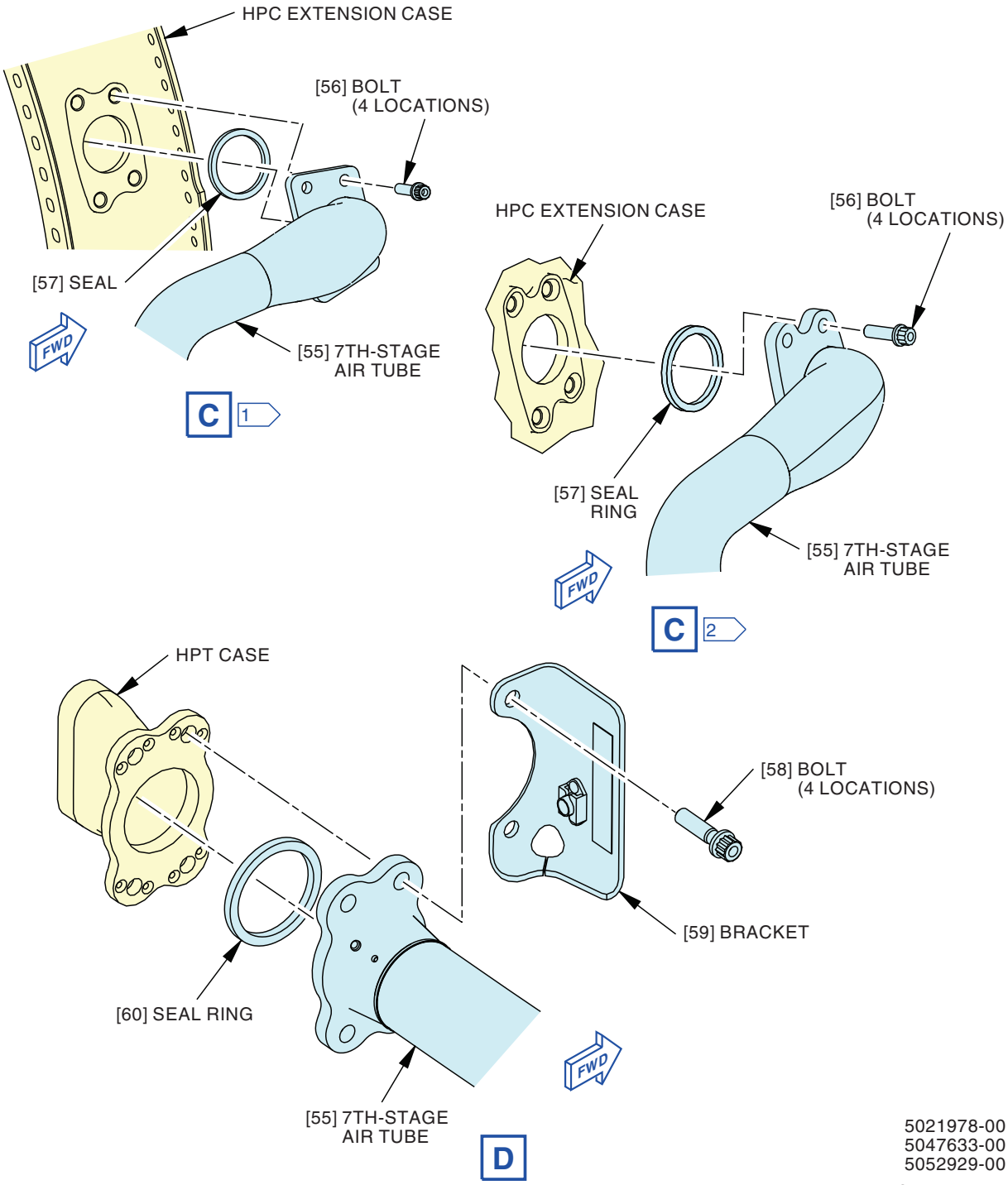
7th Stage Air Tube Installation
Figure 431/73-11-05-990-861-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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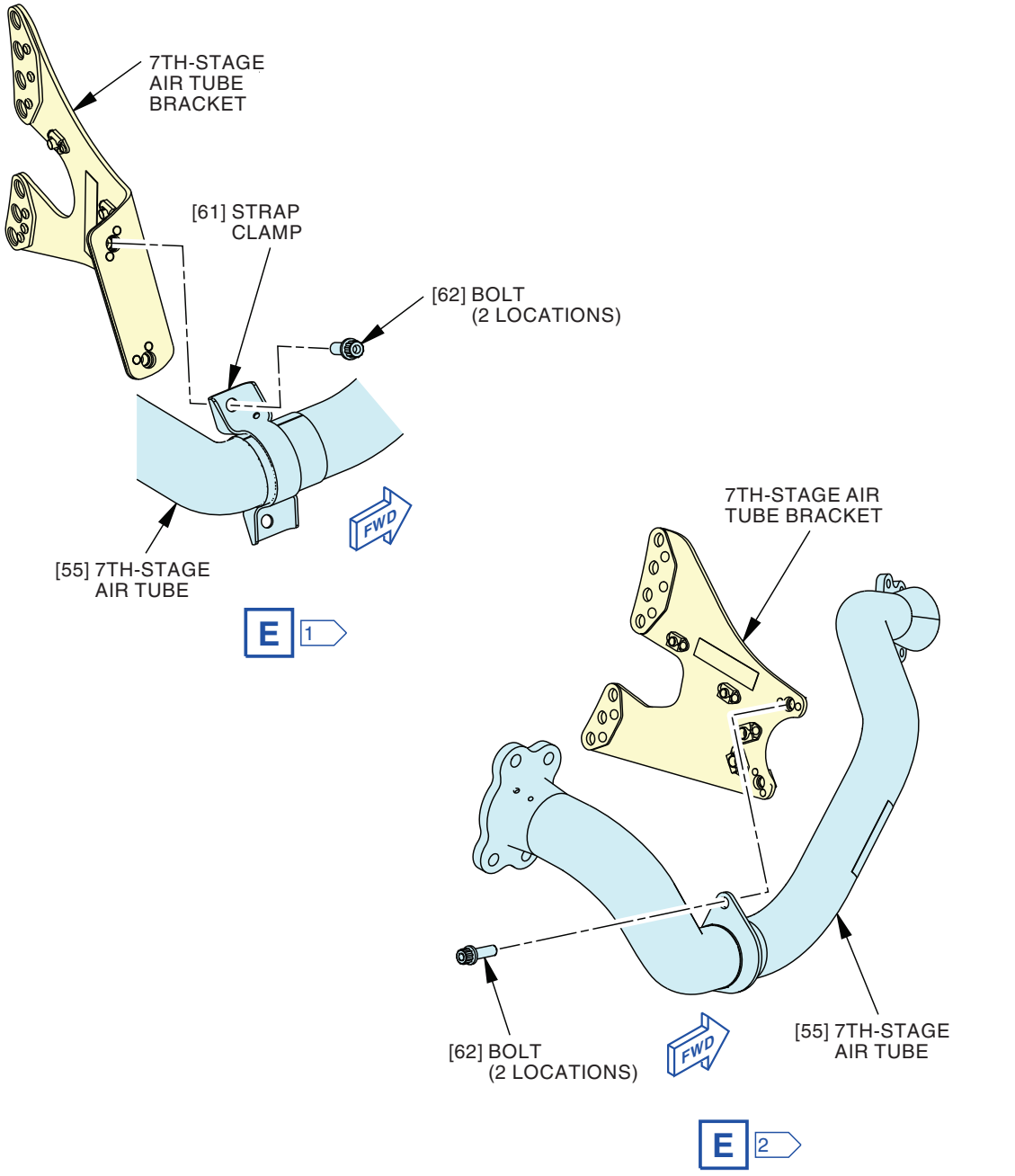
7th Stage Air Tube Installation
Figure 431/73-11-05-990-861-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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7th Stage Air Tube Installation
Figure 431/73-11-05-990-861-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(EXAMPLE)

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2947639 S0000724088_V1

Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 1 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(EXAMPLE)

A

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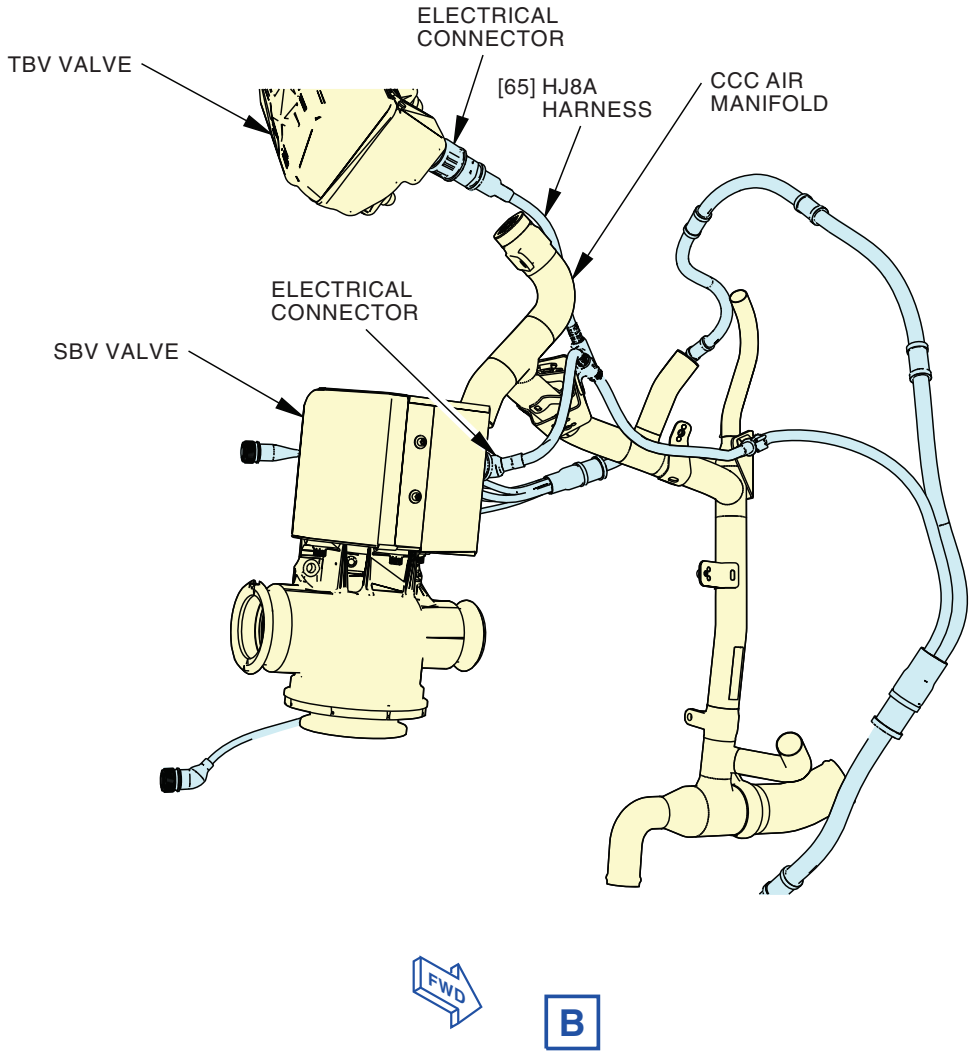
3129154 S0000860946_V1

Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 2 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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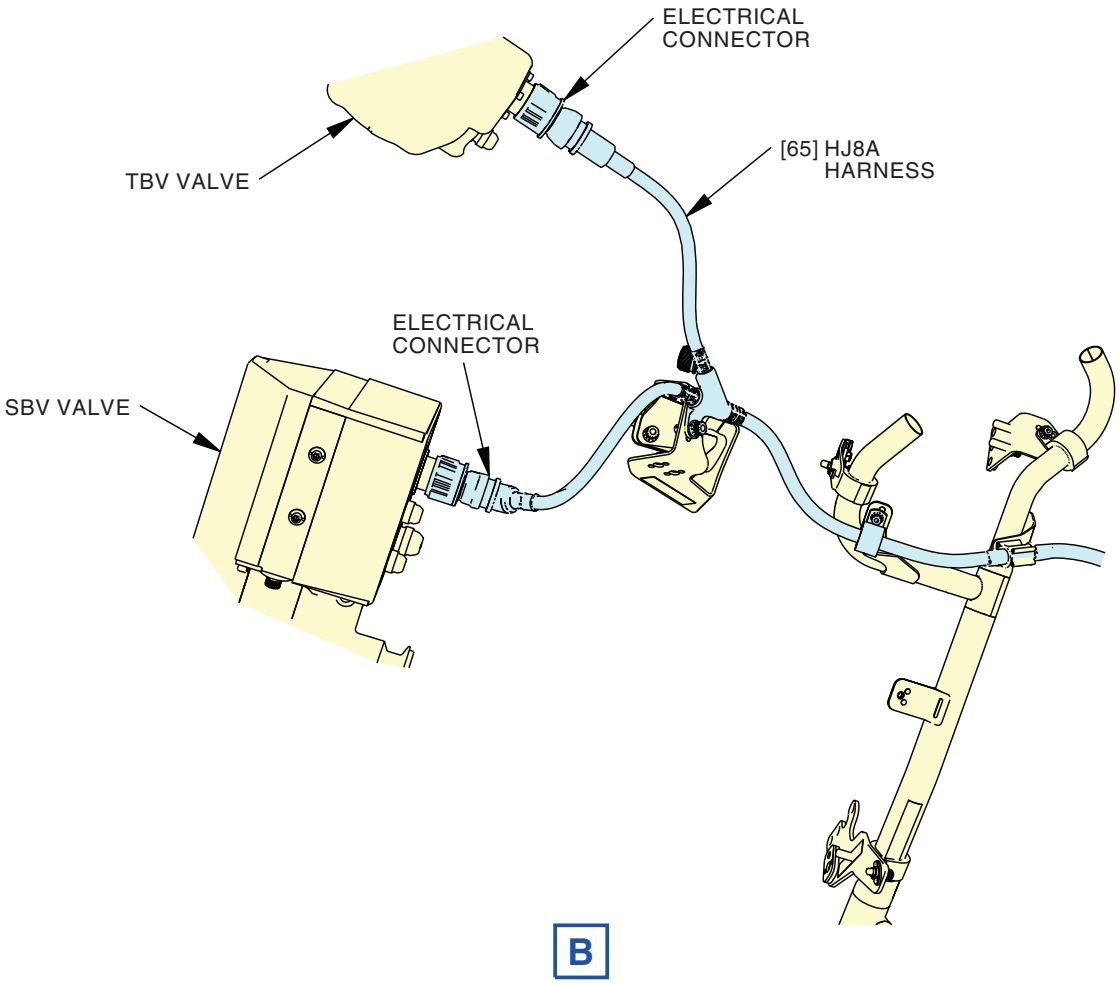
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 3 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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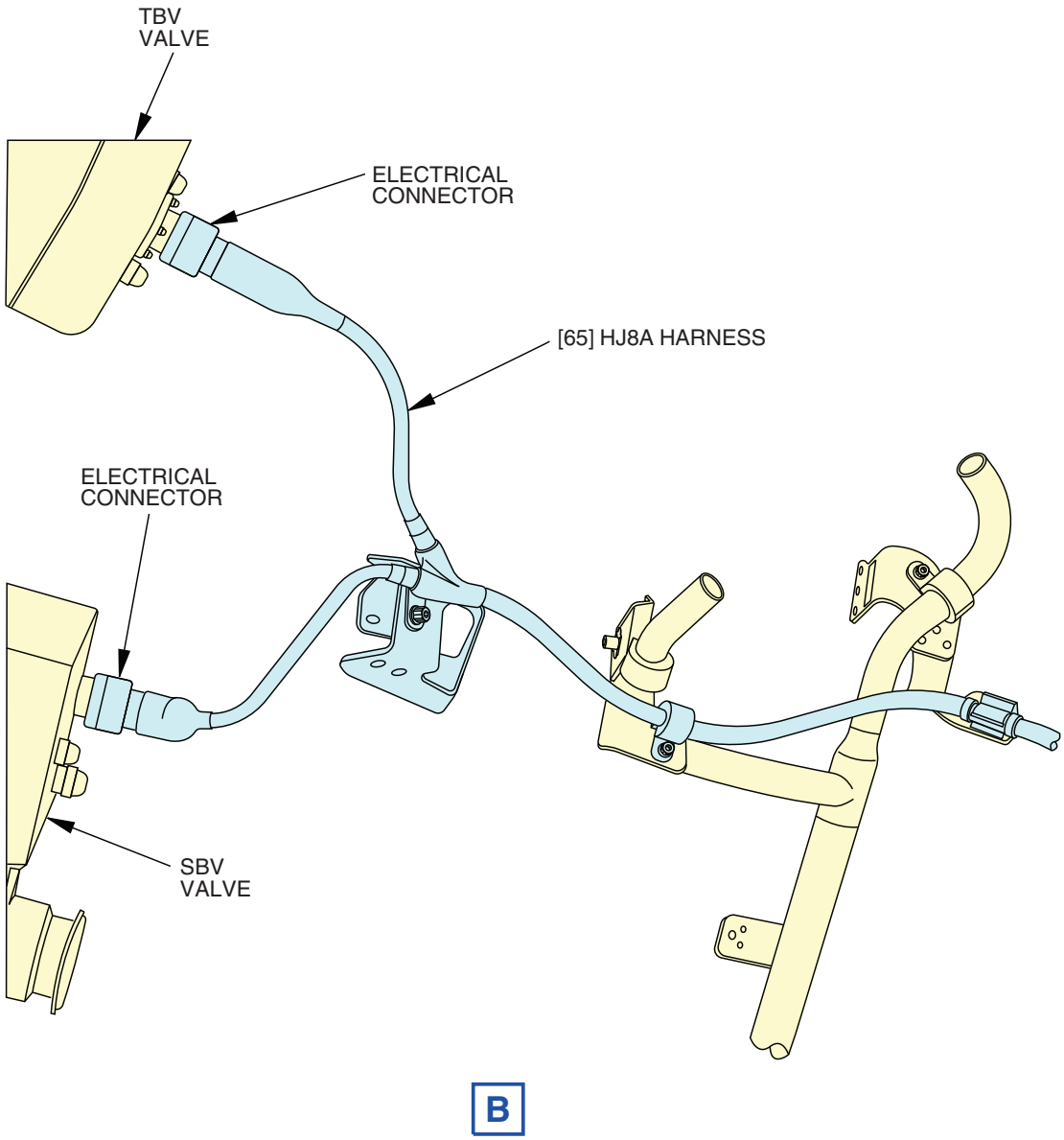
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 4 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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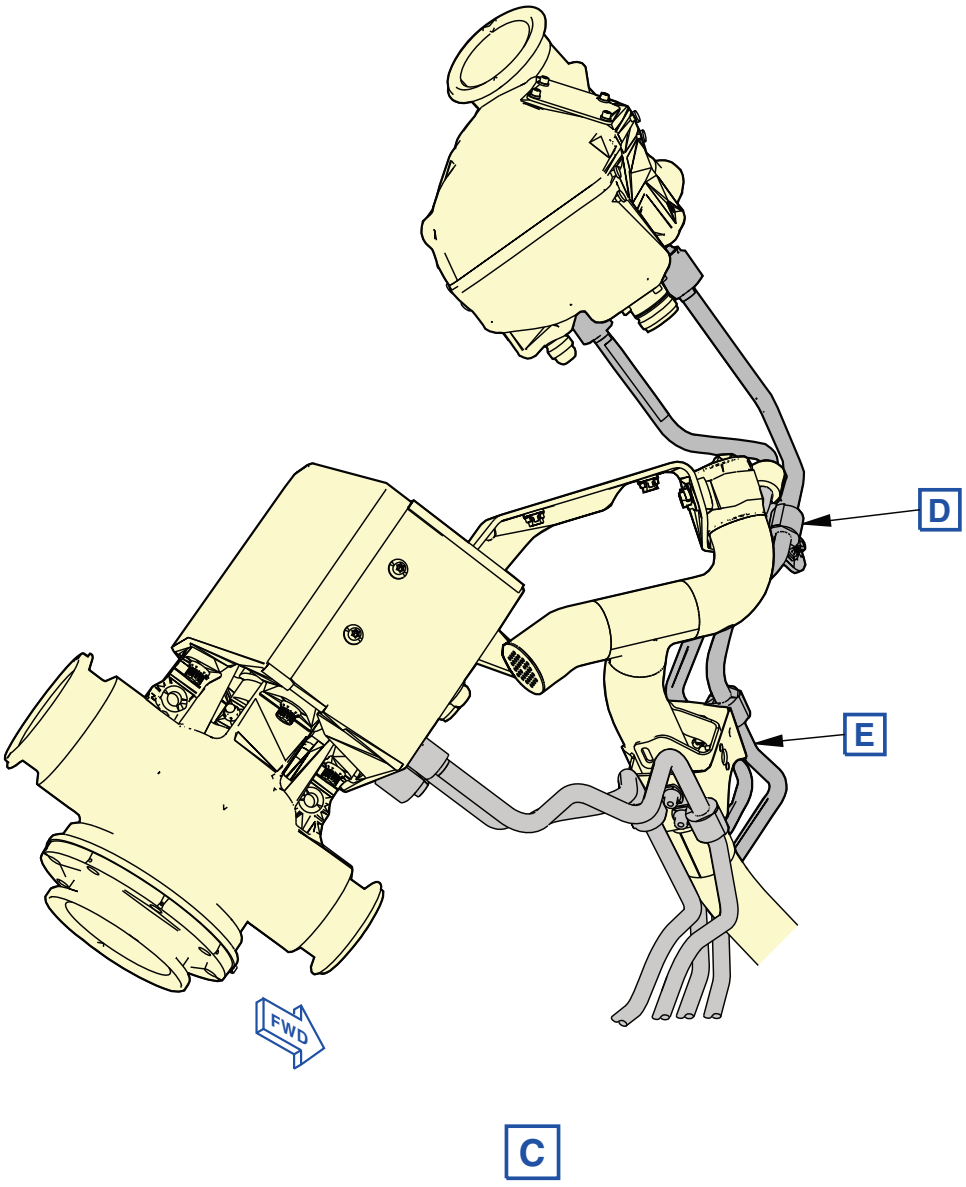
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 5 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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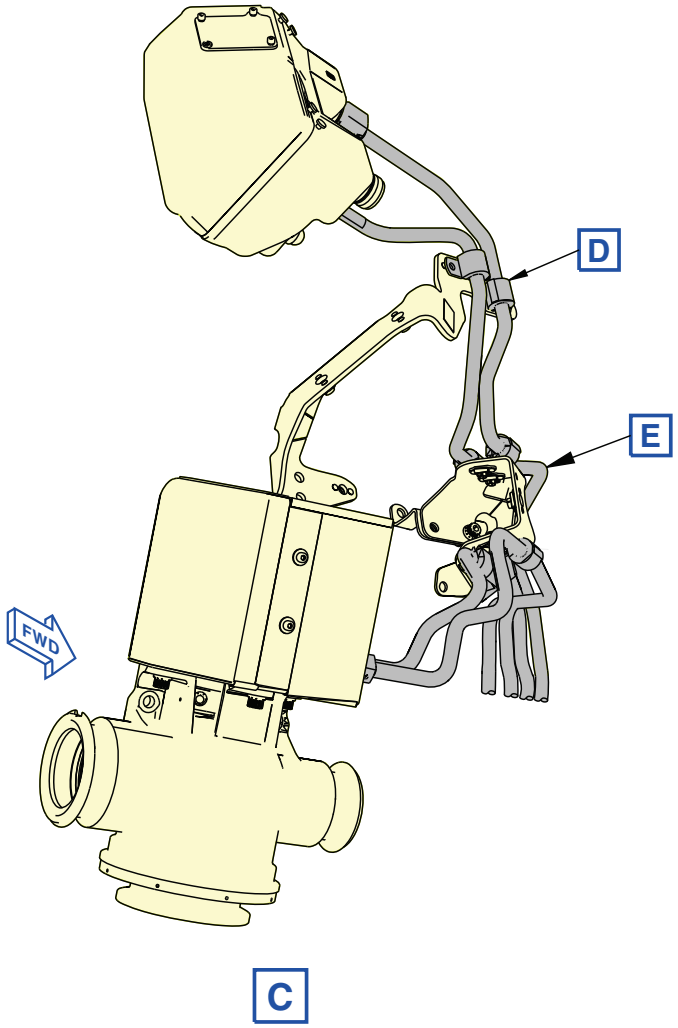
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 6 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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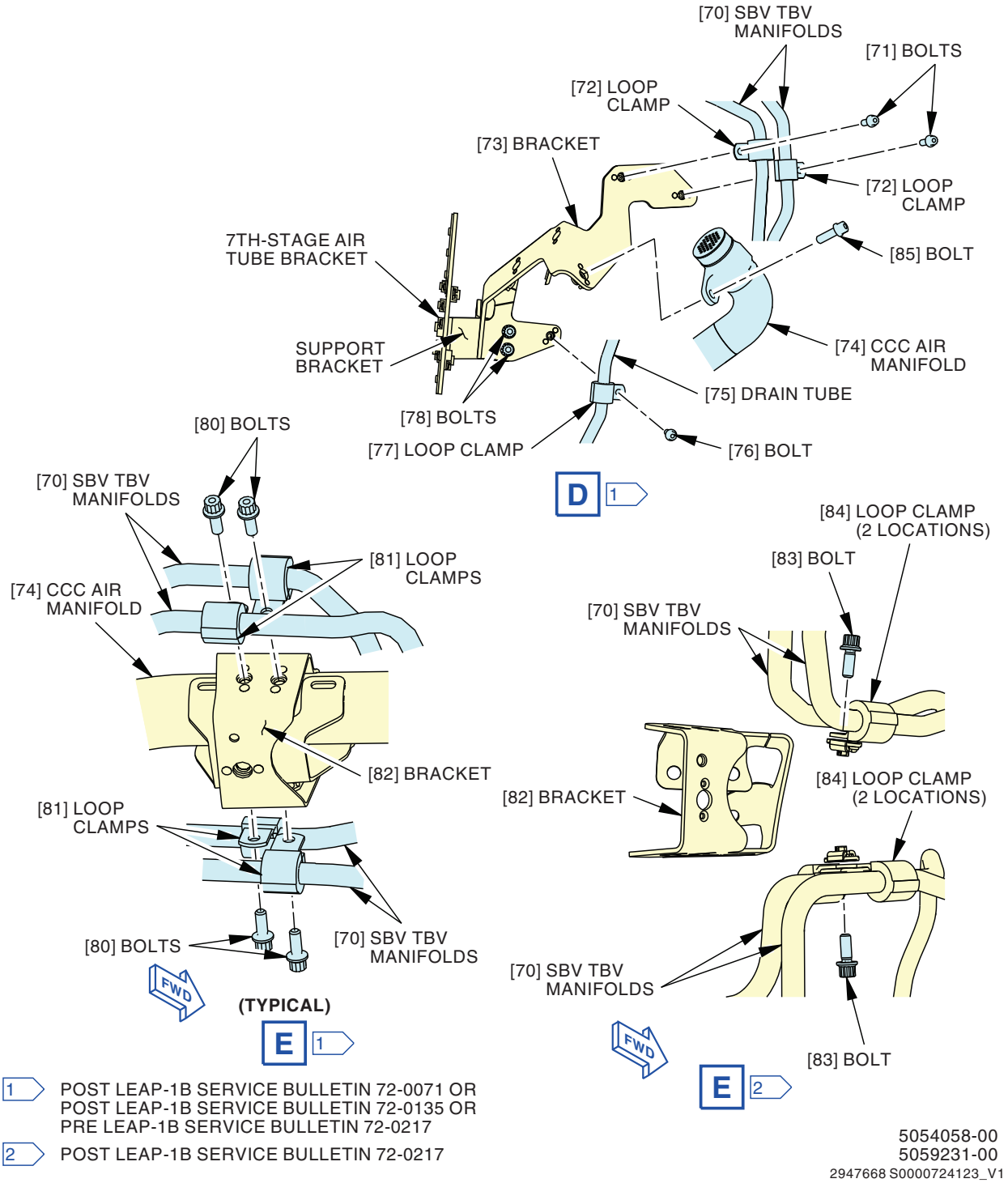
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 7 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 8 of 14)

EFFECTIVITY
PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0071 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217 OR POST LEAP-1B SB 72-0217

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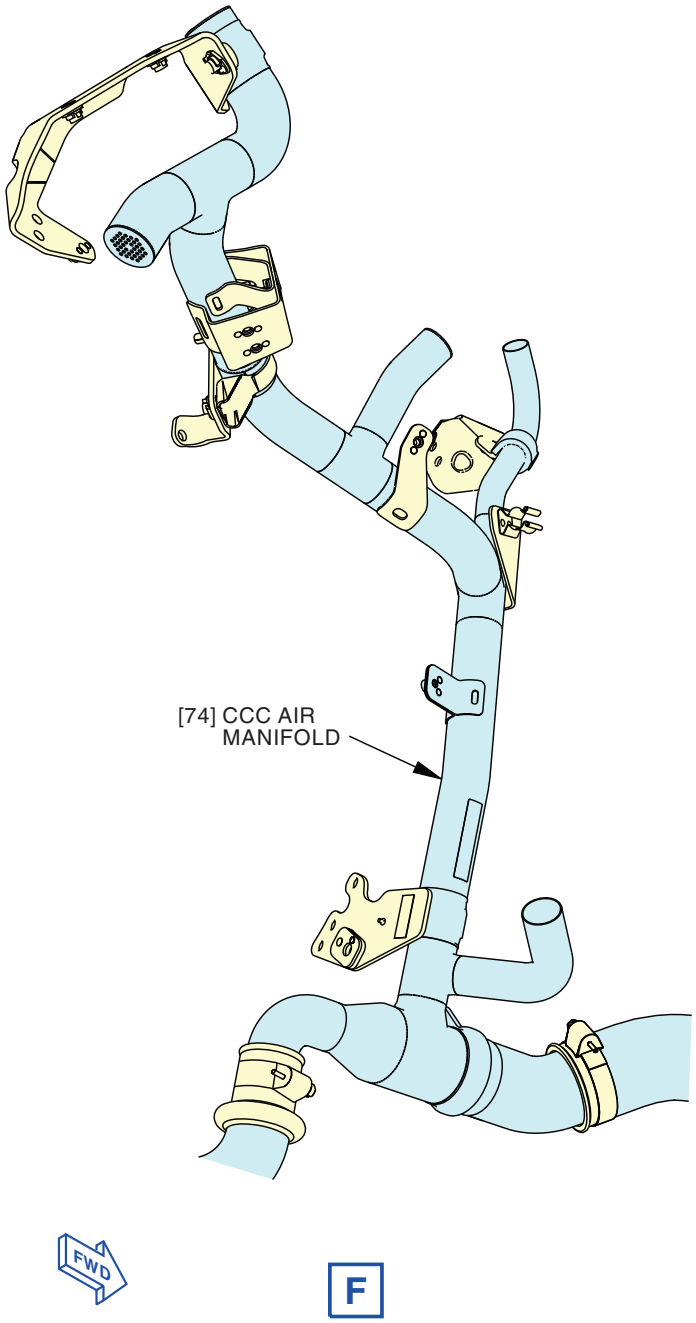
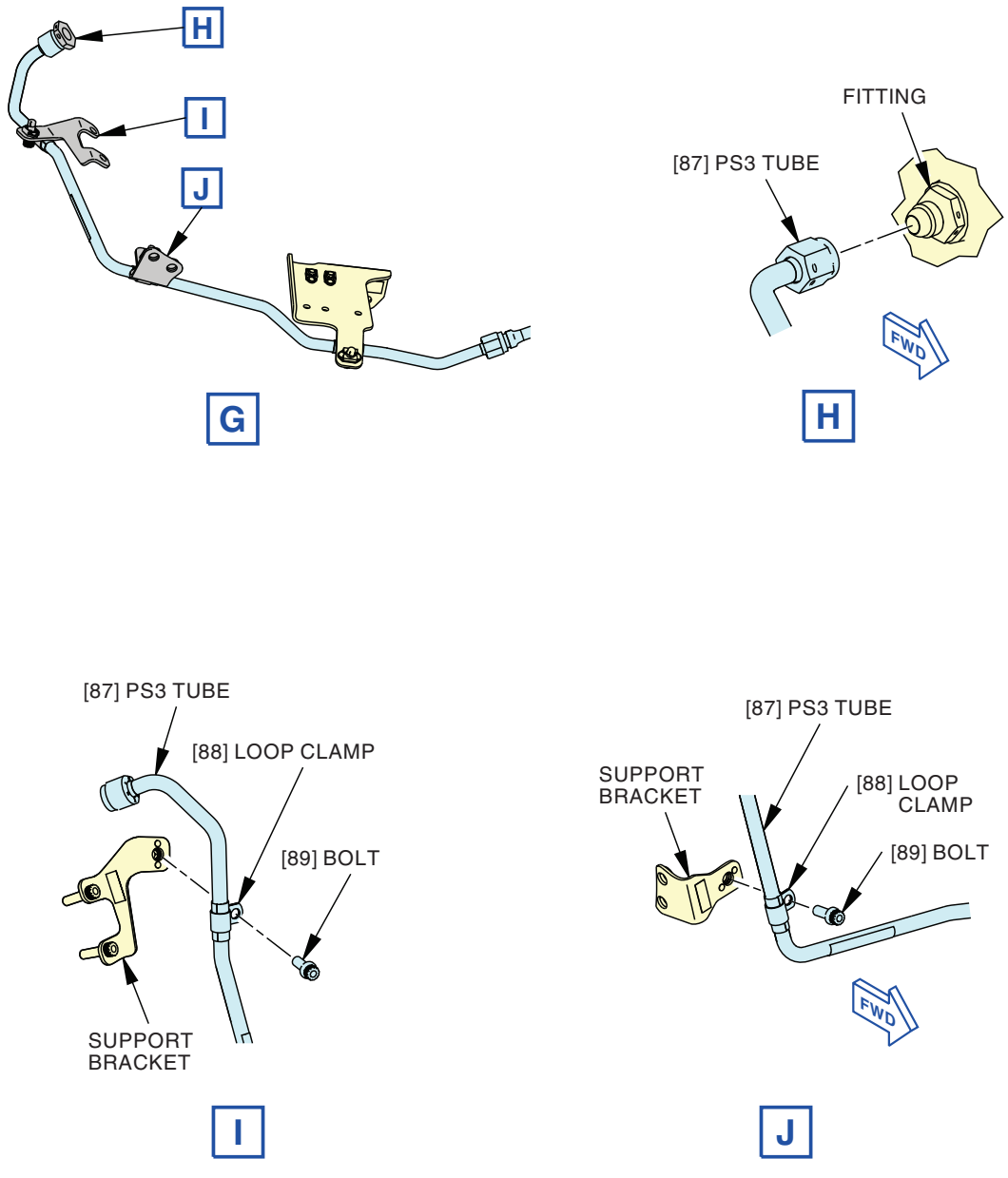


Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 9 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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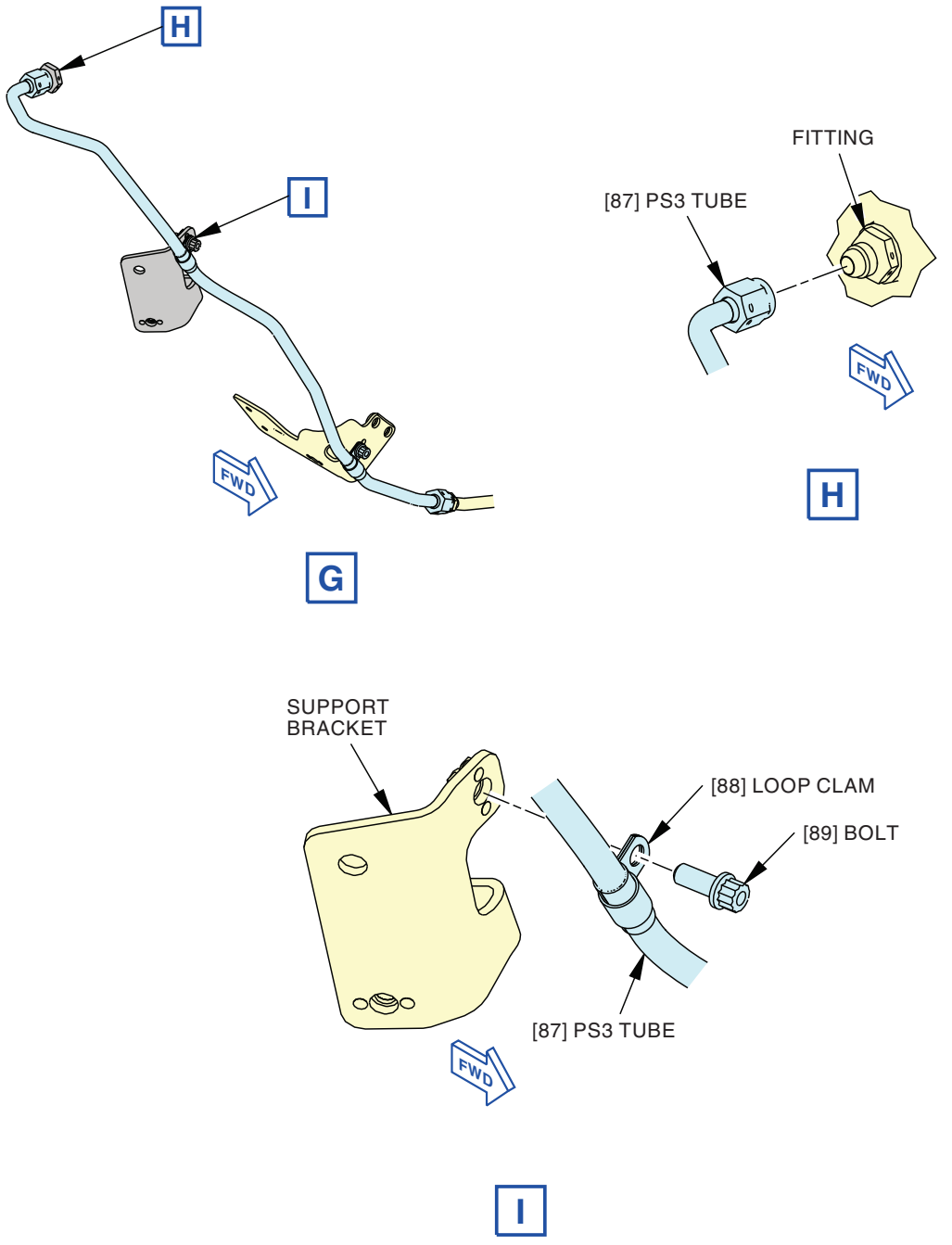
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 10 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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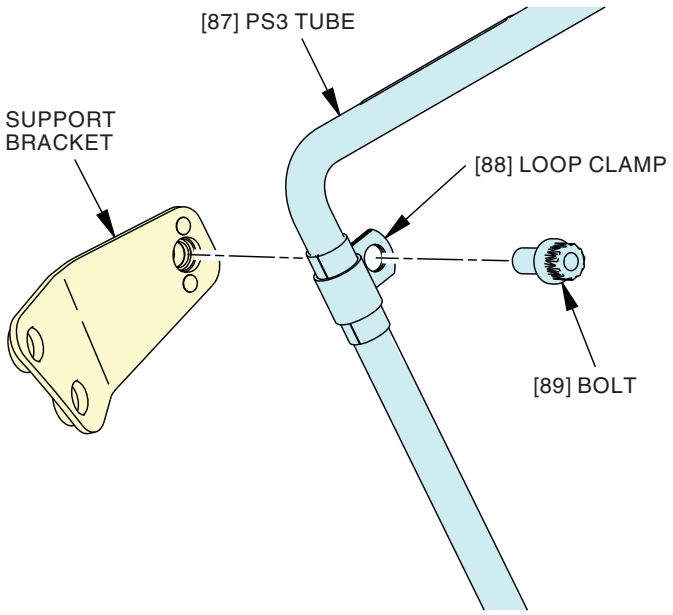
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 11 of 14)

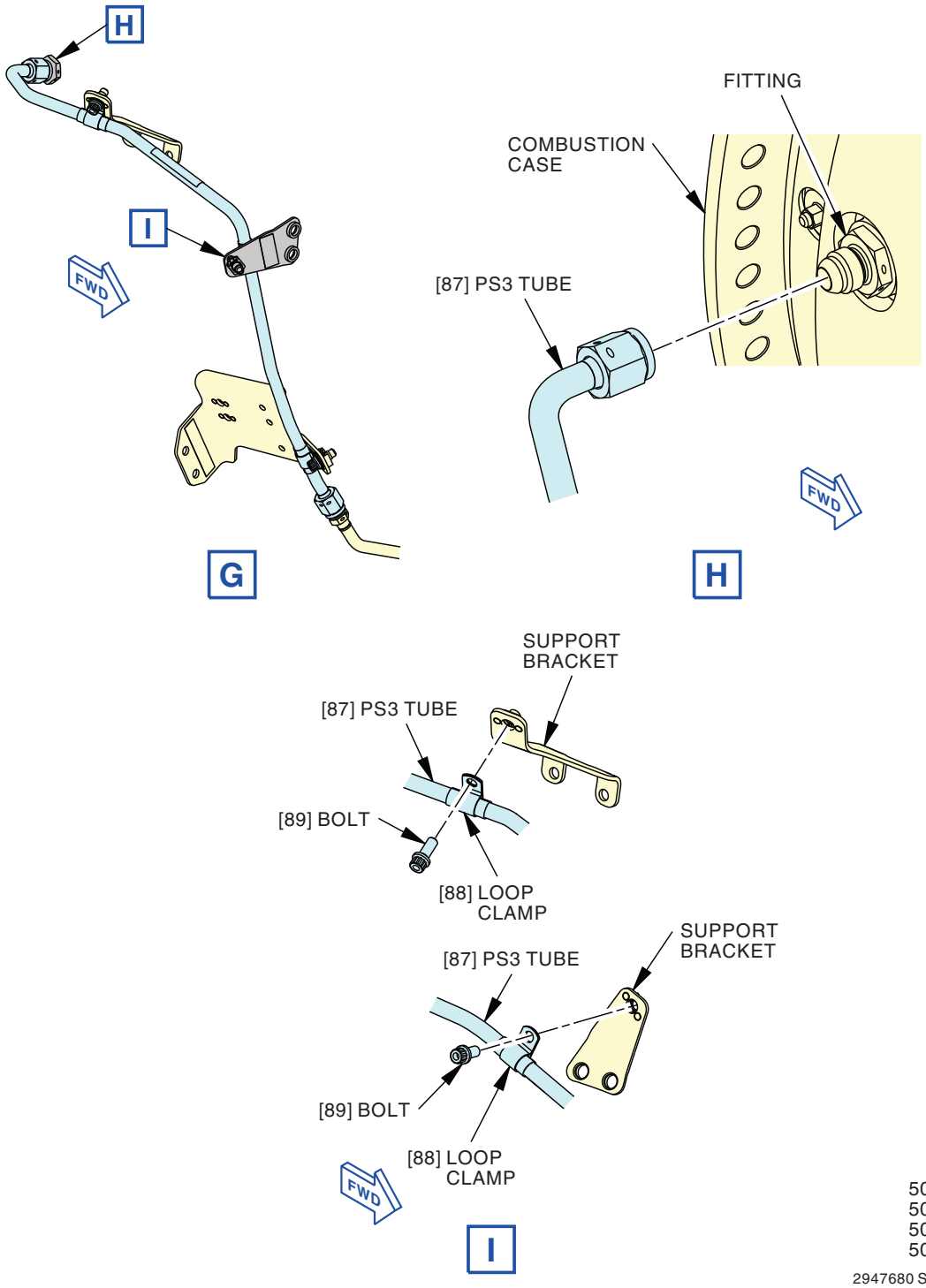
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051837-00
2947676 S0000724138_V1

Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 12 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

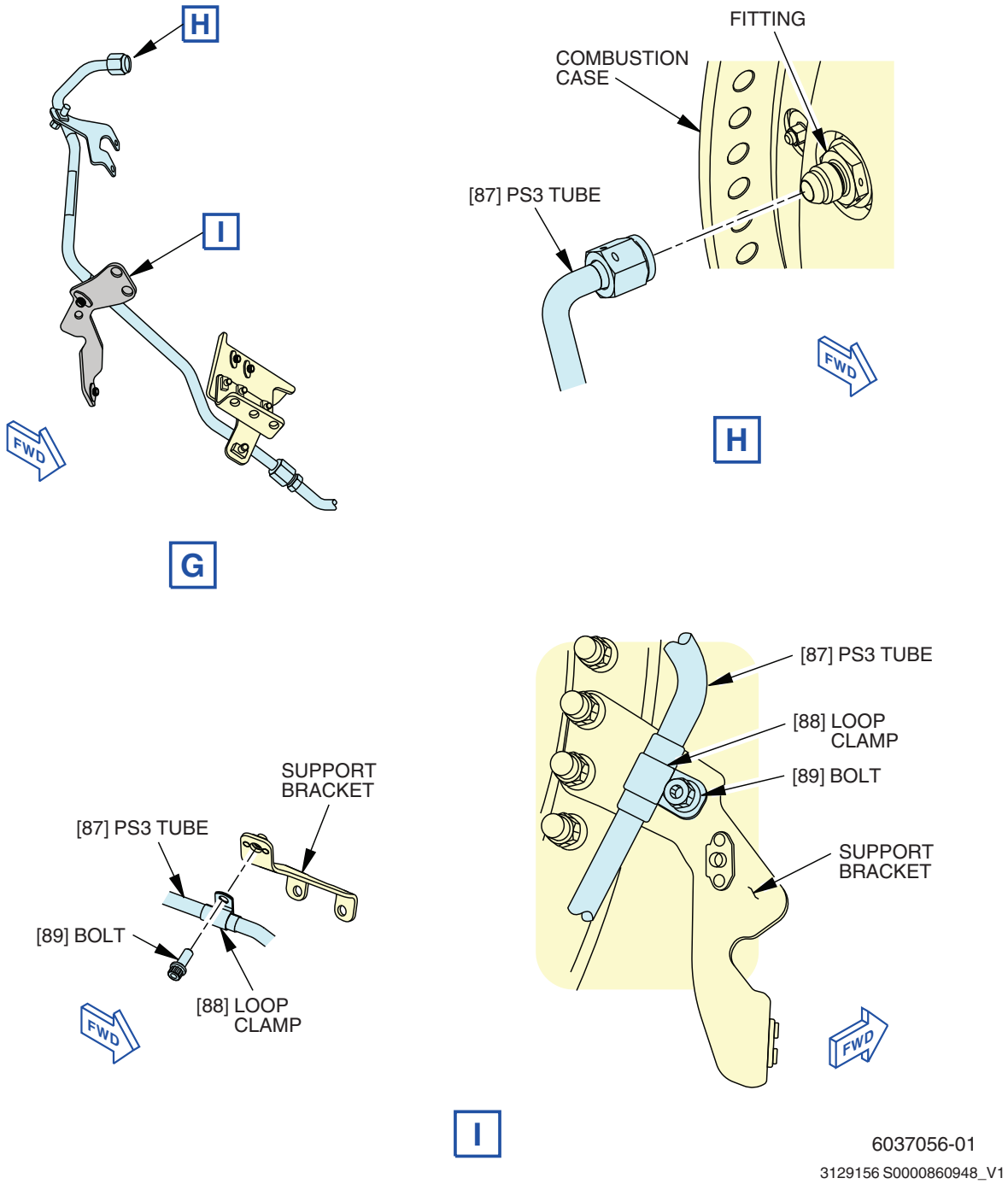


5047638-00
5047639-00
5047640-00
5047641-00

2947680 S0000724144_V1

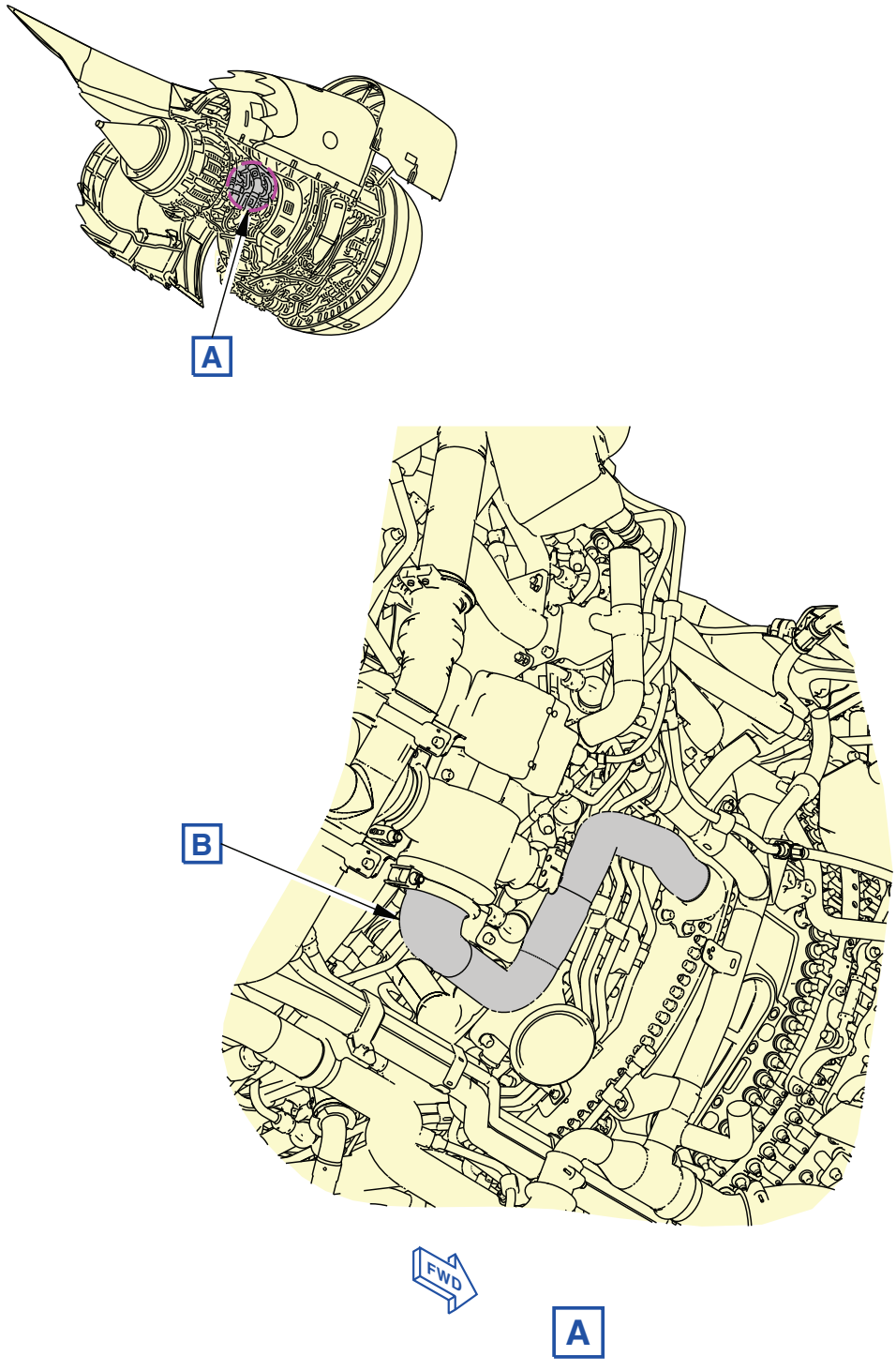
Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 13 of 14)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Harness, SBV TBV Manifold, CCC Air Manifold, and PS3 Tube Installation
Figure 432/73-11-05-990-862-G00 (Sheet 14 of 14)

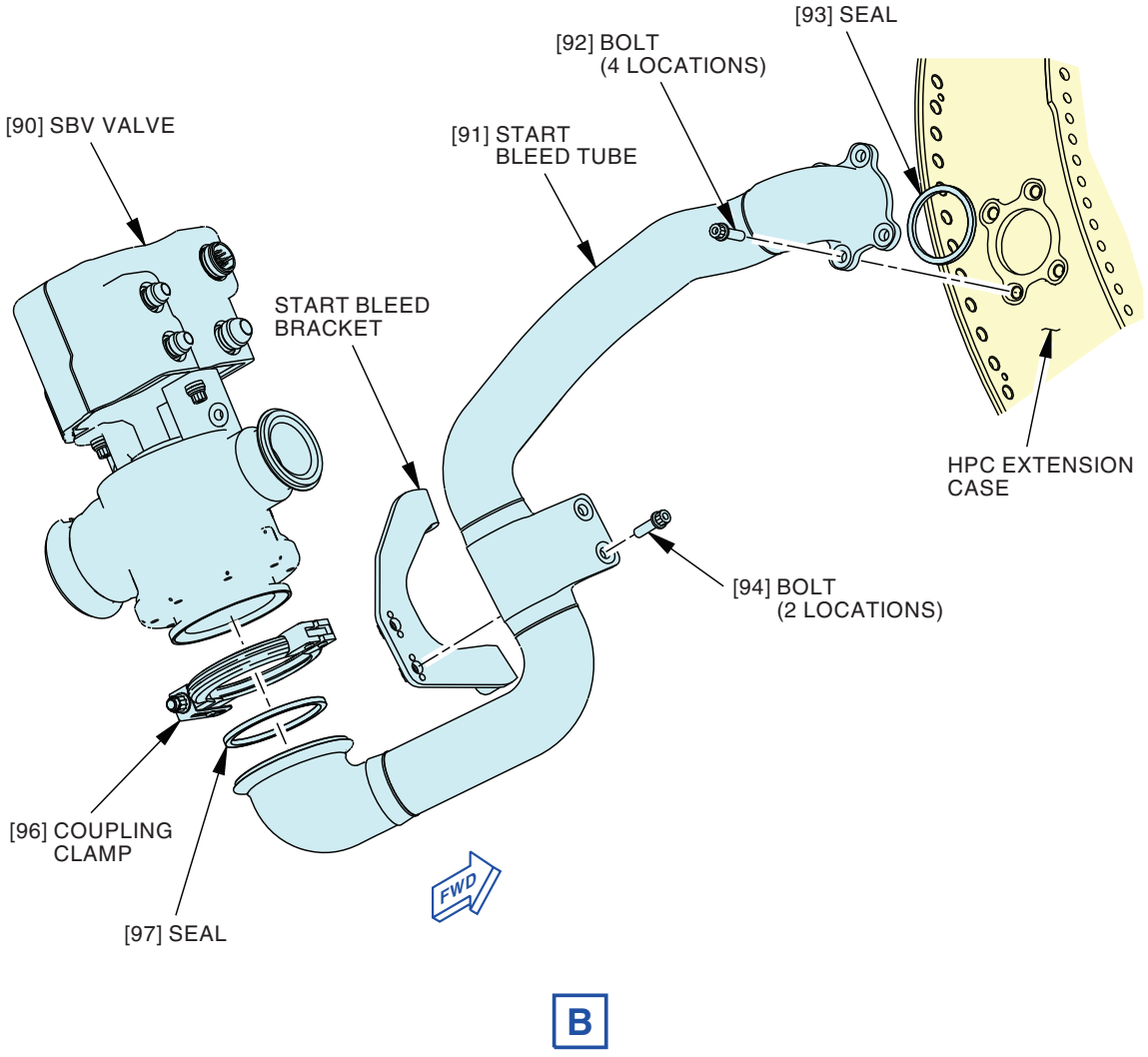
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021985-00
2947686 S0000724172_V1

Start Bleed Tube Installation
Figure 433/73-11-05-990-863-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

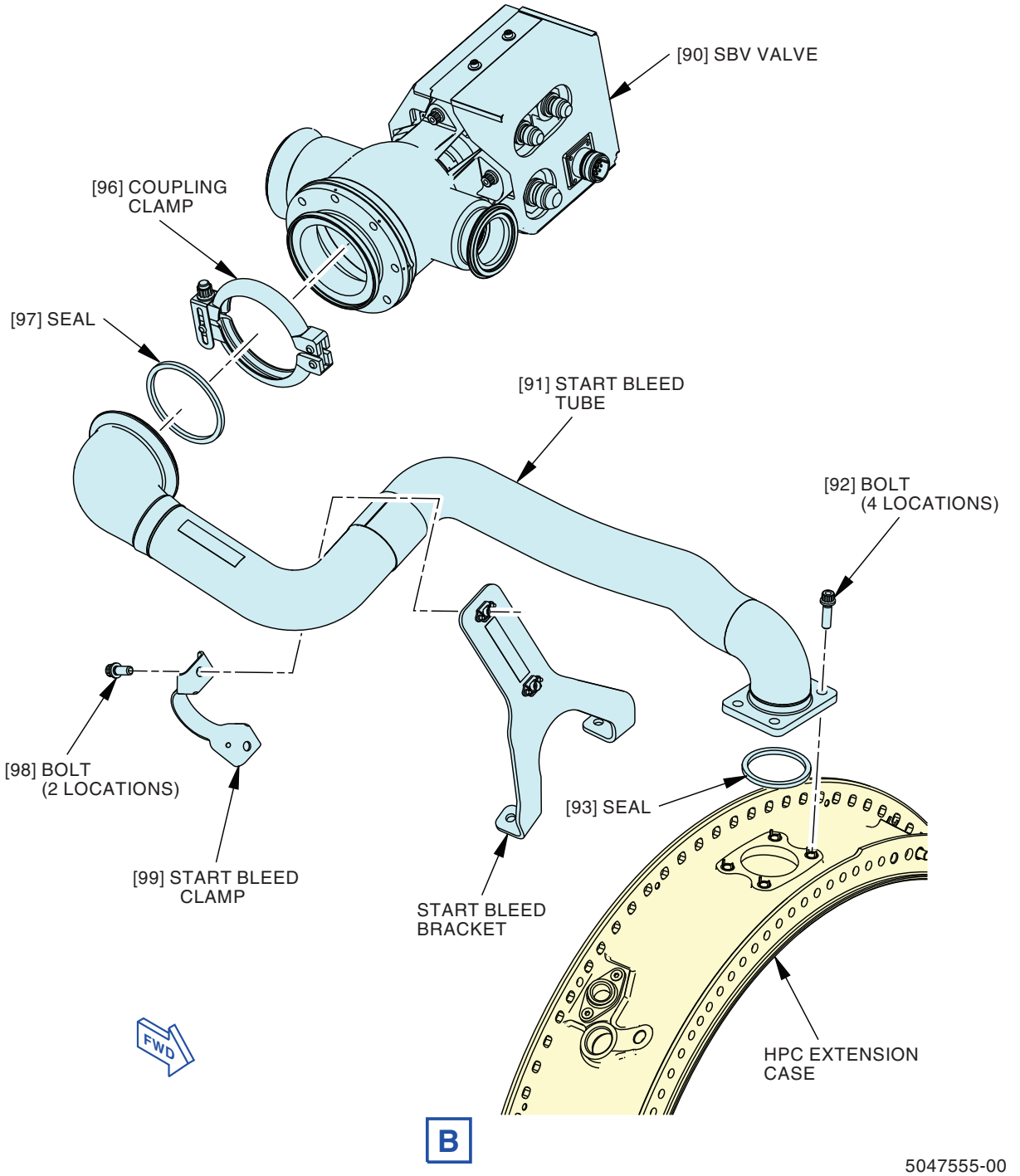


5024884-00

2947687 S0000724246_V1

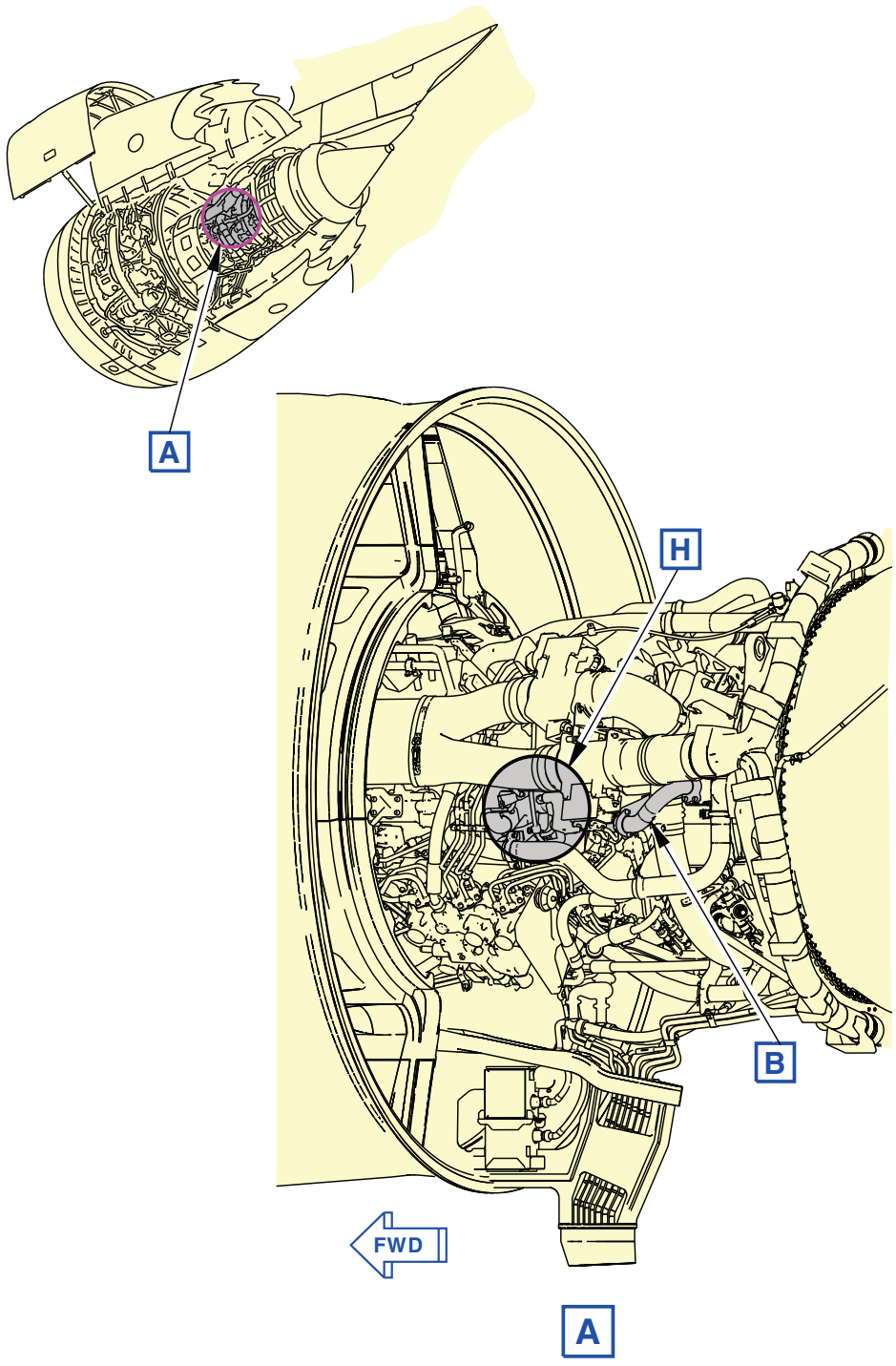
Start Bleed Tube Installation
Figure 433/73-11-05-990-863-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Start Bleed Tube Installation
Figure 433/73-11-05-990-863-G00 (Sheet 3 of 3)

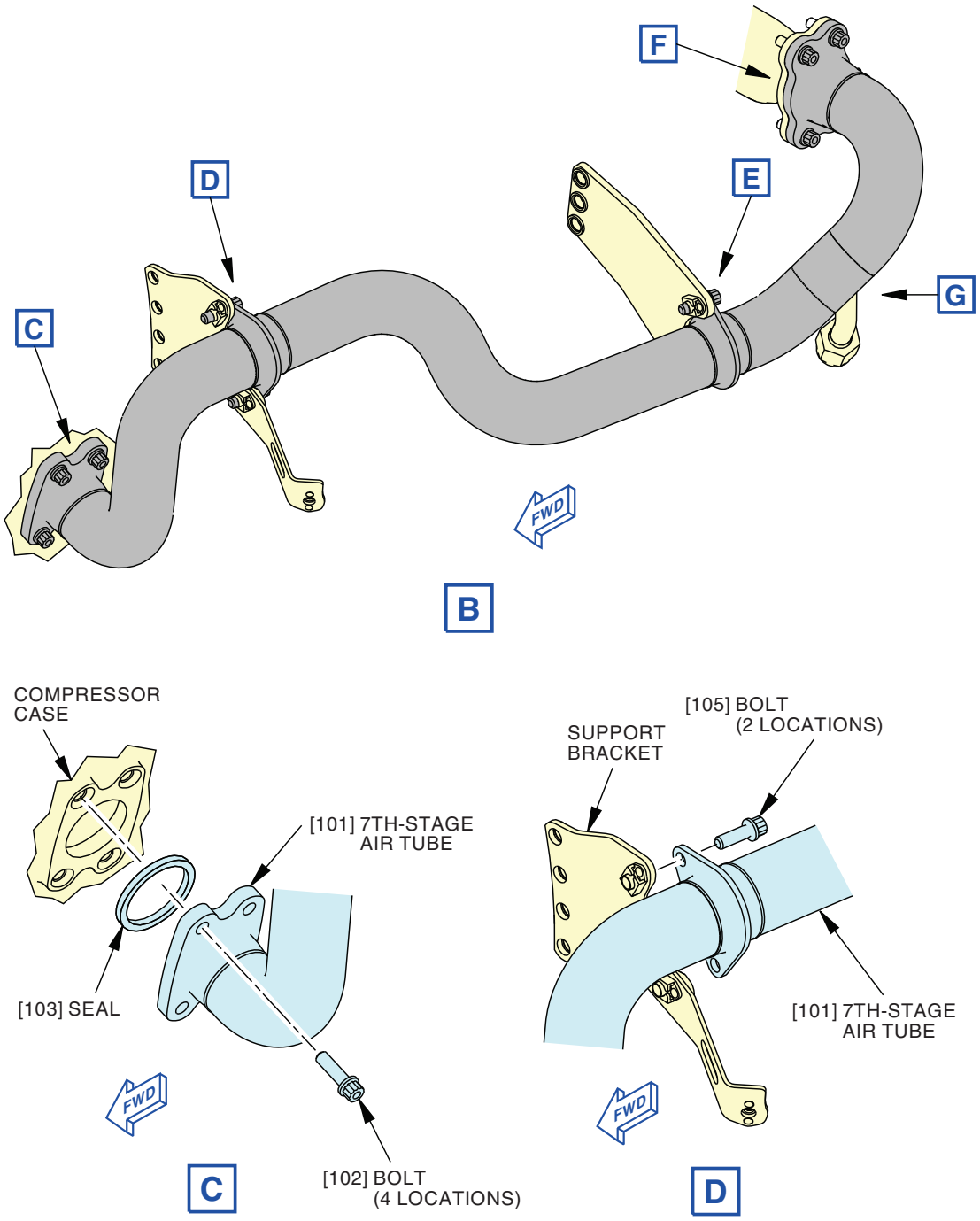
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021979-00
2947640 S0000724371_V1

7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 1 of 7)

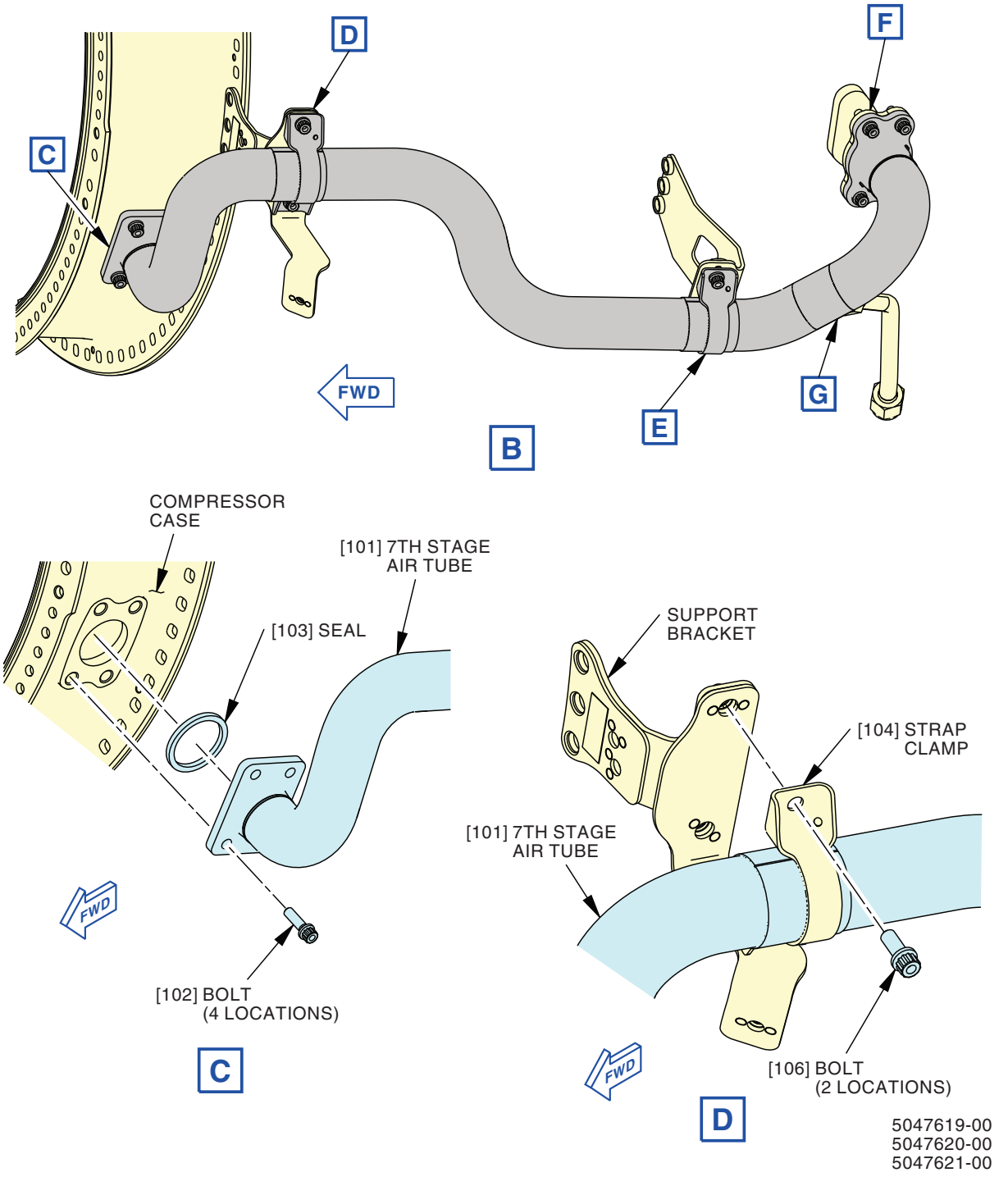
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022334-00
2947638 S0000724372_V2

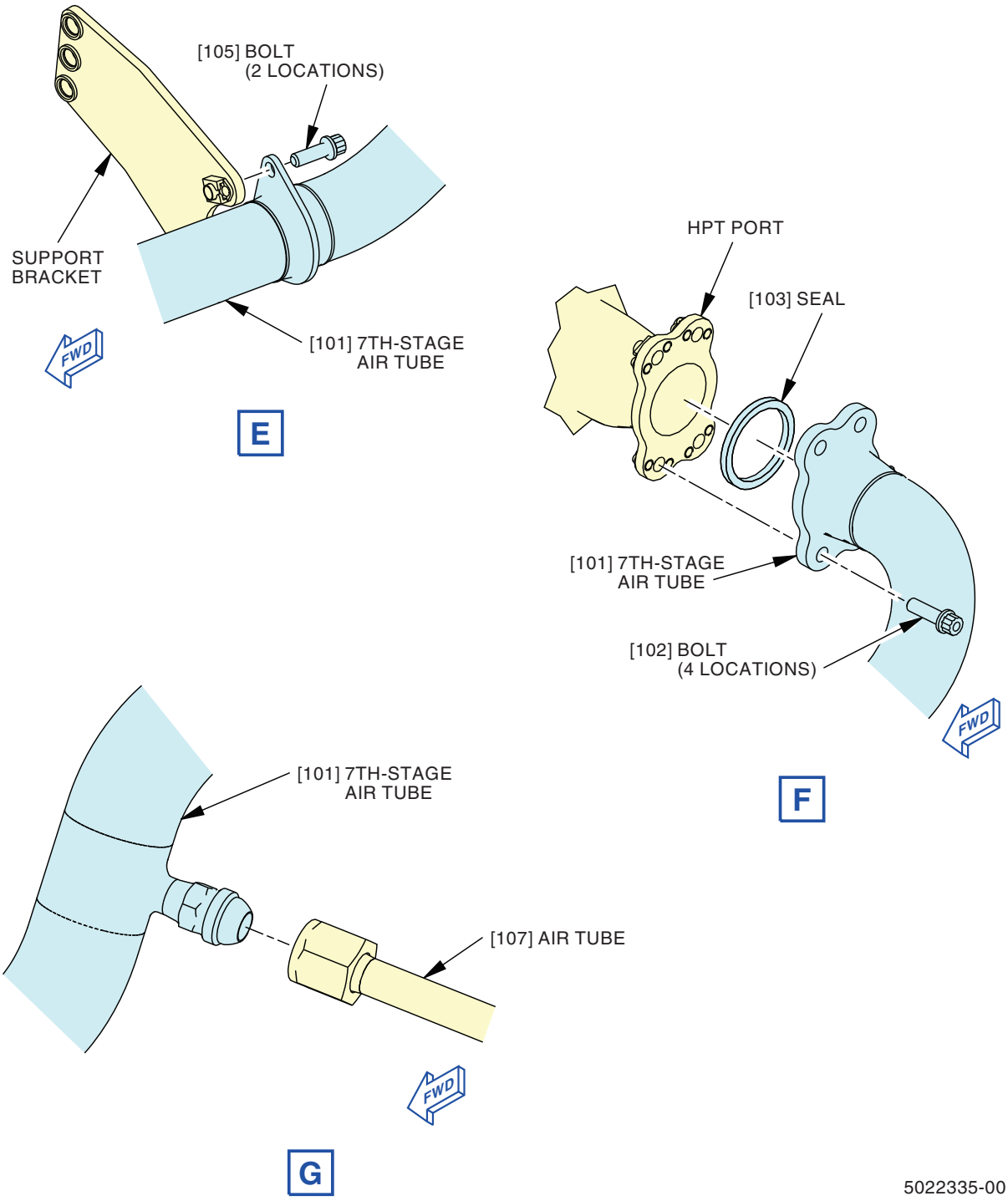
7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



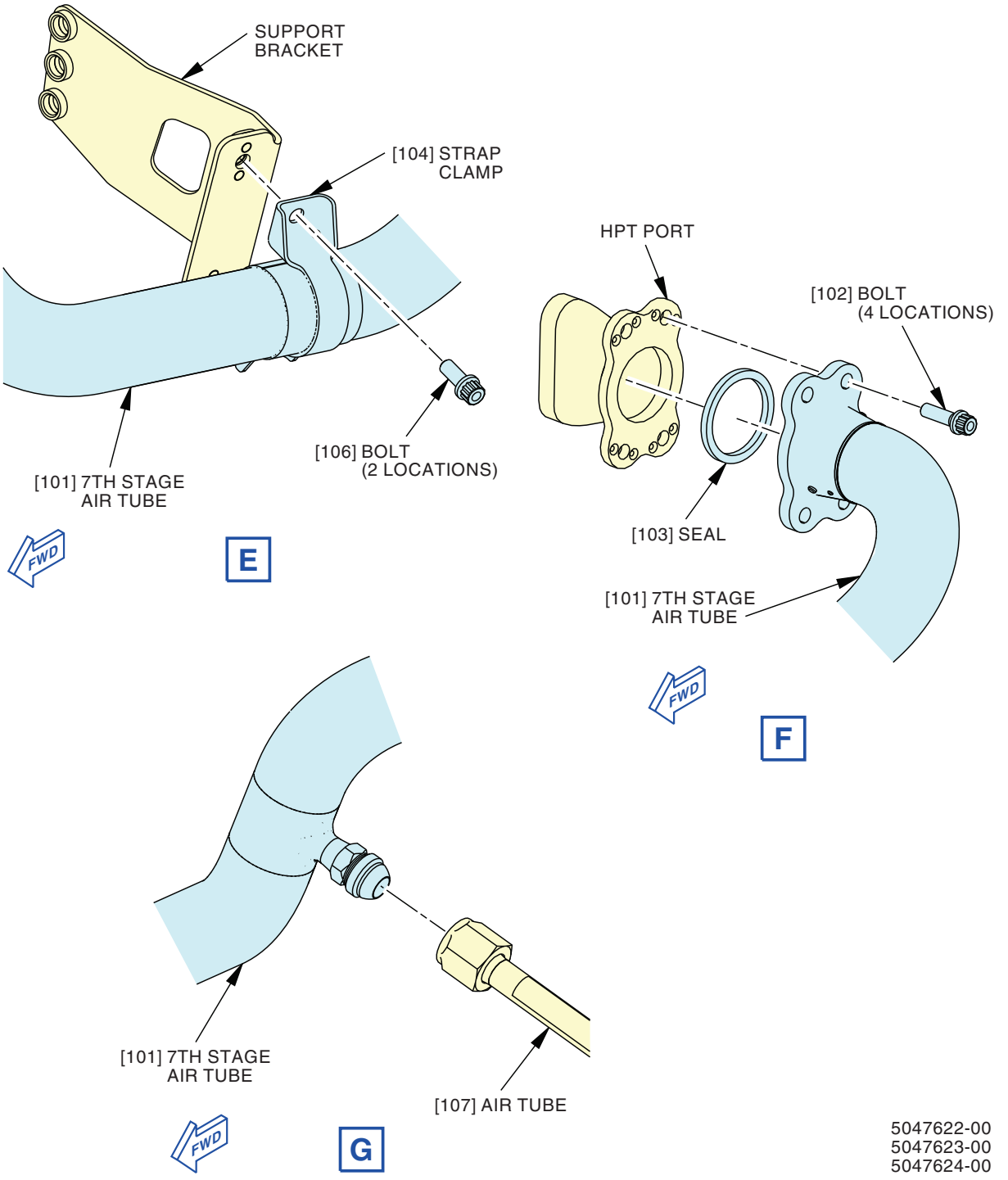
7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

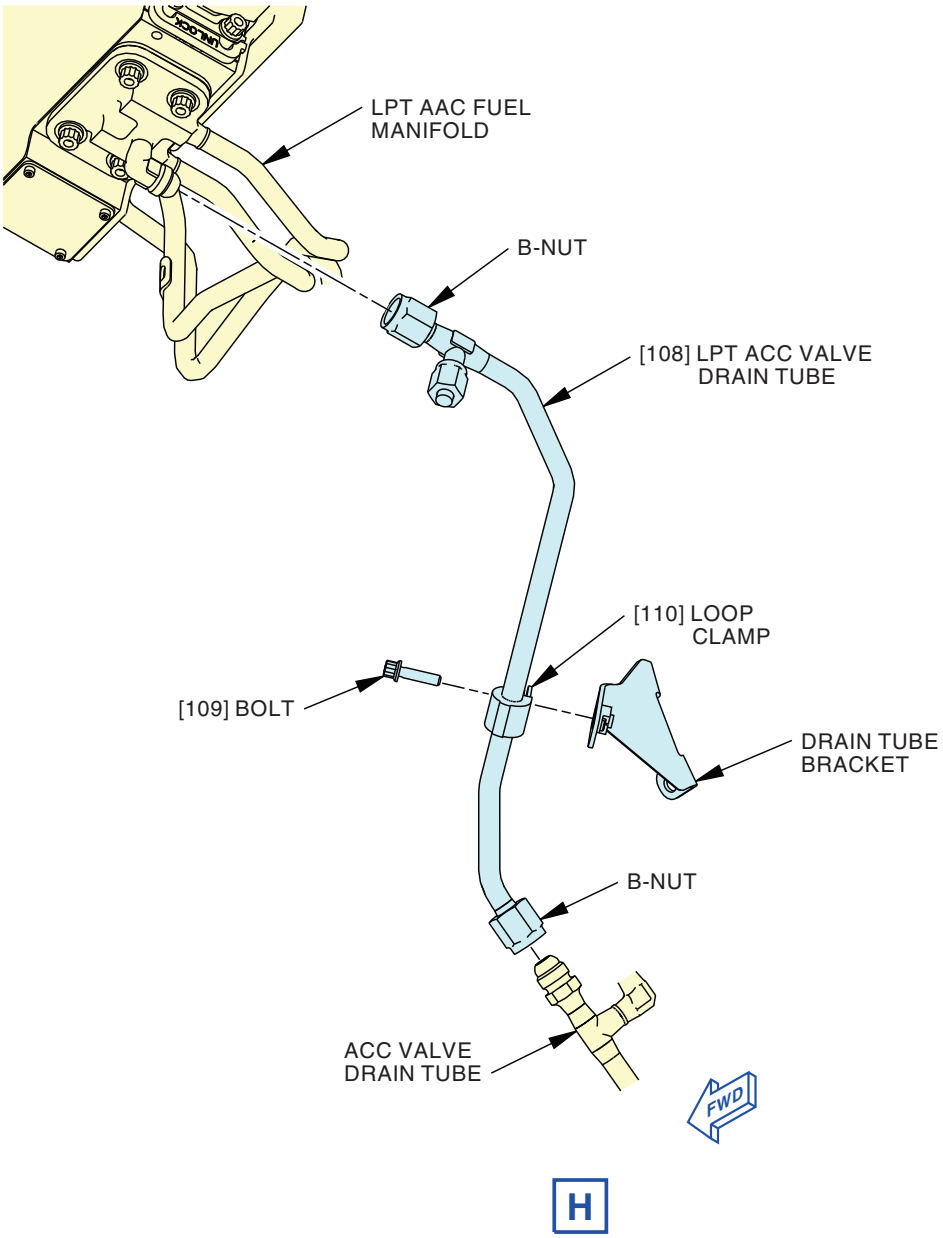


5047622-00
5047623-00
5047624-00

2947572 S0000724375_V2

7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 5 of 7)

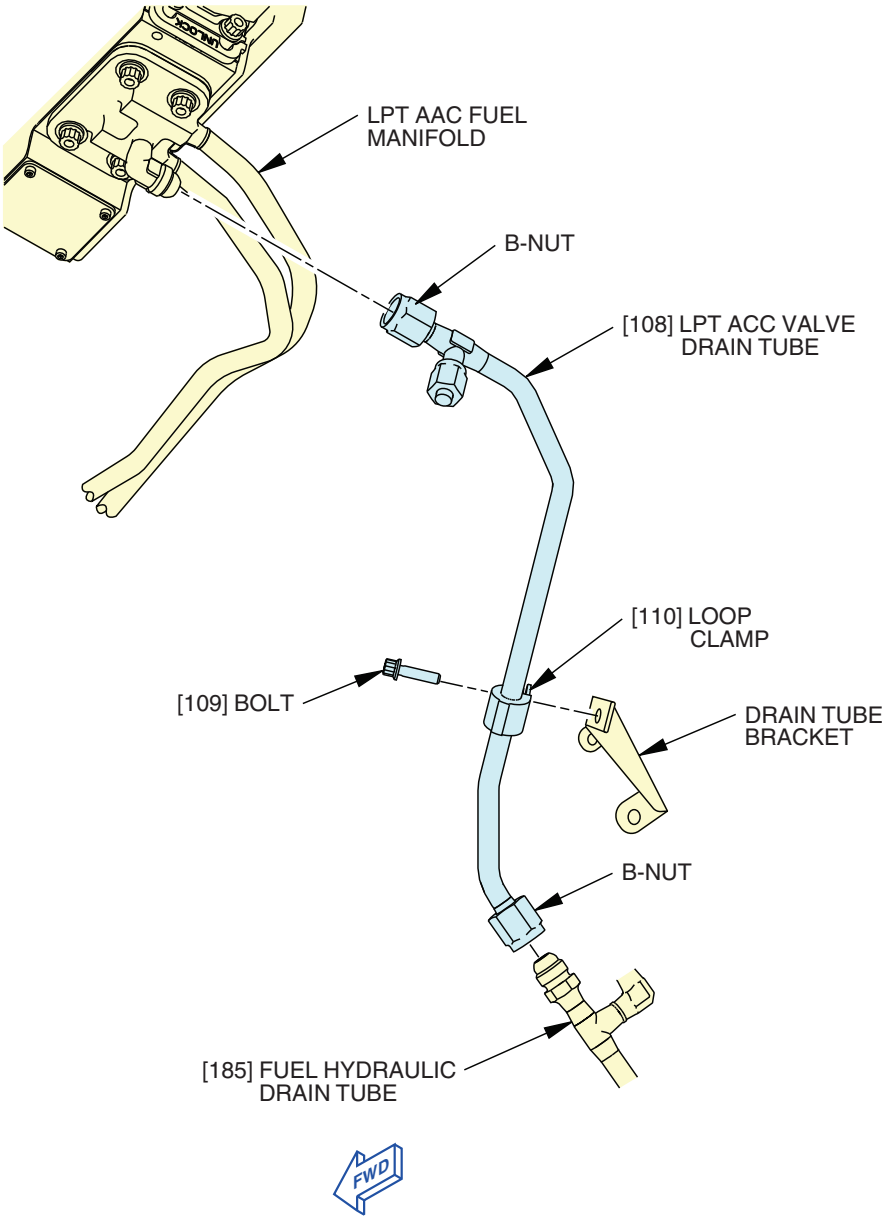
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022331-00
2947642 S0000724370_V2

7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

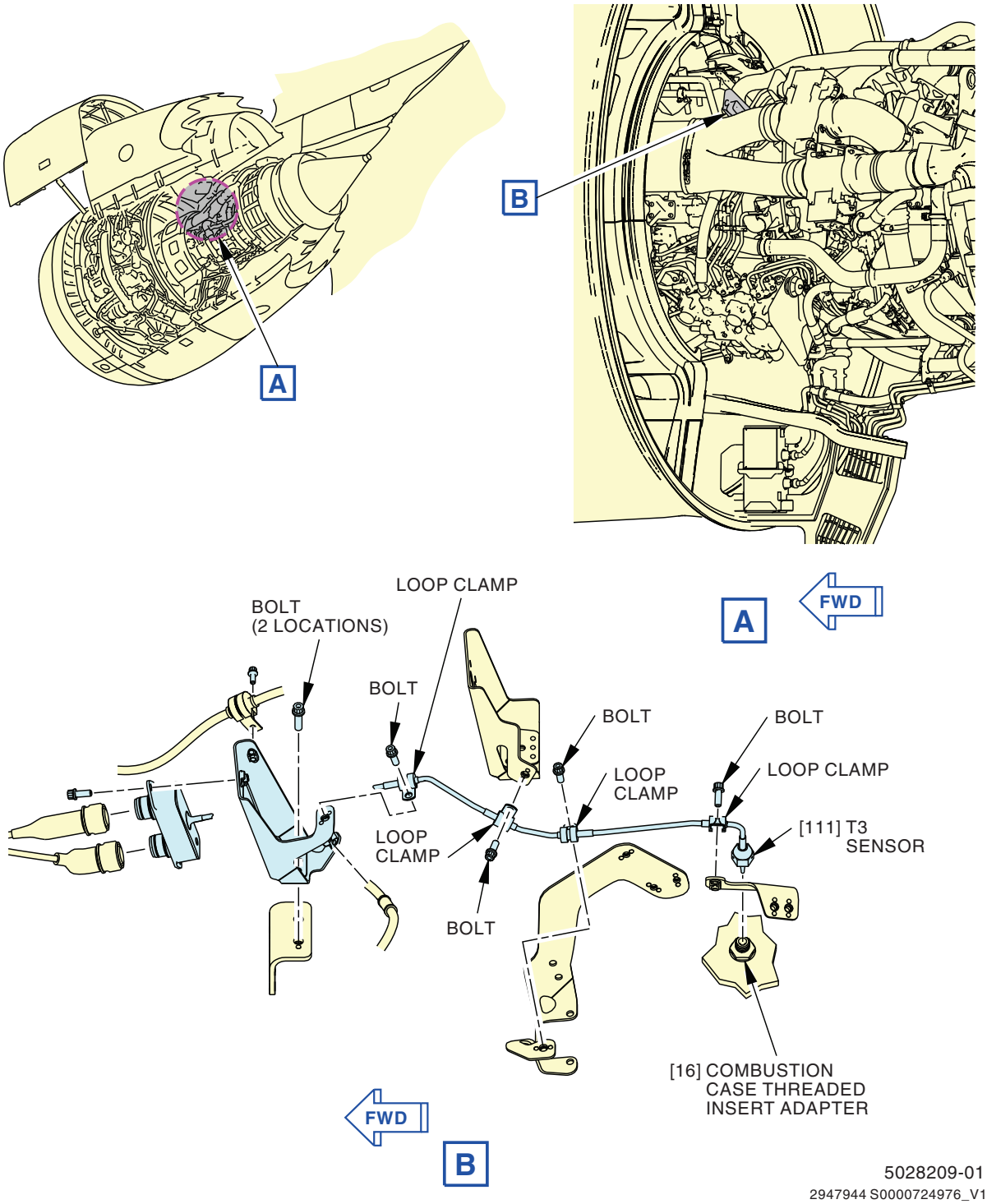


6037707-00

3129157 S0000860949_V1

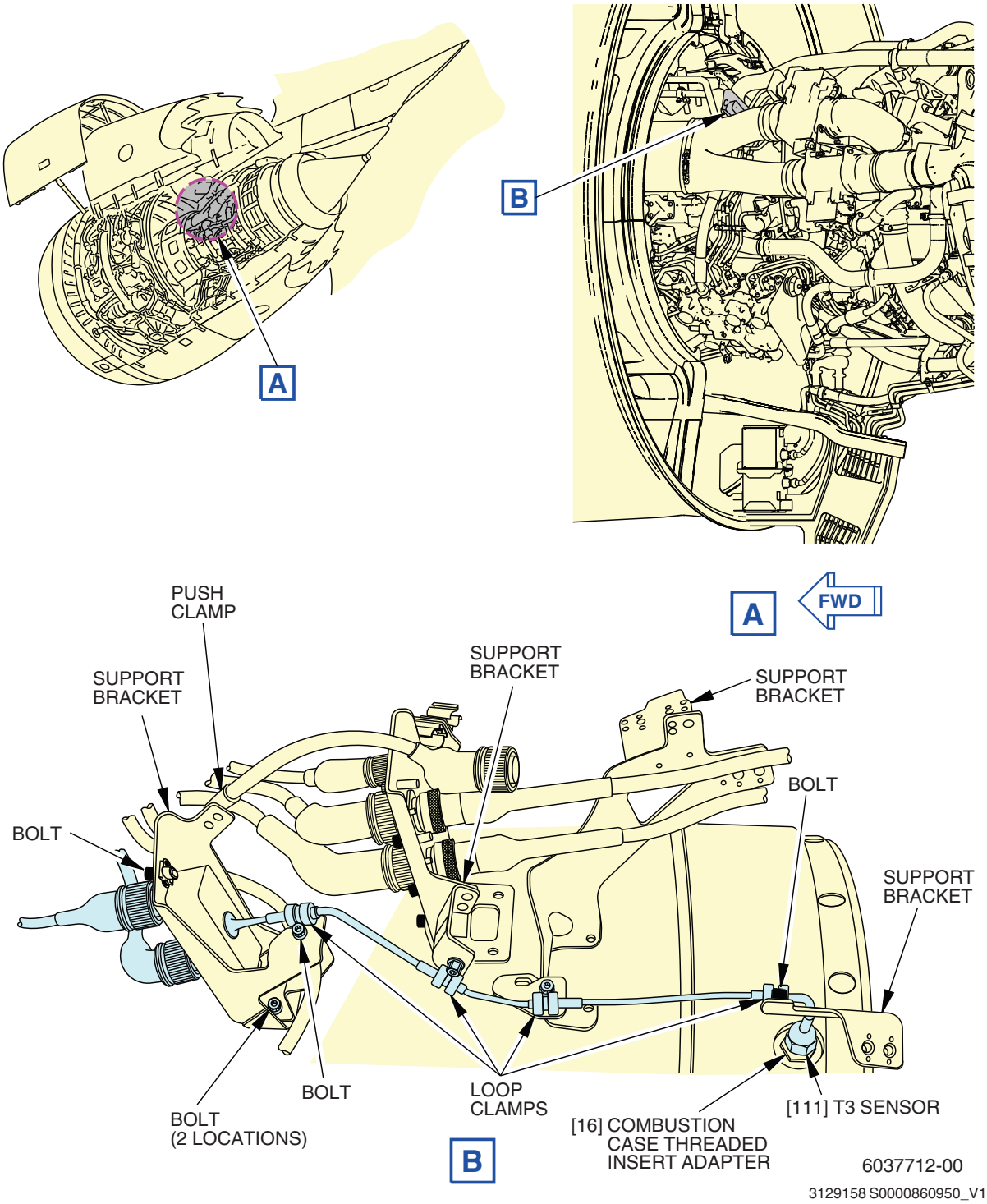
7th-Stage Air Manifold and LPT ACC Drain Tube Installation
Figure 434/73-11-05-990-864-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



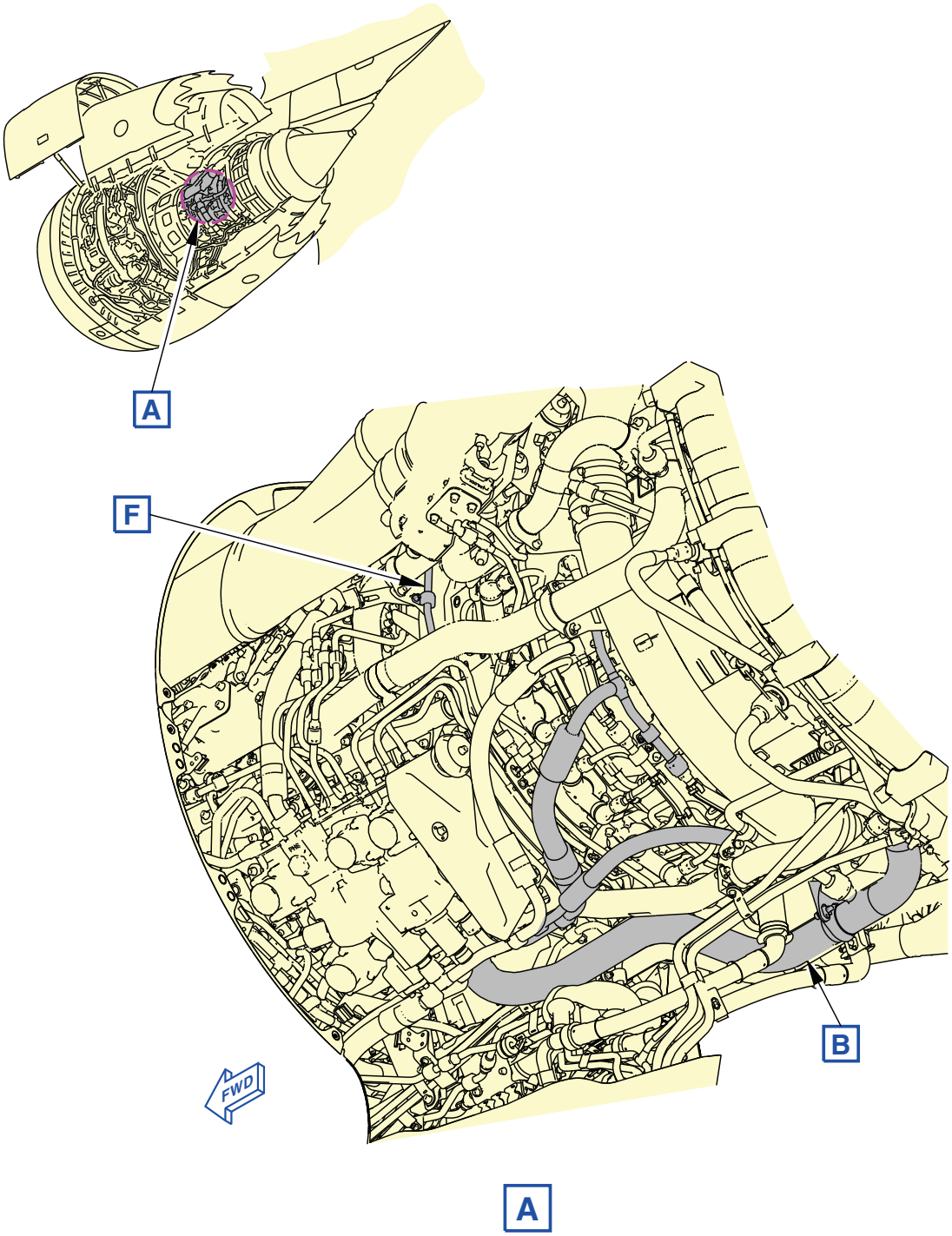
T3 Sensor Installation
Figure 435/73-11-05-990-865-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T3 Sensor Installation
Figure 435/73-11-05-990-865-G00 (Sheet 2 of 2)

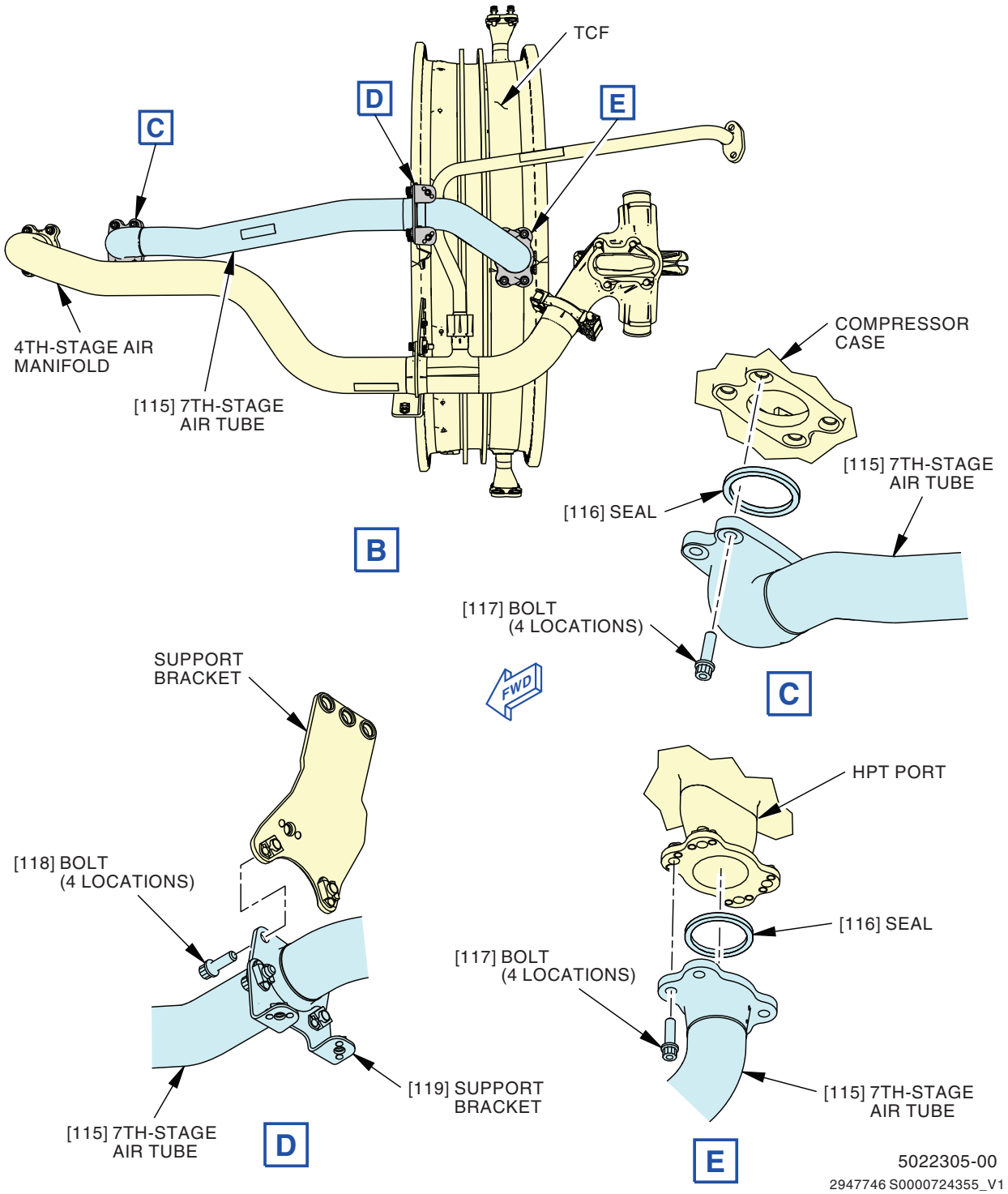
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022304-00
2947745 S0000724300_V1

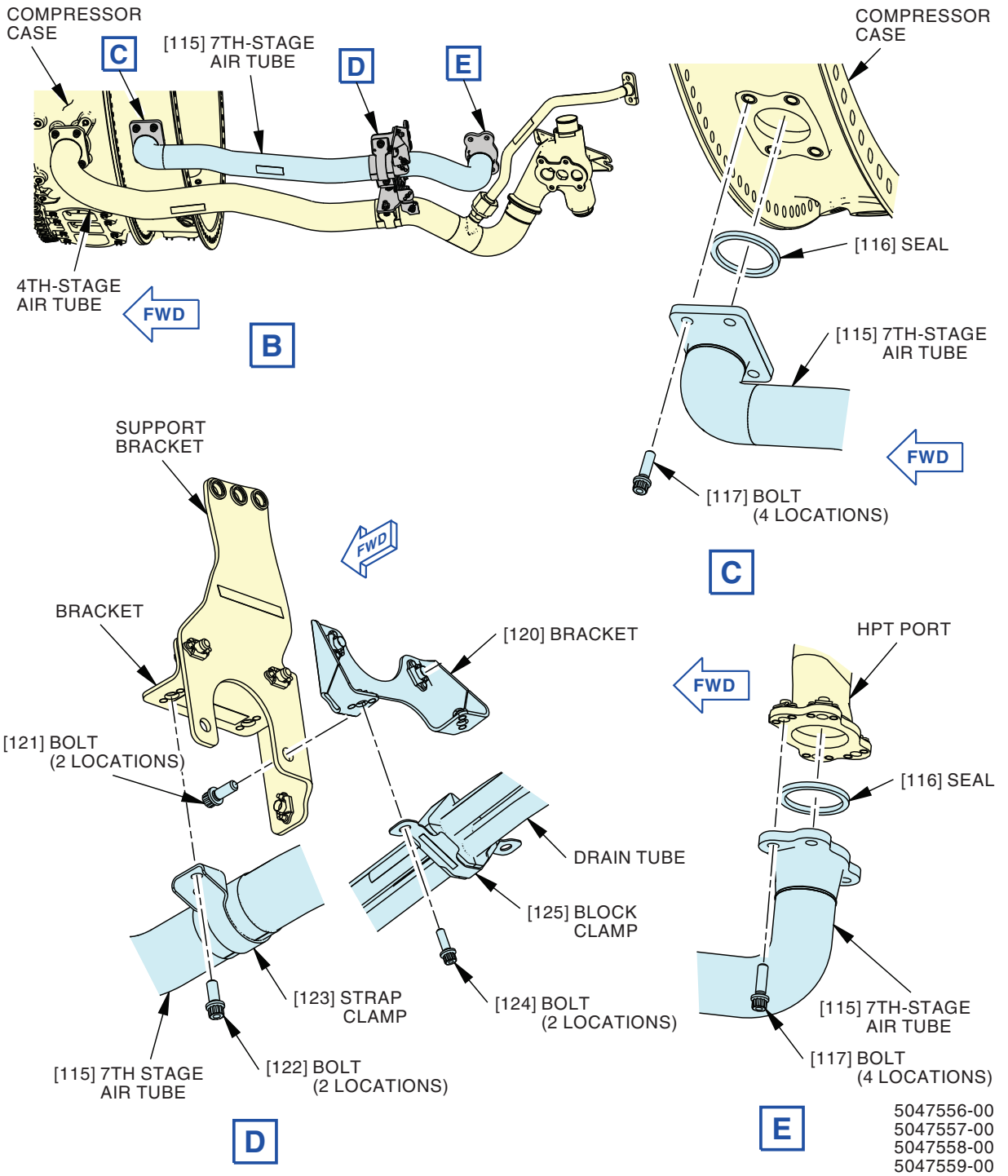
7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



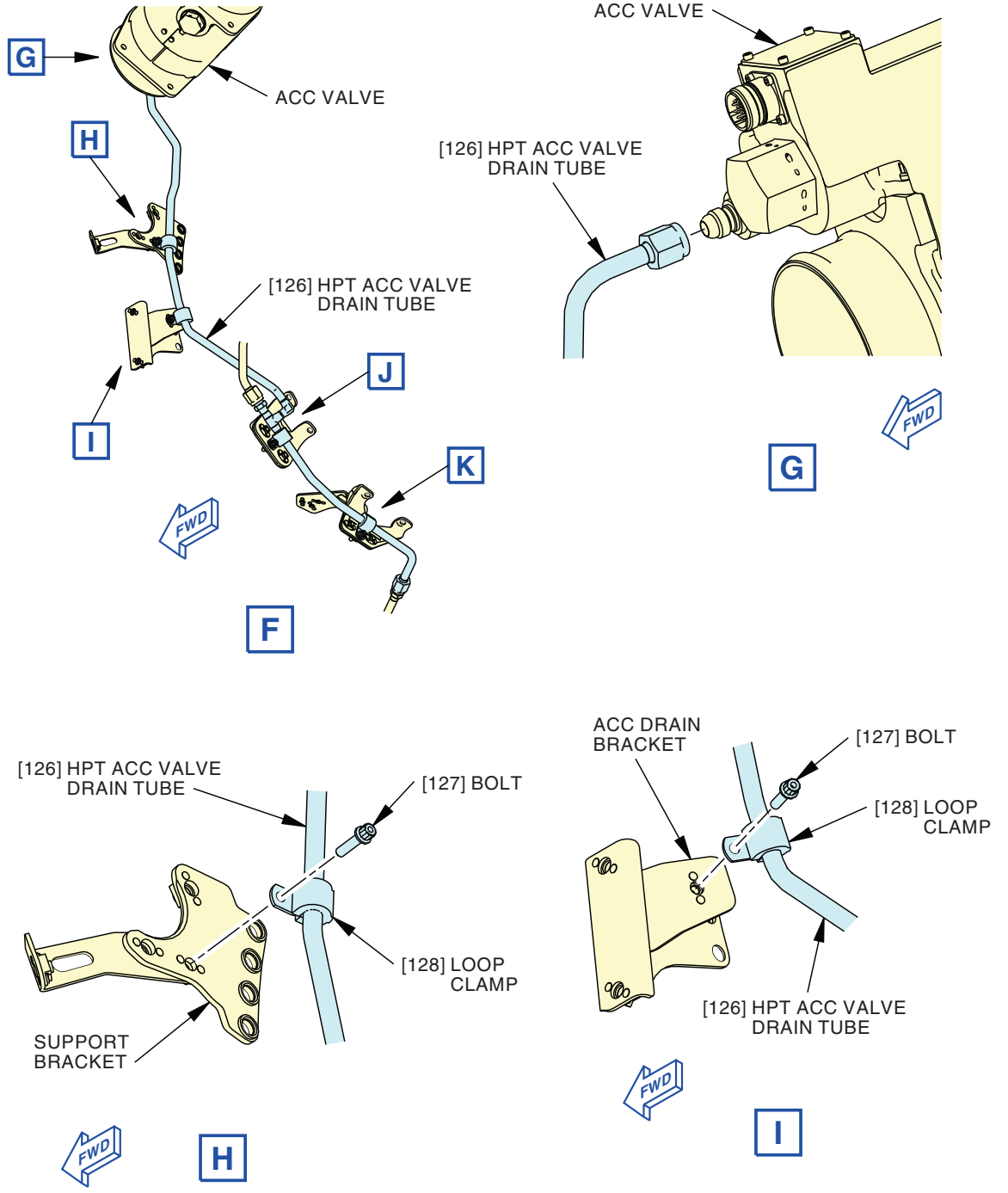
7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 3 of 7)

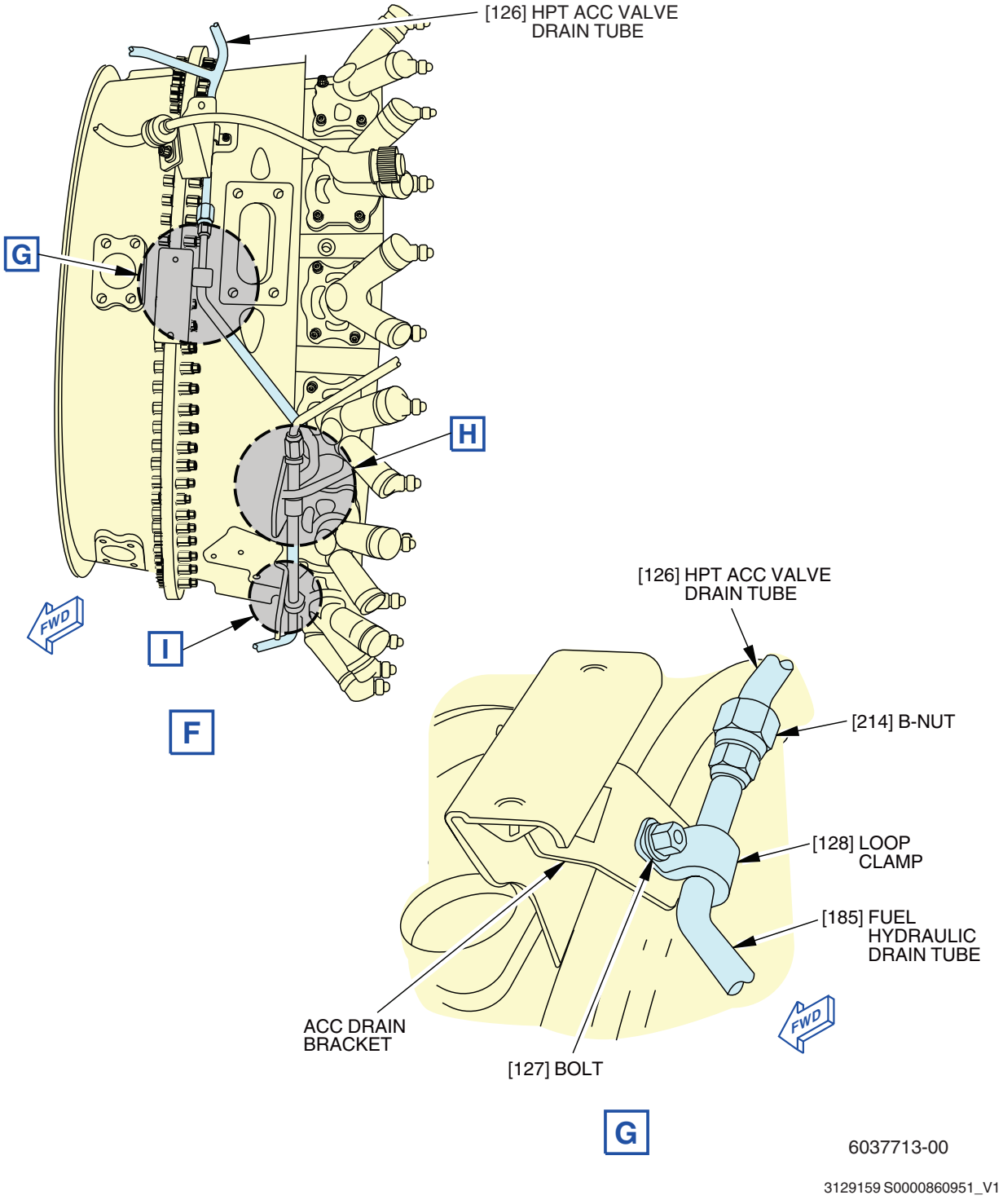
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022320-00
2947748 S0000724357_V1

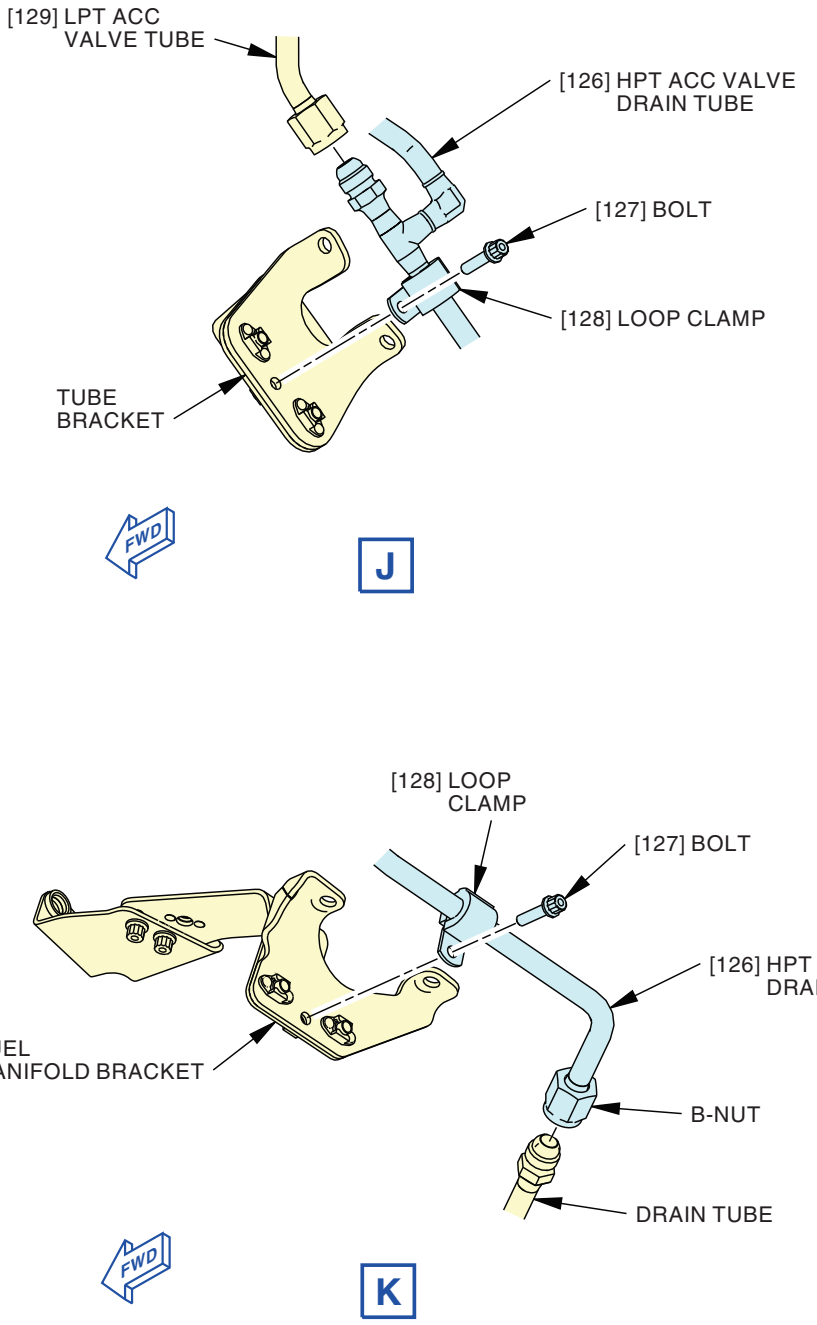
7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



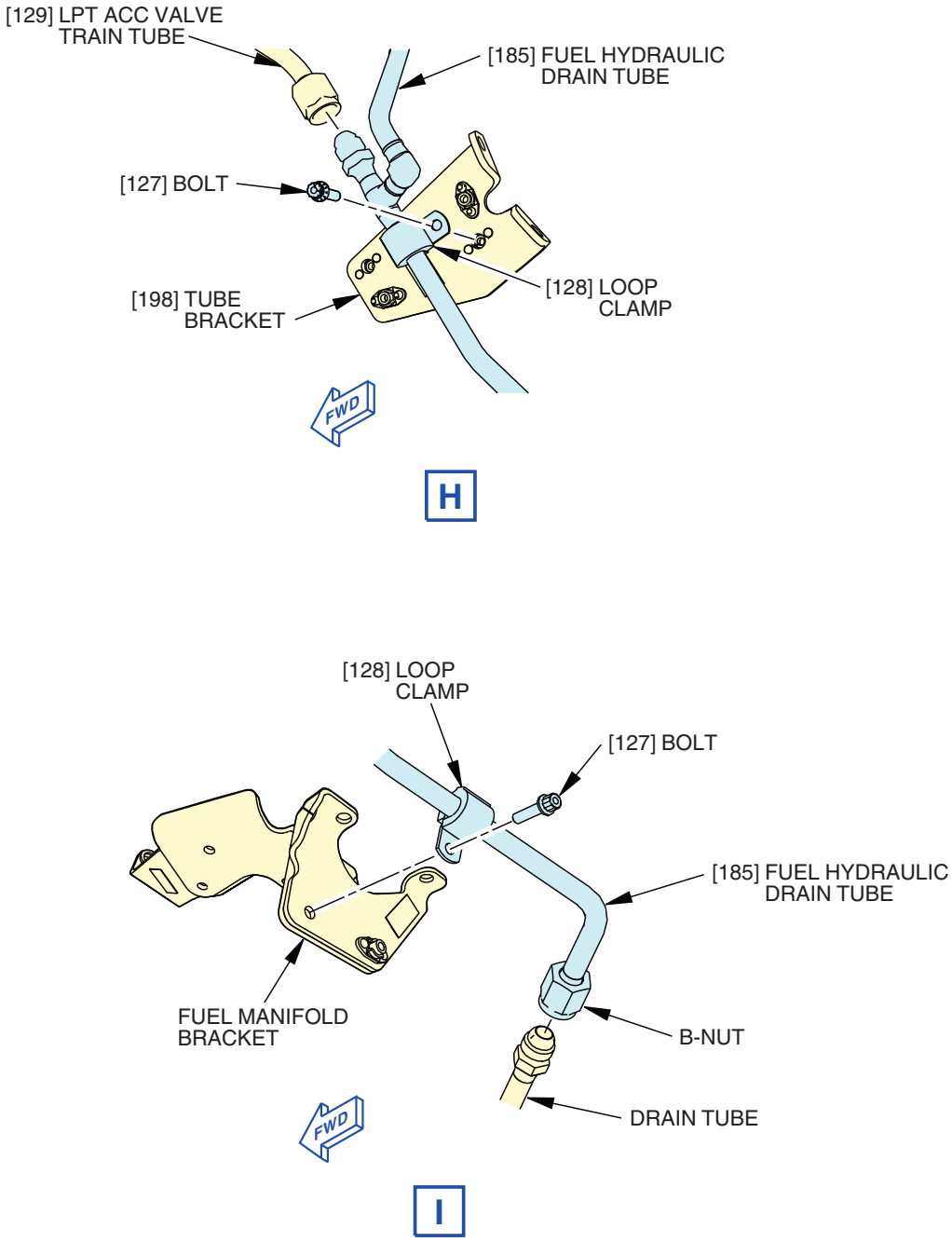
7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

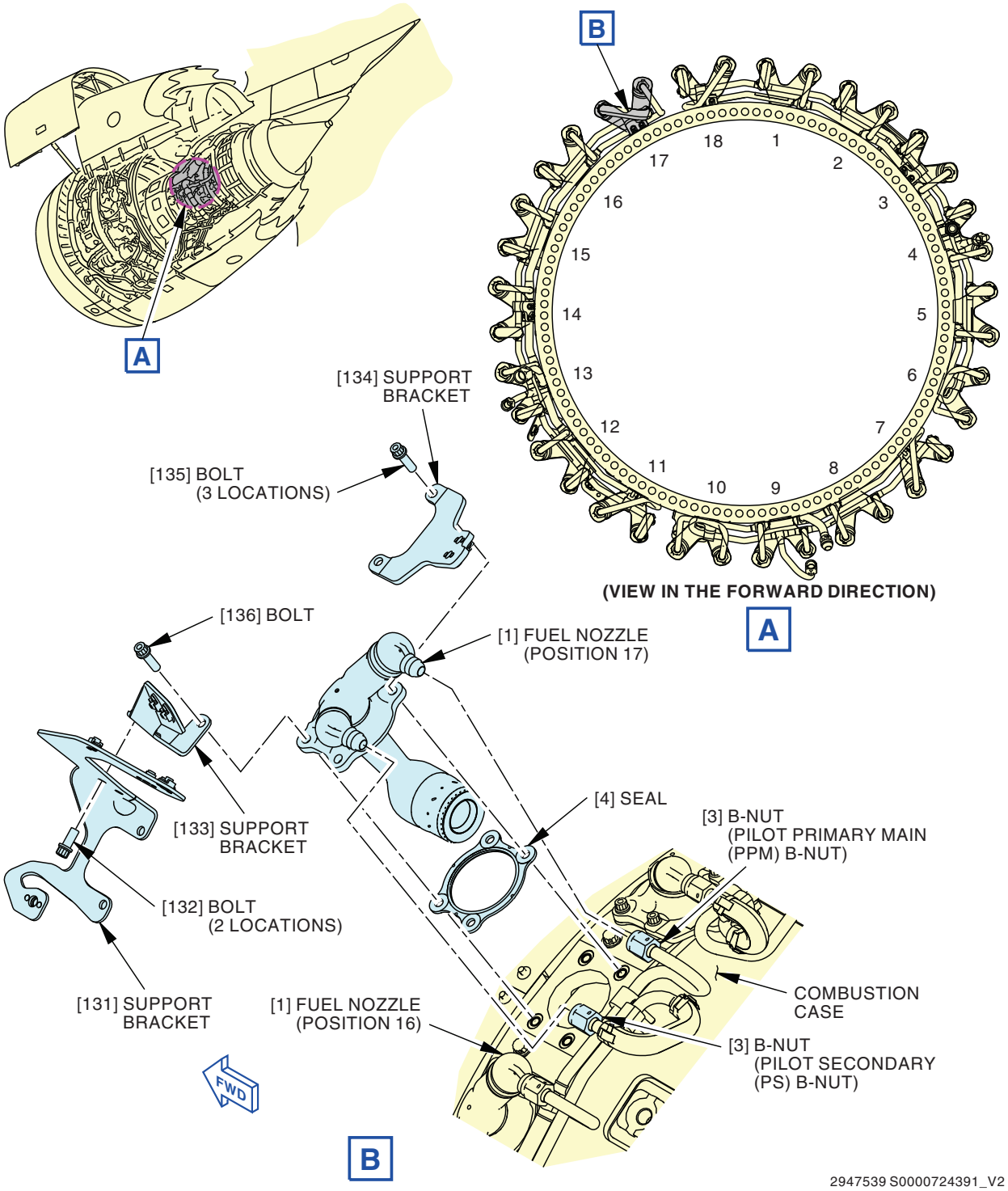


6037713-00

3129160 S0000860952_V1

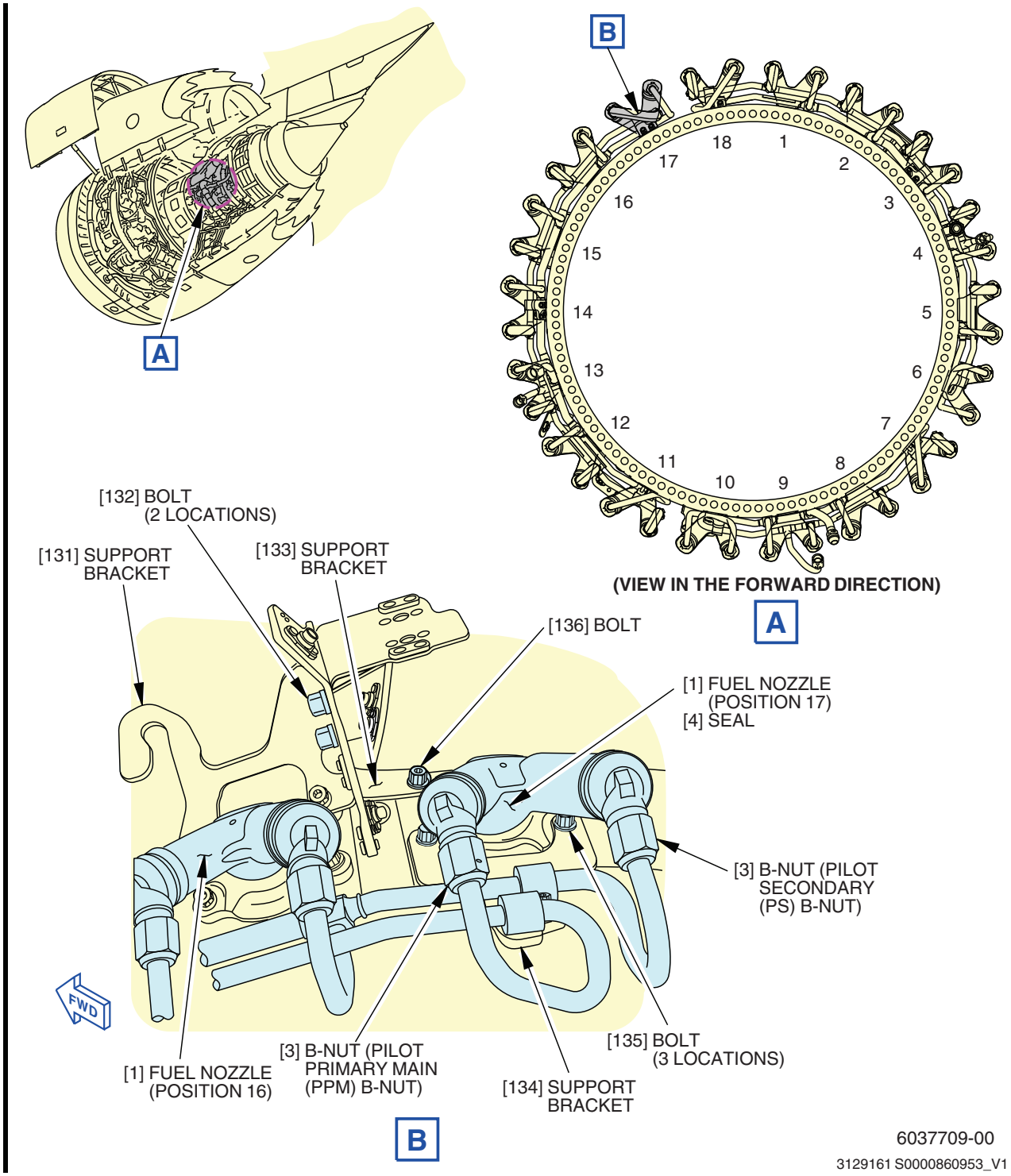
7th-stage Air Tube and HPT ACC Valve Drain Tube Installation
Figure 436/73-11-05-990-866-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



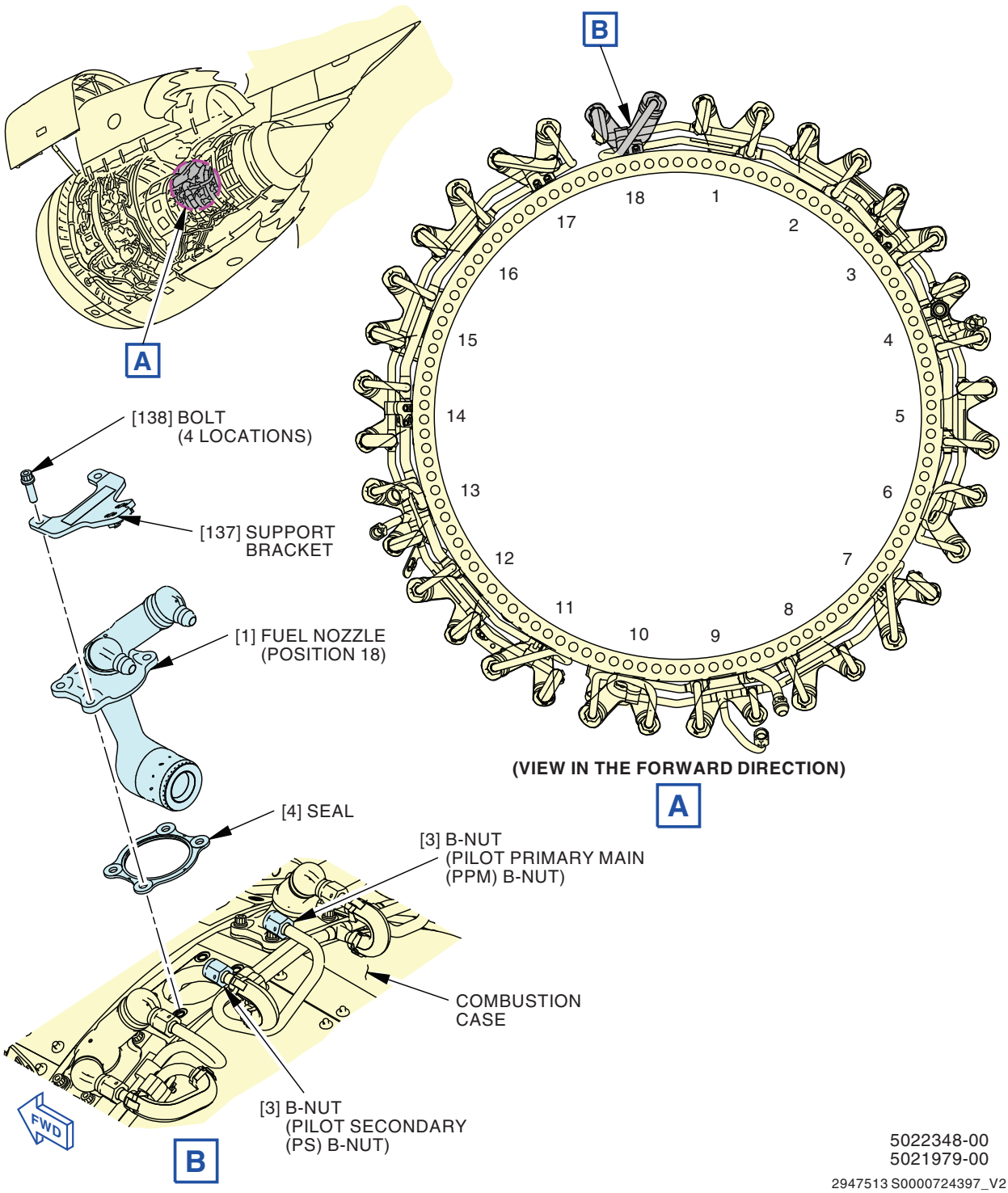
Fuel Nozzle No.17 Installation
Figure 437/73-11-05-990-867-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.17 Installation
Figure 437/73-11-05-990-867-G00 (Sheet 2 of 2)

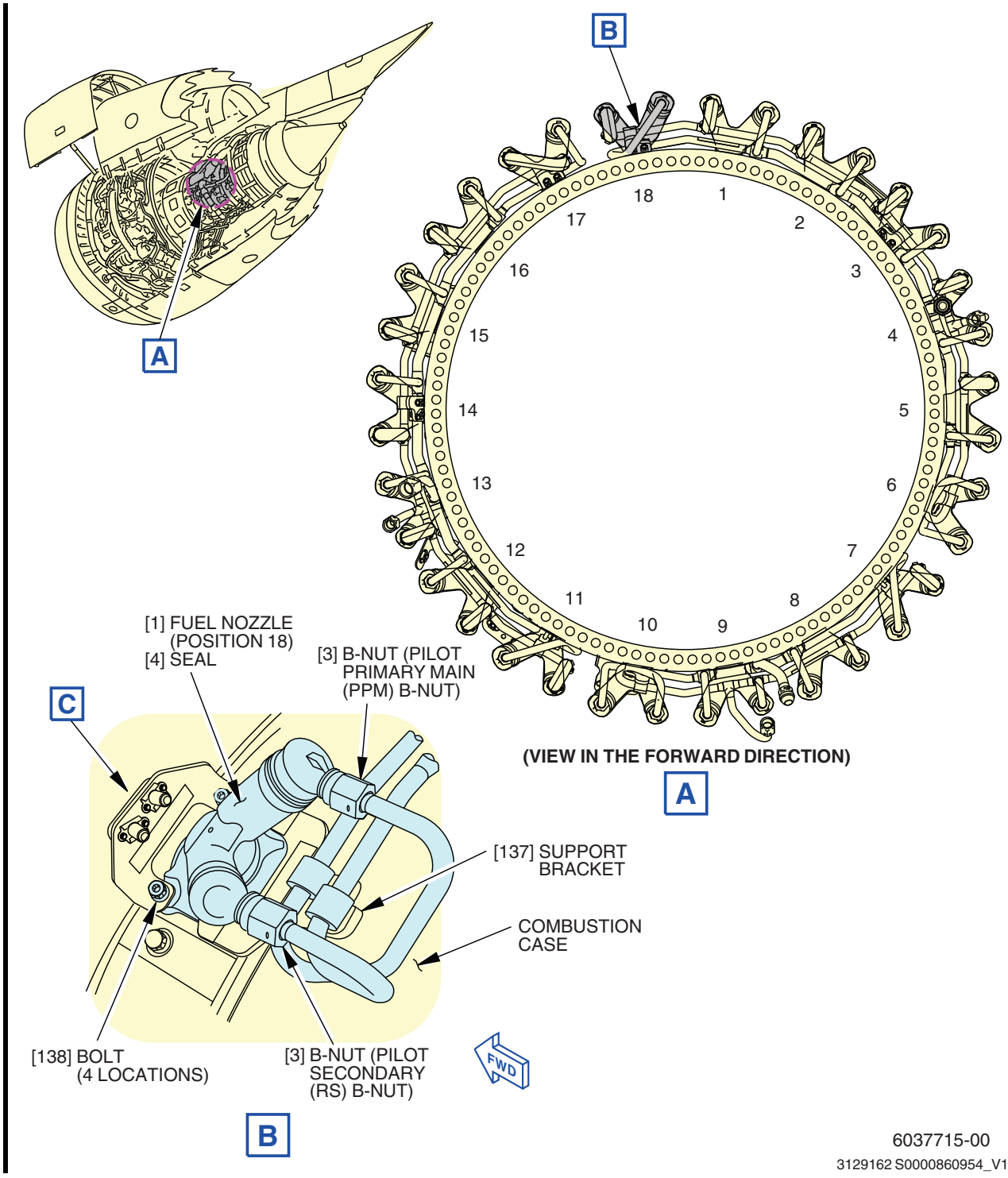
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022348-00
5021979-00
2947513 S0000724397_V2

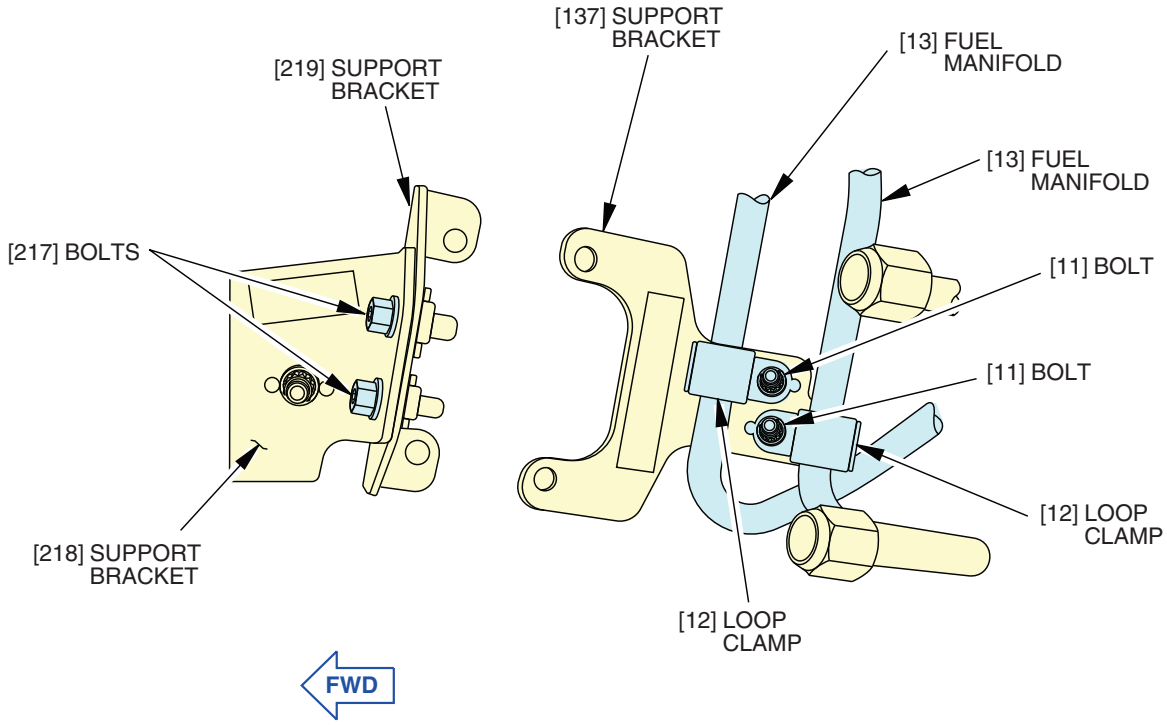
Fuel Nozzle No.18 Installation
Figure 438/73-11-05-990-868-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.18 Installation
Figure 438/73-11-05-990-868-G00 (Sheet 2 of 3)

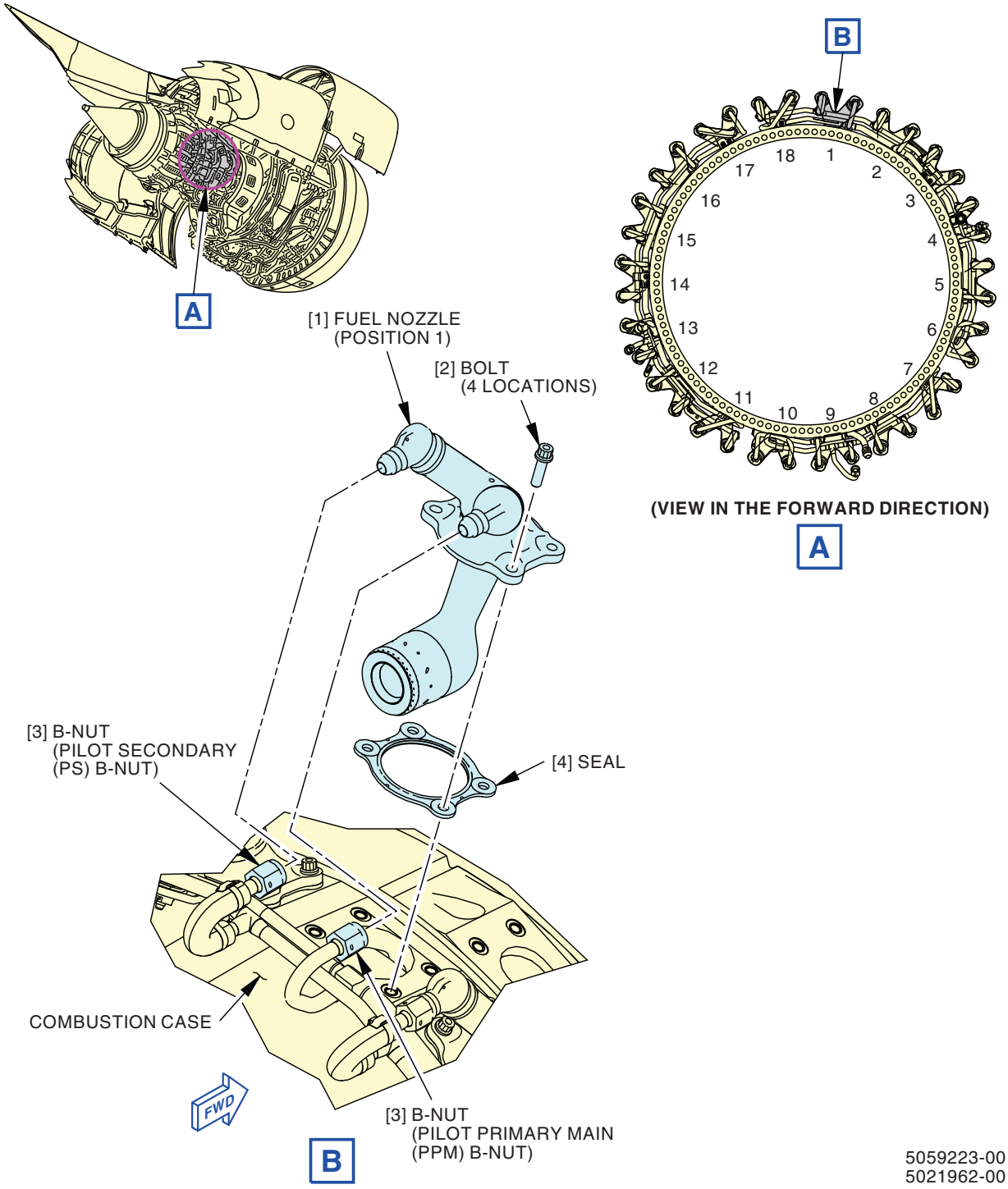
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6037715-00
3129163 S0000860955_V1

Fuel Nozzle No.18 Installation
Figure 438/73-11-05-990-868-G00 (Sheet 3 of 3)

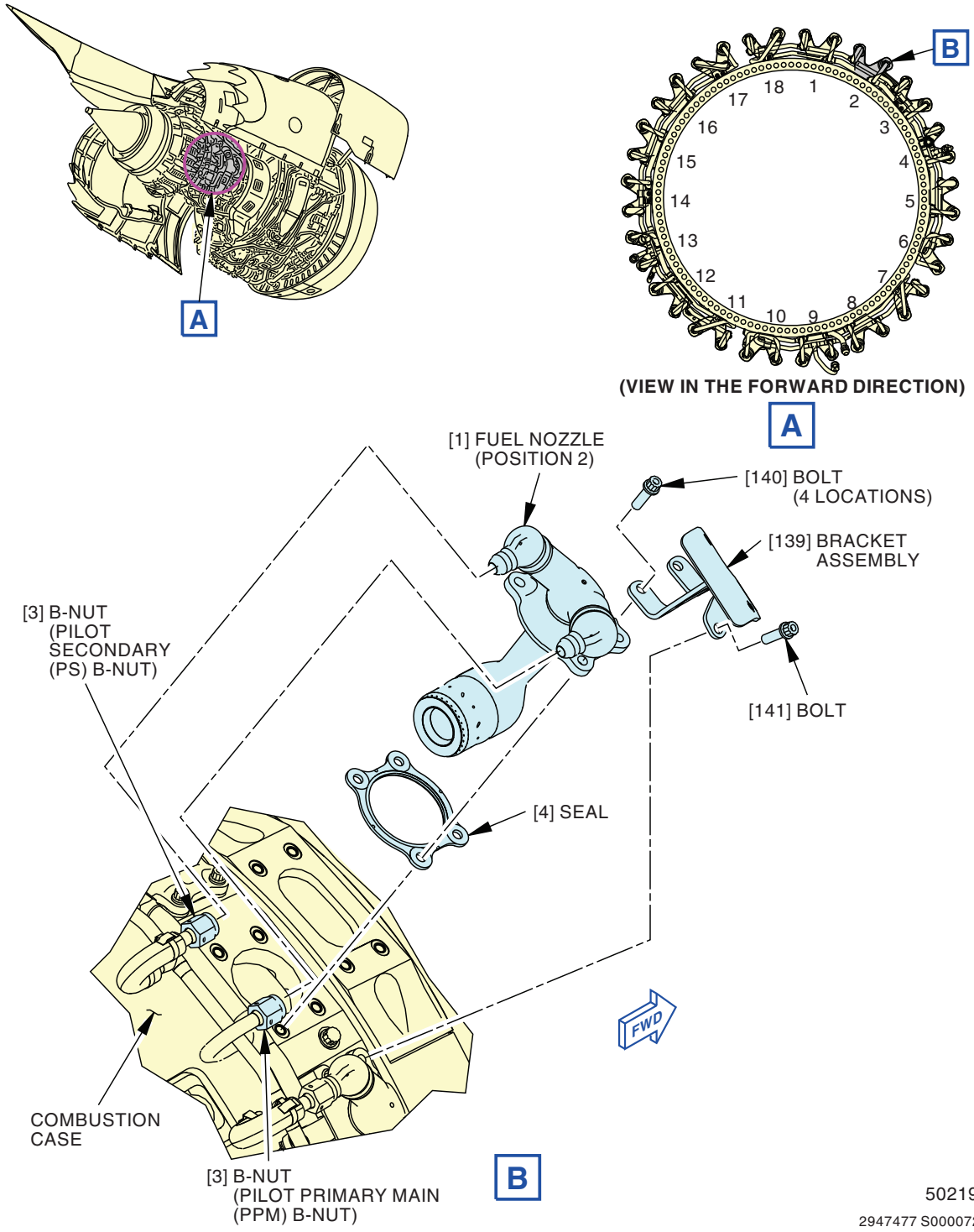
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059223-00
5021962-00
2947471 S0000723958_V2

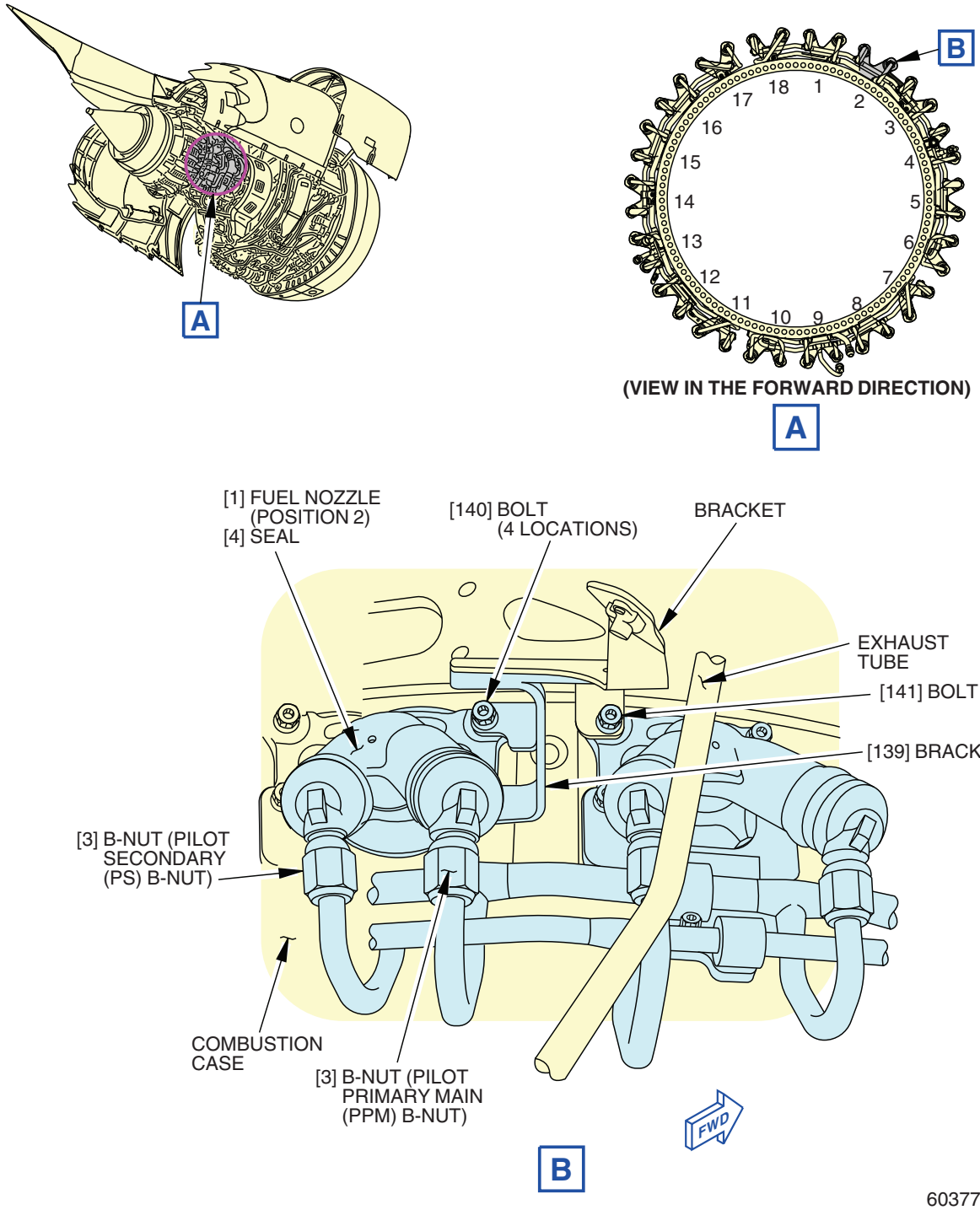
Fuel Nozzle No.1 Installation
Figure 439/73-11-05-990-869-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



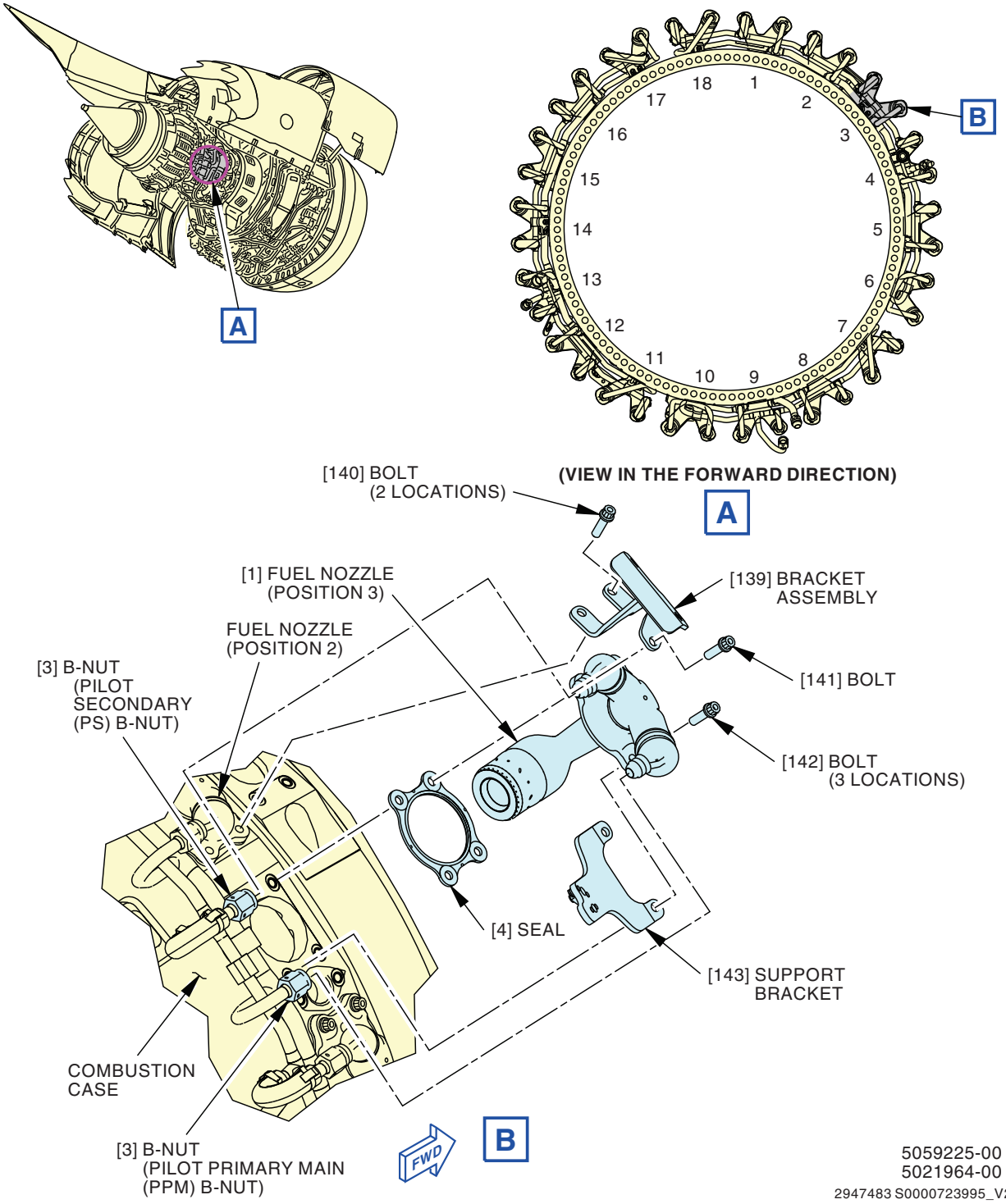
Fuel Nozzle No.2 Installation
Figure 440/73-11-05-990-870-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



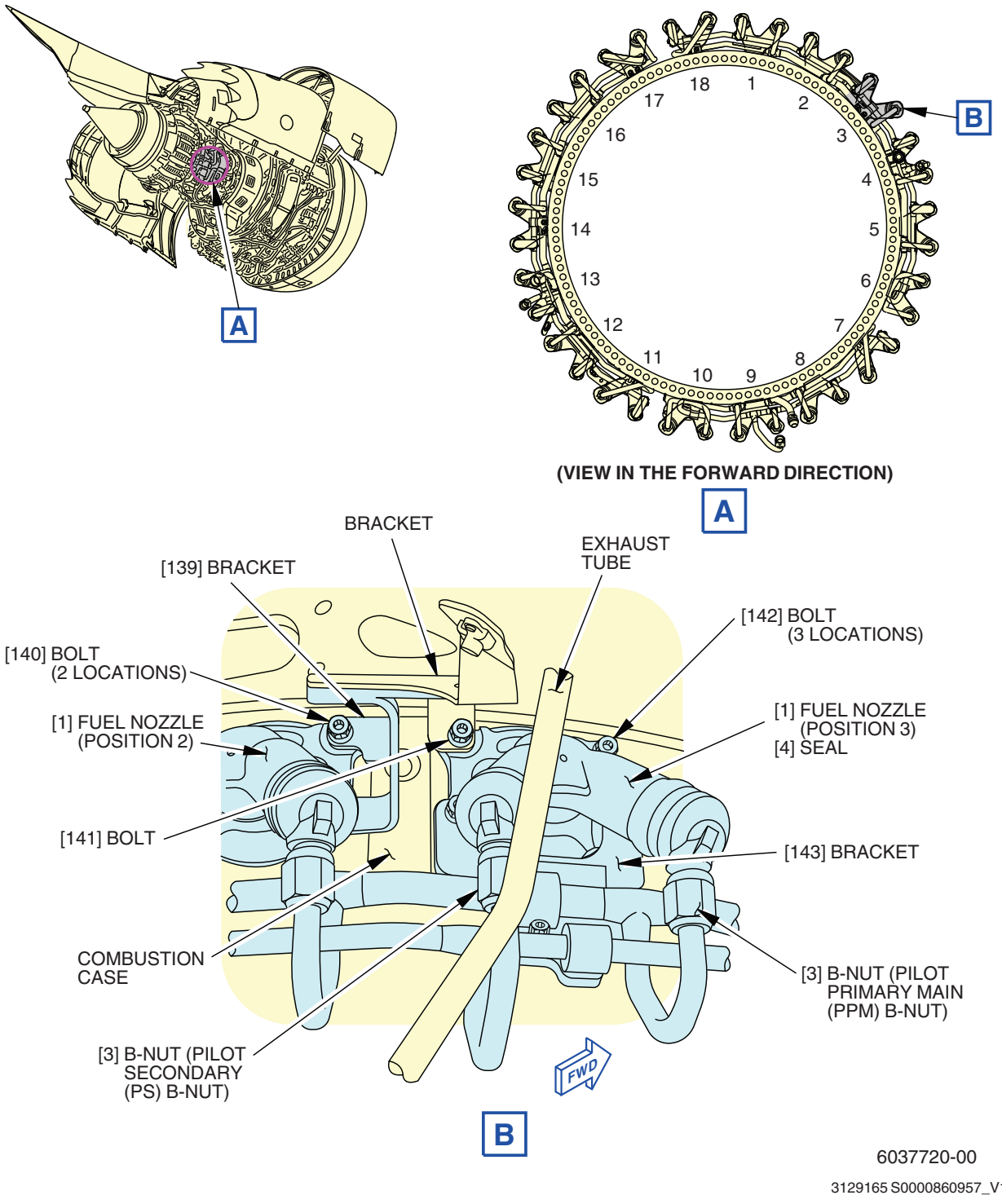
Fuel Nozzle No.2 Installation
Figure 440/73-11-05-990-870-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



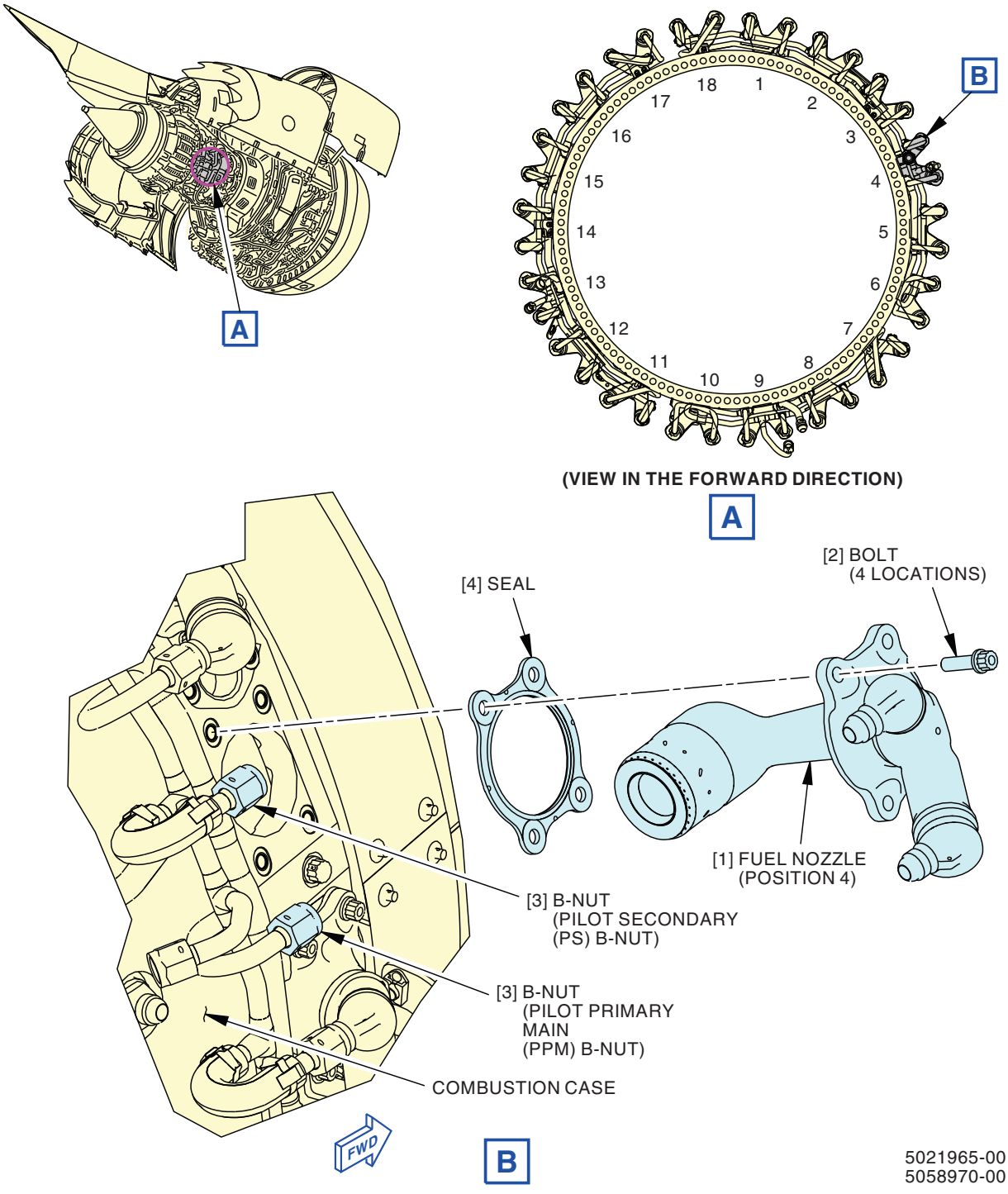
Fuel Nozzle No.3 Installation
Figure 441/73-11-05-990-871-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



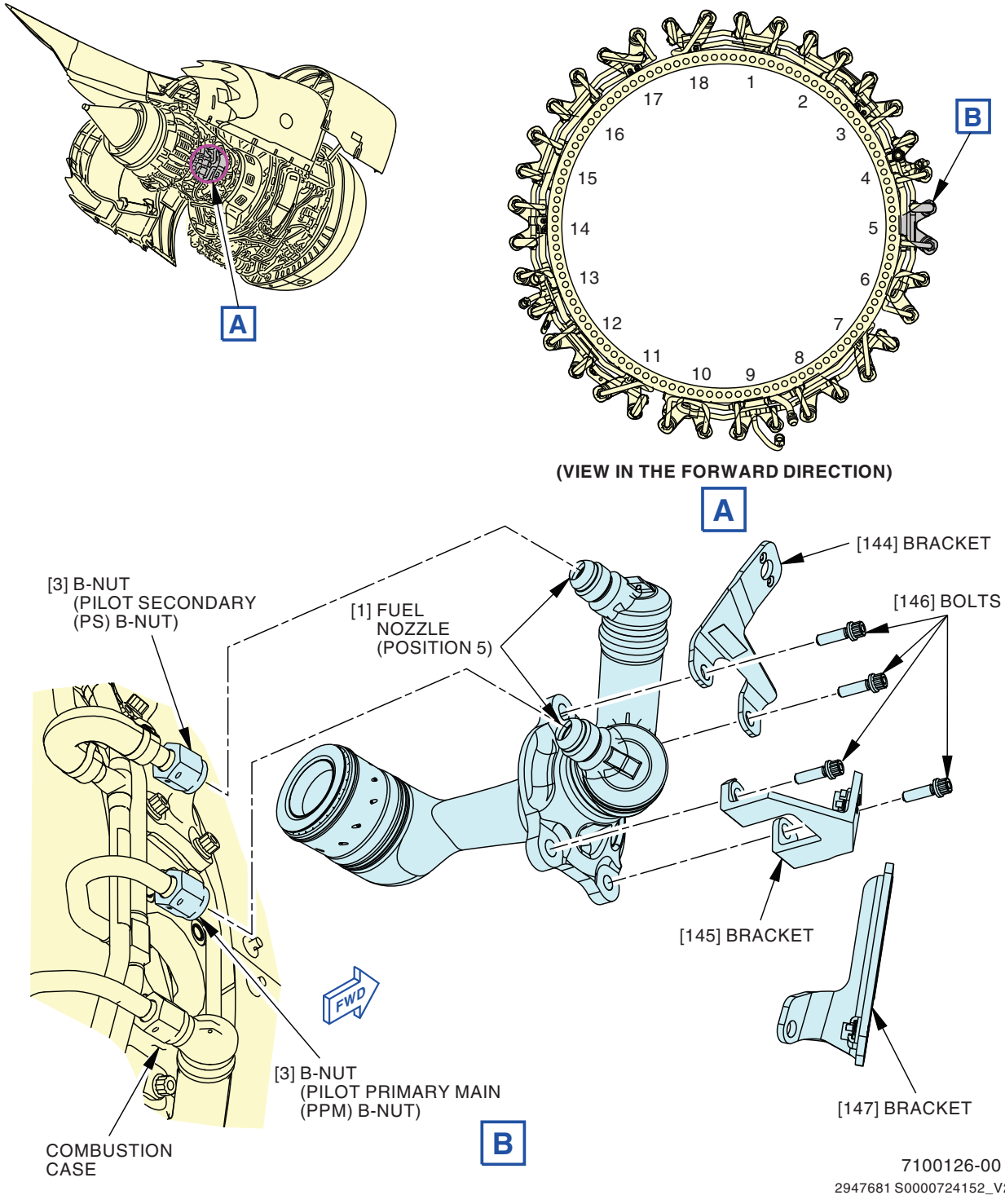
Fuel Nozzle No.3 Installation
Figure 441/73-11-05-990-871-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



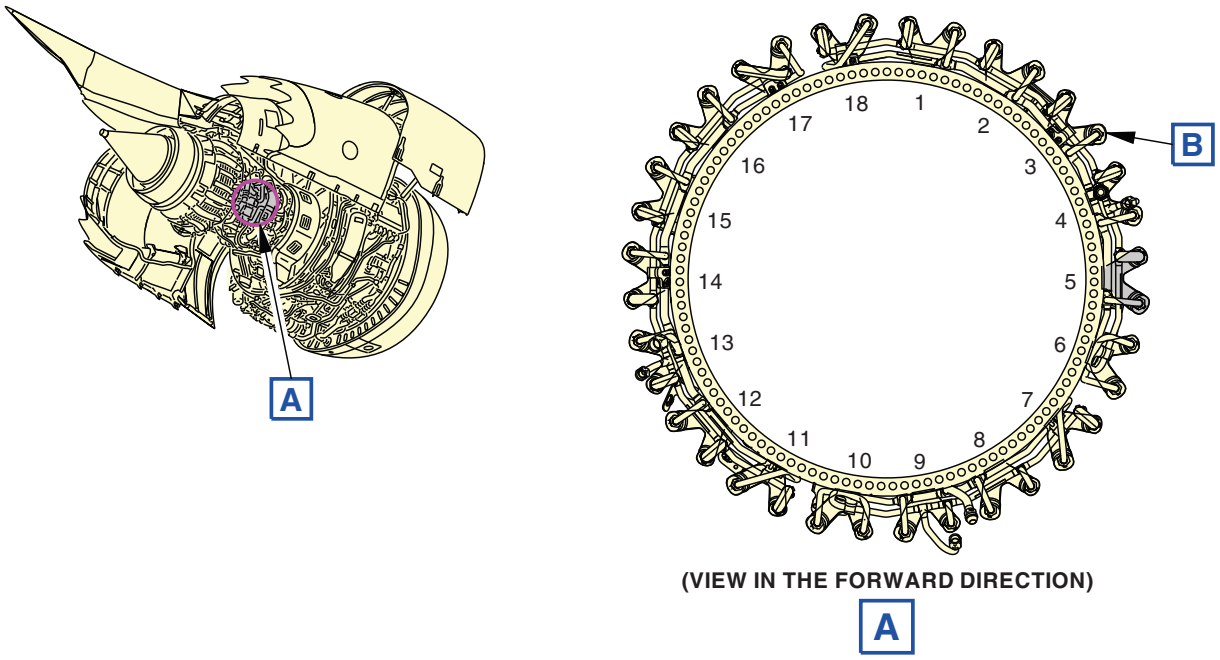
Fuel Nozzle No.4 Installation
Figure 442/73-11-05-990-872-G00

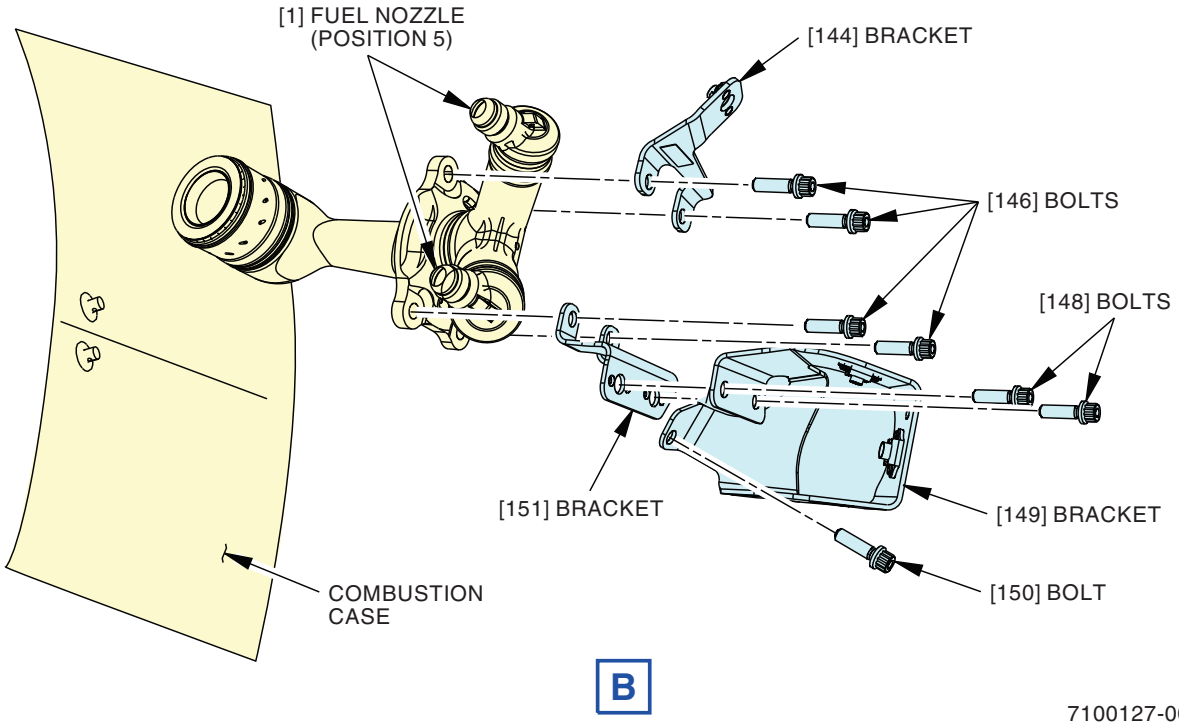
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.5 Installation
Figure 443/73-11-05-990-873-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

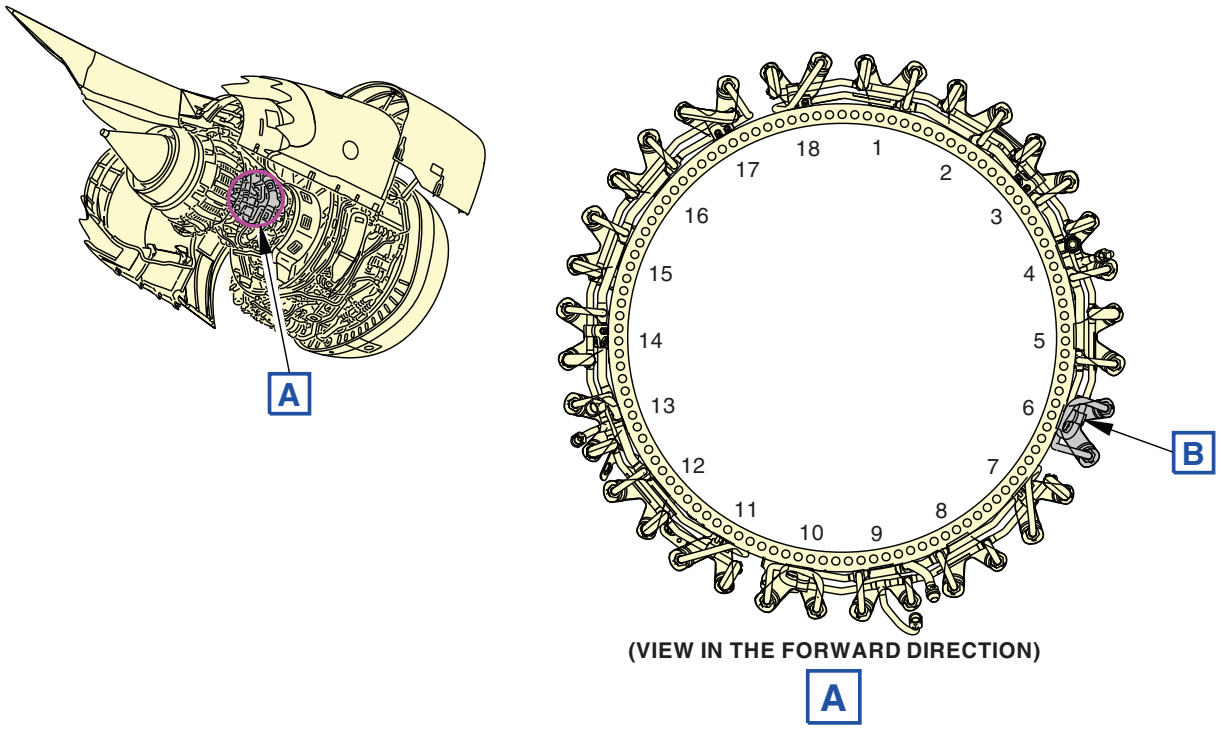




7100127-00
2947682 S0000724158_V1

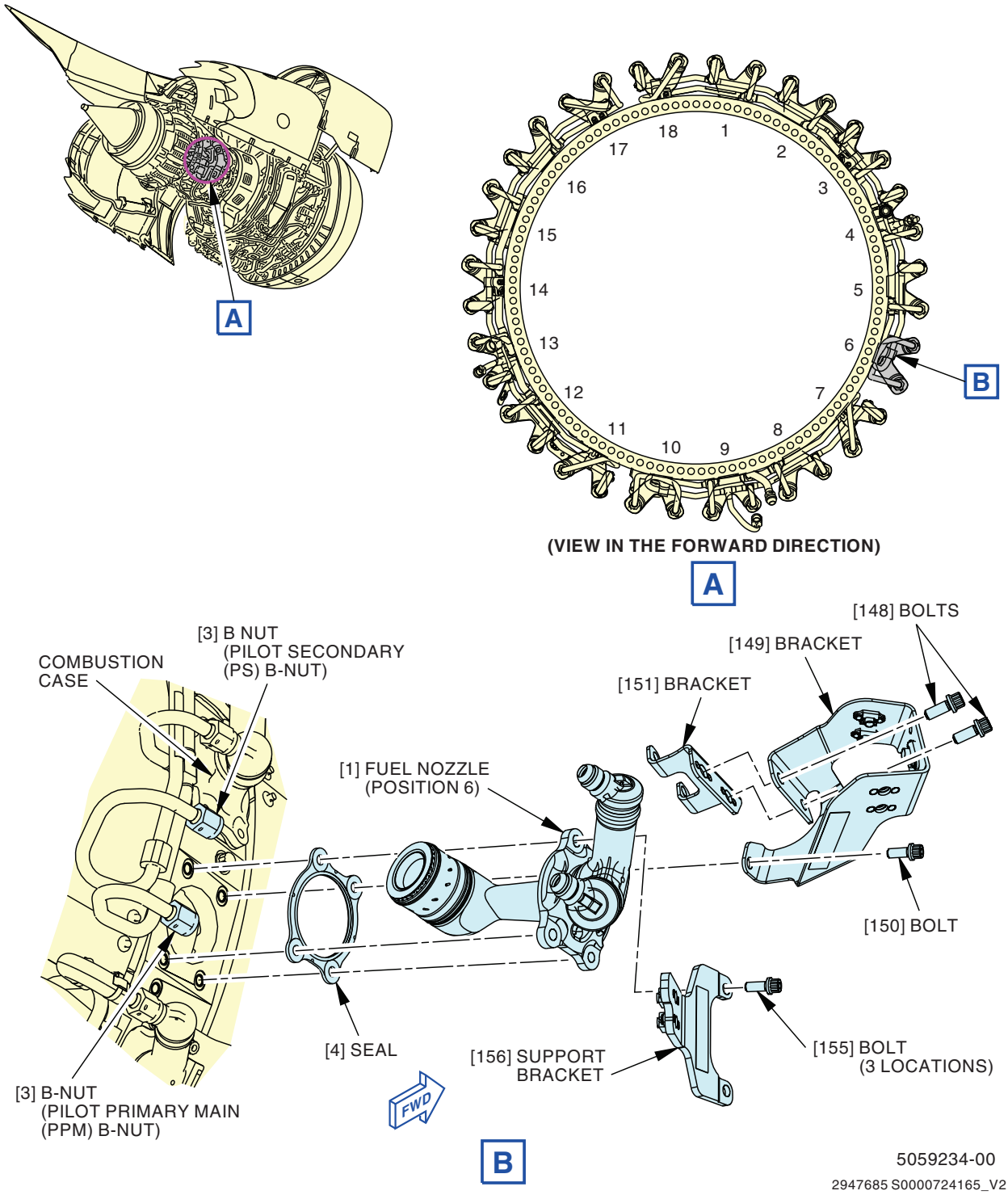
Fuel Nozzle No.5 Installation
Figure 443/73-11-05-990-873-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



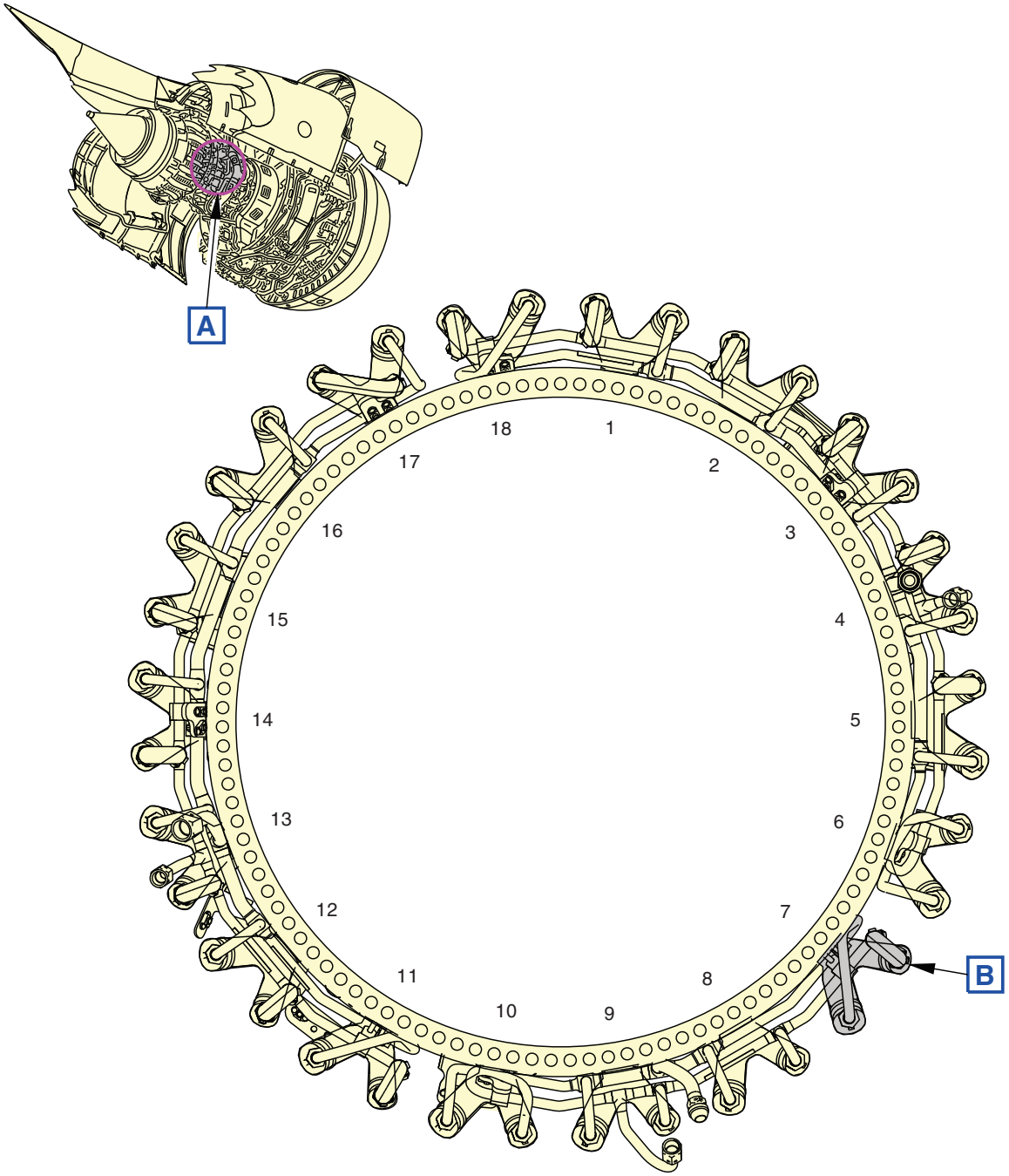
Fuel Nozzle No.6 Installation
Figure 444/73-11-05-990-874-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



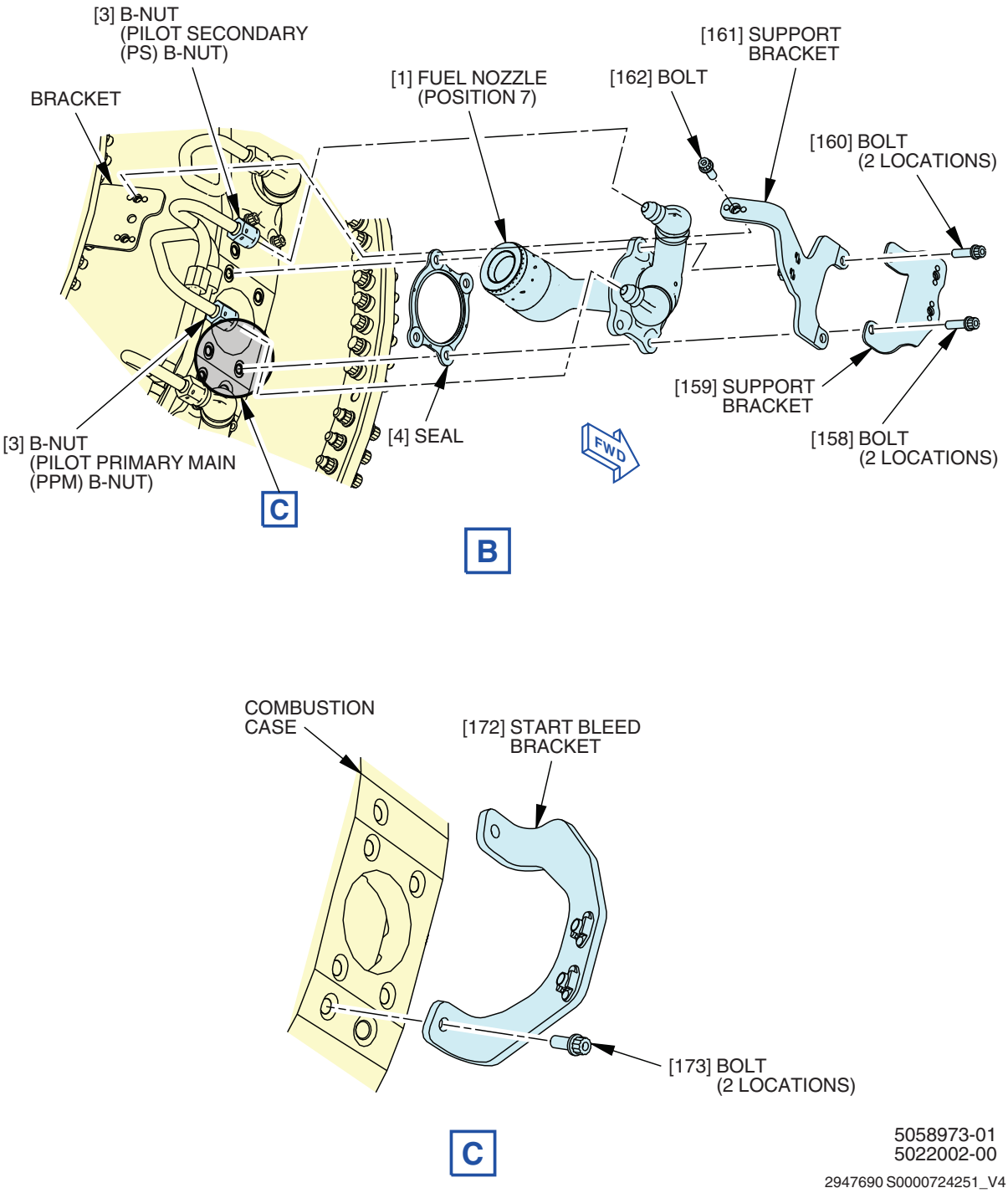
Fuel Nozzle No.6 Installation
Figure 444/73-11-05-990-874-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



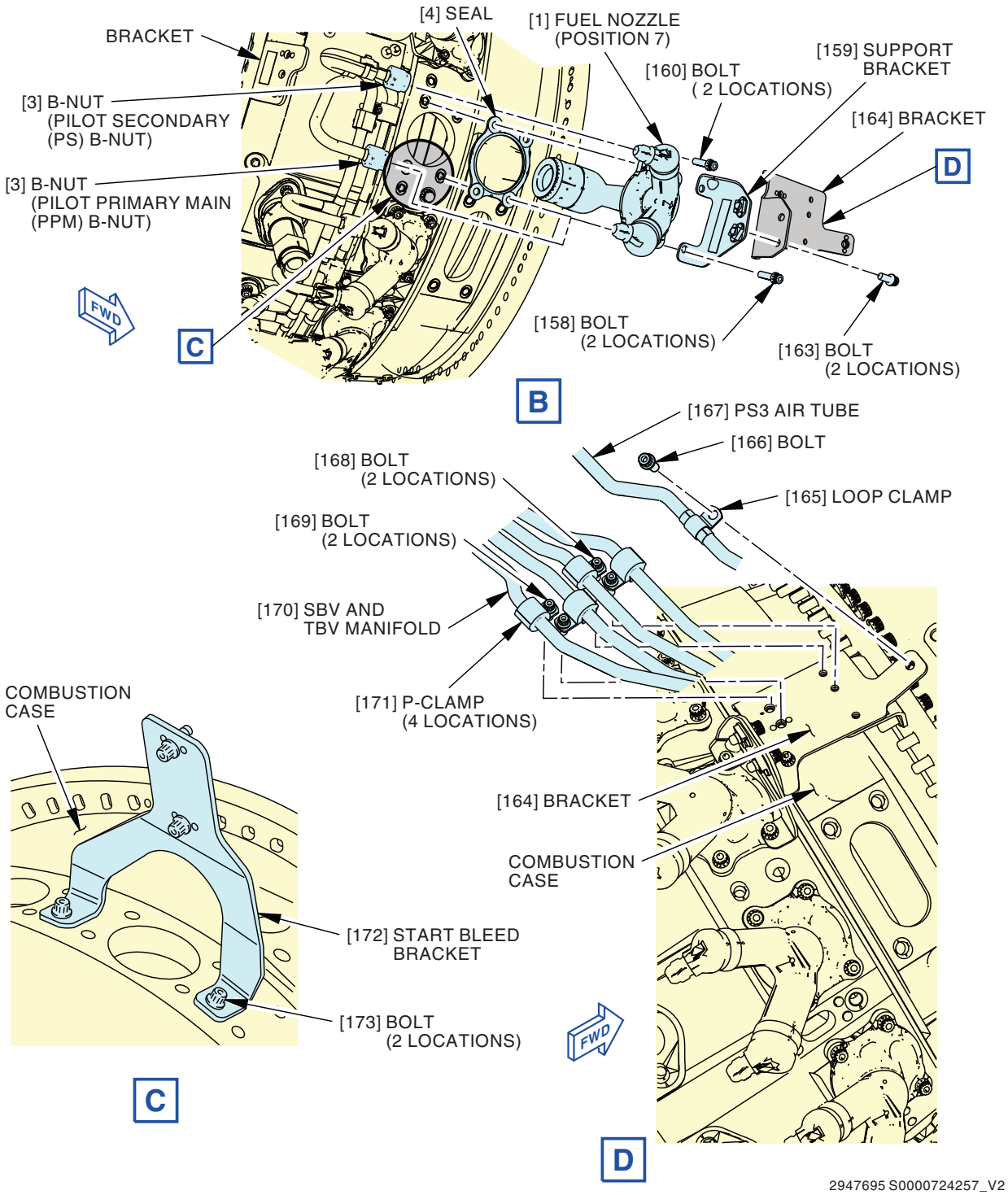
Fuel Nozzle No.7 Installation
Figure 445/73-11-05-990-875-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



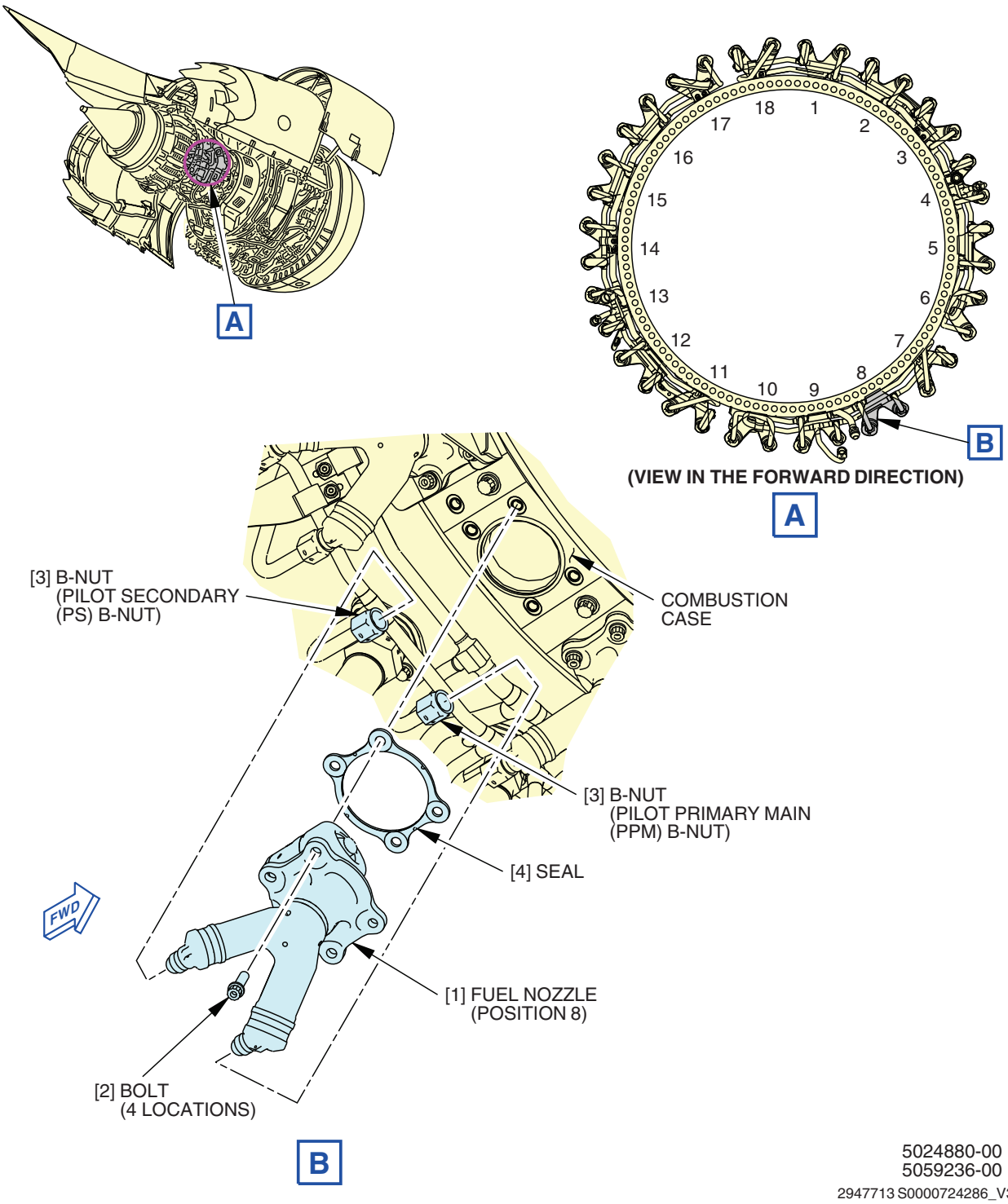
Fuel Nozzle No.7 Installation
Figure 445/73-11-05-990-875-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.7 Installation
Figure 445/73-11-05-990-875-G00 (Sheet 3 of 3)

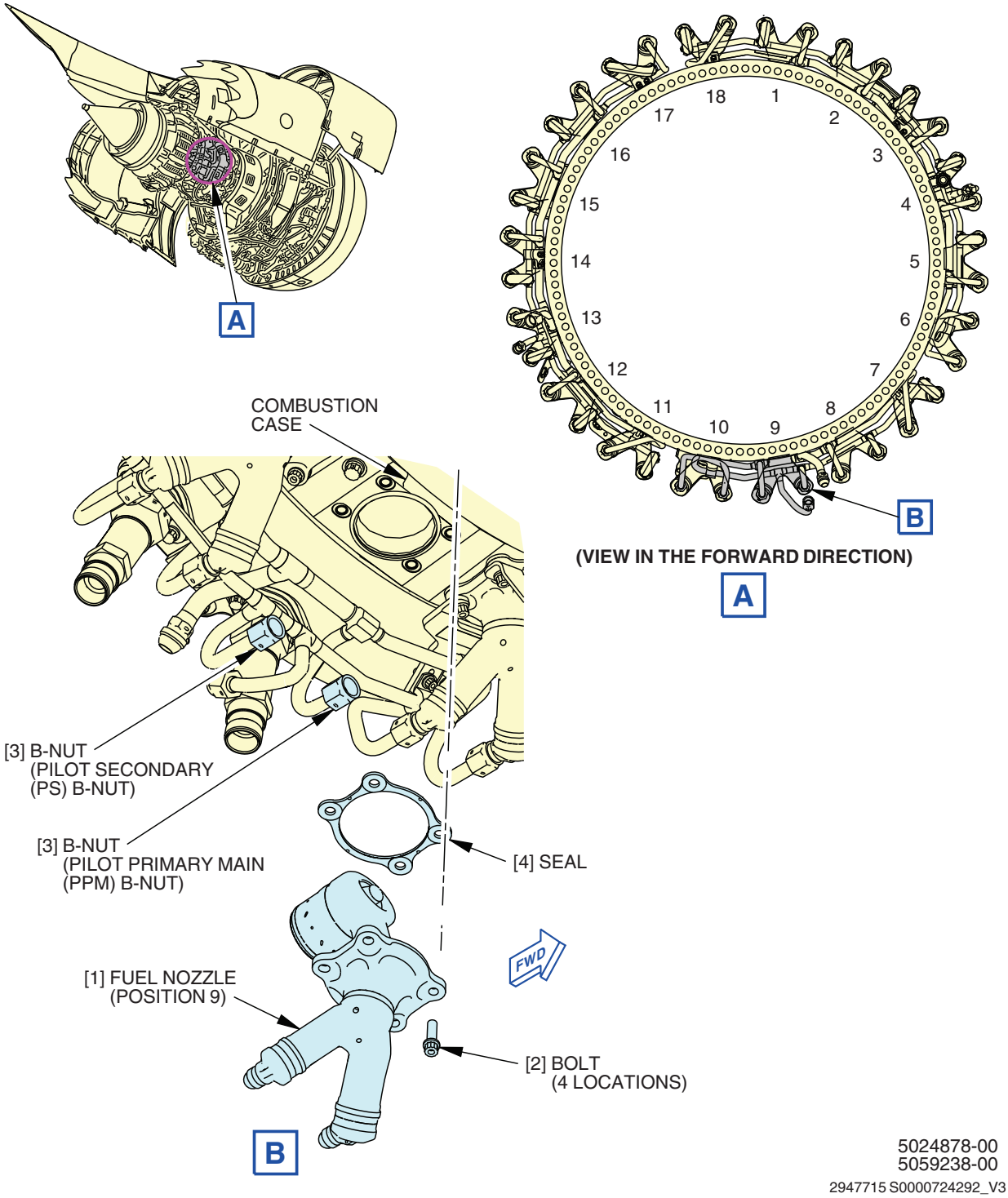
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5024880-00
5059236-00
2947713 S0000724286_V3

Fuel Nozzle No.8 Installation
Figure 446/73-11-05-990-876-G00

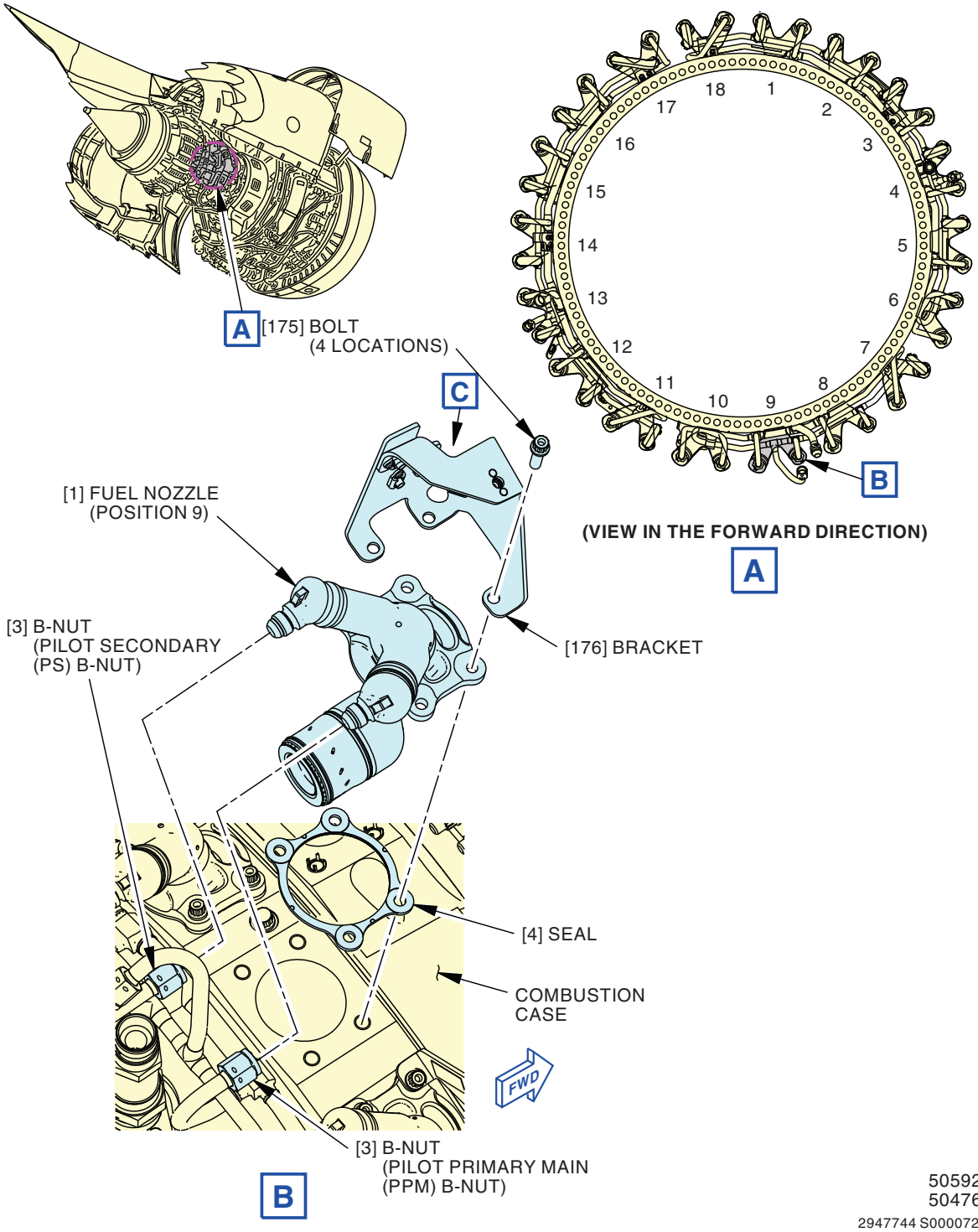
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5024878-00
5059238-00
2947715 S0000724292_V3

Fuel Nozzle No.9 Installation
Figure 447/73-11-05-990-877-G00 (Sheet 1 of 3)

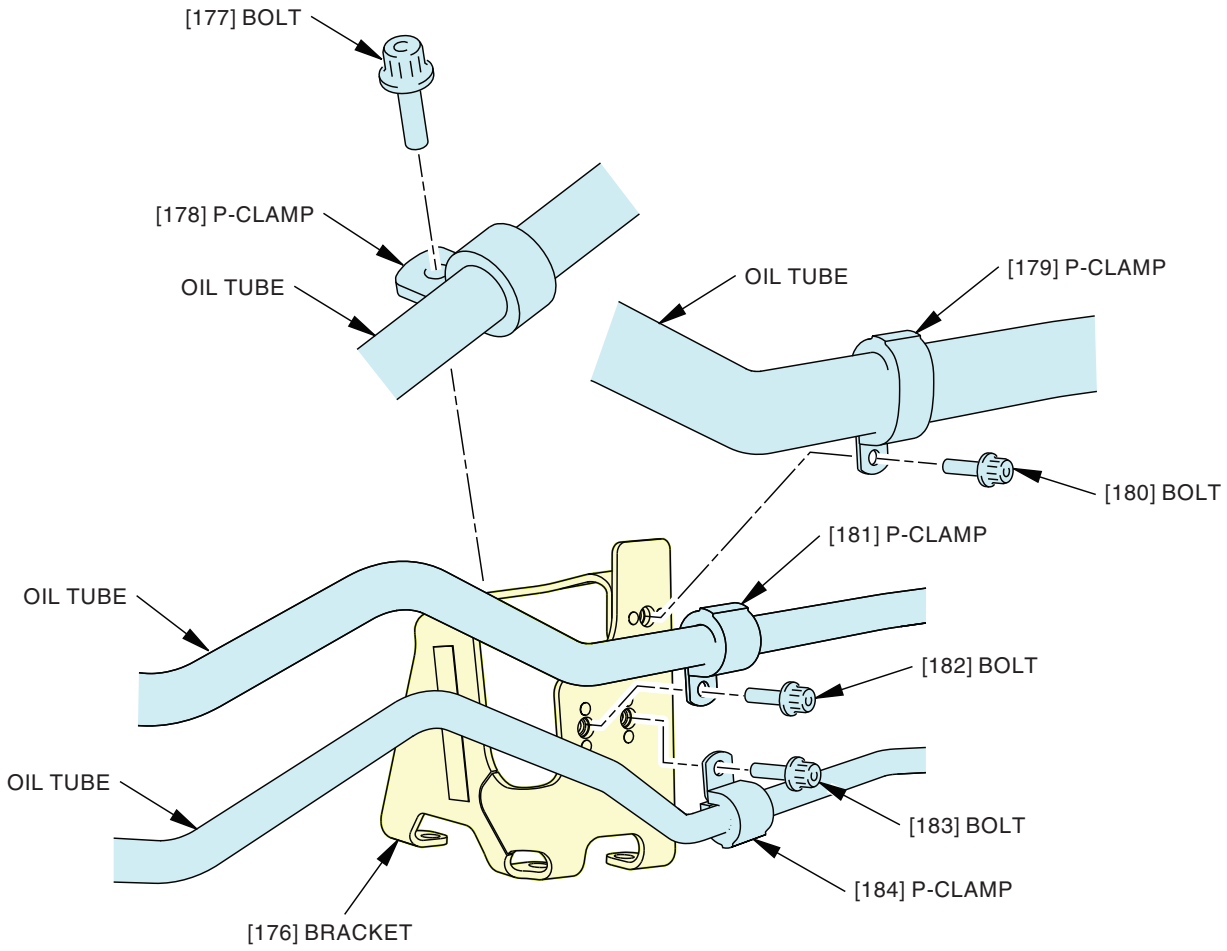
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059238-00
5047643-00
2947744 S0000724293_V2

Fuel Nozzle No.9 Installation
Figure 447/73-11-05-990-877-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

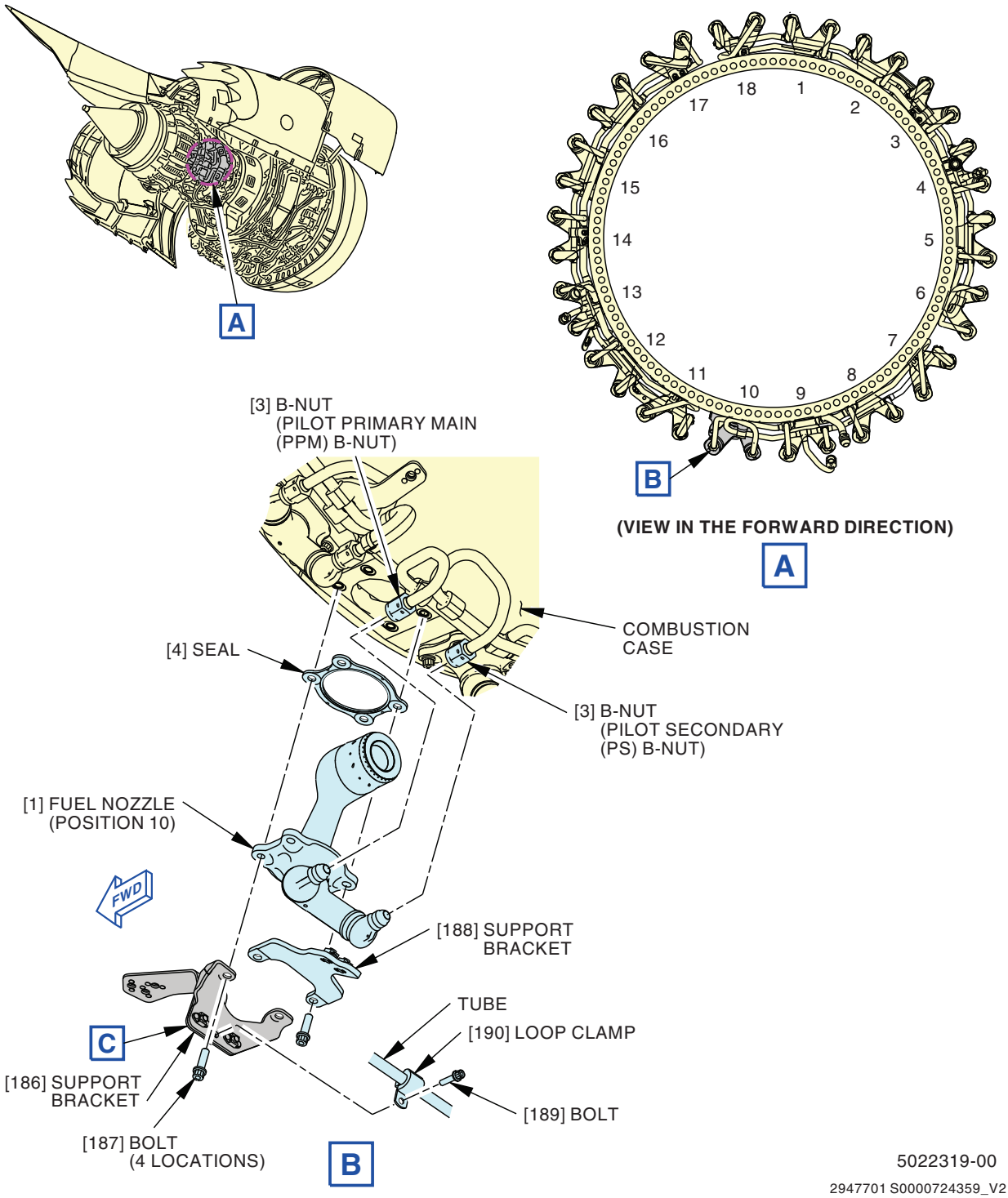


C

2947756 S0000724299_V1

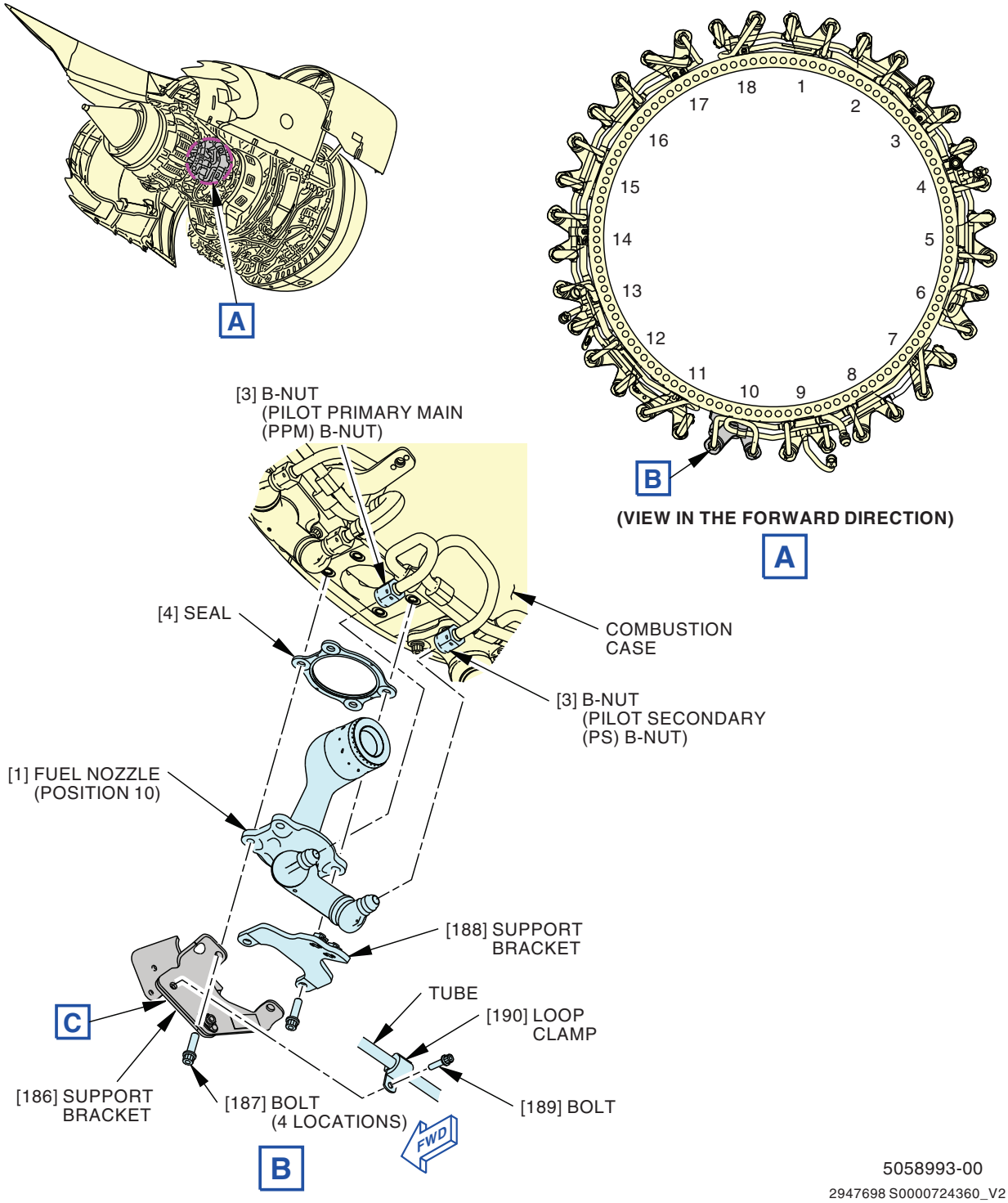
Fuel Nozzle No.9 Installation
Figure 447/73-11-05-990-877-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



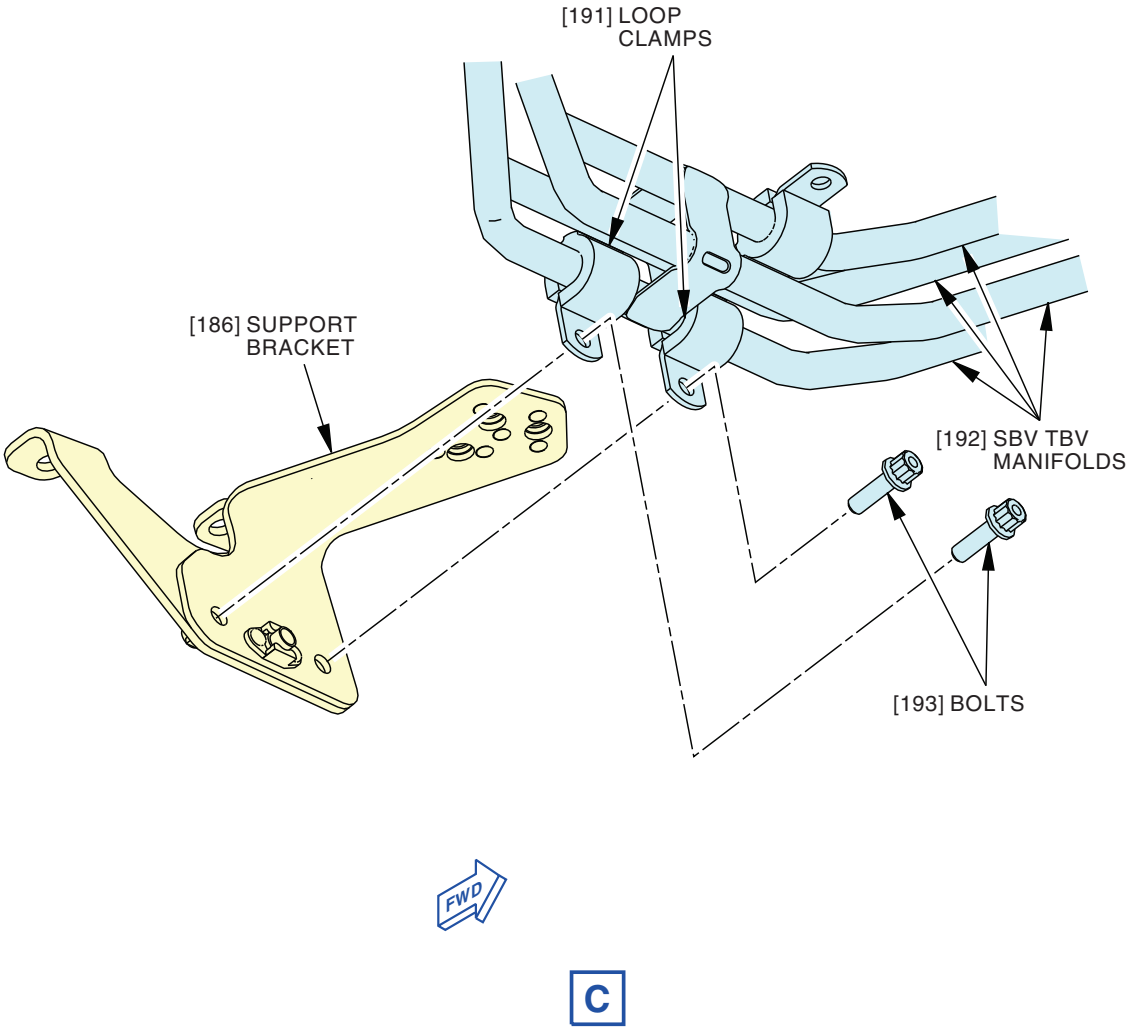
Fuel Nozzle No.10 Installation
Figure 448/73-11-05-990-878-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.10 Installation
Figure 448/73-11-05-990-878-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

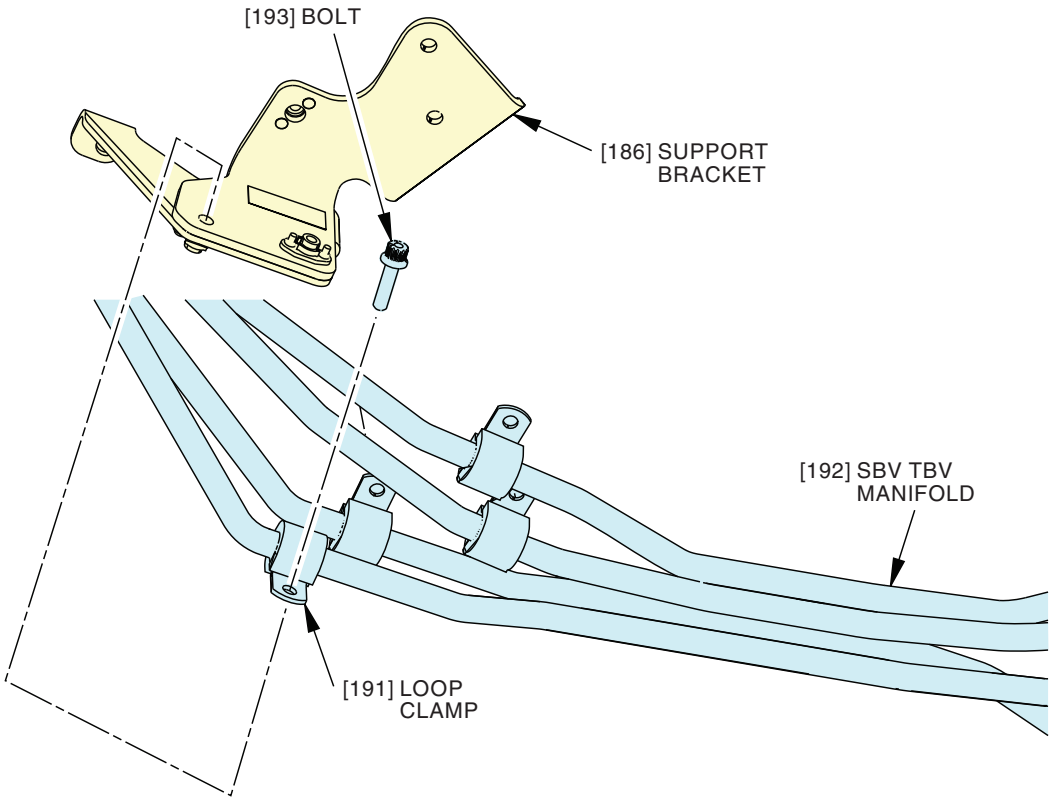


5021992-00

2947697 S0000724361_V1

Fuel Nozzle No.10 Installation
Figure 448/73-11-05-990-878-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

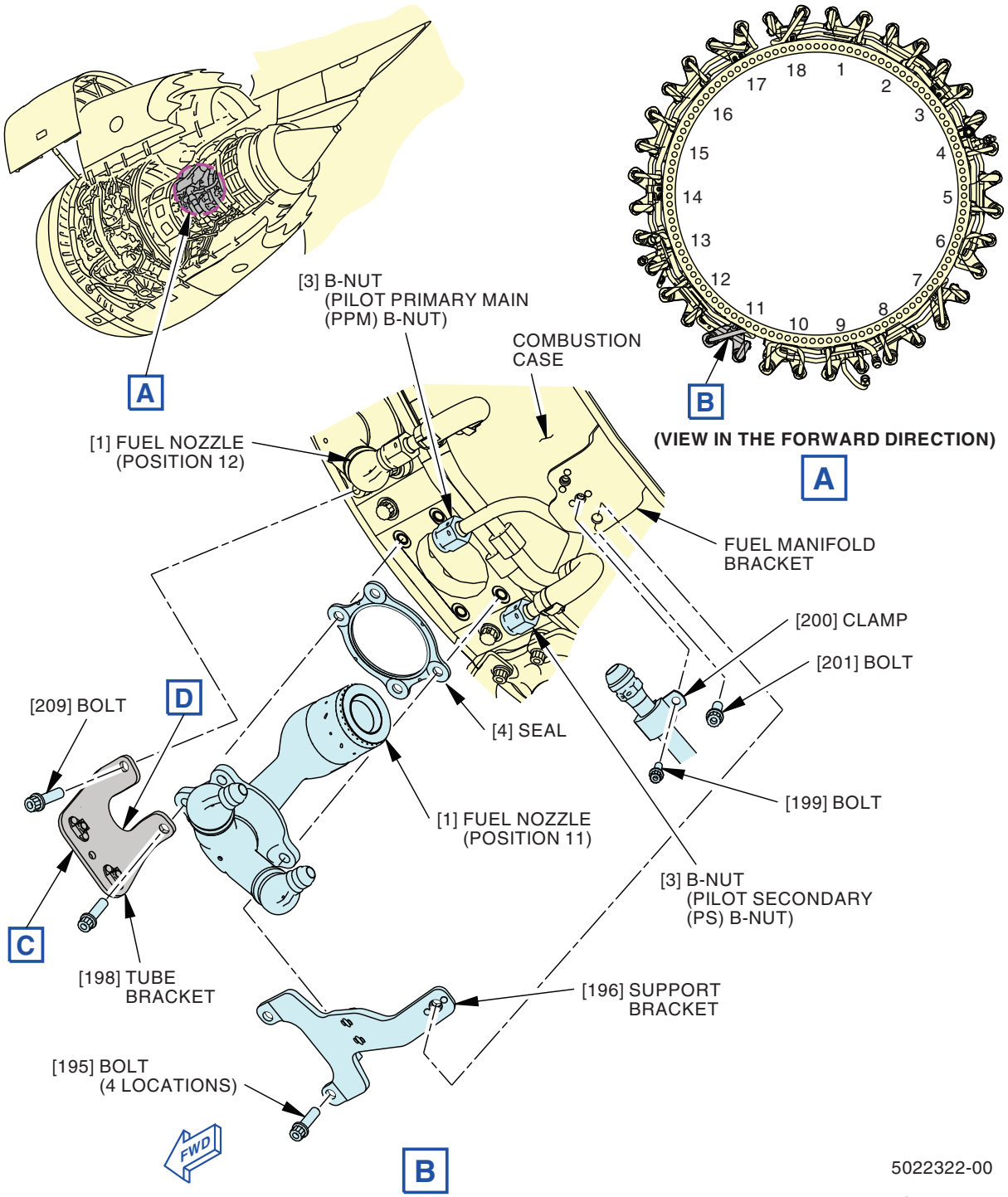


C

5051825-00
2947694 S0000724466_V2

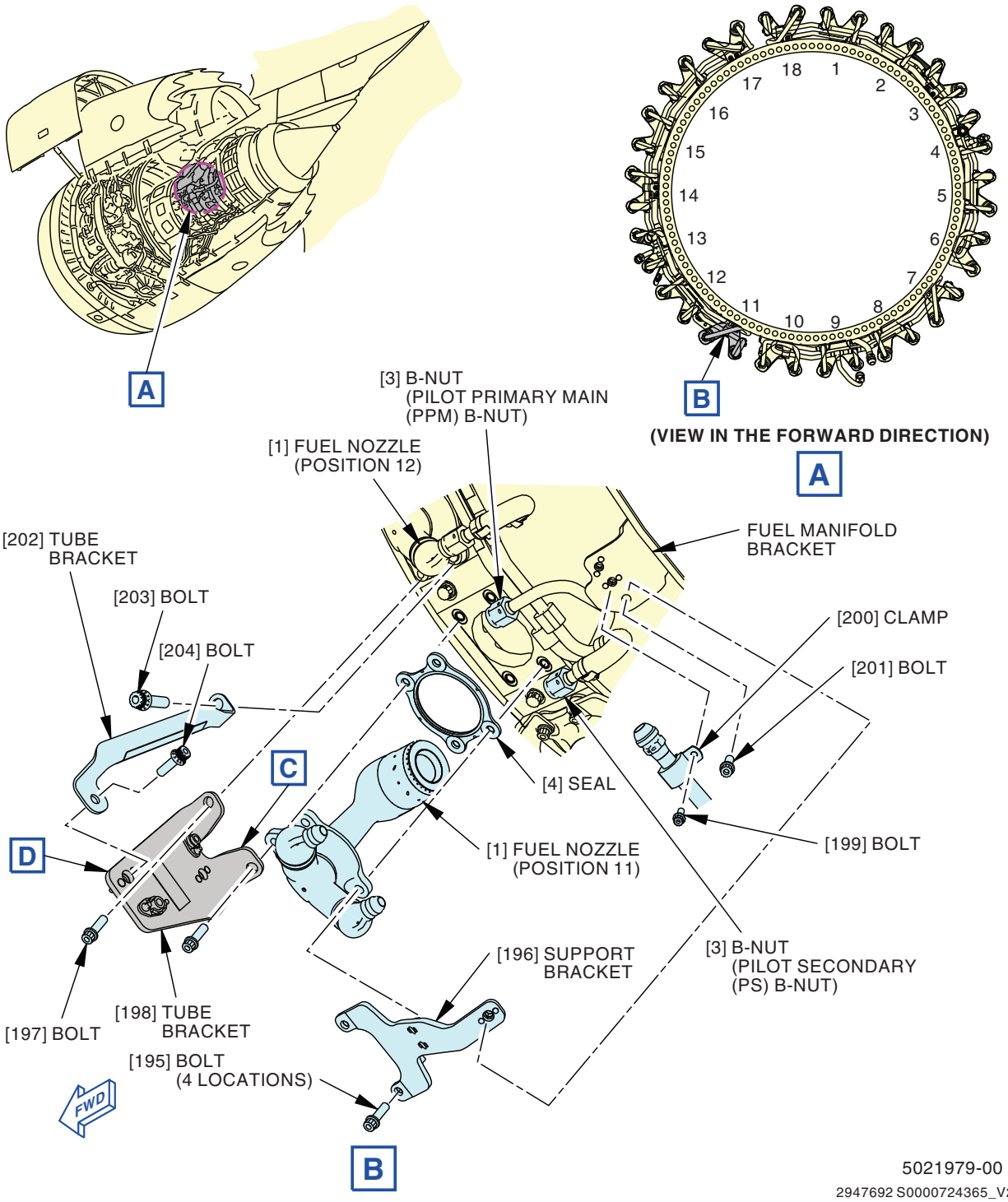
Fuel Nozzle No.10 Installation
Figure 448/73-11-05-990-878-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



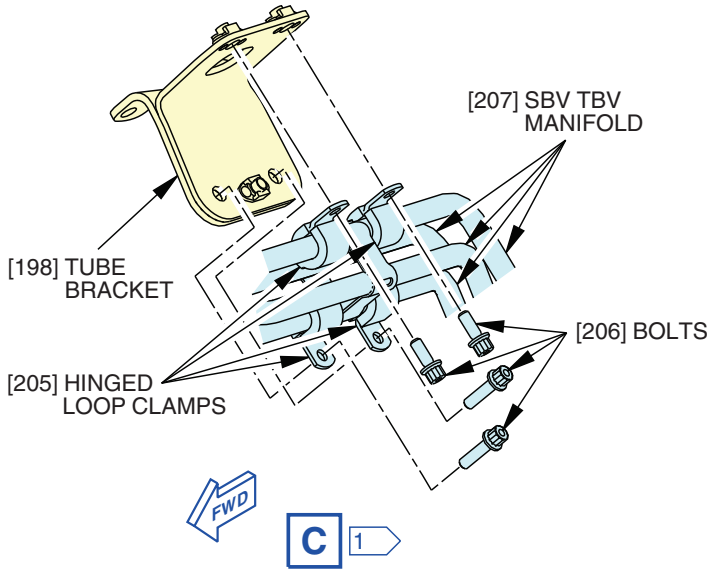
Fuel Nozzle No.11 Installation
Figure 449/73-11-05-990-879-G00 (Sheet 1 of 4)

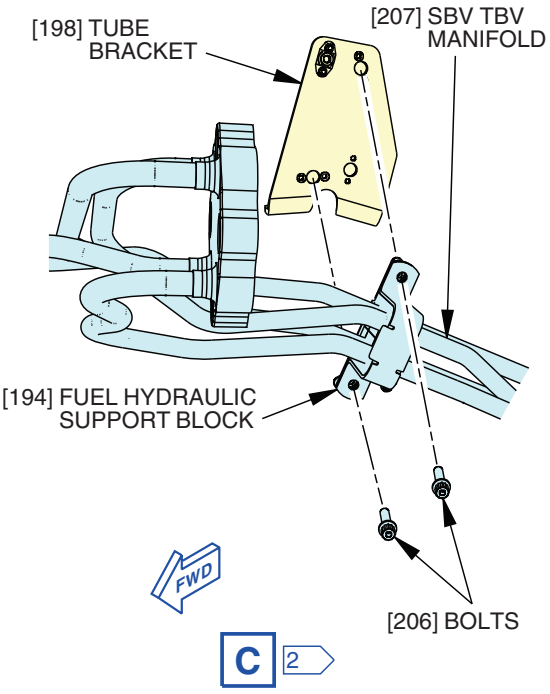
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.11 Installation
Figure 449/73-11-05-990-879-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



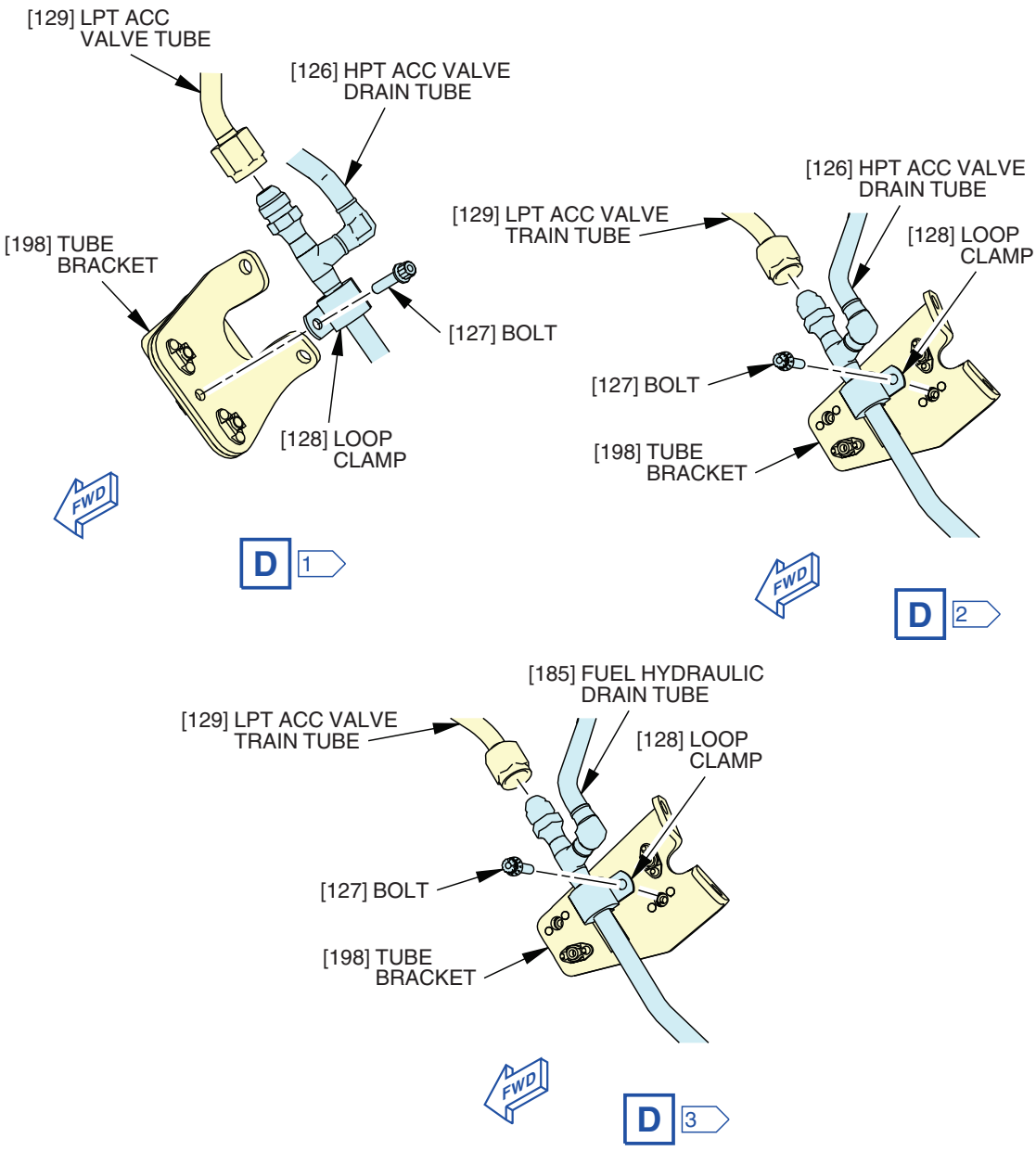


1 PRE LEAP-1B SERVICE BULLETIN 72-0071
2 POST LEAP-1B SERVICE BULLETIN 72-0071

5053481-00/ 5058996-00
2947691 S0000724366_V2

Fuel Nozzle No.11 Installation
Figure 449/73-11-05-990-879-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

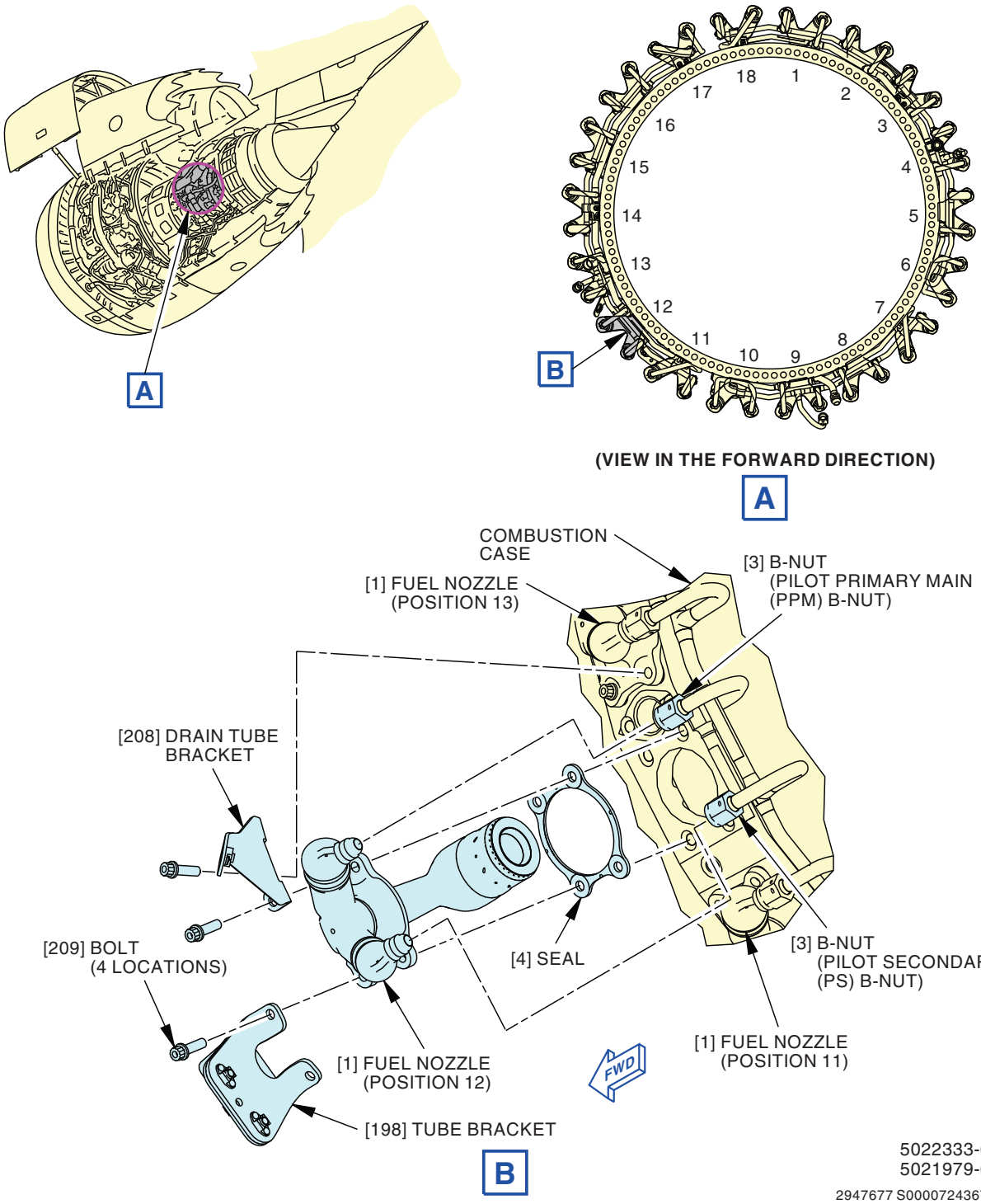


3 POST LEAP-1B SERVICE BULLETIN 72-0378

5053481-00/ 5058996-00
3129166 S0000860958_V1

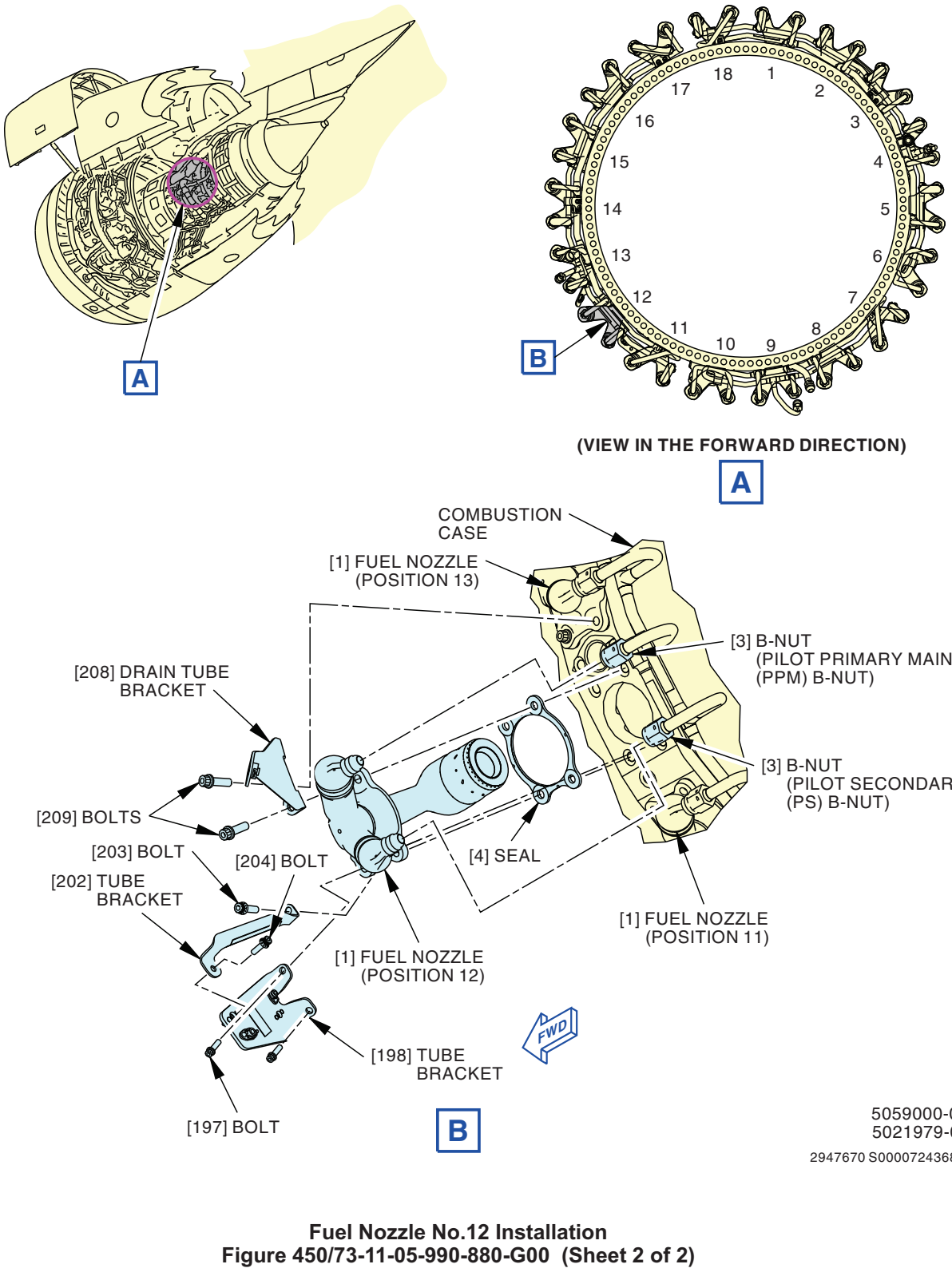
Fuel Nozzle No.11 Installation
Figure 449/73-11-05-990-879-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



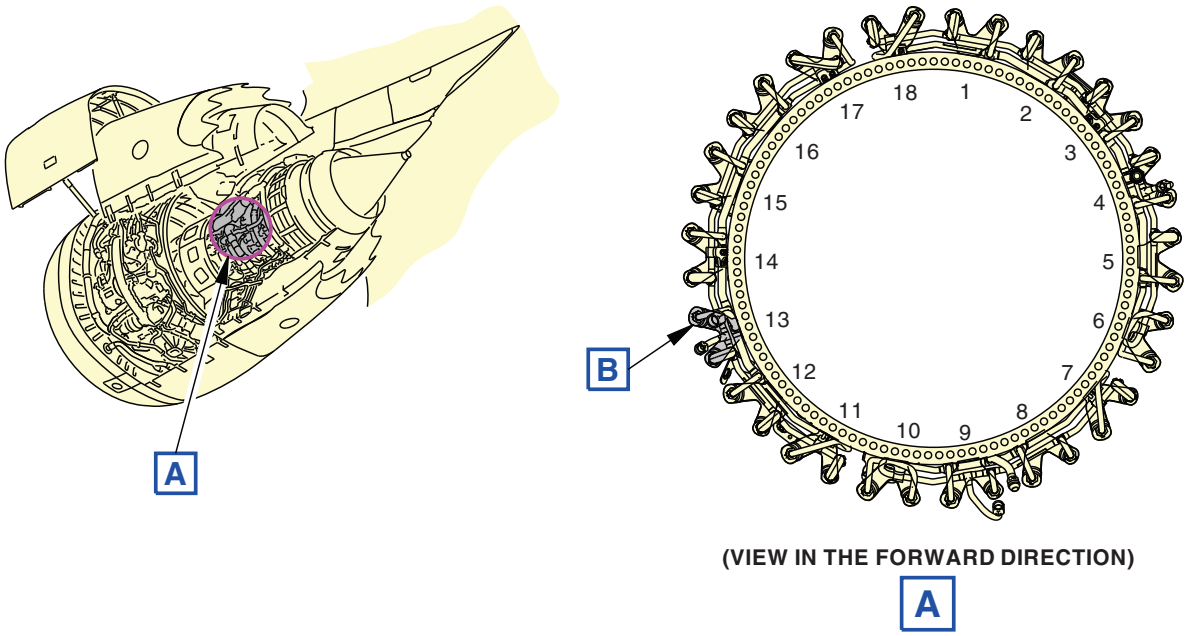
Fuel Nozzle No.12 Installation
Figure 450/73-11-05-990-880-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



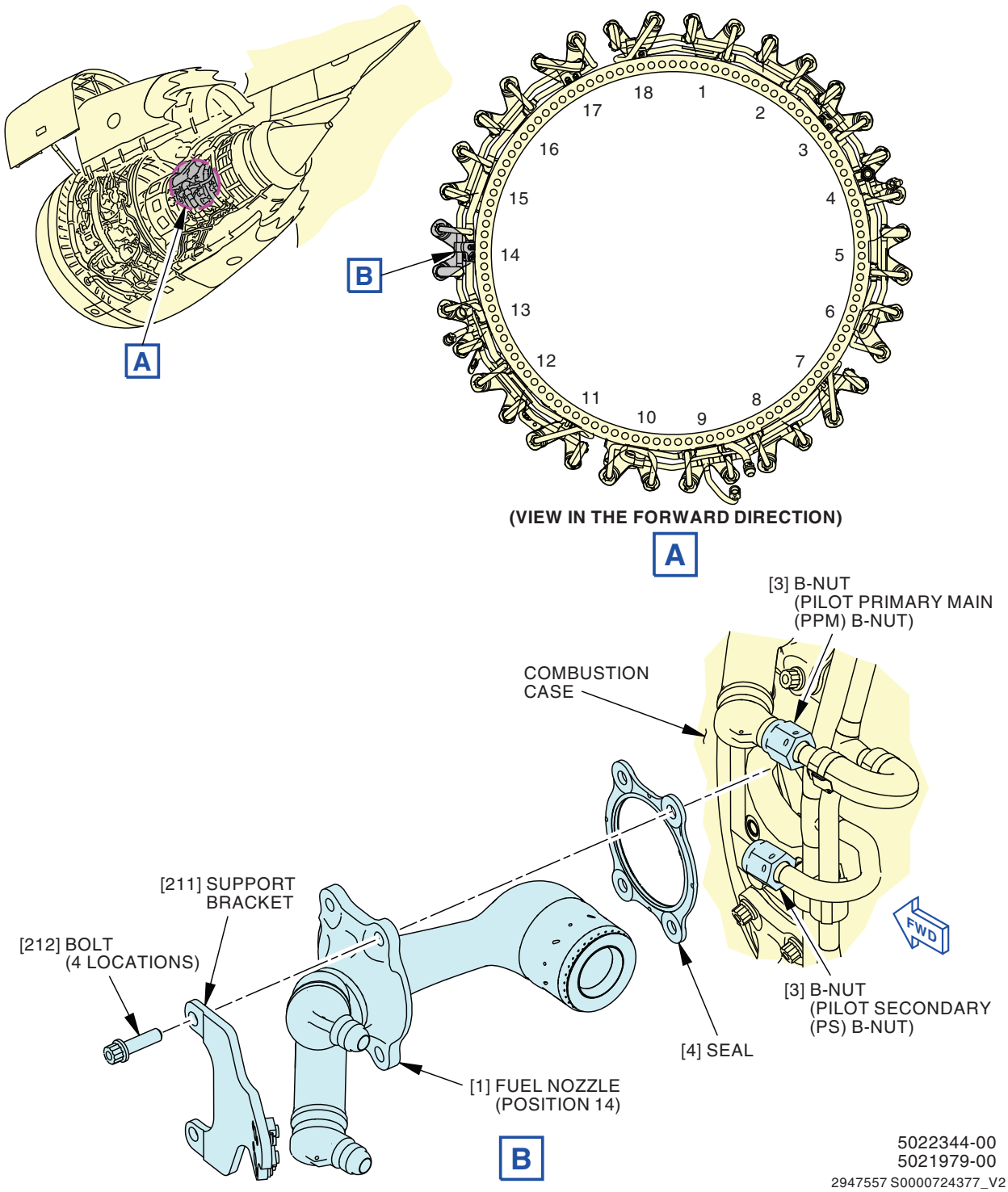
Fuel Nozzle No.12 Installation
Figure 450/73-11-05-990-880-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



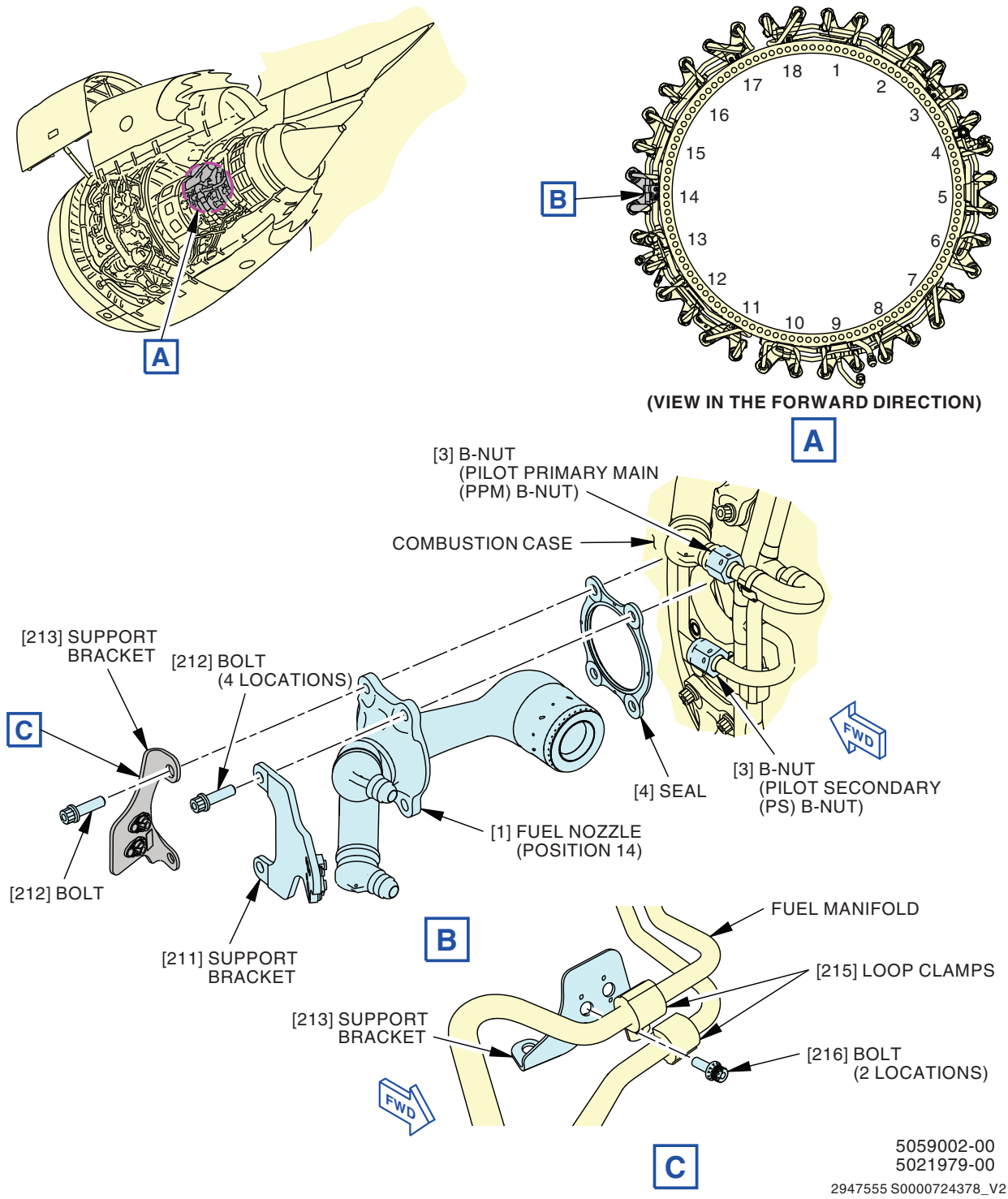
Fuel Nozzle No.13 Installation
Figure 451/73-11-05-990-881-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



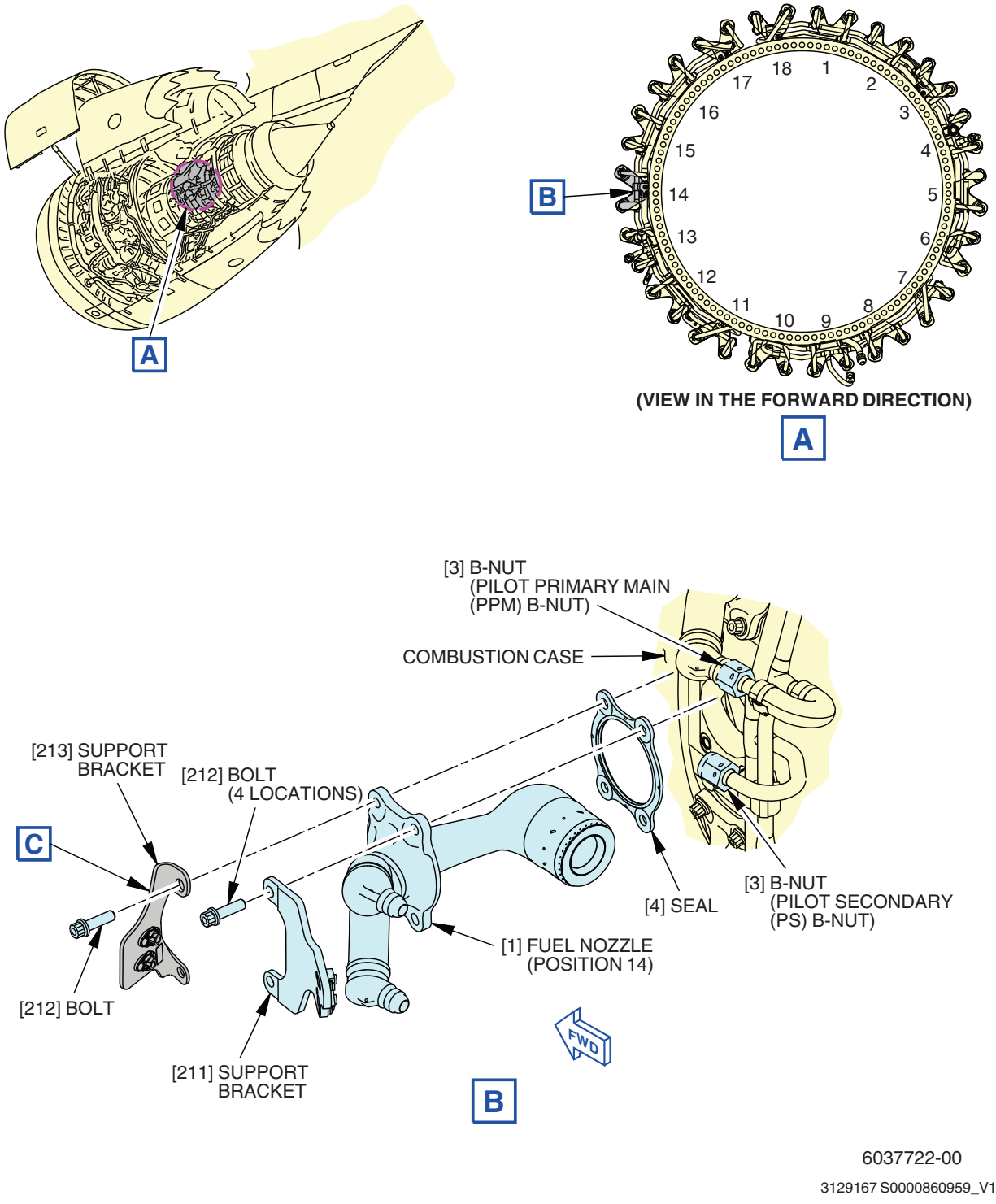
Fuel Nozzle No.14 Installation
Figure 452/73-11-05-990-882-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



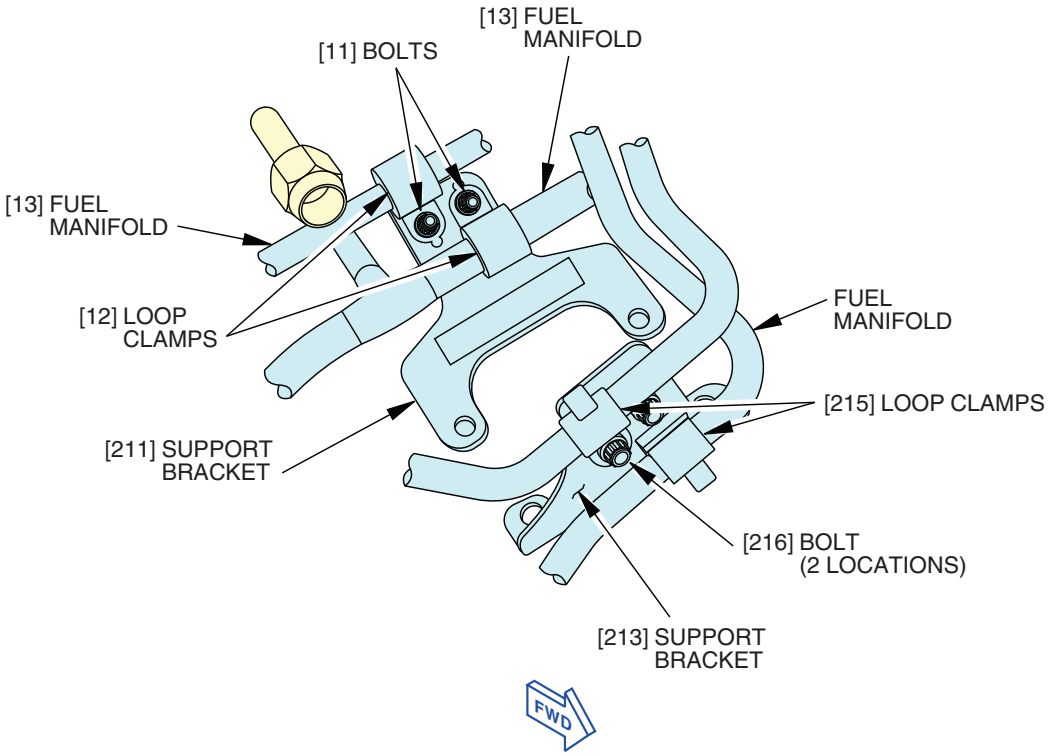
Fuel Nozzle No.14 Installation
Figure 452/73-11-05-990-882-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.14 Installation
Figure 452/73-11-05-990-882-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

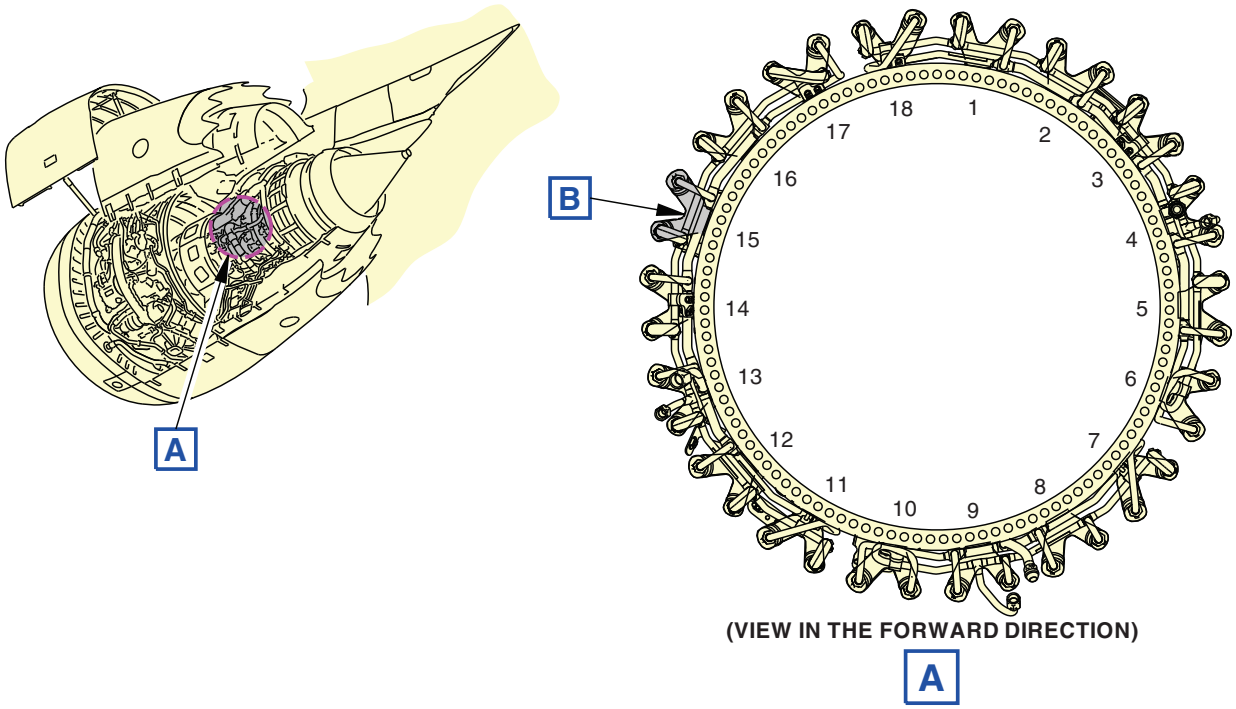


6037721-00

3129168 S0000860960_V1

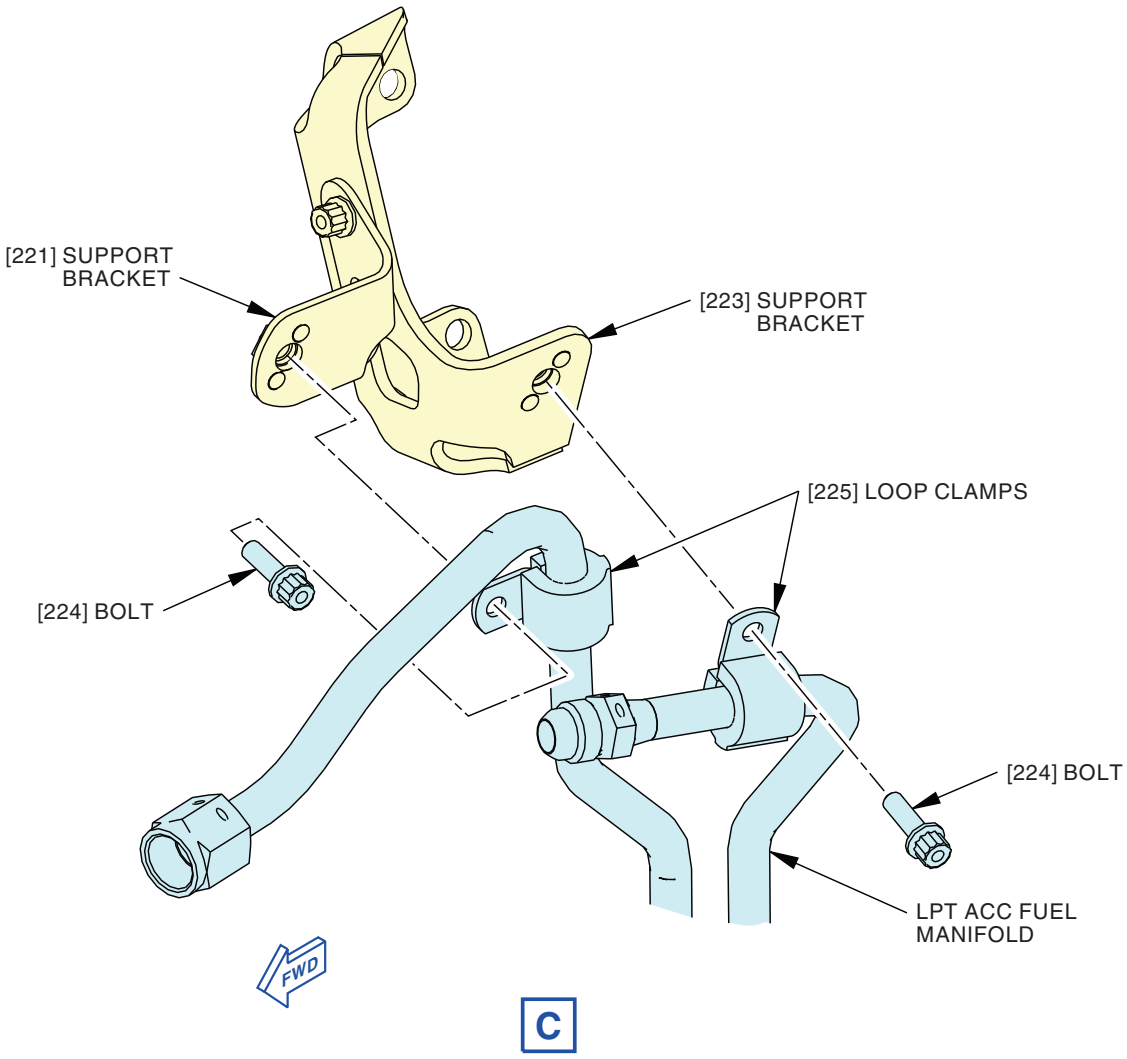
Fuel Nozzle No.14 Installation
Figure 452/73-11-05-990-882-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.15 Installation
Figure 453/73-11-05-990-883-G00 (Sheet 1 of 2)

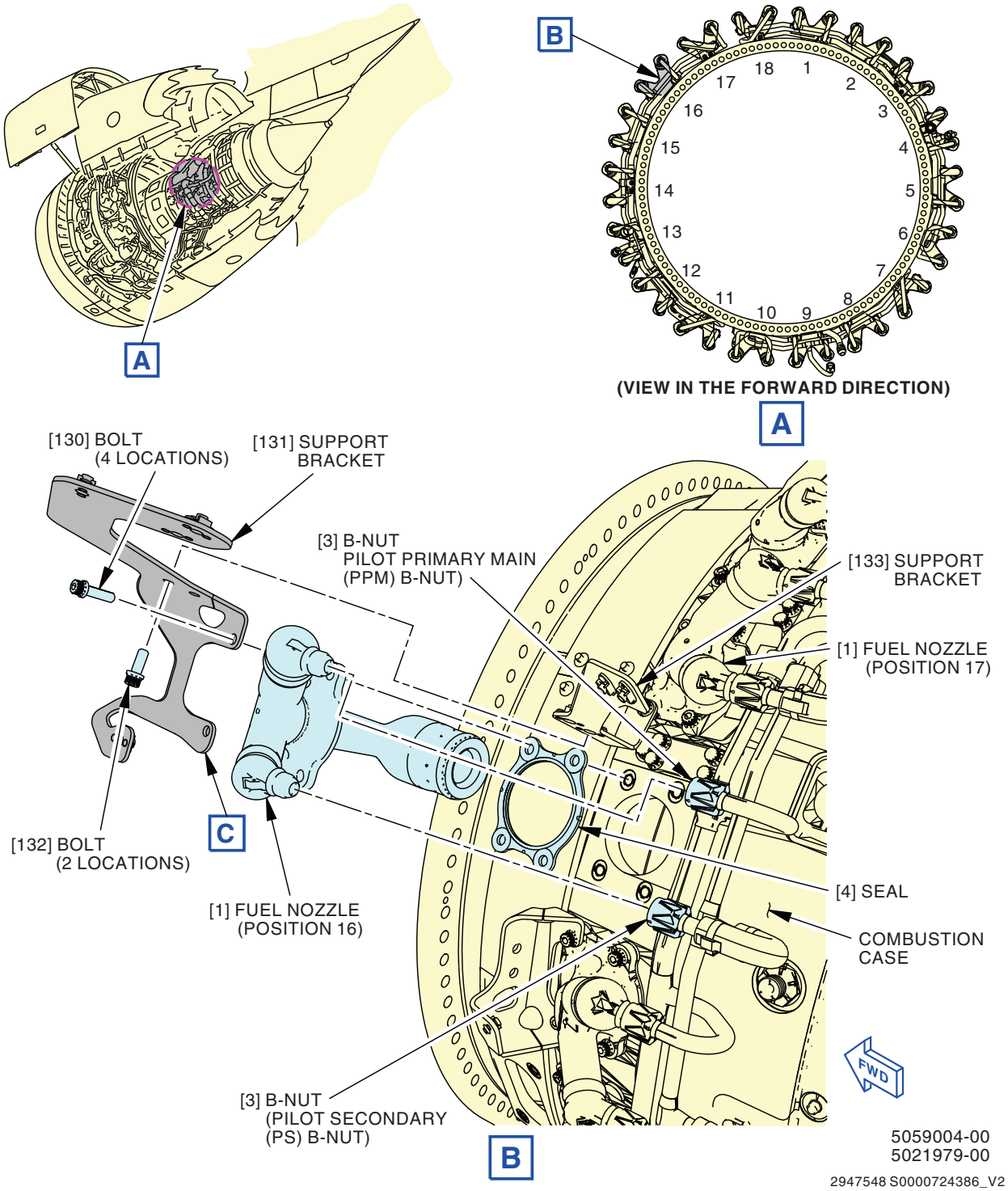
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2947552 S0000724385_V1

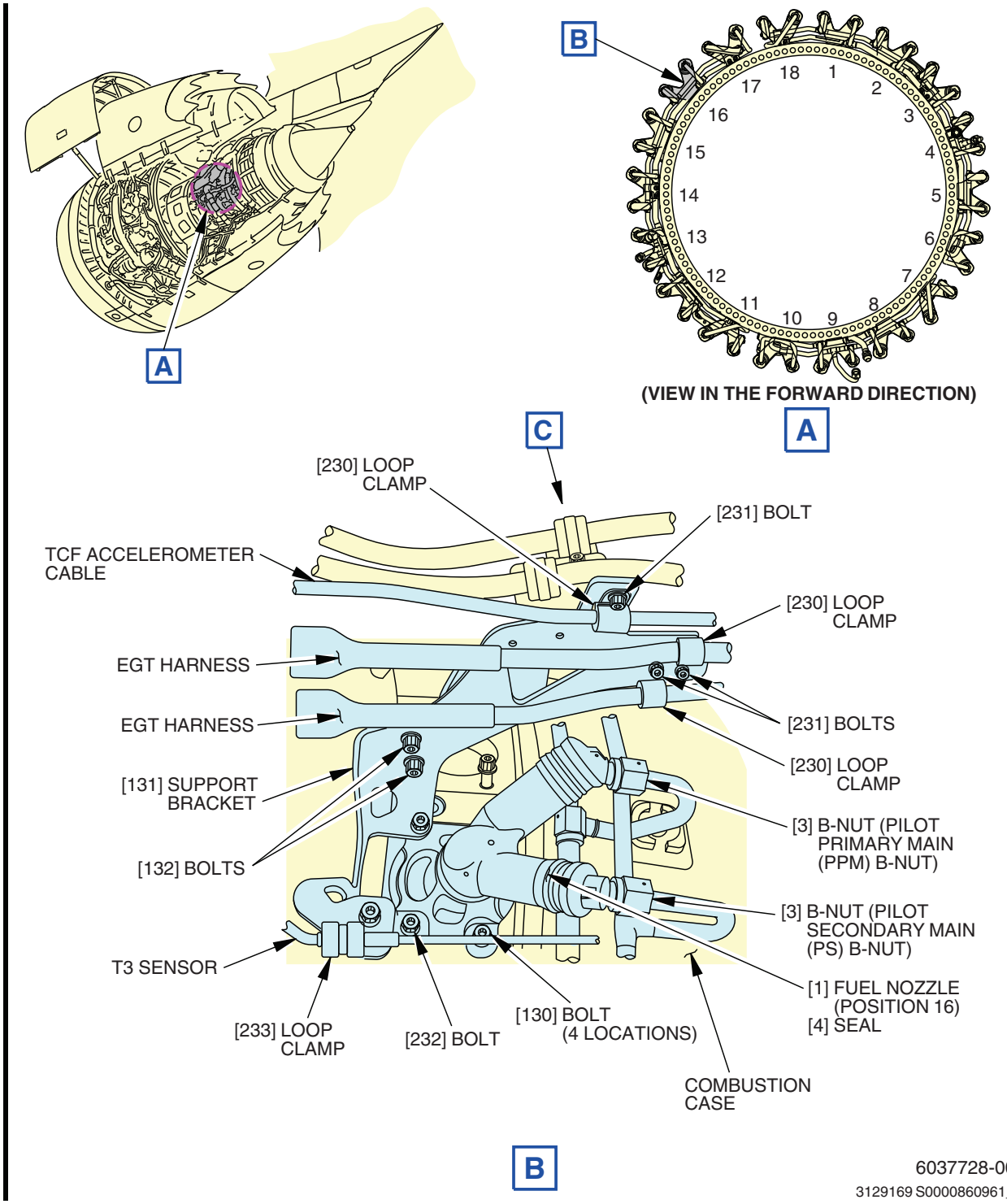
Fuel Nozzle No.15 Installation
Figure 453/73-11-05-990-883-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



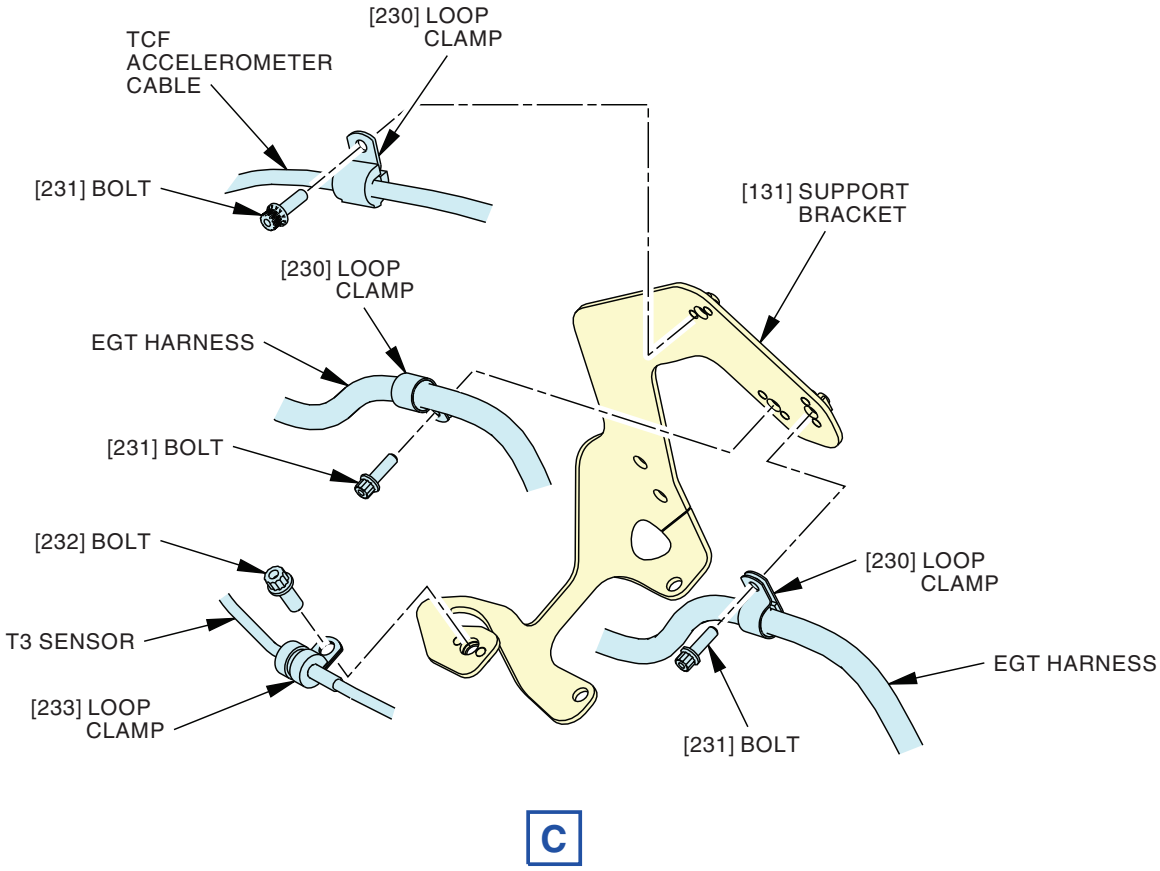
Fuel Nozzle No.16 Installation
Figure 454/73-11-05-990-884-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle No.16 Installation
Figure 454/73-11-05-990-884-G00 (Sheet 2 of 4)

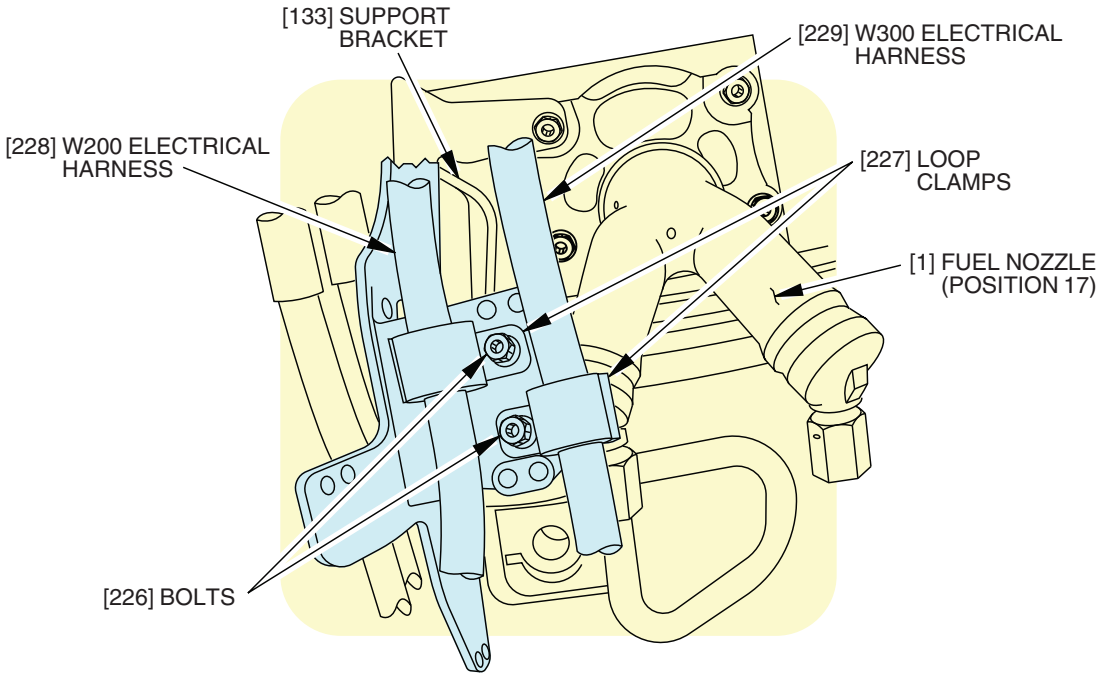
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2947546 S0000724390_V1

Fuel Nozzle No.16 Installation
Figure 454/73-11-05-990-884-G00 (Sheet 3 of 4)

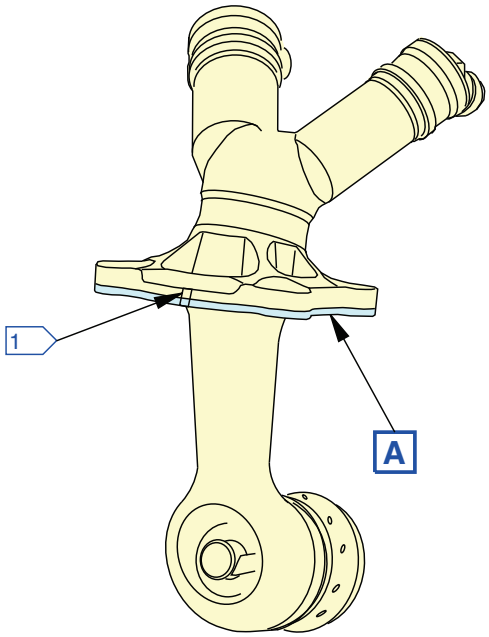
ECCN 9E991 BOEING PROPRIETARY - See title page for details

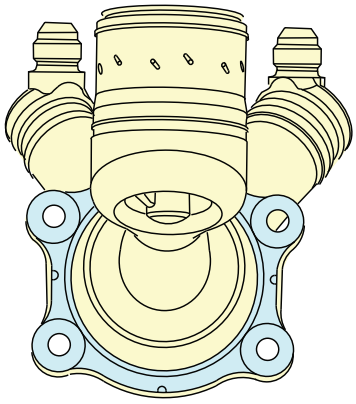


6037728-00
3129170 S0000860962_V1

Fuel Nozzle No.16 Installation
Figure 454/73-11-05-990-884-G00 (Sheet 4 of 4)

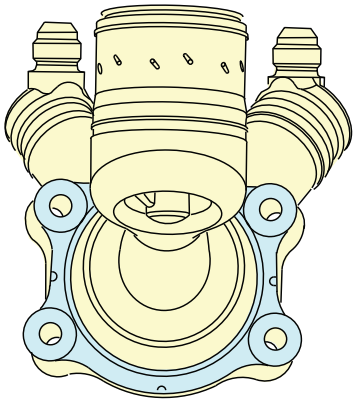
ECCN 9E991 BOEING PROPRIETARY - See title page for details





SEAL ALIGNED
(CORRECT POSITION)

A



SEAL NOT ALIGNED
(INCORRECT POSITION)

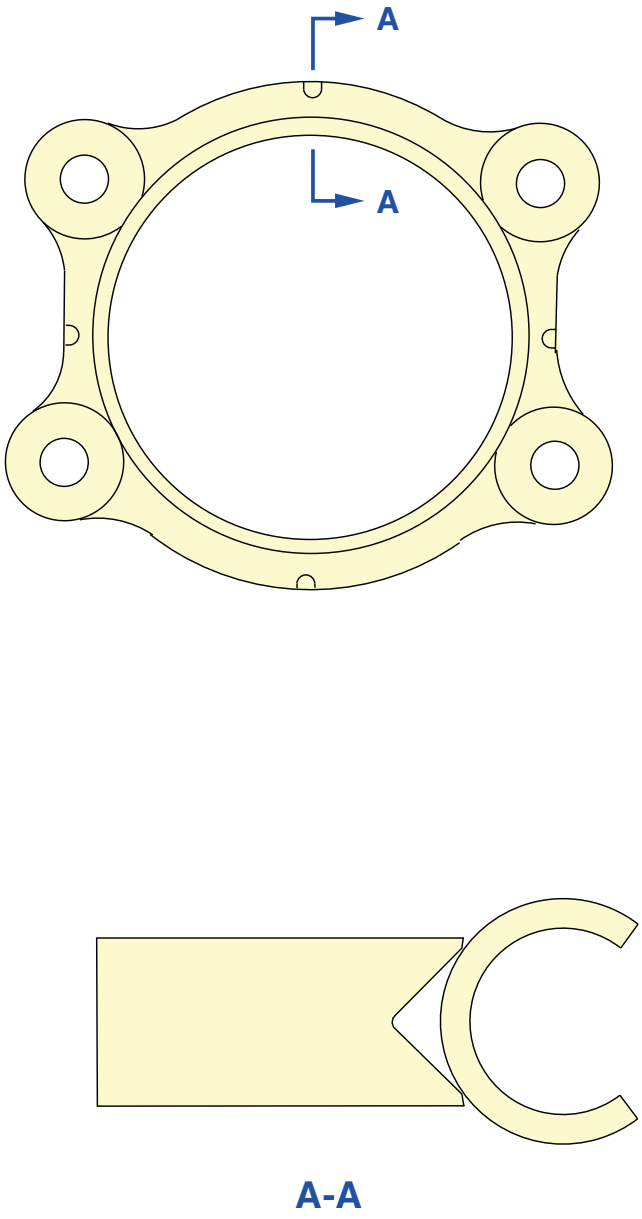
A

1 THE MARK ON THE SEAL MUST BE ALIGNED WITH THE MARK ON THE FUEL NOZZLE.

2953777 S0000731186_V1

Fuel Nozzle Seal Inspection
Figure 455/73-11-05-990-886-G00 (Sheet 1 of 2)

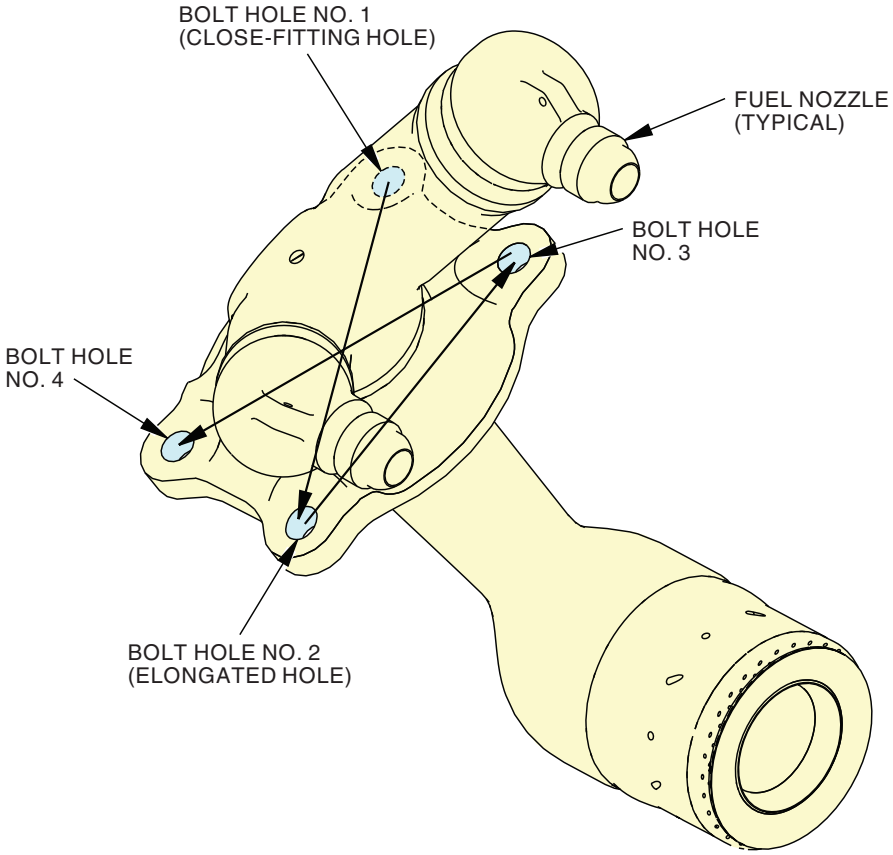
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2953822 S0000731208_V1

Fuel Nozzle Seal Inspection
Figure 455/73-11-05-990-886-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

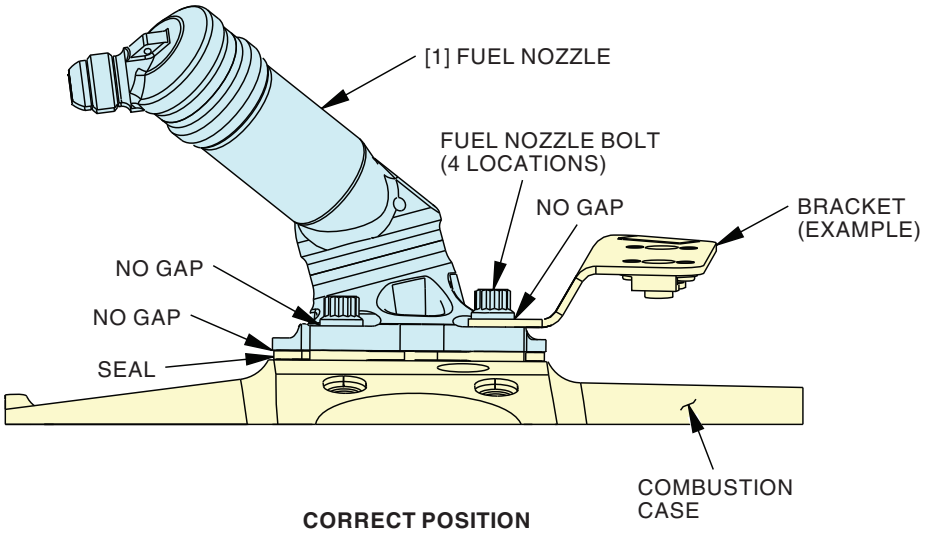


FUEL NOZZLE TORQUE SEQUENCE

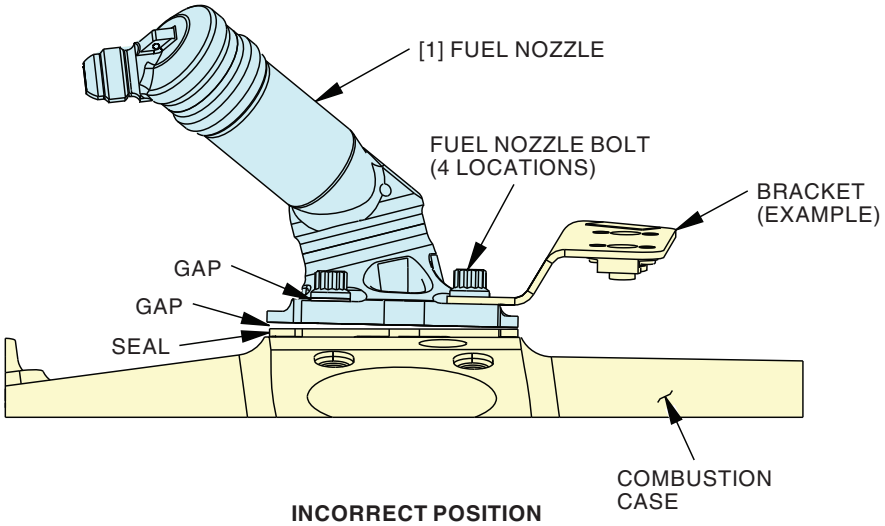
2947512 S0000724407_V2

Fuel Nozzle Torque Sequence
Figure 456/73-11-05-990-885-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CORRECT POSITION



INCORRECT POSITION

2978976 S0000757000_V1

Fuel Nozzle Seated Position Visual Inspection
Figure 457/73-11-05-990-910-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-05-400-802-G00

5. Fuel Nozzle Set Installation

(Figure 424, Figure 425, Figure 426, Figure 427, Figure 429, Figure 430, Figure 431, Figure 432, Figure 433, Figure 434, Figure 435, Figure 436, Figure 437, Figure 438, Figure 439, Figure 440, Figure 441, Figure 442, Figure 443, Figure 444, Figure 445, Figure 446, Figure 447, Figure 448, Figure 449, Figure 450, Figure 451, Figure 452, Figure 453, Figure 454, Figure 455, Figure 456, and Figure 457)

A. General

(1) This task gives the instructions to install one or more fuel nozzles.

(2) There are eighteen fuel nozzles on the combustion case.

NOTE: If you will install multiple or a full set of fuel nozzles at the same time, it is preferable that you install the fuel nozzles in sets.

(3) The fuel nozzle position numbers are in a Clockwise (CW) sequence when viewed from the rear of the engine facing forward or Aft Looking Forward (ALF).

NOTE: All bolts that attach fuel nozzle to the case have the same part number.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
36-11-01-400-801	Engine Pneumatic Duct Installation (P/B 401)
36-11-04-400-801	PRSOV Installation (P/B 401)
36-11-07-400-801	HPSOV Installation (P/B 401)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
70-51-00 P/B 201	TIGHTENING TECHNIQUES AND TORQUE VALUES - MAINTENANCE PRACTICES
71-00-00-700-808-G00	Test No. 32 - Engine Idle System Test (P/B 501)
71-00-00-700-811-G00	Test No. 35 - Erase All Channel A Faults Special Functions Test (P/B 501)
71-00-00-700-812-G00	Test No. 36 - Erase All Channel B Faults Special Functions Test (P/B 501)
71-00-00-790-801-G00	Test No. 1 - Pneumatic Leak Test (P/B 501)
71-00-00-790-803-G00	Test No. 3 - Wet Motor Leak Test (P/B 501)
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-64-400-801-G00	T3 Sensor Installation (P/B 401)
73-22-01-400-808-G00	Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation (P/B 401)
75-24-01-400-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Installation (P/B 401)
75-24-04-400-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Installation (P/B 401)
75-34-01-400-801-G00	Reverse Bleed Check Valve Assembly Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-600	Mirror - Inspection
STD-858	Tag - DO NOT OPERATE
STD-6087	Brush - Acid Resistant

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50034 [CP2619]	Compound - Antiseize - Pure Nickel Special, Never - Seez	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G50530 [CP5061]	Marker - Temporary	
G51403 [CP2166]	Tape - Polyester-Backed, Adhesive	
G51435 [CP2635]	Wax - Stick	
G51669 [CP2975]	Tape - Silicone Rubber, Glass Cloth Fabric	MIL-I-22444

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Bolt	73-11-05-01-010	PTF ALL
4	Seal	73-11-05-01-030	PTF ALL
20	Seal	75-23-00-02-310	PTF ALL
33	Metal seal ring	75-23-00-02-350	PTF ALL
41	Seal	72-00-00-25-160	PTF ALL
45	Seal	72-00-00-25-070	PTF ALL
52	Seal	75-11-00-03-230	PTF ALL
57	Seal ring	75-23-00-02-280	PTF ALL
60	Seal ring	75-23-00-02-270	PTF ALL
93	Seal	75-23-00-02-110	PTF ALL
97	Seal	75-23-00-02-100	PTF ALL
103	Seal	75-23-00-01-210	PTF ALL
116	Seal	75-23-00-01-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Installation

SUBTASK 73-11-05-420-161-G00

(1) Do the following steps:

NOTE: The steps below apply to the fuel nozzle pad bolts and any bolts that can be attached to the combustion case during the installation of the fuel nozzle.

(a) Use new fuel nozzle bolts [130], bolts [135], bolts [136], bolts [138], bolts [2], bolts [140], bolts [141], bolts [142], bolts [146], bolts [150], bolts [153], bolts [154], bolts [155], bolts [158], bolts [160], bolts [175], bolts [187], bolts [195], bolts [197], bolts [203], bolts [209], bolts [210], bolts [212], and bolts [222] at the fuel nozzles [1] (positions 1 through 18).

(b) Prepare all the 72 fuel nozzles bolts and combustion case bolts with Tiodize T8E-H grease, D50355 [CP2961], with acid resistant brush, STD-6087.

(c) Place all 18 clean fuel nozzles [1] on a bench and mark the fuel nozzles No. 1 through No.18.

NOTE: Labeling the serviceable (new or overhauled) fuel nozzles (1-18) may easy installation.

(d) Record the fuel nozzles [1] position with the corresponding fuel nozzle serial number, and the part number.

(e) Remove DO NOT OPERATE tags, STD-858, from the fuel nozzles [1].

(f) Remove 3M 851 Tape, G51403 [CP2166], from the fuel nozzles [1].

(g) Do a general visual inspection of the fuel nozzles [1].

(h) Do a general visual inspection of the seal [4] that mates to the fuel nozzles [1] to see if there is an inner C-seal (Figure 455).

1) Do an inspection for deformation or damage across mating joints.

CAUTION

TURN THE FUEL NOZZLE SEAL UNTIL BOLT HOLES ALIGN. THE ORIENTATION OF THE SEAL TURNS IN ONE DIRECTION. THE BOLT HOLES WILL NOT LINE UP IN THE OTHER DIRECTION. DO NOT USE USED FUEL NOZZLE SEALS. IF YOU DO NOT OBEY, DAMAGE TO THE FUEL NOZZLE CAN OCCUR.

(i) Install the new seal [4] on the fuel nozzles [1] (positions 1 through 18).

NOTE: It is recommended that the new set of fuel nozzles and their corresponding C-seals are marked with a match mark as a visual aid to make the alignment between the fuel nozzles and gasket seals easy during the assembly process (Figure 455).

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(j) Remove all caps and covers from the engine case and the fuel nozzles [1] being installed.

(k) Do an inspection of the new or overhauled fuel nozzles [1] inlet strainers for any foreign objects.

I. Fuel Nozzle Installation

SUBTASK 73-11-05-420-162-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT PUSH THE FUEL NOZZLE INTO THE COMBUSTION CHAMBER WITH TOO MUCH FORCE. TOO MUCH FORCE WILL CAUSE DAMAGE TO THE SWIRL CUP.

(1) Install the fuel nozzles [1] (position 1 to 18) as follows:

NOTE: The left and right sides of the engine can be worked at the same time.

(a) Do not tighten the fuel nozzle bolts until all fuel nozzles [1] are installed.

(b) Place the fuel nozzles [1] with the new seals [4] aligned into the combustion case pad with the fuel nozzle [1] tip forward.

(c) Turn the fuel nozzles [1] (positions 1 through 18) in the reverse order of the fuel nozzle [1] removal process and put the fuel nozzles [1] (positions 1 through 18) in position on the seals [4] and combustion case pad.

(d) If necessary, move the fuel nozzles [1] (positions 1 through 18) aft to engage the fuel nozzles [1] (positions 1 through 18) tip in the combustor radial mixer (cup).

(e) Align the seals [4] and fuel nozzles [1] (positions 1 through 18) bolt holes to the combustion case.

SUBTASK 73-11-05-420-164-G00

(2) Install the support brackets and fuel nozzles [1] to the combustor case on QUADRANT 1, as follows:

(a) Make sure that Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolts [135], bolts [136], bolts [138], bolts [2], bolts [140], bolts [141], and bolts [142].

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [132].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(c) Apply oil, D00599 [CP2442], on the threads of the two bolts [217].

PTF ALL

(d) Install the support bracket [133] in position on the fuel nozzle [1] (position 17), as follows (View B, Figure 437):

1) Install the bolt [136] that attaches the support bracket [133] and fuel nozzle [1] (position 17) to the combustion case.

2) Install the support bracket [134] in position on the fuel nozzle [1] (position 17).

3) Install the bolts [135] that attach the fuel nozzle [1] (position 17) and support bracket [134] to the combustion case.

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4) Install the two bolts [132] that attach the support bracket [133] to the support bracket [131].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(e) Install the support bracket [219] in the position on the fuel nozzle [1] (position 18), as follows (View C, Figure 438):

1) Install the two bolts [217] that attach the support bracket [218] to the support bracket [219] at the fuel nozzle [1] (position 18).

2) Tighten the bolts [217] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(f) Install the support bracket [137] in position on the fuel nozzle [1] (position 18), as follows (View B, Figure 438):

1) Install the four bolts [138] that attach the fuel nozzle [1] (position 18) and support bracket [137] to the combustion case.

(g) Install the fuel nozzle [1] (position 1), as follows (View B, Figure 439):

1) Install the four bolts [2] that attach the fuel nozzle [1] (position 1) to the combustion case.

(h) Install the fuel nozzle [1] (position 2), as follows (View B, Figure 440):

1) Install the bracket assembly [139] on the fuel nozzle [1] (position 2) and fuel nozzle [1] (position 3), as follows:

NOTE: The bracket assembly [139] is the bracket that span across the fuel nozzle [1] (position 2) and fuel nozzle [1] (position 3).

a) Install the two bolts [140] that attach the bracket assembly [139] to the fuel nozzle [1] (position 2).

2) Install the remaining two bolts [140] that attach the fuel nozzle [1] (position 2) to the combustion case.

(i) Install the fuel nozzle [1] (position 3), as follows (View B, Figure 441):

1) Install the bolt [141] that attaches the bracket assembly [139] to the fuel nozzle [1] (position 3).

2) Install the two bolts [142] that attach the support bracket [143] to the fuel nozzle [1] (position 3).

3) Install the remaining bolt [142] that attaches the fuel nozzle [1] (position 3) to the combustion case.

(j) Do not tighten any bolts at this time.

SUBTASK 73-11-05-420-165-G00

(3) Install the support brackets and fuel nozzles [1] to the combustion case on QUADRANT 2, as follows:

(a) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolts [2], bolts [146], bolt [150], bolt [154], bolts [153], bolts [160], bolts [155], and bolts [158].

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [166], bolt [162], bolts [163], bolts [168], bolts [169], bolts [173], bolt [177], bolt [180], bolt [182], and bolt [183].

(c) Install the fuel nozzle [1] (position 4), as follows (View B, Figure 442):

1) Install the four bolts [2] that attach the fuel nozzle [1] (position 4) to the combustion case.

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(d) Install the fuel nozzle [1] (position 5), as follows (View B, Figure 443):

1) Install the two bolts [146] that attach the bracket [144] and fuel nozzle [1] (position 5) to the combustion case.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

2) Install the two bolts [146] that attach the bracket [145] assembled with the bracket [147] and fuel nozzle [1] (position 5) to the combustion case.

NOTE: The bracket [145] and bracket [147] are two brackets assembled together that span across the fuel nozzle [1] (position 5) and fuel nozzle [1] (position 6).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

3) Install the two bolts [146] that attach the bracket [151] assembled with the bracket [149] and fuel nozzle [1] (position 5) to the combustion case.

NOTE: The bracket [151] assembled with the bracket [149] are two brackets assembled together that span across the fuel nozzle [1] (position 5) and fuel nozzle [1] (position 6).

PTF ALL

(e) Install the fuel nozzle [1] (position 6), as follows (View B, Figure 444):

PTF ALL; AIRPLANES WITH PRE LEAP-1B SB 72-0135 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217

1) Install the two bolts [153] that attach the bracket [152] and fuel nozzle [1] (position 6) to the combustion case.

2) Install the bolt [154] that attaches the bracket [147] and fuel nozzle [1] (position 6) to the combustion case.

3) Install the remaining bolt [153] that attaches the fuel nozzle [1] (position 6) to the combustion case.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

4) Install the two bolts [155] that attach the support bracket [156] and fuel nozzle [1] (position 6) to the combustion case.

5) Install the bolt [150] that attaches the bracket [149] and fuel nozzle [1] (position 6) to the combustion case.

6) Install the remaining bolt [155] that attaches the fuel nozzle [1] (position 6) to the combustion case.

PTF ALL

(f) Install the fuel nozzle [1] (position 7), as follows (View B, Figure 445):

1) Install the support bracket [159] on the fuel nozzle [1].

a) Install the two bolts [158] that attach the support bracket [159] and fuel nozzle [1] (position 7) to the combustion case.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

2) Install the two bolts [160] that attach the support bracket [161] and fuel nozzle [1] (position 7) to the combustion case.

3) Install the bolt [162] that attaches the support bracket [161] to the bracket on the combustion case.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

4) Install the two bolts [173] that attach the start bleed bracket [172] to the combustion case (View C, Figure 445).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

5) Install the two bolts [160] that attach the fuel nozzle [1] (position 7) to the combustion case.

6) Install the two bolts [173] that attach the start bleed bracket [172] to the combustion case (View C, Figure 445).

7) Install the two bolts [163] that attach the bracket [164] to the support bracket [159].

8) Install the two bolts [168] that attach the two p-clamps [171] and SBV and TBV manifolds [170] to the bracket [164] (View D, Figure 445).

9) Install the two bolts [169] that attach the two p-clamps [171] and SBV and TBV manifold [170] to the bracket [164].

10) Install the bolt [166] that attaches the loop clamp [165] and PS3 air tube [167] to the bracket [164].

PTF ALL

(g) Install the four bolts [2] that attach the fuel nozzle [1] (position 8) to the combustion case (View B, Figure 446).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(h) Install the four bolts [2] that attach the fuel nozzle [1] (position 9) to the combustion case (View B, Figure 447).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(i) Install the fuel nozzle [1] (position 9), as follows (View B, Figure 447):

1) Install the bracket [176] in position on the fuel nozzle [1] (position 9) and align with the bolt holes.

2) Install the four bolts [175] that attach the fuel nozzle [1] (position 9) and bracket [176] to the combustion case.

3) Install the bolt [177], bolt [180], bolt [182], bolt [183] with p-clamp [178], p-clamp [179], p-clamp [181], and p-clamp [184] that attach the oil tubes to the bracket [176] at the fuel nozzle [1] (position 9) (View C, Figure 447).

PTF ALL

(j) Do not tighten any bolts at this time.

SUBTASK 73-11-05-420-166-G00

(4) Install the support brackets and fuel nozzle [1] to the combustion case on QUADRANT 3, as follows:

(a) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolts [187], bolts [195], bolts [197], bolts [203], bolts [210], and bolts [209].

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [193], bolts [206], bolt [127], bolt [189], and bolt [201].

(c) Install the fuel nozzle [1] (position 10), as follows (View B, Figure 448):

1) Install the support bracket [186] and support brackets [188] in position on fuel nozzle [1] (position 10).

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2) Install the four bolts [187] that attach the fuel nozzle [1] (position 10), support bracket [186], and support brackets [188] to the combustion case.

3) Install the bolt [189] and loop clamp [190] that attaches the tube to the support bracket [186].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

4) Install the two bolts [193] with the two loop clamps [191] that attach the SBV TBV manifolds [192] to the support bracket [186] (View C, Figure 448).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

5) Install the bolt [193] with the loop clamp [191] that attaches the SBV TBV manifolds [192] to the support bracket [186] (View C, Figure 448).

PTF ALL

(d) Install the fuel nozzle [1] (position 11), as follows (View B, Figure 449):

1) Install the support bracket [196] in position on the fuel nozzle [1] (position 11).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

2) Install the tube bracket [198] in position on the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 12).

NOTE: The tube bracket [198] spans across the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 12).

3) Install the four bolts [195] that attach the fuel nozzle [1] (position 11), support bracket [196], and tube bracket [198] to the combustion case.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

4) Install the tube bracket [198] assembled with the tube bracket [202] in position on the fuel nozzle [1] (position 11) fuel nozzle [1] (position 12), as follows:

NOTE: The tube bracket [198] and tube bracket [202] span across the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 12).

a) Install the four bolts [195] that attach the support bracket [196], tube bracket [198] assembled with the tube bracket [202], and fuel nozzle [1] to the combustion case.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

5) Install the four bolts [206] with the hinged loop clamps [205] that attach the SBV TBV manifolds [207] to the tube bracket [198] (View C, Figure 449).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

6) Install the two bolts [206] with the fuel hydraulic support block [194] that attach the SBV TBV manifolds [207] to the tube bracket [198] (View C, Figure 449).

PTF ALL

7) Align the support bracket [196] to the fuel manifold bracket to and install the bolt [201].

8) Install the bolt [199] to attach the clamp [200] to the fuel manifold bracket.

9) Install the bolt [127] with the loop clamp [128] that attaches the HPT ACC valve drain tube [126] to the tube bracket [198] at the fuel nozzle [1] (position 11) (View D, Figure 449).

a) Connect the LPT ACC valve tube [129] to the HPT ACC valve drain tube [126].

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(e) Install the fuel nozzle [1] (position 12), as follows (View B, Figure 450):

1) Install the drain tube bracket [208] in position on the fuel nozzle [1] (position 12) and fuel nozzle [1] (position 13).

NOTE: The drain tube bracket [208] spans across the fuel nozzle [1] (position 12) and fuel nozzle [1] (position 13).

2) Install the bolt [209] that attaches the drain tube bracket [208] and fuel nozzle [1] (position 12) to the combustion case.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

3) Install the bolt [209] that attaches the tube bracket [198] and fuel nozzle [1] (position 12) to the combustion case.

4) Install the remaining two bolts [209] that attach the fuel nozzle [1] (position 12) to the combustion case.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

5) Install the bolt [197] and bolt [203] that attach the tube bracket [202] and tube bracket [198] to the fuel nozzle [1] (position 12).

6) Install the remaining bolt [209] that attaches the fuel nozzle [1] (position 12) to the combustion case.

PTF ALL

(f) Do not tighten any bolts at this time.

SUBTASK 73-11-05-420-167-G00

(5) Install the support brackets and fuel nozzle [1] to the combustor case on QUADRANT 4, as follows:

(a) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolts [210], bolts [212], bolts [222], and bolts [130].

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [132], bolt [224], and bolts [216].

(c) Install fuel nozzle [1] (position 13), as follows (View B, Figure 451):

1) Make sure that the drain tube bracket [208] in position on the fuel nozzle [1] (position 13) and fuel nozzle [1] (position 12).

a) Install the four bolts [210] that attach the fuel nozzle [1] (position 13) to the combustion case.

(d) Install the fuel nozzle [1] (position 14), as follows (View B, Figure 452):

1) Install the support bracket [211] in position on the fuel nozzle [1] (position 14).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

2) Install the four bolts [212] that attach the fuel nozzle [1] (position 14) and support bracket [211] to the combustion case.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

3) Install the two bolts [212] that attach the fuel nozzle [1] (position 14) and support bracket [211] to the combustion case.

4) Install the two bolts [212] that attach the fuel nozzle [1] (position 14) and support bracket [213] to the combustion case.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

5) Install the two bolts [216] with the two loop clamps [215] that attach the fuel manifolds to the support bracket [213].

PTF ALL

(e) Install the fuel nozzle [1] (position 15), as follows (View B, Figure 453):

1) Install the support bracket [223] assembled with the support bracket [221] and support bracket [220] in position on the fuel nozzle [1] (position 15).

2) Install the four bolts [222] that attach the fuel nozzle [1] (position 15) and support bracket [223] to the combustion case.

3) Install the two bolts [224] with two loop clamps [225] that attach the LPTACC fuel manifold to the support bracket [221] and support bracket [220].

(f) Install the fuel nozzle [1] (position 16), as follows (View B, Figure 454):

1) Install the support bracket [131] in position on the fuel nozzle [1] (position 16).

a) Install the two bolts [132] that attach the support bracket [131] and support bracket [133].

b) Install the four bolts [130] that attach the fuel nozzle [1] (position 16) and support bracket [131] assembled to the combustion case.

c) Install the three bolts [231], bolt [232], three loop clamps [230], and loop clamp [233] to attach TCF accelerometer cable, EGT harnesses, and T3 sensor to the support bracket [131] at the fuel nozzle [1] (position 16) (View C, Figure 454).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Apply oil, D00599 [CP2442], on the threads of the two bolts [226].

(h) Install the two bolts [226] that attach the loop clamp [227], W200 harness [228], and W300 harness [229] to the support bracket [131] at the fuel nozzle [1] (position 16) (View B and C, Figure 454).

PTF ALL

(i) Do not tighten any bolt at this time.

SUBTASK 73-11-05-400-034-G00

(6) Hand tight all the 36 fuel manifold B-nuts onto their applicable fuel nozzle [1] fittings, before you torque the fuel nozzle bolts.

(a) Do not torque B-nuts [3] at this time.

SUBTASK 73-11-05-420-168-G00

(7) Tighten the 72 fuel nozzle bolts that attach the support brackets or fuel nozzles [1] to the combustion case, as follows:

(a) Tighten the fuel nozzle bolts starting at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 87.0 in-lb (9.8 N·m) - 103.0 in-lb (11.6 N·m) (Figure 456).

(b) Repeat tighten to the fuel nozzle bolts starting at the close-fitting bolt hole (no. 1) followed by the elongated bolt hole (no. 2) and then the remaining two bolt holes (no. 3 and 4) to 129.0 in-lb (14.57 N·m) - 151.0 in-lb (17.06 N·m) (Figure 456).

NOTE: One of the four fuel nozzle bolt holes is elongated.

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SUBTASK 73-11-05-200-001-G00

(8) Do a visual inspection of the fuel nozzle bolts to make sure that the fuel nozzle bolts are fully seated on the fuel nozzle base.

(a) Use an inspection mirror, STD-600, or equivalent, and light for inspection as many locations are difficult to access and view.

NOTE: Correct and incorrect seated positions of the fuel nozzle bolts are shown in the Figure 457.

(b) If the fuel nozzle bolts are not seated correctly, or you think that they are not seated, remove the fuel nozzle bolts and install new fuel nozzle bolts with a large amount of Tiodize T8E-H grease, D50355 [CP2961].

(c) Put the verification mark line across the fuel nozzle bolts and fuel nozzle base with temporary marker, G50530 [CP5061].

SUBTASK 73-11-05-420-169-G00

(9) Install the two loosened bolts [11] with loop clamps [12] that attach the fuel manifolds [13] to the support bracket [14] at the fuel nozzles [1] (positions 17, 18, 3, 6, 7, 10, 11, and 14) (Figure 426 and Figure 427).

(a) Tighten all the bolts [11] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

SUBTASK 73-11-05-420-170-G00

(10) Install the remaining loosened bolts at the fuel nozzles [1] (positions 17, 7, 9, 10, 11, 12, 14, 15 and 16), as follows:

(a) At the fuel nozzles [1] (positions 16 and 17):

1) Tighten the bolts [132] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

2) Tighten the bolts [231] and bolt [232] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

3) Tighten the bolts [226] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

PTF ALL

(b) At the fuel nozzle [1] (position 7):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

1) Tighten the two bolts [162] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

2) Tighten the two bolts [163] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

3) Tighten the bolt [166] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

4) Tighten the two bolts [168] to 51.0 in-lb (5.76 N·m) - 59.0 in-lb (6.67 N·m).

5) Tighten the two bolts [169] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

6) Tighten the two bolts [173] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(c) At the fuel nozzle [1] (position 9):

1) Tighten the bolt [180], bolt [183], bolt [182], and bolt [177] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

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(d) At the fuel nozzle [1] (position 10):

1) Tighten the two bolts [193] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

2) Tighten the bolt [189] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(e) At the fuel nozzle [1] (position 11):

1) Tighten the bolt [201] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

2) Tighten the bolt [127] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

3) Tighten the bolt [199] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

4) Tighten the four bolts [206] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

5) Tighten the two bolts [206] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(f) At the fuel nozzles [1] (position 12):

1) Tighten the bolt [204] to 55.0 in-lb (6.21 N·m).

(g) At the fuel nozzles [1] (position 14):

1) Tighten the two bolts [216] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

(h) At the fuel nozzles [1] (position 15):

1) Tighten the two bolts [224] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) At the fuel nozzle [1] (position 16):

1) Tighten the three bolts [231] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

2) Tighten the bolt [232] to 65.0 in-lb (7.34 N·m).

SUBTASK 73-11-05-420-171-G00

CAUTION MAKE SURE THAT YOU CORRECTLY ENGAGE THE FUEL NOZZLE FITTINGS AND YOU FULLY SEAT THEM BEFORE YOU TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, DAMAGE TO THE HARDWARE AND FUEL LEAKS CAN OCCUR.

CAUTION MAKE SURE YOU CORRECTLY TIGHTEN THE B-NUTS. IF YOU DO NOT OBEY, FUEL LEAKAGE, ENGINE FIRE, ABORTED TAKE-OFF, AIR TURN BACK AND/OR IN-FLIGHT SHUT DOWN, OR OTHER DAMAGE CAN OCCUR.

CAUTION MAKE SURE THAT THE FUEL NOZZLE AND B-NUT THREADS DO NOT CONTAIN UNWANTED MATERIAL OR LUBRICANT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Tighten the 36 fuel manifold B-nut [3] to the fuel nozzle [1] pilot-primary and pilot-secondary fittings.

NOTE: False torque is a possibility, mainly in hard to reach locations. A wrench/ attachment/ extension may clash with other hardware during tightening providing a false torque.

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(a) Tighten the fuel manifold B-nuts [3] as follows:

NOTE: Fuel nozzle B-nuts [3] require sequential triple tightening in this sequence: TORQUE – LOOSEN – TORQUE – TORQUE.

NOTE: For hard to reach locations, a socket extension (not offset) and/or crowfoot extension can be used. When a crowfoot or other offset extension is used, torque calculation is necessary (PAGEBLOCK 70-51-00/201).

1) Remove any unwanted material or lubricant from the fuel nozzle and B-nut threads.

2) Triple torque the fuel nozzle B-nuts [3] one fuel nozzle at a time as follows:

a) Tighten the two B-nuts [3] to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

b) Loosen the two B-nuts [3] until you can turn them with your fingers.

c) Tighten the two B-nuts [3] again to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

d) Tighten the two B-nuts [3] for the third time to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

e) Record completion of triple tightening and torque at each fuel nozzle B-nut fitting (Table 402).

f) Paint a stripe line across the fuel nozzle B-nuts [3] and the fuel nozzle fitting with the temporary marker, G50530 [CP5061].

3) Inspection:

NOTE: It is necessary to start inspection with the fuel nozzle [1] position 1.

a) Tighten the fuel nozzle B-nuts [3] to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m), sequentially, in clockwise direction ALF.

NOTE: On each fuel nozzle [1] torque pilot secondary (PS) B-nut first followed by pilot primary main (PPM) B-nut.

NOTE: Do not break the torque of the B-nuts.

NOTE: An inspector should be a different person from the crew that does the steps to tighten fuel nozzle B-nuts [3].

Table 402/73-11-05-993-802-G00 Triple tightening and torque inspection of Pilot Secondary (PS) and Pilot Primary Main (PPM) B-nuts

FN Position	B-nut Position	Confirm First Torque (Yes/ No)	Confirm Loosen (Yes/No)	Confirm Second Torque (Yes/ No)	Record Third Torque at Each B-nut	Completion Date and Mechanic Signature	Inspection: Record Torque at Each B-nut	Inspection: Completion Date and Signature
1	PS							
	PPM							
2	PS							
	PPM							
3	PS							
	PPM							
4	PS							
	PPM							

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Table 402/73-11-05-993-802-G00 Triple tightening and torque inspection of Pilot Secondary (PS) and Pilot Primary Main (PPM) B-nuts (Continued)

FN Position	B-nut Position	Confirm First Torque (Yes/ No)	Confirm Loosen (Yes/No)	Confirm Second Torque (Yes/ No)	Record Third Torque at Each B-nut	Completion Date and Mechanic Signature	Inspection: Record Torque at Each B-nut	Inspection: Completion Date and Signature
5	PS							
	PPM							
6	PS							
	PPM							
7	PS							
	PPM							
8	PS							
	PPM							
9	PS							
	PPM							
10	PS							
	PPM							
11	PS							
	PPM							
12	PS							
	PPM							
13	PS							
	PPM							
14	PS							
	PPM							
15	PS							
	PPM							
16	PS							
	PPM							
17	PS							
	PPM							
18	PS							
	PPM							

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WARNING WHEN YOU DO WORK WITH SAFETY CABLES, MAKE SURE YOU USE SAFETY GOGGLES. IF PART OF THE SAFETY CABLE BREAKS, INJURY TO PERSONS CAN OCCUR.

(b) Secure each fuel nozzle B-nuts [3] with the safety cable, G50065 [CP8006] (TASK 70-41-00-400-802-G00), or with the safety wire, G02495 [CP8002] (TASK 70-41-00-400-801-G00).

NOTE: If lockwire is used, terminate the lockwire at the fuel nozzle end.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-172-G00

(12) Install the HPT ACC valve drain tube [126], as follows (View F, Figure 436):

(a) Connect the HPT ACC valve drain tube [126] to the ACC valve.

(b) Apply a layer of oil, D00599 [CP2442], to the threads of the four bolts [127].

1) Install the four bolts [127] and four loop clamps [128] that attach the HPT ACC valve drain tube [126] to the ACC drain bracket, fuel manifold bracket, tube bracket, and support bracket.

a) Tighten the bolts [127] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(c) Connect the HPT ACC valve drain tube [126] to the LPT ACC valve tube [129].

(d) Connect the HPT ACC valve drain tube [126] to the drain tube (View K, Figure 436).

(e) Tighten the B-nuts on HPT ACC valve drain tube [126] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-330-G00

(13) Install the fuel hydraulic drain tube [185] as follows (View F, G, H, and I, Figure 436):

(a) Connect the HPT ACC valve drain tube [126] and B-nut [214] to the fuel hydraulic drain tube [185].

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [127].

(c) Install the three bolts [127] and three loop clamps [128] that attach the HPT ACC valve drain tube [126] to the ACC drain bracket, fuel manifold bracket, and tube bracket.

1) Tighten the bolts [127] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(d) Tighten the B-nuts on the fuel hydraulic drain tube [185] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

PTF ALL

SUBTASK 73-11-05-420-173-G00

(14) Install the 7th-stage air tube [115], as follows (View B, Figure 436):

(a) Put the 7th-stage air tube [115] in position on the engine.

(b) Apply a layer of stick wax, G51435 [CP2635], to the seal [116] to hold it during installation.

(c) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], to the threads of the eight bolts [117].

(d) Install the new seals [116] on the 7th-stage air tube [115].

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(e) Install the four bolts [117] that attach the 7th-stage air tube [115] and seal [116] to the compressor case.

(f) Install the four bolts [117] that attach the 7th-stage air tube [115] and seal [116] to HPT port.

(g) Tighten the bolts [117] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0005

(h) Apply oil, D00599 [CP2442], on the threads of the four bolts [118].

(i) Install the four bolts [118] that attach the 7th-stage air tube [115] and support bracket [119] to the support bracket.

1) Tighten the bolts [118] 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(j) Apply oil, D00599 [CP2442], on the threads of the bolts [121], bolts [122], and bolts [124].

(k) Install the two bolts [121] that attach the bracket [120] to the support bracket.

(l) Install the two bolts [122] and strap clamp [123] that attach the 7th-stage air tube [115] to the bracket.

(m) Install the two bolts [124] and block clamp [125] that attach the drain tube to the bracket [120].

(n) Tighten the bolts [121] and bolts [122] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(o) Tighten the bolts [124] to 50.6 in-lb (5.72 N·m) - 59.4 in-lb (6.71 N·m).

PTF ALL

SUBTASK 73-11-05-420-174-G00

(15) Install all the bolts and loop clamps that attach the T3 sensor [111] (TASK 73-21-64-400-801-G00) (Figure 435).

(a) Connect the T3 sensor [111] to the combustor case threaded insert adapter.

(b) Tighten the T3 sensor [111] nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 73-11-05-420-176-G00

(16) Install the LPT ACC valve drain tube [108], as follows (View H, Figure 434):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [109].

(b) Install the bolt [109] and loop clamp [110] that attaches the LPT ACC valve drain tube [108] to the drain tube bracket.

1) Tighten the bolt [109] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(c) Connect the LPT ACC valve drain tube [108] to the ACC valve drain tube.

(d) Connect the LPT ACC valve drain tube [108] to the LPT ACC fuel manifold.

(e) Tighten the B-nuts on the LPT ACC valve drain tube [108] to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

SUBTASK 73-11-05-420-175-G00

(17) Install the 7th stage air tube [101], as follows (View B, Figure 434):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(a) Apply oil, D00599 [CP2442], on the threads of the two bolts [105].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(b) Apply oil, D00599 [CP2442], on the threads of the bolts [106].

PTF ALL

(c) Apply a layer of stick wax, G51435 [CP2635], to the seals [103] to hold them in position during installation.

(d) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], to the threads of the eight bolts [102].

(e) Put the 7th stage air tube [101] in position on the engine.

(f) Install the new seals [103] on the 7th stage air tube [101].

(g) Install the four bolts [102] that attach the 7th stage air tube [101] and seal [103] to the compressor case.

(h) Install the four bolts [102] that attach the 7th stage air tube [101] and seal [103] to the HPT port.

(i) Tighten the bolts [102] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(j) Connect the air tube [107] to the 7th stage air tube [101].

(k) Tighten the air tube [107] to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(l) Install the two bolts [105] that attach the 7th stage air tube [101] to the support brackets.

1) Tighten the bolts [105] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(m) Install the four bolts [106] and two strap clamps [104] that attach the 7th stage air tube [101] to the support brackets.

1) Tighten the bolts [106] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

SUBTASK 73-11-05-420-177-G00

(18) Install the start bleed tube [91], as follows (View B, Figure 433):

(a) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the bolts [92].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [98].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(c) Apply oil, D00599 [CP2442], on the threads of the two bolts [94].

PTF ALL

(d) Install the new seal [97] on the start bleed tube [91].

(e) Install the coupling clamp [96] that attach the start bleed tube [91] and seal [97] to the SBV valve [90].

1) Tighten the coupling clamp [96] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(f) Install the two bolts [98] and start bleed clamp [99] that attach the start bleed tube [91] to the start bleed bracket.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

1) Tighten the two bolts [98] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(g) Install the two bolts [94] that attach the start bleed tube [91] to the start bleed bracket.

1) Tighten the two bolts [94] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(h) Apply a layer of stick wax, G51435 [CP2635], to the seal [93] to hold it in position during installation.

(i) Install the new seal [93] on the start bleed tube [91].

(j) Install the four bolts [92] that attach the start bleed tube [91] and seal [93] to the HPC extension case.

1) Tighten the four bolts [92] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-178-G00

(19) Connect the PS3 tube [87], as follows (View G, Figure 432):

(a) Apply the layer of Tiodize T8E-H grease, D50355 [CP2961], on the threads of fitting.

NOTE: The pure nickel special compound, D50034 [CP2619], can be used as an alternative.

(b) Connect the PS3 tube [87] to the fitting on the combustion case.

(c) Apply oil, D00599 [CP2442], on the threads of the bolts [89].

1) Install the bolts [89] and loop clamps [88] that attach the PS3 tube [87] to the support brackets.

2) Tighten the bolts [89] to 60 in-lb (6.78 N·m) - 70 in-lb (7.91 N·m).

(d) Tighten the PS3 tube [87] B-nuts at the combustion case fitting, as follows:

1) Tighten the B-nut to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

2) Loosen the B-nut until you can turn it with your fingers.

3) Tighten the B-nut again to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

4) Tighten the two B-nuts for the third time to 262 in-lb (29.60 N·m) - 308 in-lb (34.80 N·m).

5) Make sure that the B-nut is fully seated.

6) If the B-nut is not fully seated, do the above steps again to tighten the PS3 tube [87] B-nuts.

7) Secure the PS3 tube [87] B-nuts with safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002] (TASK 70-41-00-400-802-G00).

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-420-179-G00

(20) Install the CCC air manifold [74] (TASK 73-11-05-400-801-G00) (View F, Figure 432).

PTF ALL

SUBTASK 73-11-05-420-180-G00

(21) Install the bracket [73], as follows (View D, Figure 432):

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PTF ALL; AIRPLANES WITH POST LEAP-1B SB 72-0071 OR POST LEAP-1B SB 72-0135 OR PRE LEAP-1B SB 72-0217

(a) Put the bracket [73] back on the engine and put it in position for installation.

(b) Apply oil, D00599 [CP2442], on the threads of the bolt [85], bolts [78], bolt [76], and bolts [71].

(c) Install the bolt [85] that attaches CCC air manifold [74] to the bracket [73].

1) Do not tighten at this time.

(d) Install the two bolts [78] that attach bracket [73] to the support bracket.

1) Do not tighten at this time.

(e) Install the bolt [76] and loop clamp [77] that attaches the drain tube [75] to the bracket [73].

1) Do not tighten at this time.

(f) Install the two bolts [71] and two loop clamps [72] that attach the SBV TBV manifolds [70] to the bracket [73].

1) Do not tighten at this time.

(g) Tighten the bolt [85] and bolts [78] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m)

(h) Tighten the bolt [76] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

(i) Tighten the bolts [71] to 51.0 in-lb (5.8 N·m) - 59.0 in-lb (6.7 N·m).

(j) Apply oil, D00599 [CP2442], on the threads of the four bolts [80].

(k) Install the four bolts [80] and four loop clamps [81] that attach the SBV TBV manifolds [70] to the bracket [82].

1) Tighten the four bolts [80] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(l) Apply oil, D00599 [CP2442], on the threads of the two bolts [83].

(m) Install the two bolts [83] and four loop clamps [84] that attach the SBV TBV manifolds [70] to the bracket [82].

1) Tighten the bolts [83] to 32.5 in-lb (3.67 N·m) - 37.5 in-lb (4.24 N·m).

PTF ALL

SUBTASK 73-11-05-420-181-G00

(22) Connect the HJ8A harness [65] (TASK 73-22-01-400-808-G00) (View B, Figure 432).

SUBTASK 73-11-05-420-182-G00

(23) Install the 7th-stage air tube [55], as follows (View B, Figure 431):

(a) Apply a layer of stick wax, G51435 [CP2635], to the new seal ring [57] and new seal ring [60] to hold them in position during installation.

(b) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the four bolts [58].

(c) Apply oil, D00599 [CP2442], on the threads of the four bolts [56] and two bolts [62].

(d) Install the new seal ring [60] at the aft end of the 7th-stage air tube [55] on the HPT case.

(e) Install the new seal ring [57] at the forward end of the 7th-stage air tube [55] on the HPC extension case.

(f) Put the 7th-stage air tube [55] in position between the 7th-stage air tube bracket and HPT case.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(g) Install the two bolts [62] that attach the strap clamp [61] and 7th-stage air tube [55] to the 7th-stage air tube bracket.

1) Tighten the two bolts [62] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0005

(h) Install the two bolts [62] that attach the 7th-stage air tube [55] to the 7th-stage air tube bracket.

1) Tighten the two bolts [62] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL

(i) Install the four bolts [58] that attach the bracket [59] to the 7th-stage air tube [55] and HPT case.

(j) Install the four bolt [56] that attach the 7th-stage air tube [55] to the HPC extension case.

(k) Tighten the bolts [58] and bolt [56] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-183-G00

(24) Install the booster anti-ice tube [37], as follows (View E, Figure 430):

(a) Support and secure the booster anti-ice tube [38] and booster anti-ice tube [39] from damage.

(b) Support and secure the booster anti-ice tube [37] from damage.

(c) Install the new seal [41] and coupling clamp [40] that attach the booster anti-ice tube [37] to the booster anti-ice tube [44].

1) Do not tighten the coupling clamp [40] at this moment.

(d) Install the two coupling clamps [46] and two new seals [45] that attach the booster anti-ice tube [37] to the booster anti-ice tube [38] and booster anti-ice tube [39].

1) Do not tighten the two coupling clamps [46] at this moment.

(e) Apply oil, D00599 [CP2442], on the threads of the bolts [42] and bolts [47].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(f) Install the four bolts [42] and two strap clamps [43] that attach the booster anti-ice tube [37] and booster anti-ice tube [38] to the support brackets.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(g) Install the two bolts [42] and strap clamp [43] that attach the booster anti-ice tube [37] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 OR PRE LEAP-1B SERVICE BULLETIN 72-0144

(h) Install the bolt [47] and loop clamp [48] that attach the booster anti-ice tube [37] and booster anti-ice tube [39] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(i) Install the two bolts [47] and strap clamp [43] that attach the booster anti-ice tube [37] and booster anti-ice tube [39] to the support bracket.

PTF ALL

(j) Tighten the coupling clamp [40] and coupling clamps [46] to 60.0 in-lb (6.78 N·m) - 70.0 in-lb (7.91 N·m).

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(k) Tighten the bolts [42] and bolts [47] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-197-G00

(25) Install the booster anti-ice tube [44], as follows (View G, Figure 430):

(a) Apply oil, D00599 [CP2442], on the threads of the bolt [50] and bolt [53].

(b) Install the new seal [52] and coupling clamp [51] that attaches the booster anti-ice tube [44] to the SBV valve at the 4:00 o'clock position (View J, Figure 430).

1) Do not tighten the coupling clamp [51] at this time.

(c) Support and tightly attach the booster anti-ice tube [44].

(d) Install the bolt [50], bolt [53], and clamps [49] that attach the booster anti-ice tube [44] to the support bracket.

1) Do not tighten the bolt [50] and bolt [53] at this time.

(e) Tighten the coupling clamp [51], bolt [50], and bolt [53] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-05-420-184-G00

(26) Install the transient bleed air tube [31], as follows (View B, Figure 430):

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

NOTE: The transient bleed air tube is referenced as the reverse bleed check valve assembly in Post LEAP-1B Service Bulletin 72-0378.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Apply a layer of stick wax, G51435 [CP2635], to the metal seal ring [33] to hold it in position during installation.

(b) Apply Tiodize T8E-H grease, D50355 [CP2961], on the threads of the four bolts [32].

(c) Install the four bolts [32] that attach the transient bleed air tube [31] and new metal seal ring [33] to the combustion case.

1) Do not tighten the bolts [32] at this time.

(d) Apply oil, D00599 [CP2442], on the threads of the two bolts [34].

(e) Install the two bolts [34] that attach the transient bleed air tube [31] to the support bracket.

1) Do not tighten the bolts [34] at this time.

(f) Install the coupling clamp [35] and new seal [20] on the transient bleed valve [36].

(g) Tighten the coupling clamp [35] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(h) Tighten the bolts [32] and bolts [34] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(i) Install the reverse bleed check valve assembly (TASK 75-34-01-400-801-G00).

PTF ALL

SUBTASK 73-11-05-420-185-G00

(27) Install the wye duct [28], as follows (Figure 429):

(a) Put the wye duct [28] onto the core module to interface with the High-Pressure Turbine ACC inlet duct at the 9:30 o'clock position.

1) If the wye duct [28] already has silicone rubber tape, G51669 [CP2975], in the clamping area, proceed to step (b).

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2) If the wye duct [28] does not have silicone rubber tape, G51669 [CP2975], in the clamping area, apply as follows:

a) Apply the two adjacent rows of silicone rubber tape, G51669 [CP2975].

NOTE: Each row will be three plies thick of silicone rubber tape, G51669 [CP2975], 0.02 in. (0.51 mm) x 1.0 in. (25.4 mm) width to the wye duct [28] directly aft of the forward edge bead of the wye duct [28].

(b) Apply oil, D00599 [CP2442], on the threads of the two bolts [30].

(c) Attach the wye duct [28] to the HPTACC inlet duct with the bolts [30] and saddle clamps [29].

(d) Torque the bolts [30] to 51 in-lb (5.8 N·m) - 59 in-lb (6.7 N·m).

(e) Measure the gap between the saddle clamps [29] with the feeler gage (View D, Figure 429).

1) Measure each gap in a location that is between the centerline of the bolt and the bend radii for the relief holes.

NOTE: The gaps must be measured at the forward and aft sides of both bolts.

2) Acceptable gap between the saddle clamps [29] adjacent to the wye duct [28] is to 0.045 in. (1.143 mm) - 0.075 in. (1.905 mm).

3) If measured gap exceeds 0.075 in. (1.905 mm), remove the bolts [30] from the saddle clamps [29].

a) Remove one ply of silicone rubber tape, G51669 [CP2975], from wye duct [28].

b) Repeat steps (c), (d), and (e).

4) If measured gap is less than 0.045 in. (1.143 mm), remove the bolts [30] from the saddle clamps [29].

a) Add one ply of silicone rubber tape, G51669 [CP2975], to wye duct [28].

b) Repeat steps (c), (d), and (e).

SUBTASK 73-11-05-420-186-G00

(28) Install the LPTACC valve [27] (TASK 75-24-01-400-801-G00).

(a) If necessary, remove the length from the hose clamp tab over 1 in. (25.4 mm) from the hose clamp screw housing.

1) Remove all burrs from the hose clamp tabs.

SUBTASK 73-11-05-020-195-G00

(29) Install the HPTACC valve [26] (TASK 75-24-04-400-801-G00).

(a) If necessary, remove the length from the hose clamp tab over 1 in. (25.4 mm) from the hose clamp screw housing.

1) Remove all burrs from the hose clamp tabs.

SUBTASK 73-11-05-420-187-G00

(30) Install the upper CDP duct assembly (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-188-G00

(31) Install the lower CDP duct assembly (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-189-G00

(32) Install the High Pressure Shutoff Valve (HPSOV) [24] (TASK 36-11-07-400-801).

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SUBTASK 73-11-05-420-190-G00

(33) Install the mix manifold [22] (TASK 36-11-01-400-801).

SUBTASK 73-11-05-420-191-G00

(34) Install the Pressure Regulating and Shutoff Valve (PRSOV) [21] (TASK 36-11-04-400-801).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-05-420-325-G00

(1) Install the thrust reverser compression rod No. 2 (Figure 424).

SUBTASK 73-11-05-860-018-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-420-192-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-05-420-193-G00

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 73-11-05-420-194-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

K. Fuel Nozzle Installation Test

SUBTASK 73-11-05-420-323-G00

(1) Do the tests as follows:

(a) Do an idle check (TASK 71-00-00-790-804-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

a) Electronic Engine Control (EEC) can find fuel system faults after fuel system maintenance because of residual air in fuel lines, do these steps:

<1> Do an engine idle run (TASK 71-00-00-790-804-G00).

<a> Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table, note 7.a.

b) Check the Onboard Maintenance Function (OMF) for any new fuel system faults.

<1> If the faults are found, clear the faults (TASK 71-00-00-700-811-G00 and TASK 71-00-00-700-812-G00).

<a> Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table, note 7.b.

(b) Do a pneumatic leak test (TASK 71-00-00-790-801-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

NOTE: This is necessary for High Pressure Shutoff Valve (HPSOV), Environmental Control System (ECS) ducts, Pressure Regulating and Shutoff Valve (PRSOV), and Transient Bleed Valve (TBV) duct installation.

a) After the tests are done, follow the applicable inspections:

<1> Do a visual inspection of the engine after idle leak check and do as follows:

<a> Do a visual inspection of the fuel lines and connections and check for fuel leaks.

<b> Do a visual inspection of the air ducts, tubes, and seals to check for hot air leakage indication and/or damage.

(c) Do an idle test menu mode (TASK 71-00-00-700-808-G00).

1) Refer to the Power Plant Test Reference Table, TASK 71-00-00-800-804-G00.

NOTE: Use Power Plant Test table.

SUBTASK 73-11-05-420-196-G00

(2) If you opened the EEC air signal manifolds, do this step:

(a) Do the test for the air ducts and air tubes (TASK 71-00-00-800-804-G00).

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SUBTASK 73-11-05-700-004-G00

(3) If it is necessary, do this step: Test No. 3 - Wet Motor Leak Test, TASK 71-00-00-790-803-G00.

END OF TASK

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FUEL NOZZLE AND FUEL MANIFOLD - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fuel nozzle and fuel manifold check.

TASK 73-11-05-200-801-G00

2. Fuel Nozzle and Fuel Manifold Leak Check

(Figure 601, Figure 602, Figure 603, and Figure 604)

A. General

(1) This task provides the instructions on how to examine the fuel nozzle and fuel manifold for leakage.

B. References

Reference	Title
71-00-00-790-804-G00	Test No. 4 - Idle Leak Test (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-05 P/B 401	FUEL NOZZLE - REMOVAL/INSTALLATION
73-11-05-000-801-G00	Fuel Nozzle Removal (P/B 401)
73-11-05-000-802-G00	Fuel Nozzle Set Removal (P/B 401)
73-11-05-400-801-G00	Fuel Nozzle Installation (P/B 401)
73-11-05-400-802-G00	Fuel Nozzle Set Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14801	Adapter Set - Manifold Pressure Check Part #: 956A3209G03 Supplier: 58828 Opt Part #: 956A3209G01 Supplier: 58828 Opt Part #: 956A3209G02 Supplier: 58828
SPL-14802	Cart- Fuel Manifold Pressure Check Part #: 956A1256G01 Supplier: 58828
SPL-14835	Actuator - Portable Hydraulic, VSV Part #: 856A1084G04 Supplier: 58828 Part #: 9465M11P01 Supplier: 58828 Part #: 9471M46G01 Supplier: 58828
SPL-26358	Adapter Set - Pressure Check, Fuel Manifold Part #: 956A1631G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

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(Continued)

Reference	Description
STD-1054	Container - Fuel Resistant, 5-Gallon (19-Liter)

D. Consumable Materials

Reference	Description	Specification
D00623 [CP5066]	Oil - Fuel System, Corrosion Preventive	MIL-PRF-6081, Grade 1010
G50380	LEAK TEC - 16-OX, THIN FILM BUBBLE TESTING SOLUTIONS	MIL-PRF-25567

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Check

SUBTASK 73-11-05-860-012-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-05-860-013-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-010-001-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fuel Nozzle and Fuel Manifold Leak Check (Preferred Method)

SUBTASK 73-11-05-680-004-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Drain the three fuel manifold sections as follows (TASK 73-11-05-000-801-G00 or TASK 73-11-05-000-802-G00):

NOTE: There are no fuel manifold drain plugs on the engine. Each fuel manifold tube is independent. You must drain each one at its lowest connection point on the manifold section. Each manifold section has two separate manifold tubes.

NOTE: The fuel manifolds can have a small amount of pressure remaining after you drain the fuel manifold tube.

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the fuel manifold drain point.

(b) Loosen the B-nuts at the fuel manifold drain points and drain the fuel into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 73-11-05-480-002-G00

(2) Prepare the hydraulic actuator, SPL-14835, and adapter set, SPL-26358, as follows:

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(a) Fill the reservoir of the hydraulic actuator, SPL-14835, with oil, D00623 [CP5066].

(b) Close the inlet valve [32] and outlet valve [34] on the regulator assembly [30].

(c) Connect the supply line of the hydraulic actuator, SPL-14835, to the inlet nipple [33] on the regulator assembly [30].

(d) Connect the return line of the hydraulic actuator, SPL-14835, to the outlet nipple [35] on the regulator assembly [30].

(e) Open the inlet valve [32] on the regulator assembly [30].

1) Make sure that the outlet valve [34] on the regulator assembly [30] is closed.

SUBTASK 73-11-05-790-005-G00

(3) Do these steps to examine for leaks on the Pilot Primary and Main Enriched (PPME) fuel manifold:

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the PPME inlet fuel hose [10].

(b) Disconnect the PPME inlet fuel hose [10] from the SCU [22] (TASK 73-11-05-000-801-G00).

1) Let the fuel drain into the 5-gallon (19-liter) fuel resistant container, STD-1054.

(c) Install the PPME adapter [36] on the PPME inlet fuel hose [10] as follows:

1) Remove the cap and dust cover from the PPME adapter [36].

2) Tightly connect the PPME adapter [36] to the PPME inlet fuel hose [10].

(d) Tightly connect the hose coupler [31] on the regulator assembly [30] to the PPME adapter [36].

(e) Use the hydraulic actuator, SPL-14835, to apply pressure to the PPME fuel manifold as follows:

1) Make sure that the directional valve on the hydraulic pump is set to a position that supplies fluid flow to the inlet nipple [33] on the regulator assembly [30].

2) Make sure that the outlet valve [34] on the regulator assembly [30] is closed.

3) Pressurize the PPME fuel manifold to 40 psig (276 kPa) - 43 psig (296 kPa).

4) When the pressure becomes stable, close the inlet valve [32] on the regulator assembly [30].

(f) Keep the pressure in the PPME fuel manifold for 5-6 minutes as follows:

1) Monitor the pressure gauge on the regulator assembly [30].

2) If the pressure does not stay stable, find and make a record of the source(s) of the leakage as follows:

a) Examine the connections on the PPME fuel manifold.

b) Examine the fuel nozzles inside of the combustion chamber.

(g) Open the outlet valve [34] on the regulator assembly [30] to release the pressure in the PPME fuel manifold.

(h) If leakage was found at a connection, do these steps:

1) Loosen the B-nut that has the leak.

2) Tighten the B-nut to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

3) Loosen the B-nut until you can turn it with your fingers.

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4) Tighten the B-nut again to the specified torque.

5) Tighten the B-nut a third time to the specified torque.

6) Apply pressure to the PPME fuel manifold for 5-6 minutes and recheck for leaks as specified in steps above.

(i) If leakage was found at a fuel nozzle, do these steps:

1) Replace the fuel nozzle that has the leak (PAGEBLOCK 73-11-05/401).

2) Apply pressure to the PPME fuel manifold for 5-6 minutes and recheck for leaks as specified in steps above.

(j) Disconnect the hose coupler [31] on the regulator assembly [30] from the PPME adapter [36].

(k) Remove the PPME adapter [36] from the PPME inlet fuel hose [10].

(l) Connect the PPME inlet fuel hose [10] to the SCU [22] (TASK 73-11-05-400-801-G00).

SUBTASK 73-11-05-790-006-G00

(4) Do the steps that follow to examine for leaks on the Pilot Primary and Main (PPM) fuel manifolds:

NOTE: The PPM fuel manifolds can also be identified as the Pilot Primary and Main Non-Enriched (PPMNE) fuel manifolds.

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the PPM fuel supply hose [11].

(b) Disconnect the PPM fuel supply hose [11] from the SCU [22] (TASK 73-11-05-000-801-G00).

1) Let the fuel drain into the 5-gallon (19-liter) fuel resistant container, STD-1054.

(c) Install the PPM adapter [37] on the PPM fuel supply hose [11] as follows:

1) Remove the cap and dust cover from the PPM adapter [37].

2) Tightly connect the PPM adapter [37] to the PPM fuel supply hose [11].

(d) Tightly connect the hose coupler [31] on the regulator assembly [30] to the PPM adapter [37].

(e) Use the hydraulic actuator, SPL-14835, to apply pressure to the PPM fuel manifolds as follows:

1) Make sure that the directional valve on the hydraulic pump is set to a position that supplies fluid flow to the inlet nipple [33] on the regulator assembly [30].

2) Make sure that the outlet valve [34] on the regulator assembly [30] is closed.

3) Pressurize the PPM fuel manifolds to 40 psig (276 kPa) - 43 psig (296 kPa).

4) When the pressure becomes stable, close the inlet valve [32] on the regulator assembly [30].

(f) Keep the pressure in the PPM fuel manifolds for 5 - 6 minutes as follows:

1) Monitor the pressure gauge on the regulator assembly [30].

2) If the pressure does not stay stable, find and make a record of the source(s) of the leakage as follows:

a) Examine the connections on the PPM fuel manifolds.

b) Examine the fuel nozzles inside of the combustion chamber.

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(g) Open the outlet valve [34] on the regulator assembly [30] to release the pressure in the PPM fuel manifolds.

(h) If leakage was found at a connection, do these steps:

1) Loosen the B-nut that has the leak.

2) Tighten the B-nut to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

3) Loosen the B-nut until you can turn it with your fingers.

4) Tighten the B-nut again to the specified torque.

5) Tighten the B-nut a third time to the specified torque.

6) Apply pressure to the PPM fuel manifolds for 5-6 minutes and recheck for leaks as specified in steps above.

(i) If leakage was found at a fuel nozzle, do these steps:

1) Replace the fuel nozzle that has the leak (PAGEBLOCK 73-11-05/401).

2) Apply pressure to the PPM fuel manifolds for 5-6 minutes and recheck for leaks as specified in steps above.

(j) Disconnect the hose coupler [31] on the regulator assembly [30] from the PPM adapter [37].

(k) Remove the PPM adapter [37] from the PPM fuel supply hose [11].

(l) Connect the PPM fuel supply hose [11] to the SCU [22] (TASK 73-11-05-400-801-G00).

SUBTASK 73-11-05-790-007-G00

(5) Do these steps to examine for leaks on the Pilot Secondary (PSEC) fuel manifolds:

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the PSEC fuel supply hose [12].

(b) Disconnect the PSEC fuel supply hose [12] from the SCU [22] (TASK 73-11-05-000-801-G00).

1) Let the fuel drain into the 5-gallon (19-liter) fuel resistant container, STD-1054.

(c) Install the PSEC adapter [38] on the PSEC fuel supply hose [12] as follows:

1) Remove the cap and dust cover from the PSEC adapter [38].

2) Tightly connect the PSEC adapter [38] to the PSEC fuel supply hose [12].

(d) Tightly connect the hose coupler [31] on regulator assembly [30] to the PSEC adapter [38].

(e) Use the hydraulic actuator, SPL-14835, to apply pressure to the PSEC fuel manifolds as follows:

1) Make sure that the directional valve on the hydraulic pump is set to a position that supplies fluid flow to the inlet nipple [33] on the regulator assembly [30].

2) Make sure that the outlet valve [34] on the regulator assembly [30] is closed.

3) Pressurize the PSEC fuel manifolds to 40 psig (276 kPa) - 43 psig (296 kPa).

4) When the pressure becomes stable, close the inlet valve [32] on the regulator assembly [30].

(f) Keep the pressure in the PSEC fuel manifolds for 5 - 6 minutes as follows:

1) Monitor the pressure gauge on the regulator assembly [30].

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2) If the pressure does not stay stable, find and make a record of the source(s) of the leakage as follows:

a) Examine the connections on the PSEC fuel manifolds.

b) Examine the fuel nozzles inside of the combustion chamber.

(g) Open the outlet valve [34] on the regulator assembly [30] to release the pressure in the PSEC fuel manifolds.

(h) If leakage was found at a connection, do these steps:

1) Loosen the B-nut that has the leak.

2) Tighten the B-nut to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

3) Loosen the B-nut until you can turn it with your fingers.

4) Tighten the B-nut again to the specified torque.

5) Tighten the B-nut a third time to the specified torque.

6) Apply pressure to the PSEC fuel manifolds for 5-6 minutes and recheck for leaks as specified in steps above.

(i) If leakage was found at a fuel nozzle, do these steps:

1) Replace the fuel nozzle that has the leak (PAGEBLOCK 73-11-05/401).

2) Apply pressure to the PSEC fuel manifolds for 5-6 minutes and recheck for leaks as specified in steps above.

(j) Disconnect the hose coupler [31] on the regulator assembly [30] from the PSEC adapter [38].

(k) Remove the PSEC adapter [38] from the PSEC fuel supply hose [12].

(l) Connect the PSEC fuel supply hose [12] to the SCU [22] (TASK 73-11-05-400-801-G00).

SUBTASK 73-11-05-080-001-G00

(6) Disassemble the hydraulic pressure test equipment as follows:

(a) Put the caps and dust covers back on the PPME adapter [36], PPM adapter [37], and PSEC adapter [38].

(b) Open the inlet valve [32] and outlet valve [34] on the regulator assembly [30].

(c) Allow the remaining fluid in the regulator assembly [30] to flow back into the reservoir of the hydraulic actuator, SPL-14835.

(d) Disconnect the supply line of the hydraulic actuator, SPL-14835, from the inlet nipple [33] on the regulator assembly [30].

(e) Disconnect the return line of the hydraulic actuator, SPL-14835, from the outlet nipple [35] on the regulator assembly [30].

(f) Put the dust covers back on the regulator assembly [30].

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I. Fuel Nozzle and Fuel Manifold Leak Check (Alternative Method)

SUBTASK 73-11-05-680-002-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Drain the three fuel manifold sections as follows (TASK 73-11-05-000-801-G00 or TASK 73-11-05-000-802-G00):

NOTE: There are no fuel manifold drain plugs on the engine. Each fuel manifold tube is independent. You must drain each one at its lowest connection point on the manifold section. Each manifold section has two separate manifold tubes.

NOTE: The fuel manifolds can have a small amount of pressure remaining after you drain the fuel manifold tube.

(a) Put a 5-gallon (19-liter) fuel resistant container, STD-1054, below the fuel manifold drain point.

(b) Loosen the B-nut at the fuel manifold drain points and drain the fuel into the 5-gallon (19-liter) fuel resistant container, STD-1054.

SUBTASK 73-11-05-790-002-G00

(2) Do the steps that follow to examine for leaks on the right side fuel manifolds.

NOTE: As you view the manifolds aft looking forward.

(a) Remove the caps from the Adapter Set, SPL-14801, as follows:

1) Remove the cap [15] from the tube assembly [13].

2) Remove the cap [14] from the tube assembly [21].

3) Remove the cap [20] from the male connector [19] on the gage adapter [18].

(b) Make sure that you install the plug [16] tightly into tube assembly [17].

(c) Connect the tube assembly [21] to the PSEC RS fuel manifold [2] as follows:

1) Disconnect the PSEC fuel tube [7] from the PSEC RS fuel manifold [2].

2) Tightly connect the tube assembly [21] to the PSEC RS fuel manifold [2].

(d) Connect the tube assembly [13] to the PPM RS fuel manifold [1] as follows:

1) Disconnect the PPM fuel tube [8] from the PPM RS fuel manifold [1].

2) Tightly connect the tube assembly [13] to the PPM RS fuel manifold [1].

(e) Tightly connect the hose from the nitrogen cart, SPL-14802, to the male connector [19].

(f) Use the nitrogen cart, SPL-14802, to apply pressure to the fuel manifolds as follows:

1) Make sure that you filter the inert gas to 10 micron maximum with a non-bypass metallic or ceramic mesh filter.

2) Apply a pressure of 10 psig (69 kPa) to the fuel manifolds.

3) Make sure that the check valves in the fuel nozzles do not open at this pressure.

4) Slowly increase the pressure to 15 psig (103 kPa).

5) Make sure that the check valves in the fuel nozzles do not open as the pressure increases.

6) If a check valve opens, replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

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7) Use the nitrogen cart, SPL-14802, to apply a pressure of 40 psig (276 kPa) - 50 psig (345 kPa) to the manifolds.

(g) Let the system pressure stabilize.

(h) Use Leak Tec 16-OX (leak detection compound), G50380, at the manifold to the fuel nozzle B-nuts of the PSEC RS fuel manifold [2] and PPM RS fuel manifold [1].

(i) Make sure that there is no leakage.

(j) If you find leakage on a connection, do these steps:

1) Turn off the pressure from the nitrogen cart, SPL-14802.

2) Loosen the B-nut that have the leakage.

3) Tighten the connection to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

4) Loosen the two B-nut until you can turn them with your fingers.

5) Tighten the connection again to the specified torque.

6) Tighten the connection a third time to the specified torque.

(k) Use Leak Tec 16-OX (leak detection compound), G50380, at the fuel nozzle assemblies where they attach to the combustion case.

(l) Make sure that there are no leaks.

(m) If there are leaks at the fuel nozzle assemblies, do these steps:

1) Turn off the pressure from the nitrogen cart, SPL-14802.

2) Replace the fuel nozzle assembly that has the leak (PAGEBLOCK 73-11-05/401).

(n) Turn off the pressure from the nitrogen cart, SPL-14802.

(o) Disconnect the Adapter Set, SPL-14801, from the nitrogen cart, SPL-14802.

(p) Disconnect the tube assembly [13] from the PPM RS fuel manifold [1].

(q) Connect the PPM RS fuel manifold [1] to the PPM fuel tube [8] as follows:

1) Tighten the B-nut to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

2) Loosen the B-nut until you can turn it with your fingers.

3) Tighten the B-nut again to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

4) Tighten the B-nut a final time to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

(r) Disconnect the tube assembly [21] from the PSEC RS fuel manifold [2].

(s) Connect the PSEC RS fuel manifold [2] to the PSEC fuel tube [7] as follows:

1) Tighten the B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

2) Loosen the B-nut until you can turn it with your fingers.

3) Tighten the B-nut again to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

4) Tighten the B-nut a final time to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

SUBTASK 73-11-05-211-001-G00

(3) Do the steps that follow to examine for leaks on the left side fuel manifolds.

NOTE: As you view the manifolds aft looking forward.

(a) Remove the caps from the Adapter Set, SPL-14801, as follows:

1) Remove the cap [15] from the tube assembly [13].

2) Remove the cap [14] from the tube assembly [21].

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3) Remove the cap [20] from the male connector [19] on the gage adapter [18].

(b) Make sure that you install the plug [16] tightly into the tube assembly [17].

(c) Connect the tube assembly [21] to the PSEC LS fuel manifold [5] as follows:

1) Disconnect the PSEC fuel supply hose [12] from the PSEC LS fuel manifold [5].

2) Tightly connect the tube assembly [21] to the PSEC LS fuel manifold [5].

(d) Connect the tube assembly [13] to the PPM LS fuel manifold [6] as follows:

1) Disconnect the PPM fuel supply hose [11] from the PPM LS fuel manifold [6].

2) Tightly connect the tube assembly [13] to the PPM LS fuel manifold [6].

(e) Tightly connect the hose from the nitrogen cart, SPL-14802, to the male connector [19].

(f) Use the nitrogen cart, SPL-14802, to apply pressure to the fuel manifolds as follows:

1) Make sure that you filter the inert gas to 10 micron maximum with a non-bypass metallic or ceramic mesh filter.

2) Apply a pressure of 10 psig (69 kPa) to the fuel manifolds.

3) Make sure that the check valves in the fuel nozzles do not open at this pressure.

4) Slowly increase the pressure to 15 psig (103 kPa).

5) Make sure that the check valves in the fuel nozzles do not open as the pressure increases.

6) If a check valve opens, replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

7) Use the nitrogen cart, SPL-14802, to apply a pressure of 40 psig (276 kPa) - 50 psig (345 kPa) to the manifolds.

(g) Let the system pressure stabilize.

(h) Use Leak Tec 16-OX (leak detection compound), G50380, at the manifold to the fuel nozzle B-nuts of the manifolds PSEC LS fuel manifold [5] and PPM LS fuel manifold [6].

(i) Make sure that there is no leakage.

(j) If you find leakage on a connection, do these steps:

1) Turn off the pressure from the nitrogen cart, SPL-14802.

2) Loosen the B-nut that have the leakage.

3) Tighten the connection to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

4) Loosen the B-nut until you can turn them with your fingers.

5) Tighten the connection again to the specified torque.

6) Tighten the connection a third time to the specified torque.

(k) Use Leak Tec 16-OX (leak detection compound), G50380, at the fuel nozzle assemblies where they attach to the combustion case.

(l) Make sure that there are no leaks.

(m) If there are leaks at the fuel nozzle assemblies, do these steps:

1) Turn off the pressure from the nitrogen cart, SPL-14802.

2) Replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

(n) Turn off the pressure from the nitrogen cart, SPL-14802.

(o) Disconnect the Adapter Set, SPL-14801, from the nitrogen cart.

(p) Disconnect the tube assembly [21] from the PSEC LS fuel manifold [5].

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(q) Connect the PSEC LS fuel manifold [5] to the PSEC fuel supply hose [12] as follows:
1) Tighten the B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).
2) Loosen the B-nut until you can turn it with your fingers.
3) Tighten the B-nut again to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).
4) Tighten the B-nut a final time to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

(r) Disconnect the tube assembly [13] from the PPM LS fuel manifold [6].

(s) Connect the PPM LS fuel manifold [6] to the PPM fuel supply hose [11] as follows:
1) Tighten the B-nut to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).
2) Loosen the B-nut until you can turn it with your fingers.
3) Tighten the B-nut again to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).
4) Tighten the B-nut a final time to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

SUBTASK 73-11-05-211-002-G00

(4) Do the steps that follow to examine for leaks on the bottom fuel manifolds.

NOTE: As you view the manifolds aft looking forward.

(a) Remove the plug and caps from the Adapter Set, SPL-14801, as follows:
1) Remove the plug [16] from the tube assembly [17].
2) Remove the cap [14] from the tube assembly [21].
3) Remove the cap [20] from the male connector [19] on the gage adapter [18].

(b) Make sure that you install the cap [15] tightly on the tube assembly [13].

(c) Connect the tube assembly [21] to the PSEC bottom fuel manifold [4] as follows:
1) Disconnect the PSEC fuel tube [9] from the PSEC bottom fuel manifold [4].
2) Tightly connect the tube assembly [21] to the PSEC bottom fuel manifold [4].

(d) Connect the tube assembly [17] to the PPM bottom fuel manifold [3] as follows:
1) Disconnect the PPME inlet fuel hose [10] from the PPM bottom fuel manifold [3].
2) Tightly connect the tube assembly [17] to the PPM bottom fuel manifold [3].

(e) Tightly connect the hose from the nitrogen cart, SPL-14802, to the male connector [19].

(f) Use the nitrogen cart, SPL-14802, to apply pressure to the fuel manifolds as follows:
1) Make sure that you filter the inert gas to 10 micron maximum with a non-bypass metallic or ceramic mesh filter.
2) Apply a pressure of 10 psig (69 kPa) to the fuel manifolds.
3) Make sure that the check valves in the fuel nozzles do not open at this pressure.
4) Slowly increase the pressure to 15 psig (103 kPa).
5) Make sure that the check valves in the fuel nozzles do not open as the pressure increases.
6) If a check valve opens, replace the fuel nozzle (PAGEBLOCK 73-11-05/401).
7) Use the nitrogen cart, SPL-14802, to apply a pressure of 40 psig (276 kPa) - 50 psig (345 kPa) to the manifolds.

(g) Let the system pressure stabilize.

(h) Use Leak Tec 16-OX (leak detection compound), G50380, at the manifold to the fuel nozzle B-nuts of the PPM bottom fuel manifold [3] and PSEC bottom fuel manifold [4].

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(i) Make sure that there is no leakage.

(j) If you find leakage on a connection, loosen the B-nut and do these steps:

1) Turn off the pressure from the nitrogen cart.
2) Loosen the B-nut that have the leakage.
3) Tighten the connection to the specified torque (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).
4) Loosen the B-nut until you can turn them with your fingers.
5) Tighten the connection again to the specified torque.
6) Tighten the connection a third time to the specified torque.

(k) Use Leak Tec 16-OX (leak detection compound), G50380, at the fuel nozzle assemblies where they attach to the combustion case.

(l) Make sure that there are no leaks.

(m) If there are leaks at the fuel nozzle assemblies, do these steps:

1) Turn off the pressure from the nitrogen cart, SPL-14802.
2) Replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

(n) Turn off the pressure from the nitrogen cart, SPL-14802.

(o) Disconnect the Adapter Set, SPL-14801, from the nitrogen cart, SPL-14802.

(p) Disconnect the tube assembly [21] from the PSEC bottom fuel manifold [4].

(q) Connect the PSEC bottom fuel manifold [4] to the PSEC fuel tube [9] as follows:

1) Tighten the B-nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).
2) Loosen the B-nut until you can turn it with your fingers.
3) Tighten the B-nut again to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).
4) Tighten the B-nut a final time to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

(r) Disconnect the tube assembly [17] from the PPM bottom fuel manifold [3].

(s) Connect the PPM bottom fuel manifold [3] to the PPME inlet fuel hose [10] and triple tighten as follows:

1) Tighten the B-nut to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).
2) Loosen the B-nut until you can turn it with your fingers.
3) Tighten the B-nut again to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).
4) Tighten the B-nut a final time to 460.0 in-lb (51.97 N·m) - 540.0 in-lb (61.01 N·m).

(t) Remove the nitrogen cart, SPL-14802, from the manifold fitting.

(u) Remove the Norgen general purpose filter.

SUBTASK 73-11-05-480-001-G00

(5) Put all caps and plug on the Adapter Set, SPL-14801.

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J. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-05-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-05-860-016-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-860-017-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

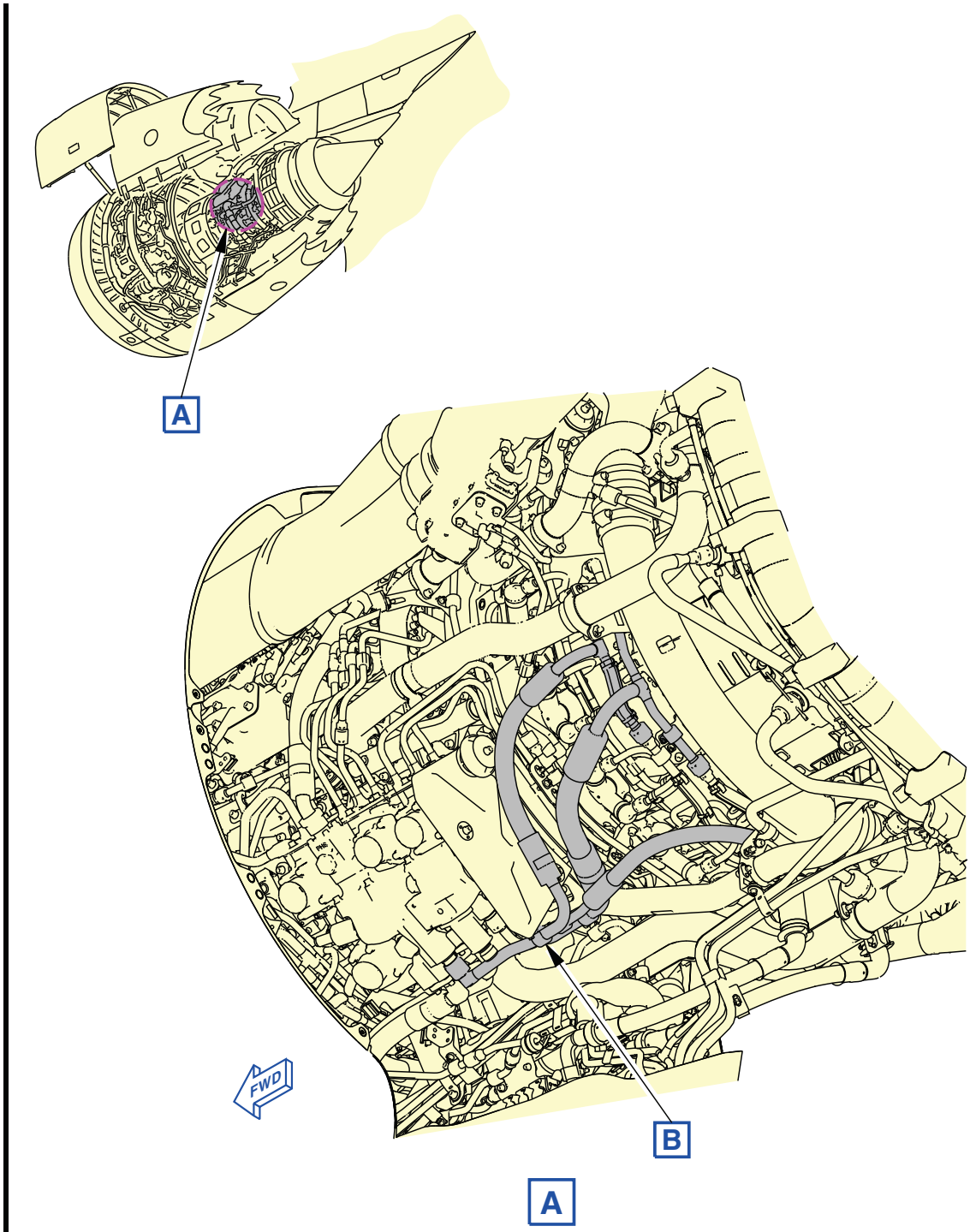
(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-05-790-003-G00

(4) Do this task: Test No. 4 - Idle Leak Test, TASK 71-00-00-790-804-G00.

END OF TASK

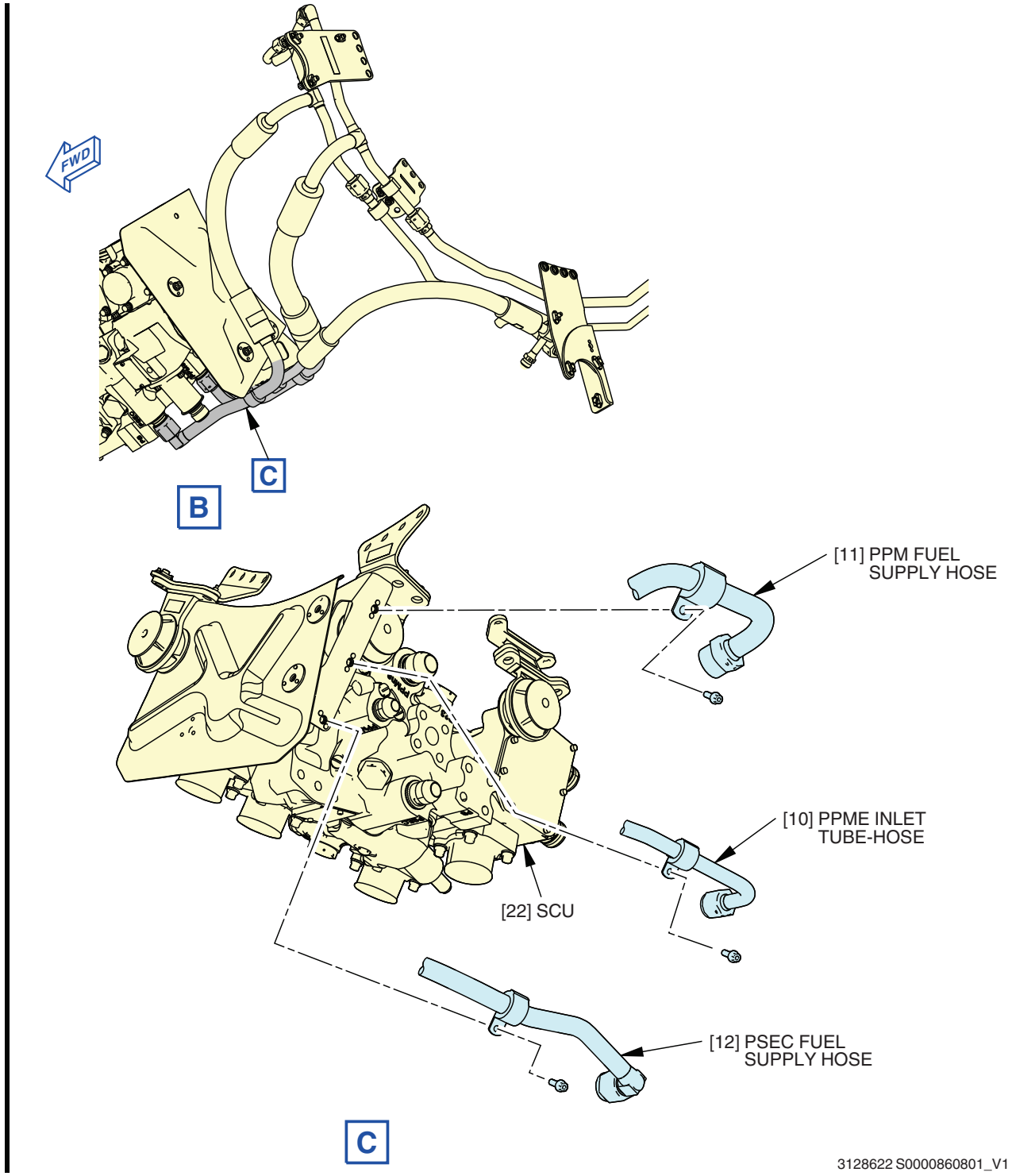
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022304-00
3128621 S0000860800_V1

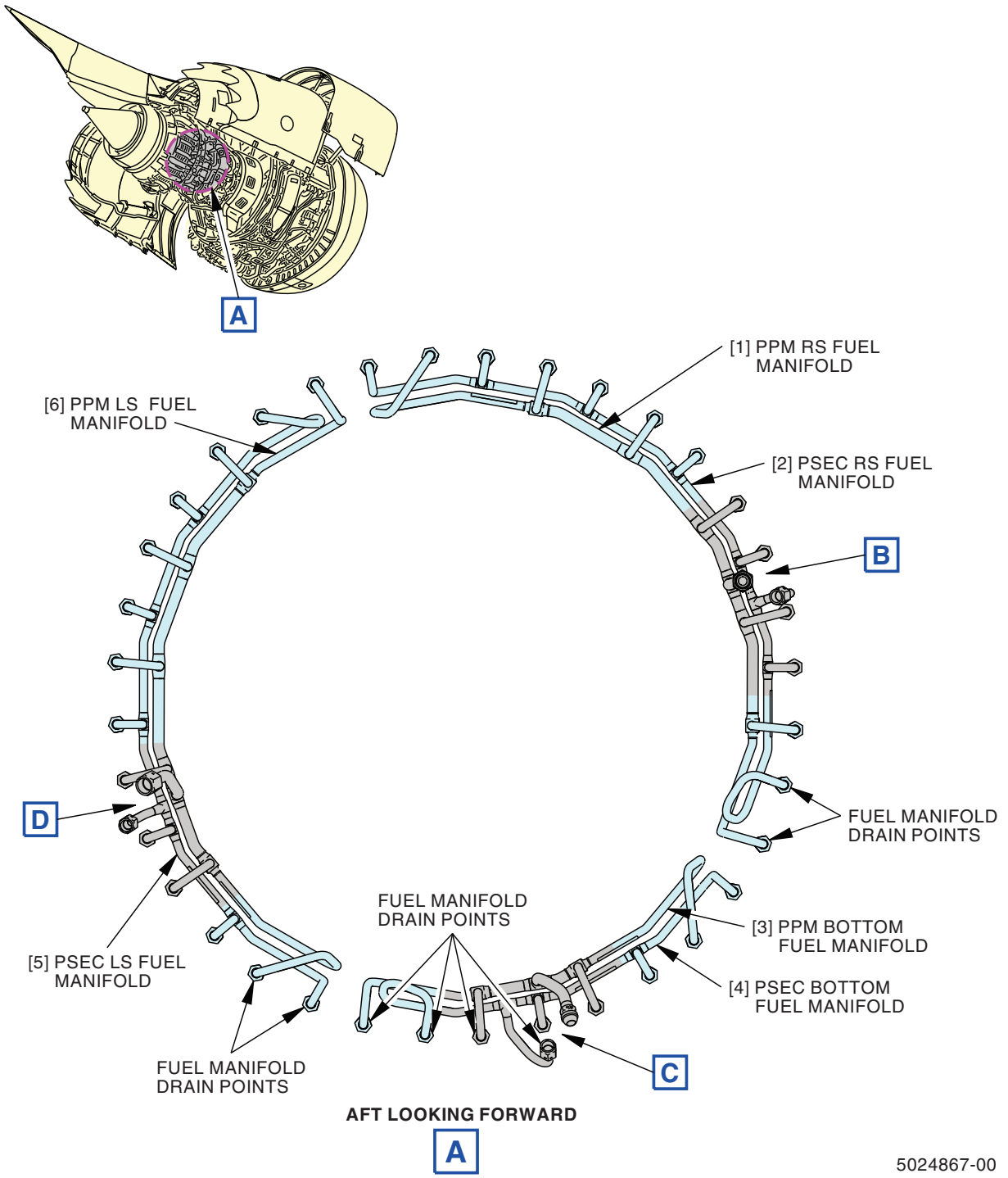
Fuel Nozzle and Fuel Manifold Leak Check
Figure 601/73-11-05-990-911-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



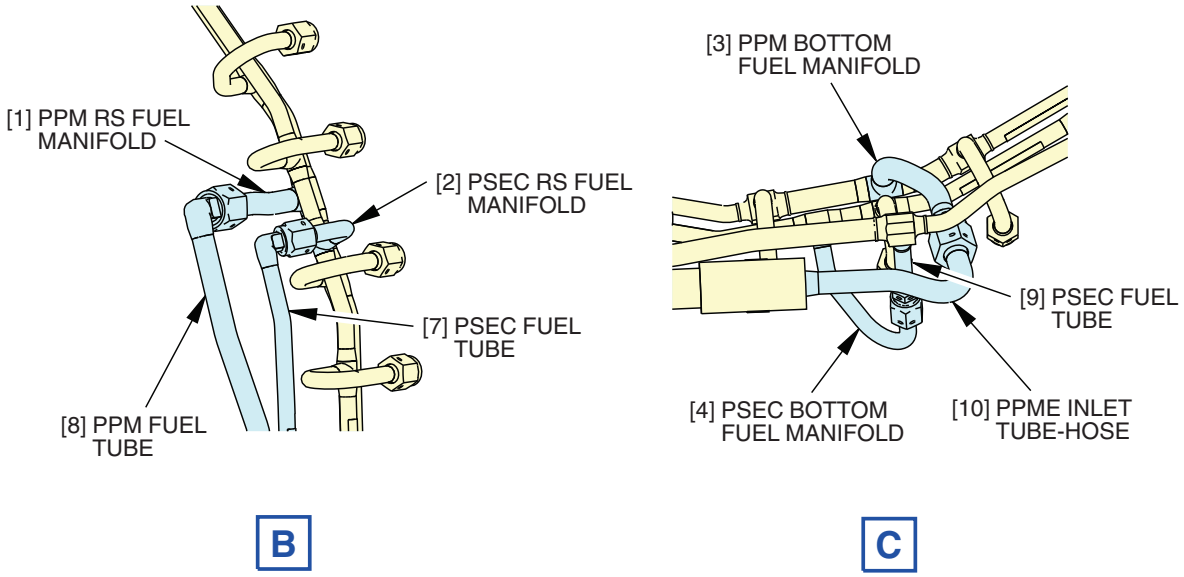
Fuel Nozzle and Fuel Manifold Leak Check
Figure 601/73-11-05-990-911-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle and Fuel Manifold Leak Check
Figure 602/73-11-05-990-829-G00 (Sheet 1 of 2)

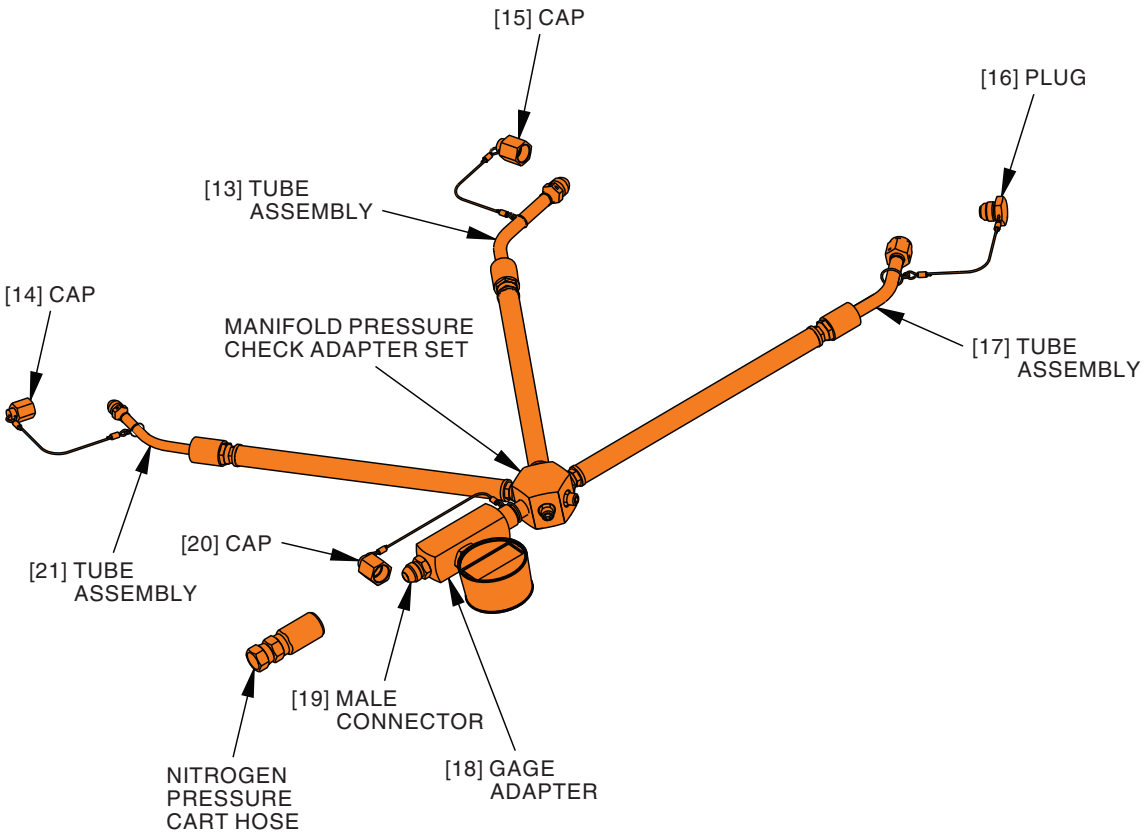
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5024868-00
2473015 S0000578791_V2

Fuel Nozzle and Fuel Manifold Leak Check
Figure 602/73-11-05-990-829-G00 (Sheet 2 of 2)

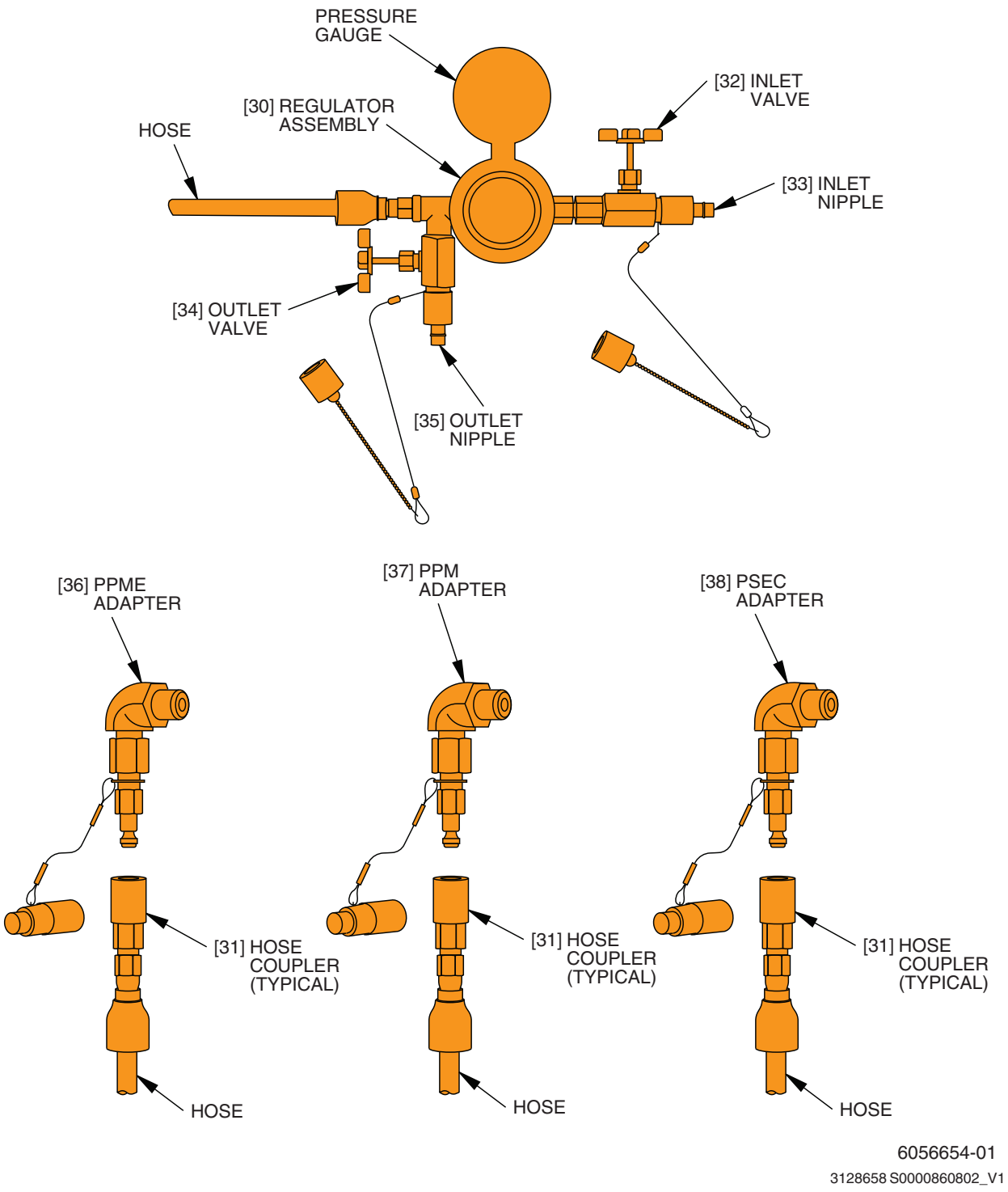
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2473018 S0000578793_V1

Manifold Pressure Check Adapter Set (Nitrogen Cart)
Figure 603/73-11-05-990-830-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Manifold Pressure Check Adapter Set (Hydraulic Pump)
Figure 604/73-11-05-990-912-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPLIT CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the split control unit

(2) An installation of the split control unit.

TASK 73-11-06-000-801-G00

2. Split Control Unit Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the Split Control Unit (SCU).

(2) A minimum of two persons are required to handle the SCU if the jack stand and the SCU adapter support equipment are not used.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
73-21-67-000-801-G00	Fuel Manifold Pressure Transducer Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14501	Adapter- Support Part #: 956A1614G01 Supplier: 58828
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

SUBTASK 73-11-06-860-014-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-06-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-06-010-001-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Split Control Unit Removal

SUBTASK 73-11-06-010-002-G00

(1) Remove the shroud segment 4 (TASK 72-23-04-000-801-G00).

SUBTASK 73-11-06-010-004-G00

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use a teflon-jawed pliers, STD-664, to disconnect the SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3] from the SCU [1] (TASK 70-00-01-910-803-G00).

(a) Disconnect the SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3] from the SCU [1].

(b) Install the protective covers on the electrical receptacles on the SCU [1], SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3].

SUBTASK 73-11-06-010-005-G00

(3) Do this task: Fuel Manifold Pressure Transducer Removal, TASK 73-21-67-000-801-G00.

SUBTASK 73-11-06-010-007-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Disconnect the P23 fuel tube-hose [15] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the P23 fuel tube-hose [15].

(b) Remove the four bolts [14] that attach the P23 fuel tube-hose [15] to the SCU [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove and discard the gasket seal [16].

SUBTASK 73-11-06-010-011-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Disconnect the P8 fuel tube-hose [18] as follows:

(a) Remove the HJ5A electrical harness [46] from the clip on the support bracket [45].

(b) Put a fuel resistant container (5 gal), STD-201, below the P8 fuel tube-hose [18].

(c) Remove the four bolts [17] that attach the P8 fuel tube-hose [18] and support bracket [45] to the SCU [1].

(d) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(e) Remove and discard the gasket seal [19].

SUBTASK 73-11-06-030-001-G00

(6) Remove the bolt [27] that attaches the loop clamp [28] and secondary fuel manifold hose [20] to the support bracket [29].

SUBTASK 73-11-06-020-002-G00

(7) Remove the bolt [25] that attaches the loop clamp [26] and primary enriched fuel tube-hose [21] to the support bracket [29].

SUBTASK 73-11-06-020-003-G00

(8) Remove the bolt [24] that attaches the loop clamp [23] and primary non-enriched fuel manifold hose [22] to the support bracket [29].

SUBTASK 73-11-06-010-008-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(9) Disconnect the secondary fuel manifold hose [20] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the secondary fuel manifold hose [20].

(b) Cut the safety wire/cable and disconnect the secondary fuel manifold hose [20] from the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 73-11-06-010-009-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(10) Disconnect the primary enriched fuel tube-hose [21] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put a fuel resistant container (5 gal), STD-201, below the primary enriched fuel tube-hose [21].

(b) Cut the safety wire/cable and disconnect the primary enriched fuel tube-hose [21] from the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 73-11-06-010-010-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(11) Disconnect the primary non-enriched fuel manifold hose [22] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the primary non-enriched fuel manifold hose [22].

(b) Cut the safety wire/cable and disconnect the primary non-enriched fuel manifold hose [22] from the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 73-11-06-680-001-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(12) Drain the PHS fuel tube-hose [13] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the PHS fuel tube-hose [13].

(b) Disconnect the PHS fuel tube-hose [13] from the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

SUBTASK 73-11-06-680-002-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(13) Disconnect the fuel hydraulics manifold [11] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the fuel hydraulics manifold [11].

(b) Remove the four bolts [12] that attach the fuel hydraulics manifold [11] to the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

CAUTION

DO NOT USE A SHARP OR POINTED TOOL TO REMOVE A SEAL. DAMAGE TO THE SEALING SURFACE OF A COMPONENT CAN CAUSE AN ENGINE FAILURE AND DAMAGE TO THE EQUIPMENT.

(d) Remove and discard the gasket seal [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-06-020-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(14) Disconnect the fuel hydraulics manifold [4] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the fuel hydraulics manifold [4].

(b) Remove the four bolts [5] that attach the fuel hydraulics manifold [4] to the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove and discard the gasket seal [10].

SUBTASK 73-11-06-010-006-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(15) Disconnect the fuel hydraulics manifold [6] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the fuel hydraulics manifold [6].

(b) Remove the four bolts [7] that attach the fuel hydraulics manifold [6] to the SCU [1].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove and discard the gasket seal [8].

SUBTASK 73-11-06-010-013-G00

(16) Remove the bolt [33] that attaches the aft skirt heat shield [32] to the SCU [1].

(a) Do not remove any additional bolts.

SUBTASK 73-11-06-010-016-G00

(17) Remove the bolt [31] that attaches the support bracket [29] to the SCU [1].

(a) Do not remove any additional bolts.

SUBTASK 73-11-06-010-014-G00

(18) Remove the aft skirt heat shield [32] as follows:

(a) Remove the bolt [35] and two washers [36] that attach the aft skirt heat shield [32] and support bracket [30] to the SCU [1].

(b) Remove the aft skirt heat shield [32] with the attached support bracket [29] from the SCU [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-06-020-004-G00

WARNING

FOLLOW THE INSTRUCTIONS INCLUDED WITH THE JACK STAND, SPL-14500, BEFORE YOU USE THE TOOL. IF YOU DO NOT FOLLOW THE INSTRUCTIONS, DAMAGE TO THE EQUIPMENT OR INJURY TO PERSONNEL CAN OCCUR.

(19) If necessary, install the SCU/SVA adapter, SPL-14501, on the jack stand, SPL-14500.

NOTE: The SCU/SVA weight is 38.1 lb (17.3 kg).

NOTE: If the two-person alternative method is used, you do not have to install the adapter, SPL-14501, and jack stand, SPL-14500.

NOTE: To extend the SCU/SVA adapter, SPL-14501, turn the adjusting wheel clockwise.

SUBTASK 73-11-06-020-005-G00

(20) If necessary, move the jack stand, SPL-14500, below the SCU [1].

SUBTASK 73-11-06-010-012-G00

(21) If necessary, attach the SCU/SVA adapter, SPL-14501, to the SCU [1].

SUBTASK 73-11-06-010-017-G00

(22) Remove the bolt [40] and two washers [41] that attach the SCU [1] to the support bracket [39].

SUBTASK 73-11-06-010-018-G00

(23) Remove the bolt [38] and two washers [42] that attach the SCU [1] to the support bracket [37].

SUBTASK 73-11-06-010-019-G00

(24) Remove the two bolts [43] that attach the SCU [1] to the support bracket [44].

SUBTASK 73-11-06-010-020-G00

(25) If necessary, move the SCU [1] away from the engine with the jack stand, SPL-14500.

SUBTASK 73-11-06-010-021-G00

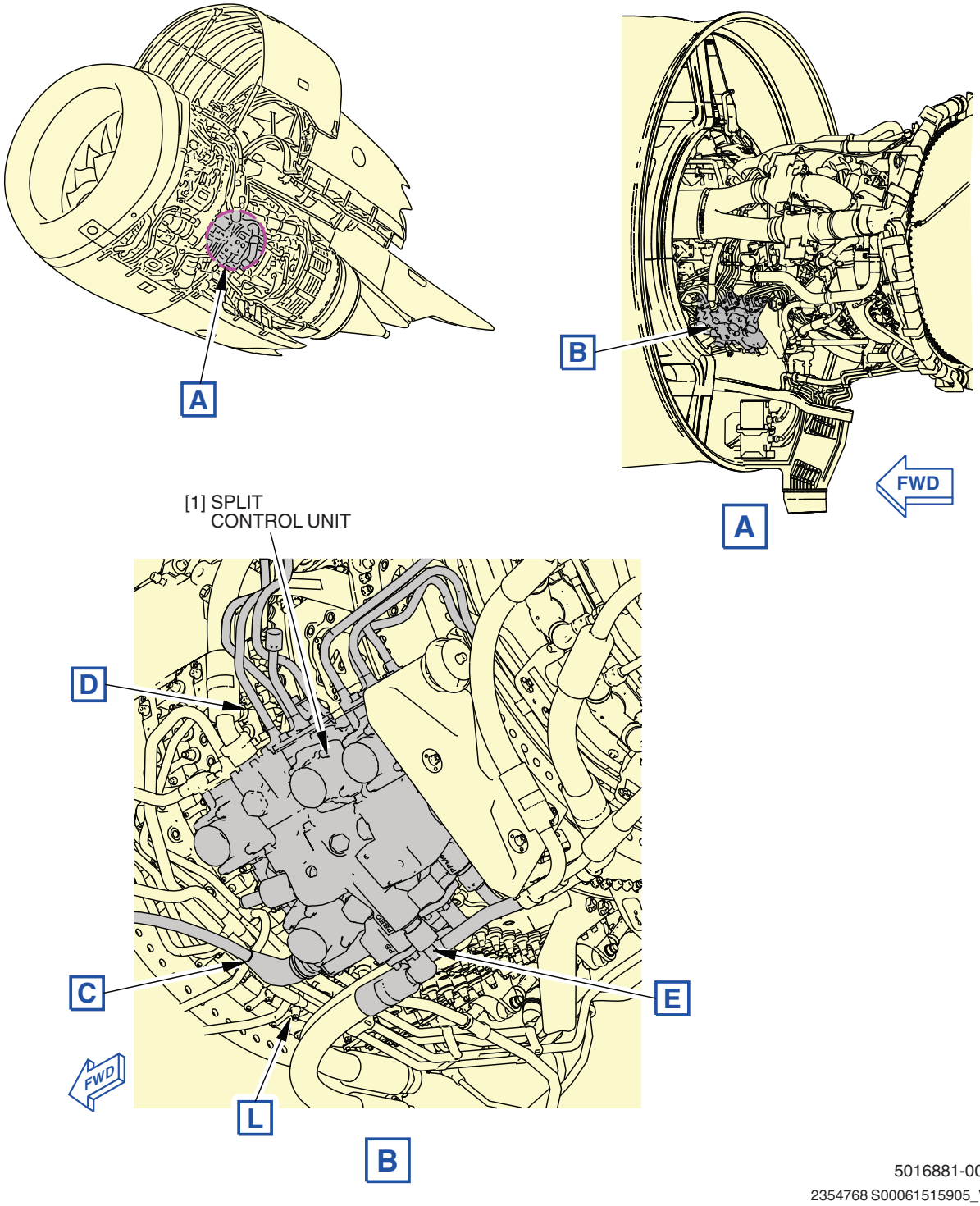
(26) If necessary, remove the SCU [1] from the SCU/SVA adapter, SPL-14501.

SUBTASK 73-11-06-010-022-G00

(27) Install protective covers on all the open tubes, hoses and open ports and fittings on the SCU [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5016881-00
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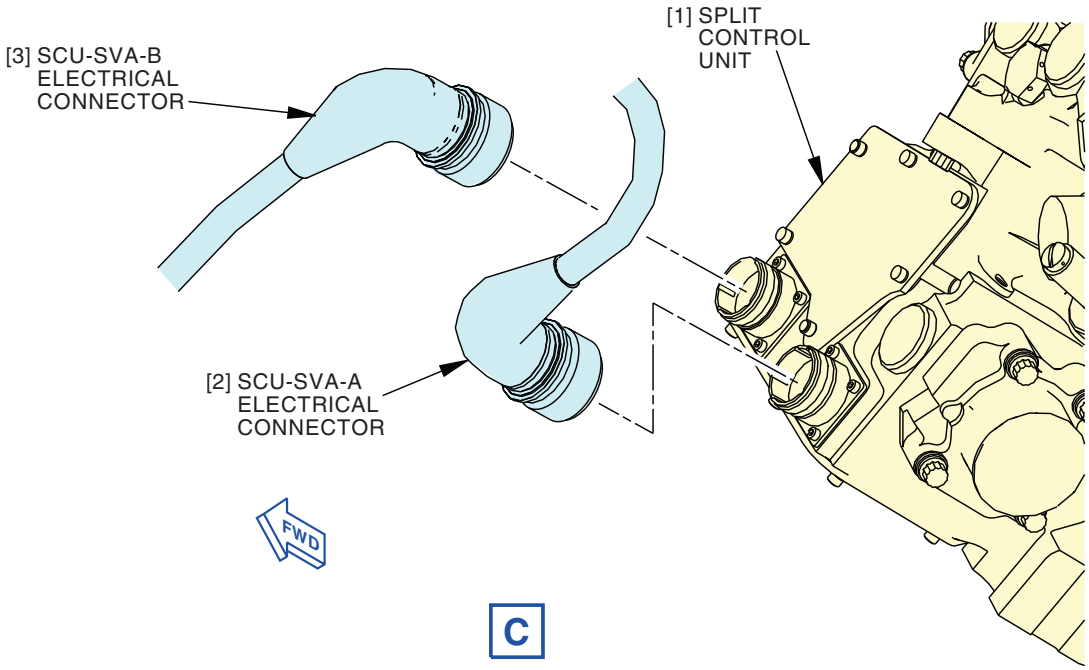
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 1 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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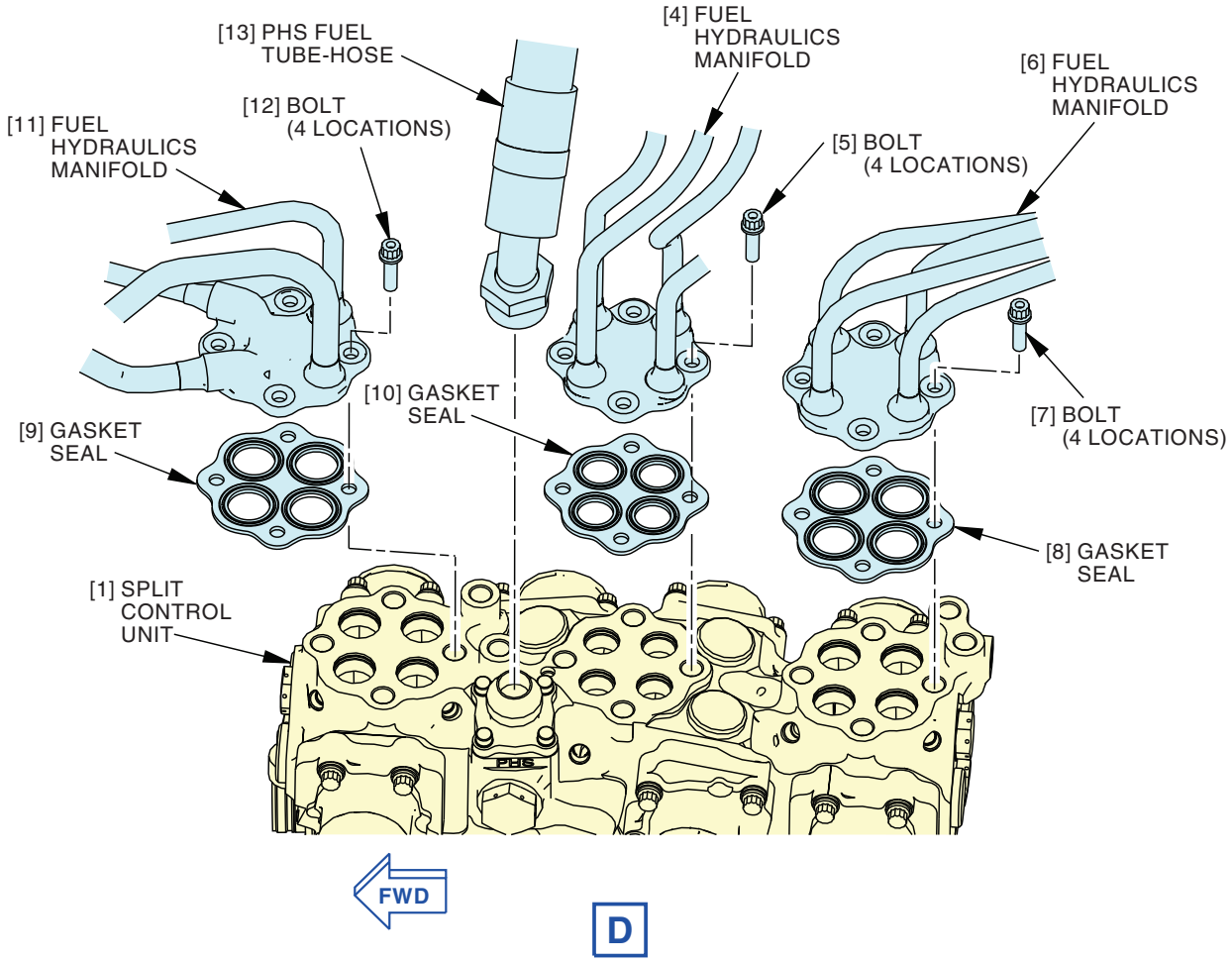
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 2 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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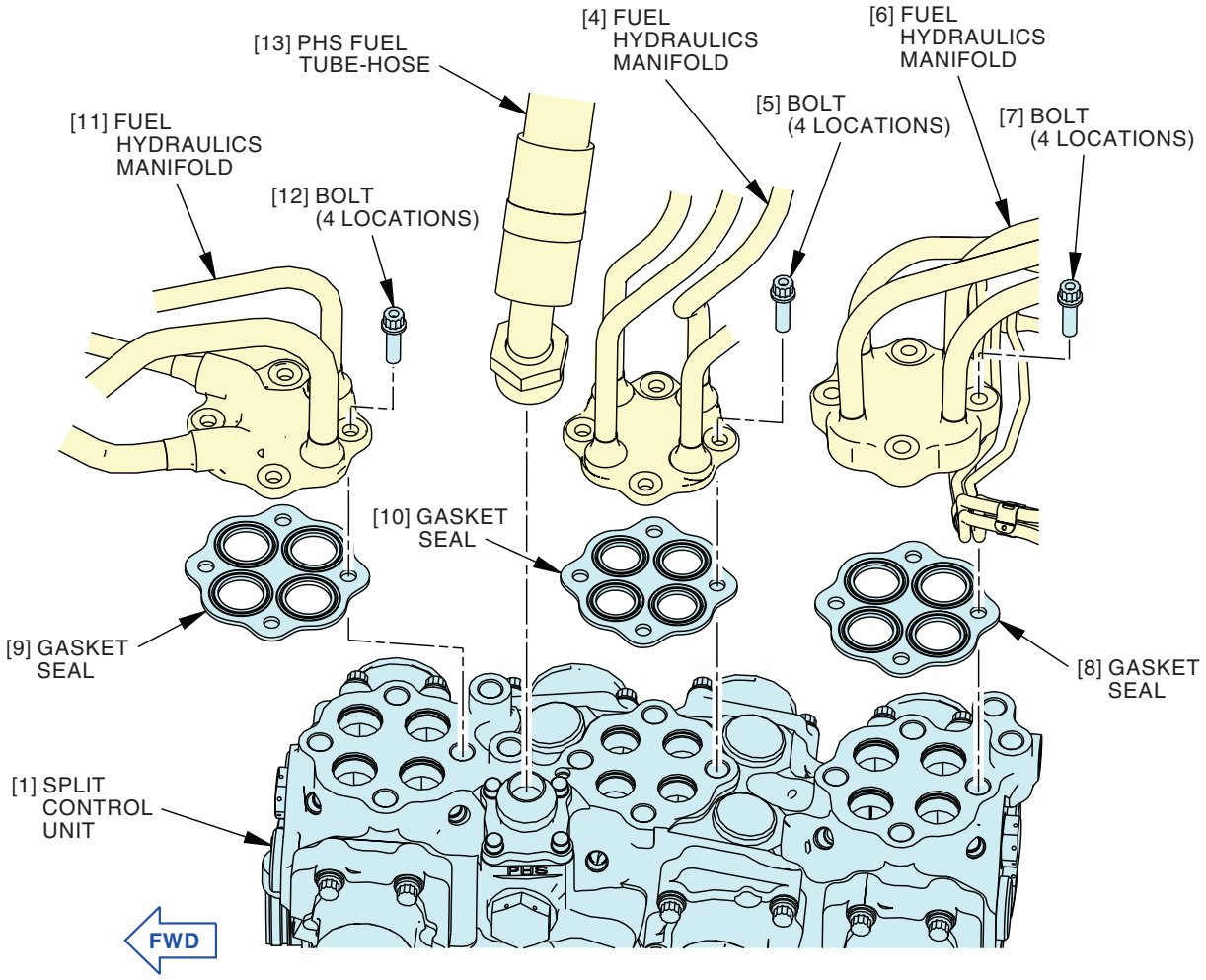
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 3 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059011-00
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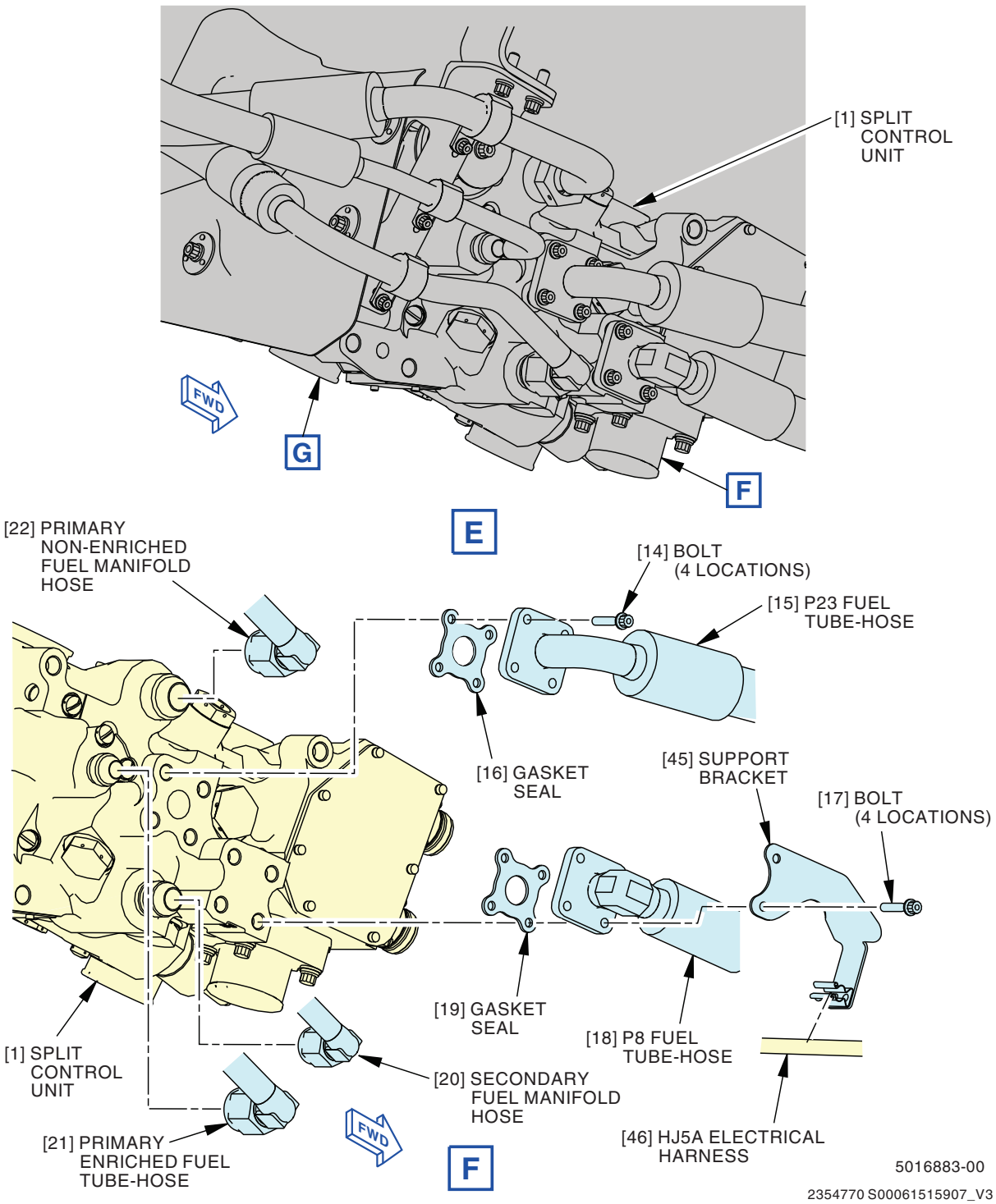
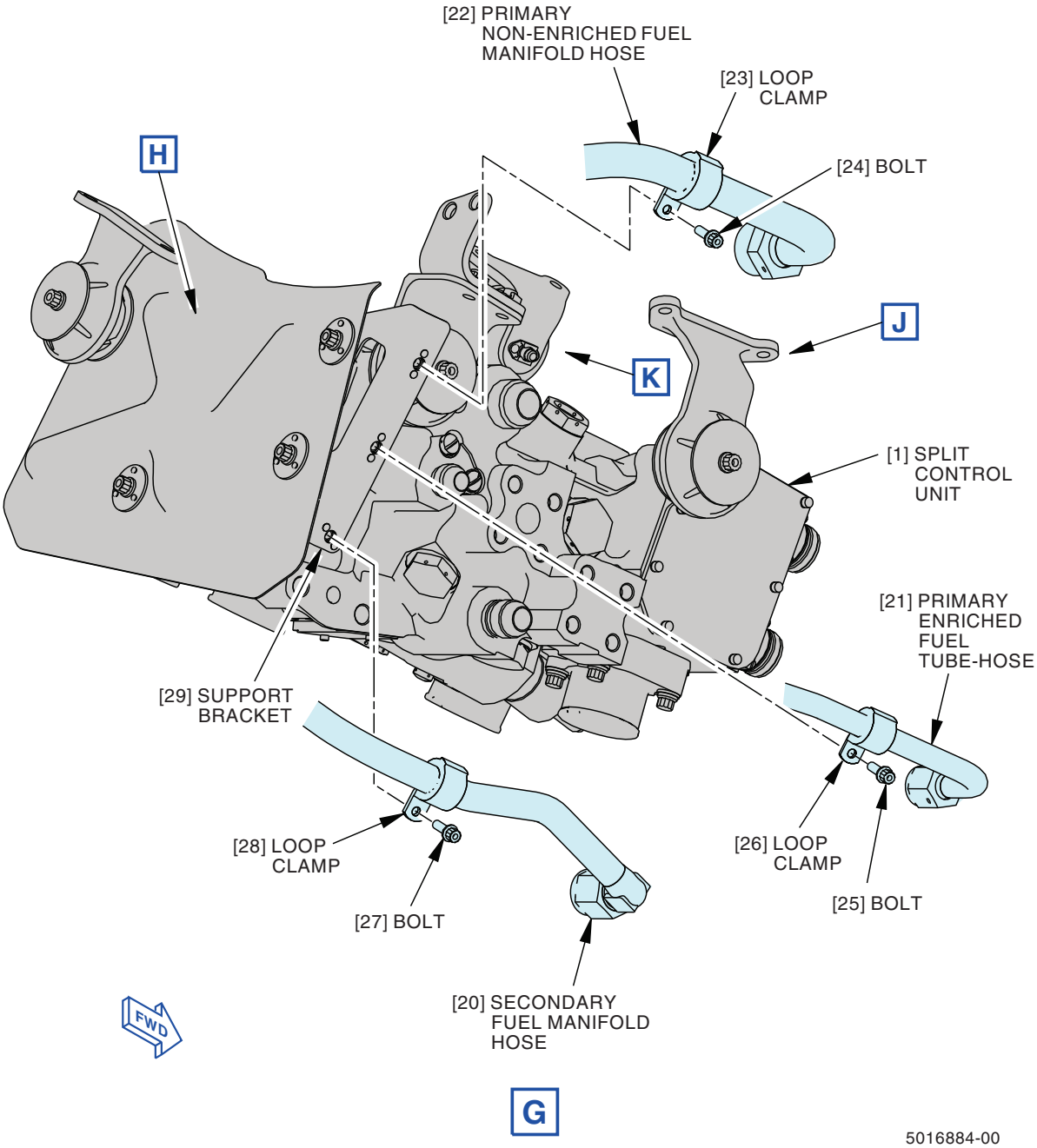
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 4 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



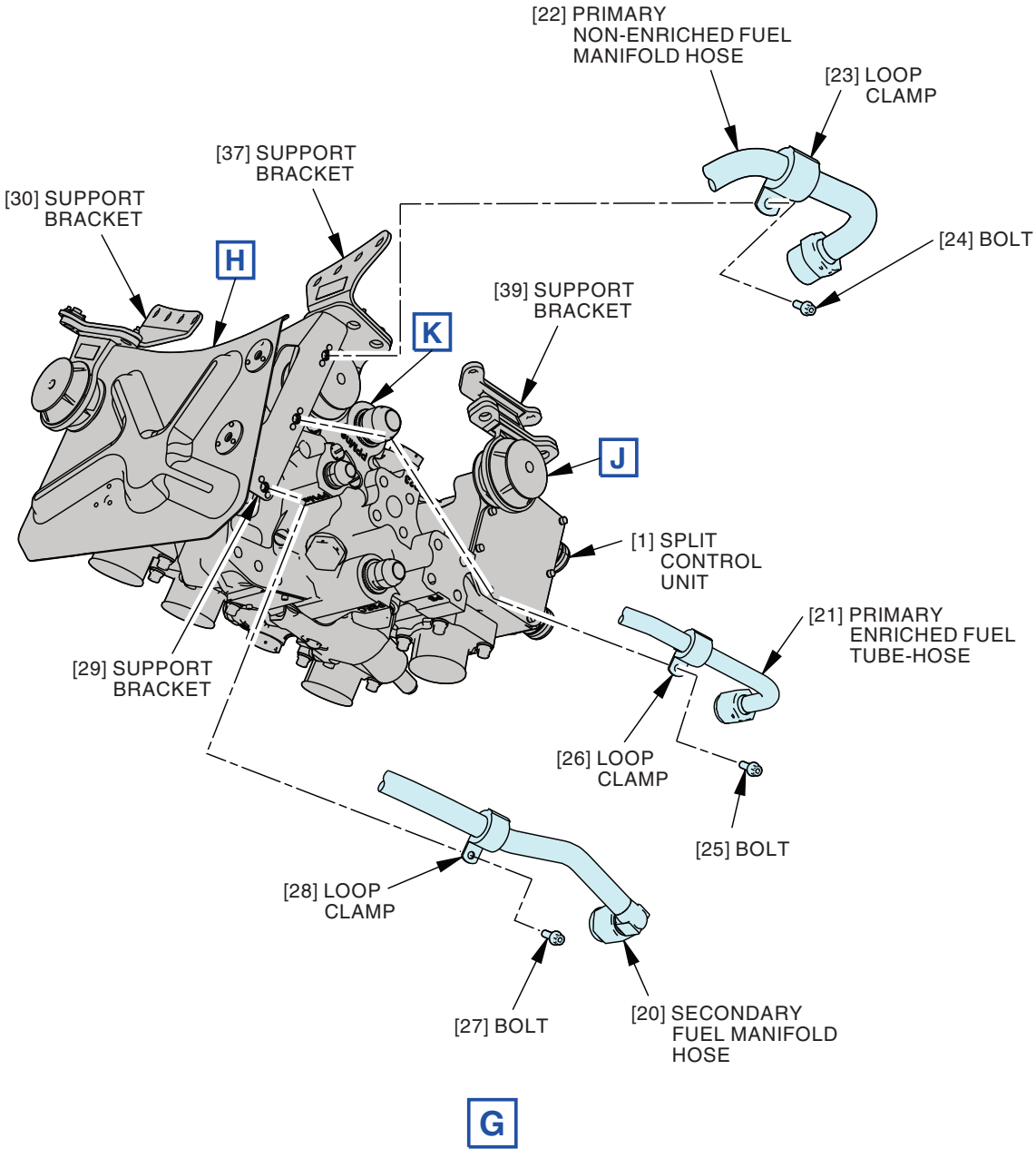
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 5 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 6 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5050830-00

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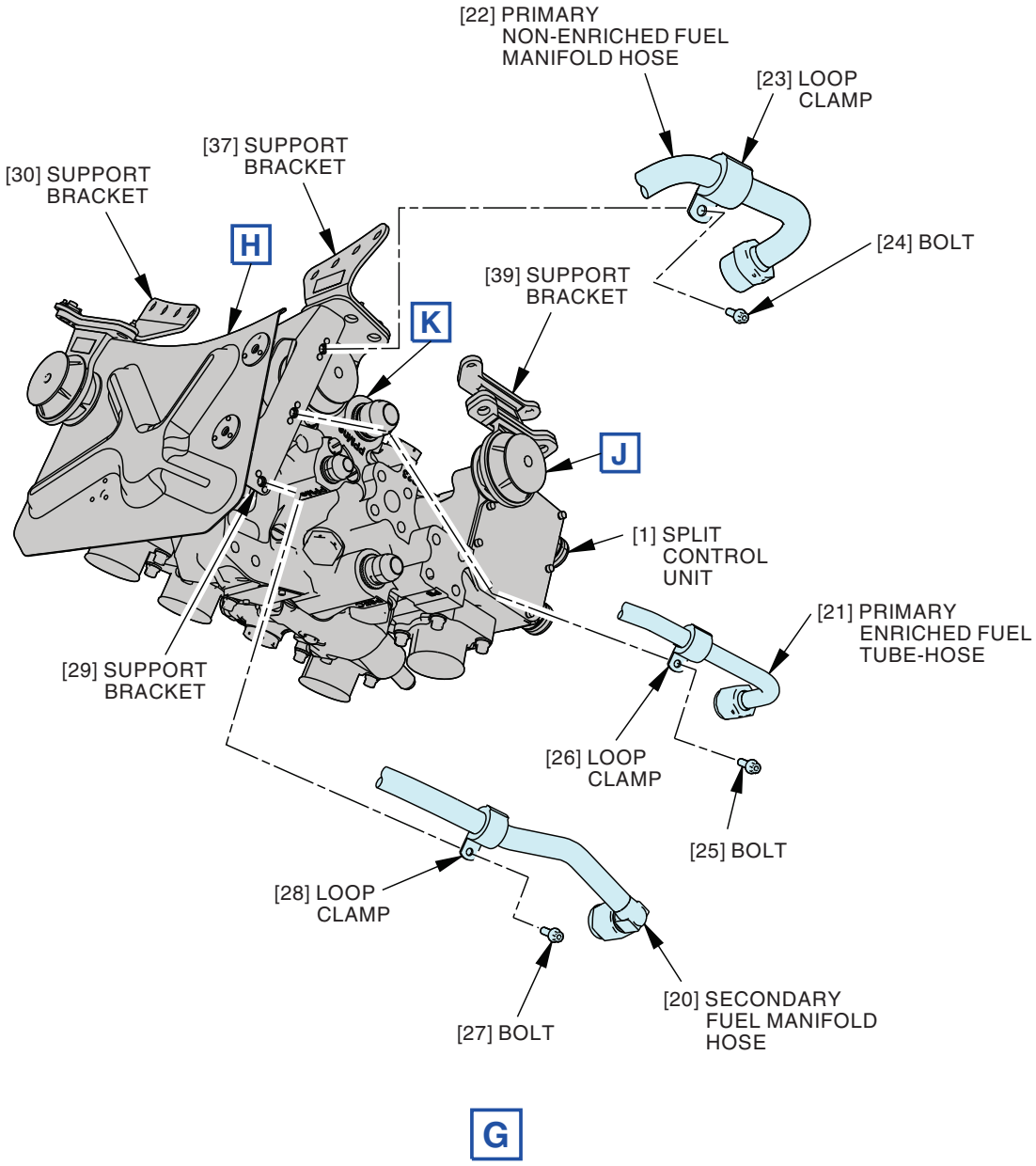
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 7 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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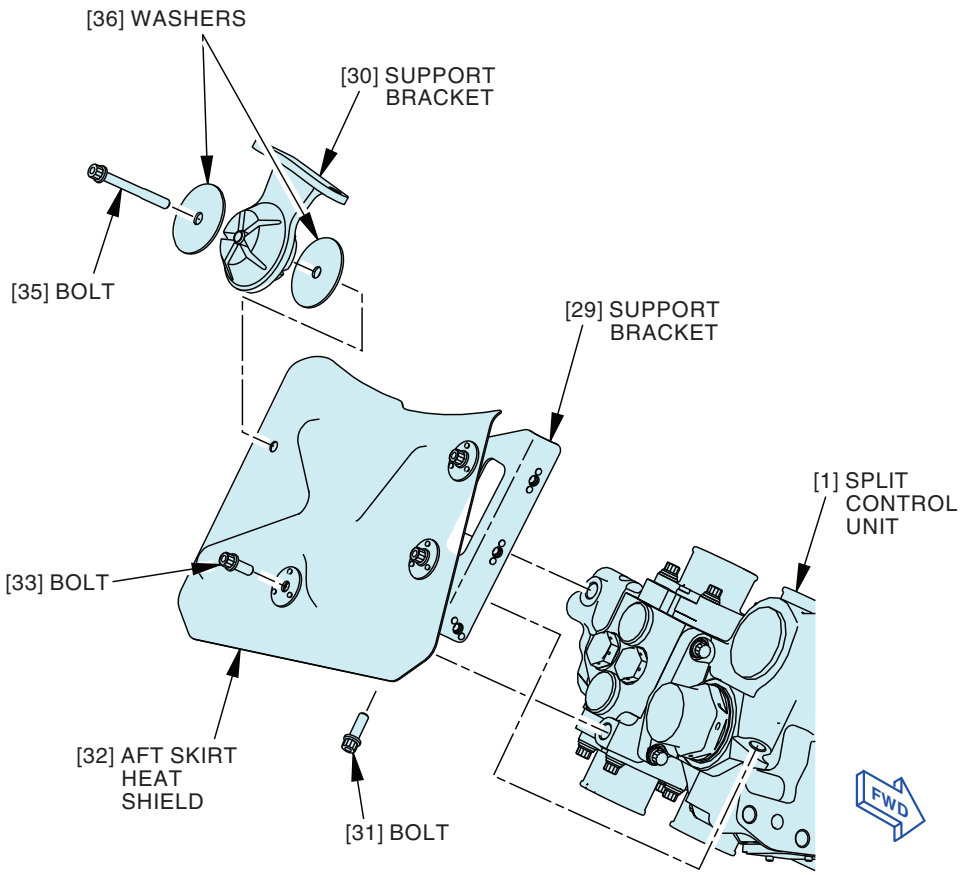
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 8 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5016889-00
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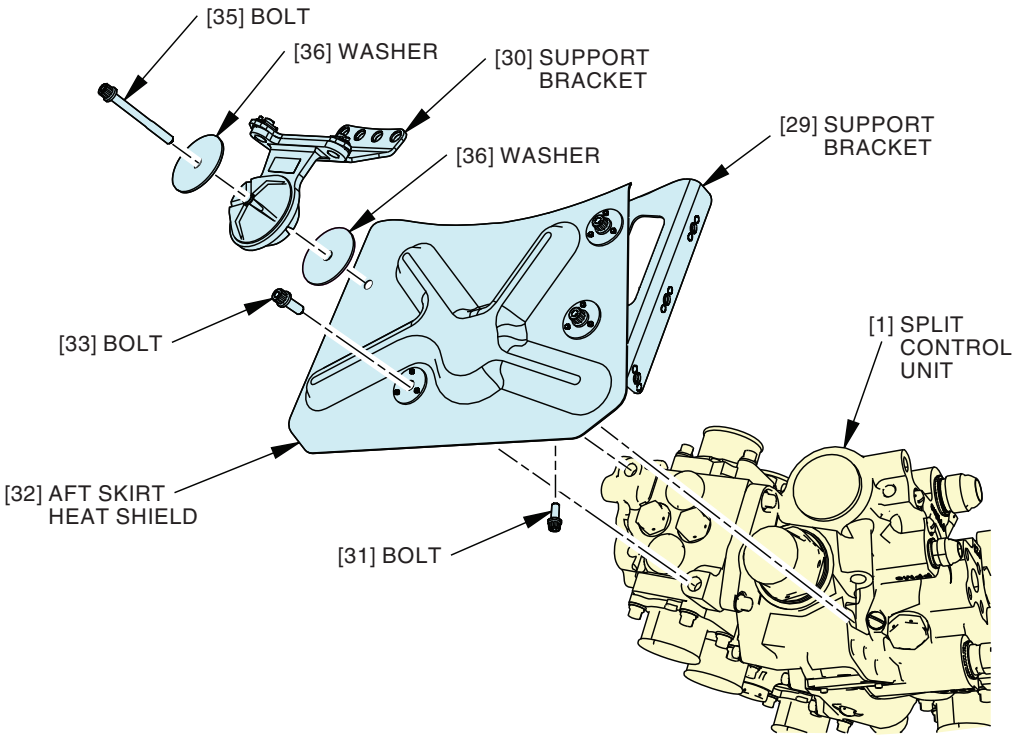
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 9 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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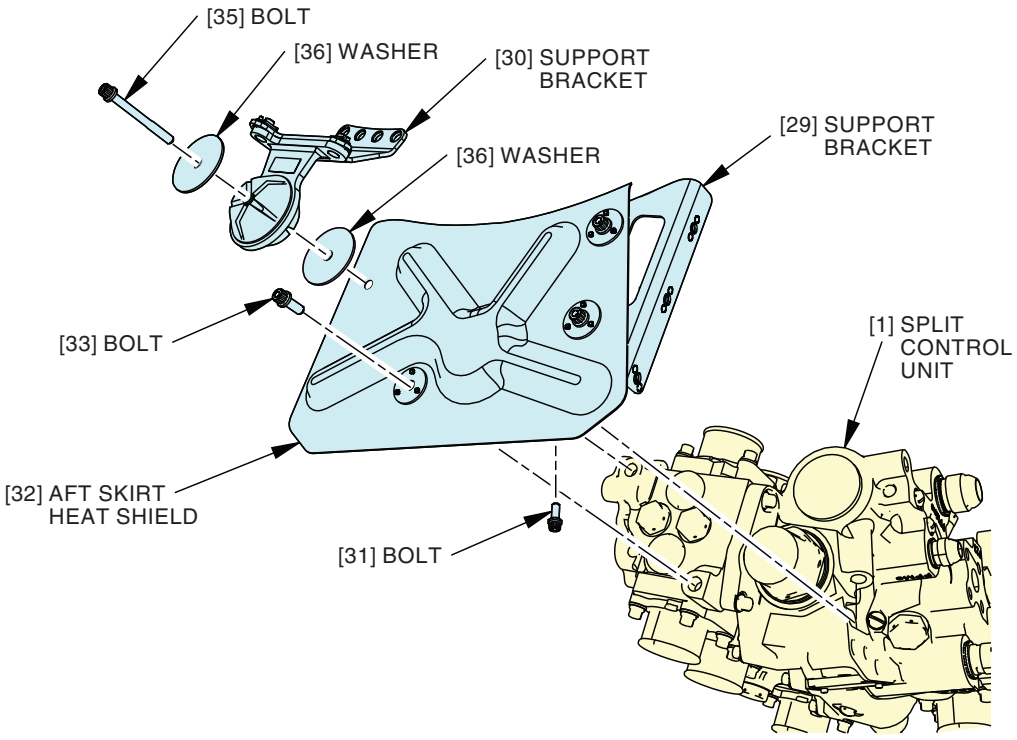
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 10 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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5050835-00
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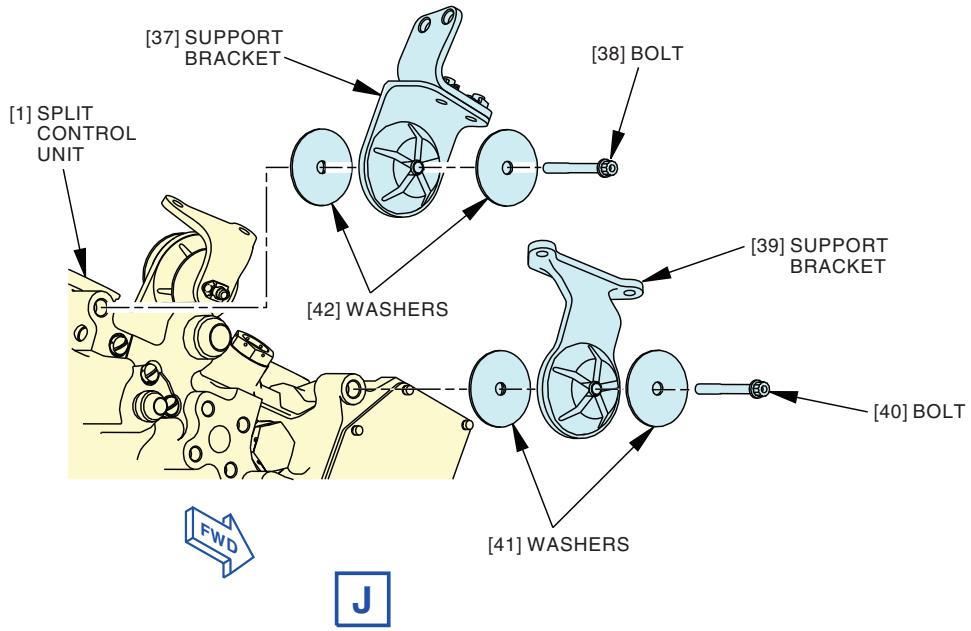
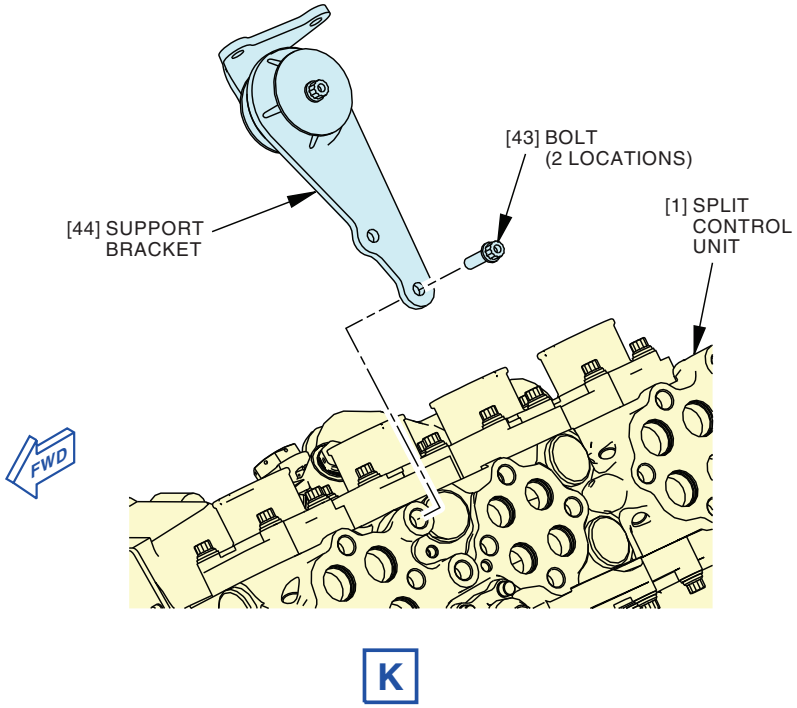
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 11 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





5016891-00
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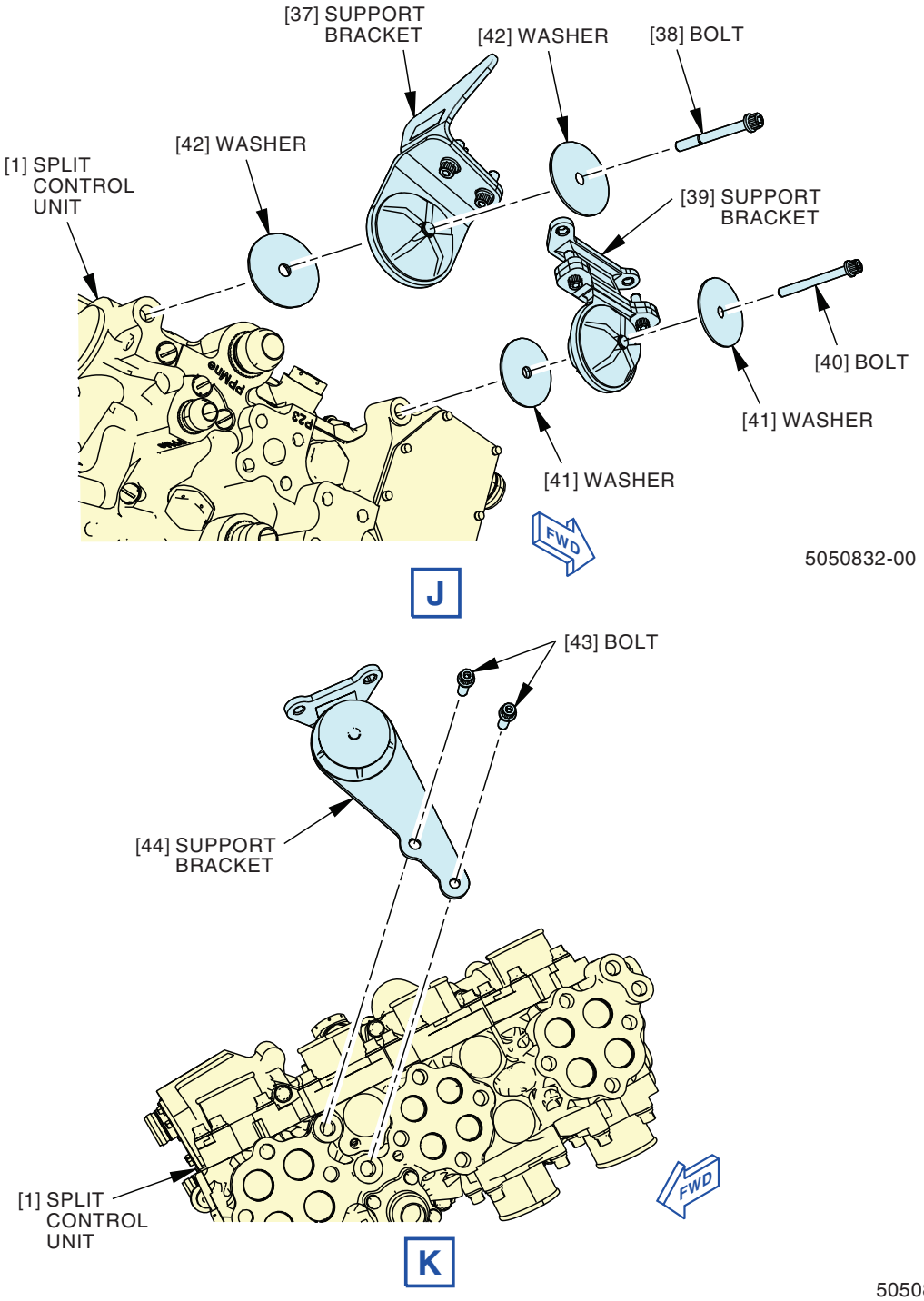
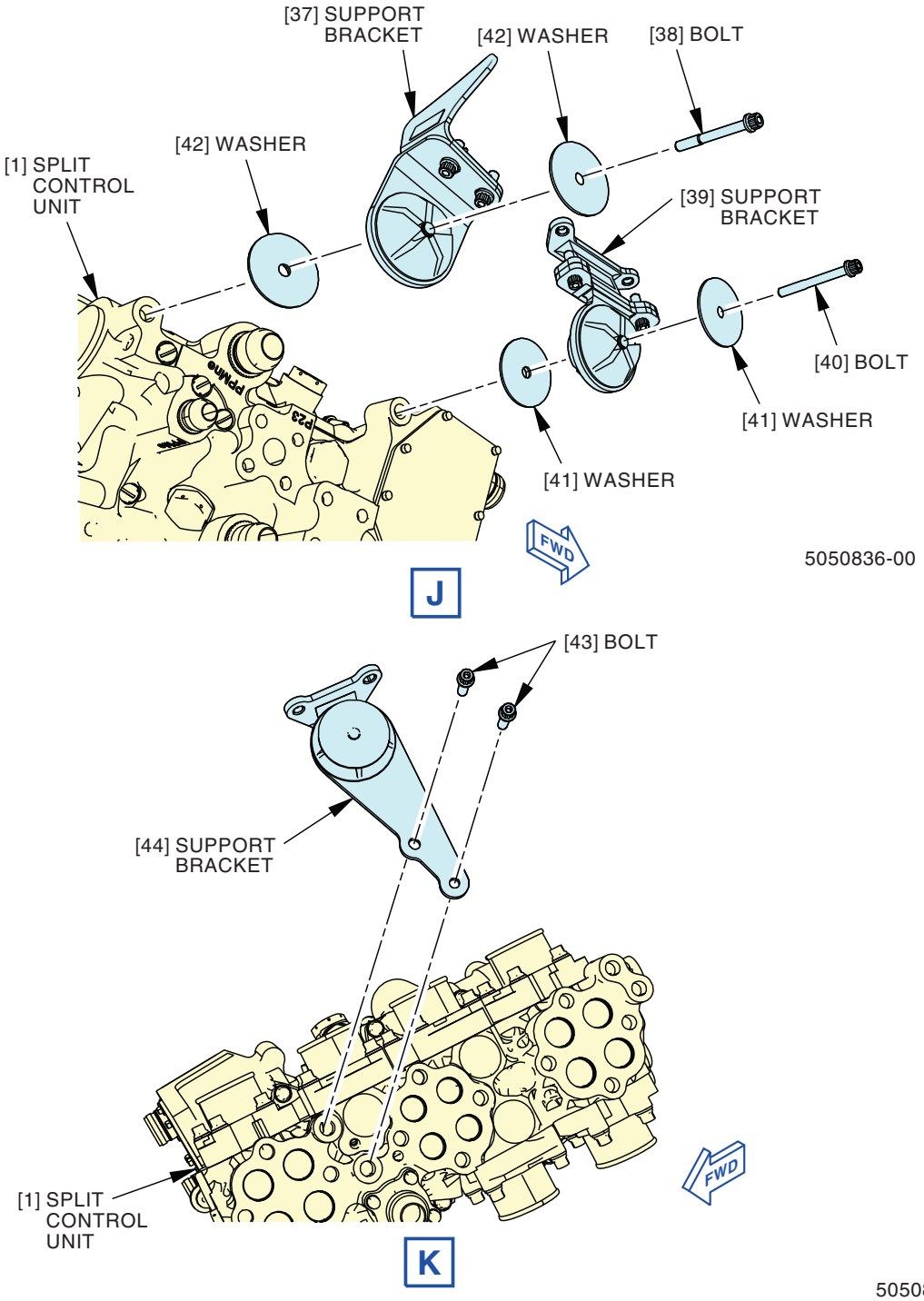
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 12 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



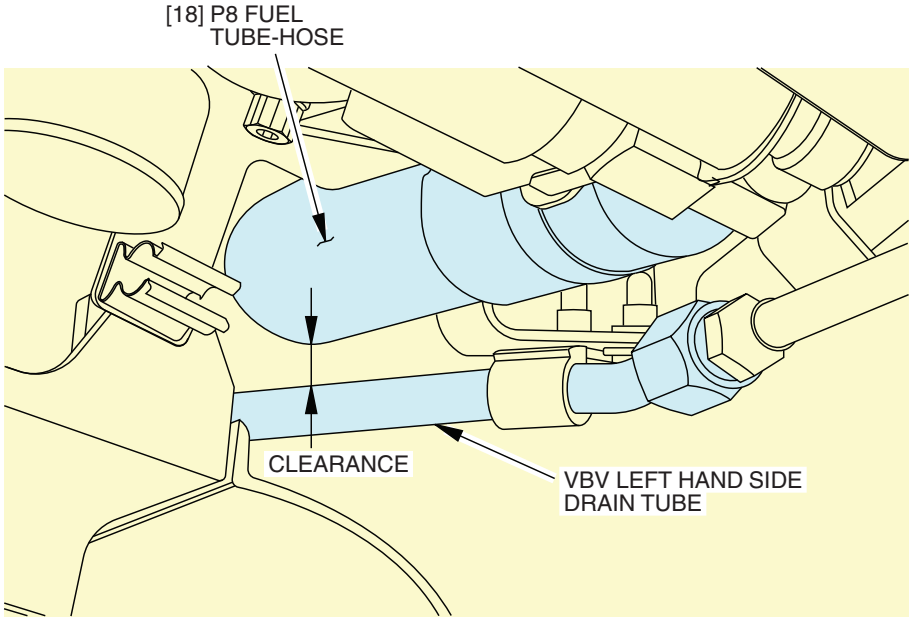
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 13 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 14 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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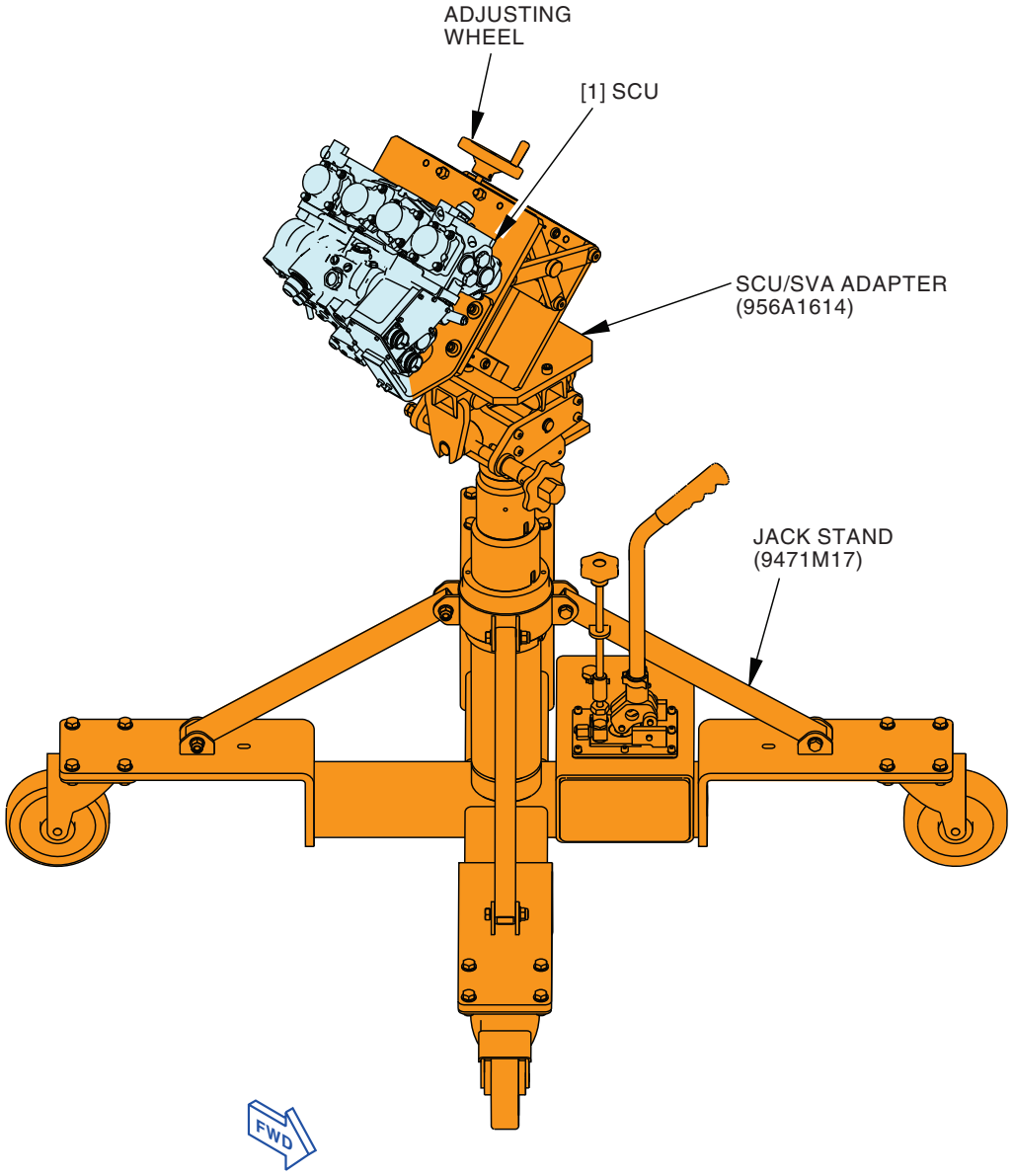
Split Control Unit Installation
Figure 401/73-11-06-990-802-G00 (Sheet 15 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5017361-00
2354775 S00061515912_V3

Split Control Unit LRU Fixture Installation
Figure 402/73-11-06-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-06-400-801-G00

3. Split Control Unit Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the Split Control Unit (SCU).

(2) A minimum of two persons are required to handle the SCU if the jack stand and the SCU adapter support equipment are not used.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)
73-21-09-700-801-G00	Data Entry Plug Configuration Check (P/B 601)
73-21-67-400-801-G00	Fuel Manifold Pressure Transducer Installation (P/B 401)
75-24-01-000-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14501	Adapter- Support Part #: 956A1614G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	SCU	73-11-06-01-010	PTF ALL
8	Gasket seal	73-11-00-20-140	PTF ALL
9	Gasket seal	73-11-00-20-330	PTF ALL
10	Gasket seal	73-11-00-20-260	PTF ALL
16	Gasket seal	72-00-00-22-460	PTF ALL
19	Gasket seal	73-11-00-05-290	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Installation

SUBTASK 73-11-06-860-011-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-06-860-012-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-06-860-013-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Split Control Unit Installation

SUBTASK 73-11-06-210-002-G00

(1) Check the SCU [1], Low Pressure Turbine Active Clearance Control (LPTACC) valve part numbers, and Data Entry Plug (DEP) configuration to be installed as follows:

PTF ALL; POST LEAP-1B SERVICE BULLETIN 73-0013

NOTE: It is not necessary to do the part number check.

PTF ALL

(a) Examine and record the part number of the installed LPTACC valve.

1) To get access to the LPTACC valve, do this task: Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal, TASK 75-24-01-000-801-G00.

(b) Examine and record the part number of the SCU [1] to be installed.

(c) Examine and record the hardware configuration data entry plug pin configuration from the DEP, do this task: Data Entry Plug Configuration Check, TASK 73-21-09-700-801-G00.

(d) Install the DEP again (TASK 73-21-09-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION THE USE OF ANY SCU AND LPTACC VALVE PART NUMBER AND DEP CONFIGURATION COMBINATION NOT LISTED BELOW IS NOT PERMITTED. THIS CAN LEAD TO INCORRECT COMPONENT AND ENGINE PERFORMANCE AND TOO MUCH ENGINE WEAR.

(e) Make sure that the SCU [1], LPTACC part numbers, and DEP configuration agree with the combinations given in Table 401.

Table 401/73-11-06-993-801-G00 SCU and LPTACC Valve Part Number and DEP Configuration Combination

Allowed Configurations	SCU	LPTACC Valve	Engine Configuration Plug
Pre LEAP-1B Service Bulletin 73-0013	8062-1151	8910-274	Make sure that the DEP hardware configuration is correct (TASK 73-21-09-700-801-G00).
Post LEAP-1B Service Bulletin 73-0013	8062-1180	8910-309	

(f) If the part numbers do not agree, replace the SCU [1] with a unit that agrees with the installed LPTACC valve part number and DEP configuration.

SUBTASK 73-11-06-480-001-G00

WARNING FOLLOW THE INSTRUCTIONS INCLUDED WITH THE JACK STAND, SPL-14500, BEFORE YOU USE THE TOOL. IF YOU DO NOT FOLLOW THE INSTRUCTIONS, DAMAGE TO THE EQUIPMENT OR INJURY TO PERSONNEL CAN OCCUR.

(2) If necessary, install the SCU/SVA adapter, SPL-14501, on the jack stand, SPL-14500.

NOTE: The SCU/SVA weight is 38.1 lb (17.3 kg).

NOTE: If the two-person alternative method is used, you do not have to install the adapter, SPL-14501, and jack stand, SPL-14500.

NOTE: To retract the SCU/SVA adapter, SPL-14501, turn the adjusting wheel counter-clockwise.

SUBTASK 73-11-06-480-002-G00

(3) If necessary, install the SCU [1] on the SCU/SVA adapter, SPL-14501.

SUBTASK 73-11-06-080-001-G00

(4) Remove the protective covers from the open tubes, hoses, ports, and fittings on the SCU [1].

SUBTASK 73-11-06-640-001-G00

(5) Lubricate the threads of these bolts with engine oil, D00599 [CP2442]:

(a) four bolts [5]

(b) four bolts [7]

(c) four bolts [12]

(d) bolt [14]

(e) bolt [17]

(f) bolt [25]

(g) bolt [24]

(h) bolt [27]

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(i) bolt [31]

(j) bolt [33]

(k) bolt [35]

(l) bolt [38]

(m) bolt [40]

(n) two bolts [43].

SUBTASK 73-11-06-560-001-G00

(6) If necessary, move the SCU [1] under engine with the jack stand, SPL-14500.

(a) Align the SCU [1] with the support bracket [44].

SUBTASK 73-11-06-410-001-G00

(7) Install and loosely tighten the two bolts [43] that attach the SCU [1] to the support bracket [44].

SUBTASK 73-11-06-410-002-G00

(8) Install and loosely tighten the bolt [38] and two washers [42] that attach the SCU [1] to the support bracket [37].

SUBTASK 73-11-06-410-003-G00

(9) Install and loosely tighten the bolt [40] and two washers [41] that attach the SCU [1] to the support bracket [39].

SUBTASK 73-11-06-080-002-G00

(10) If necessary, disconnect the SCU/SVA adapter, SPL-14501, from the SCU [1].

SUBTASK 73-11-06-010-025-G00

(11) If necessary, move the jack stand, SPL-14500, away from the engine.

SUBTASK 73-11-06-410-006-G00

(12) Put the aft skirt heat shield [32] in position on the SCU [1].

SUBTASK 73-11-06-410-008-G00

(13) Install and loosely tighten the bolt [35] and two washers [36] that attach the aft skirt heat shield [32] and support bracket [30] to the SCU [1].

SUBTASK 73-11-06-410-005-G00

(14) Install and loosely tighten the bolt [31] that attaches the support bracket [29] to the SCU [1].

SUBTASK 73-11-06-410-009-G00

(15) Install and loosely tighten the bolt [33] that attaches the aft skirt heat shield [32] to the SCU [1].

SUBTASK 73-11-06-420-004-G00

(16) Tighten the two bolts [43] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-420-005-G00

(17) Tighten the bolt [38] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-420-006-G00

(18) Tighten the bolt [40] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-420-007-G00

(19) Tighten the bolt [35] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-420-008-G00

(20) Tighten the bolt [31] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-420-009-G00

(21) Tighten the bolt [33] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

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SUBTASK 73-11-06-640-003-G00

(22) Lubricate the new gasket seal [8], new gasket seal [9], and new gasket seal [10] with vaseline, D00672 [CP5070].

SUBTASK 73-11-06-410-020-G00

(23) Install the new gasket seal [8] between the fuel hydraulics manifold [6] and SCU [1].

SUBTASK 73-11-06-410-021-G00

(24) Install the four bolts [7] that attach the fuel hydraulics manifold [6] to the SCU [1].

(a) Tighten the four bolts [7] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-410-022-G00

(25) Install a new gasket seal [10] between the fuel hydraulics manifold [4] and SCU [1].

SUBTASK 73-11-06-400-001-G00

(26) Install the four bolts [5] that attach the fuel hydraulics manifold [4] to the SCU [1].

(a) Tighten the four bolts [5] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-410-023-G00

(27) Install a new gasket seal [9] between the fuel hydraulics manifold [11] and SCU [1].

SUBTASK 73-11-06-410-024-G00

(28) Install the four bolts [12] that attach the fuel hydraulics manifold [11] to the SCU [1].

(a) Tighten the four bolts [12] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

SUBTASK 73-11-06-410-025-G00

(29) Connect the PHS fuel tube-hose [13] to the SCU [1] and triple tighten as follows:

(a) Tighten the PHS fuel tube-hose [13] B-nut to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(b) Loosen the B-nut until it turns freely.

(c) Tighten the B-nut again to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(d) Tighten the B-nut for a third time to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(e) Make sure that the B-nut is fully seated.

(f) If the B-nut is not fully seated, do the above steps again.

SUBTASK 73-11-06-410-010-G00

(30) Install the bolt [24] that attaches the loop clamp [23] and primary non-enriched fuel manifold hose [22] to the support bracket [29].

(a) Tighten the bolt [24] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

SUBTASK 73-11-06-410-015-G00

(31) Connect and loosely tighten the primary non-enriched fuel manifold hose [22] to the SCU [1].

SUBTASK 73-11-06-410-011-G00

(32) Install the bolt [25] that attaches the loop clamp [26] and primary enriched fuel tube-hose [21] to the support bracket [29].

(a) Tighten the bolt [25] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

SUBTASK 73-11-06-410-016-G00

(33) Connect and loosely tighten the primary enriched fuel tube-hose [21] to the SCU [1].

SUBTASK 73-11-06-420-011-G00

(34) Triple tighten the primary enriched fuel tube-hose [21] B-nut as follows:

(a) Tighten the B-nut to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

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(b) Loosen the B-nut until it turns freely.

(c) Tighten the B-nut again to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

(d) Tighten the B-nut for a third time to 460 in-lb (51.97 N·m) - 540 in-lb (61.01 N·m).

(e) Make sure that the B-nut is fully seated.

(f) If the B-nut is not fully seated, do the above steps again.

(g) Secure the B-nut with safety safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], to the SCU [1] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-06-410-012-G00

(35) Install the bolt [27] that attaches the loop clamp [28] and secondary fuel manifold hose [20] to the support bracket [29].

(a) Tighten the bolt [27] to 32 in-lb (3.62 N·m) - 38 in-lb (4.29 N·m).

SUBTASK 73-11-06-410-017-G00

(36) Connect and loosely tighten the secondary fuel manifold hose [20] to the SCU [1].

SUBTASK 73-11-06-420-012-G00

(37) Triple tighten the secondary fuel manifold hose [20] B-nut as follows:

(a) Tighten the B-nut to 662.4 in-lb (74.84 N·m) - 777.6 in-lb (87.86 N·m).

(b) Loosen the B-nut until it turns freely.

(c) Tighten the B-nut again to 662.4 in-lb (74.84 N·m) - 777.6 in-lb (87.86 N·m).

(d) Tighten the B-nut for a third time to 662.4 in-lb (74.84 N·m) - 777.6 in-lb (87.86 N·m).

(e) Make sure that the B-nut is fully seated.

(f) If the B-nut is not fully seated, do the above steps again.

(g) Secure the B-nut with safety safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], to the SCU [1] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-06-420-010-G00

(38) Triple tighten the primary non-enriched fuel manifold hose [22] B-nut as follows:

(a) Tighten the B-nut to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(b) Loosen the B-nut until it turns freely.

(c) Tighten the B-nut again to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(d) Tighten the B-nut for a third time to 938.4 in-lb (106.02 N·m) - 1101.6 in-lb (124.46 N·m).

(e) Make sure that the B-nut is fully seated.

(f) If the B-nut is not fully seated, do the above steps again.

(g) Secure the B-nut with safety safety cable, G50065 [CP8006], or safety wire, G02495 [CP8002], to the SCU [1] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-06-410-018-G00

(39) Install a new gasket seal [19] between the P8 fuel tube-hose [18] and SCU [1].

SUBTASK 73-11-06-410-019-G00

(40) Install the four bolts [17] that attach the P8 fuel tube-hose [18] and support bracket [45] to the SCU [1].

(a) Make sure that there is a minimum clearance of 0.125 in. (3.175 mm) between the P8 fuel tube-hose [18] and the Variable Bleed Valve (VBV) left hand side drain tube.

1) If you do not get the minimum clearance, do the following:

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a) Remove the P8 fuel tube-hose [18] (TASK 73-11-06-000-801-G00).

b) Install the P8 fuel tube-hose [18].

c) Repeat until the minimum clearance is 0.125 in. (3.175 mm).

(b) Tighten the four bolts [17] to 51.0 in-lb (5.8 N·m) - 59.0 in-lb (6.7 N·m).

SUBTASK 73-11-06-410-013-G00

(41) Install a new gasket seal [16] between the P23 fuel tube-hose [15] and SCU [1].

SUBTASK 73-11-06-410-014-G00

(42) Install the four bolts [14] that attach the P23 fuel tube-hose [15] to the SCU [1].

(a) Tighten the four bolts [14] to 51.0 in-lb (5.76 N·m) - 59.0 in-lb (6.67 N·m).

SUBTASK 73-11-06-420-013-G00

(43) Install the HJ5A electrical harness [46] in the clip of the support bracket [45].

SUBTASK 73-11-06-420-002-G00

(44) Do this task: Fuel Manifold Pressure Transducer Installation, TASK 73-21-67-400-801-G00.

SUBTASK 73-11-06-420-003-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(45) Use the teflon-jawed pliers, STD-664, to connect the SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3] to the SCU [1] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacles on the SCU [1], SCU-SVA-A electrical connector [2], and SCU-SVA-B electrical connector [3].

(b) Connect the SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3] to the SCU [1].

1) Tighten the SCU-SVA-A electrical connector [2] and SCU-SVA-B electrical connector [3].

SUBTASK 73-11-06-410-027-G00

(46) Install the shroud segment 4 (TASK 72-23-04-400-801-G00).

SUBTASK 73-11-06-160-002-G00

(47) Clean the work area and clear the tools and other items.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-06-860-004-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

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SUBTASK 73-11-06-860-005-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-06-410-029-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

K. Split Control Unit Installation Test

SUBTASK 73-11-06-730-002-G00

(1) Do the applicable tests given in the Power Plant Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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SPLIT CONTROL UNIT - INSPECTION/CHECK

1. General

A. This procedure has one task.

(1) Split Control Unit Inspection

TASK 73-11-06-210-801-G00

2. Split Control Unit Inspection

Figure 601

A. General

(1) This task gives the instructions to inspect the Split Control Unit (SCU).

(a) The SCU is located on the left side of the engine core.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-06-000-801-G00	Split Control Unit Removal (P/B 401)
73-11-06-400-801-G00	Split Control Unit Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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F. Prepare for the Inspection

SUBTASK 73-11-06-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-06-860-008-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-06-010-026-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. SCU Inspection

SUBTASK 73-11-06-210-001-G00

(1) Inspect the Split Control Unit (SCU):

(a) Look for signs of fuel leaks between the SCU and the fuel connections.

1) If you see signs of fuel leaks, do the steps in these tasks to replace the gaskets (Split Control Unit Removal, TASK 73-11-06-000-801-G00 and Split Control Unit Installation, TASK 73-11-06-400-801-G00).

(b) Make sure that the SCU is not loose.

1) If the SCU is loose, tighten the bolts that attach the SCU to the engine (Split Control Unit Installation, TASK 73-11-06-400-801-G00).

(c) Look for cracks in the SCU.

1) If you see cracks, replace the SCU (Split Control Unit Removal, TASK 73-11-06-000-801-G00 and Split Control Unit Installation, TASK 73-11-06-400-801-G00).

(d) Make sure that the electrical connectors are not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-06-410-030-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

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1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-06-860-009-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-06-860-010-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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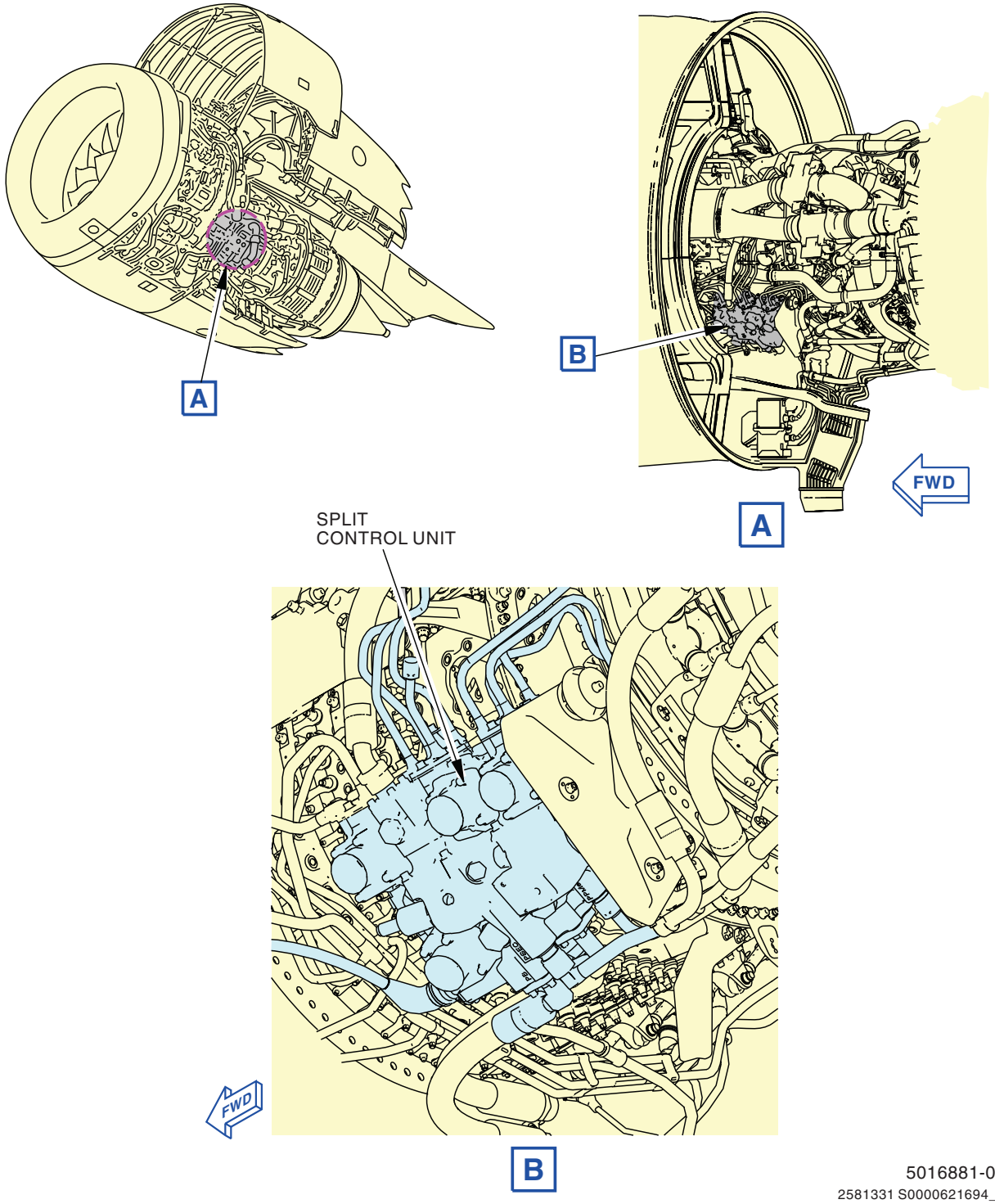

Split Control Unit Inspection
Figure 601/73-11-06-990-804-G00

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MAIN FUEL SUPPLY HOSE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) A removal of the main fuel supply hose

(2) An installation of the main fuel supply hose.

TASK 73-11-07-000-801-G00

2. Main Fuel Supply Hose Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the main fuel supply hose.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Preformed packing	73-11-01-01-050	PTF ALL
12	Preformed packing	73-11-01-01-070	PTF ALL
15	Preformed packing	73-11-01-01-070	PTF ALL
16	Preformed packing	73-11-03-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 73-11-07-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-07-860-002-G00

(2) For the applicable engine, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 73-11-07-860-003-G00

(3) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-07-860-016-G00

(4) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED light, on the P5 forward overhead panel, is on dim.

(c) Make sure that the applicable ENG VALVE CLOSED light, on the P5 forward overhead panel, is on dim.

SUBTASK 73-11-07-010-002-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Main Fuel Supply Hose Removal

SUBTASK 73-11-07-680-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

(1) Drain the main fuel pump as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the main fuel pump.

(b) Remove the drain plug [9] from the main fuel pump as follows:

1) Cut and discard the safety wire [8] from the drain plug [9].

2) Remove the drain plug [9] from the main fuel pump.

3) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

4) Remove and discard the preformed packing [7] from drain plug [9].

(c) Install the drain plug [9] on the main fuel pump as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [9] and the new preformed packing [7] with clean oil, D00599 [CP2442].

3) Install the preformed packing [7] on the drain plug [9].

4) Install the drain plug [9] in a correct position on the main fuel pump.

5) Tighten the drain plug [9] to 13 in-lb (1.5 N·m) - 16 in-lb (1.8 N·m).

6) Install the safety wire [8] on the drain plug [9].

(d) Remove the drain plug [11] from the main fuel pump as follows:

1) Cut and discard the safety wire [10] from the drain plug [11].

2) Remove the drain plug [11] from the main fuel pump.

3) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

4) Remove and discard the preformed packing [12] from the drain plug [11].

(e) Install the drain plug [11] on the main fuel pump as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [11] and the new preformed packing [12] with clean oil, D00599 [CP2442].

3) Install the preformed packing [12] on the drain plug [11].

4) Install the drain plug [11] in a correct position on the main fuel pump.

5) Tighten the drain plug [11] to 190 in-lb (21.5 N·m) - 210 in-lb (23.7 N·m).

6) Install the safety wire [10] on the drain plug [11].

(f) Remove the drain plug [14] from the main fuel pump as follows:

1) Cut and discard the safety wire [13] from the drain plug [14].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Remove the drain plug [14] from the main fuel pump.

3) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

4) Remove and discard the preformed packing [15] from the drain plug [14].

(g) Install the drain plug [14] on the main fuel pump as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [14] and the new preformed packing [15] with clean oil, D00599 [CP2442].

3) Install the preformed packing [15] on the drain plug [14].

4) Install the drain plug [14] in a correct position on the main fuel pump.

5) Tighten the drain plug [14] to 190 in-lb (21.5 N·m) - 210 in-lb (23.7 N·m).

6) Install the safety wire [13] on the drain plug [14].

SUBTASK 73-11-07-680-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

(2) Drain the fuel supply system at the fuel filter bowl as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the fuel filter bowl.

(b) Remove the drain plug [17] from the fuel filter bowl as follows:

1) Cut and remove the safety wire [18] from the drain plug [17].

2) Remove the drain plug [17] from the fuel filter bowl.

3) Remove and discard the preformed packing [16] from the groove of the drain plug [17].

4) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(c) Install the drain plug [17] on the fuel filter bowl as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate a new preformed packing [16] with oil, D00599 [CP2442].

3) Install the preformed packing [16] onto the groove of the drain plug [17].

CAUTION MAKE SURE THAT YOU INSTALL THE PREFORMED PACKING. IF YOU DO NOT INSTALL THE PACKING, THE ENGINE OIL LEVEL CAN DECREASE DURING OPERATION. IF THE OIL LEVEL IS TOO LOW, IT CAN CAUSE A SHUTDOWN OF THE ENGINE.

4) Install the drain plug [17] in a correct position on the fuel filter bowl.

5) Tighten the drain plug [17] to 140 in-lb (15.8 N·m) - 150 in-lb (16.9 N·m).

6) Install the safety wire [18] on the drain plug [17].

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SUBTASK 73-11-07-010-001-G00

(3) Remove the two bolts [2] and saddle clamp [3] that attach the main fuel supply hose [1] to the upper fan case bracket.

(a) Keep the two bolts [2] and saddle clamp [3] for the subsequent installation.

SUBTASK 73-11-07-020-001-G00

(4) Remove the two bolts [2A] and saddle clamp [3] that attach the main fuel supply hose [1] to the lower fan case bracket.

(a) Keep the two bolts [2A] and saddle clamp [3] for the subsequent installation.

SUBTASK 73-11-07-030-001-G00

(5) Disconnect the main fuel supply hose [1] from the main fuel pump as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the main fuel pump.

(b) Loosen the four bolts [4] that attach the main fuel supply hose [1] to the main fuel pump inlet port.

1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(c) Remove the four bolts [4] and four washers [5].

(d) Remove the main fuel supply hose [1] from the main fuel pump.

(e) Remove and discard the gasket seal [6].

SUBTASK 73-11-07-030-002-G00

(6) Disconnect the main fuel supply hose [1] from the strut fitting as follows:

(a) Use two wrenches to disconnect the main fuel supply hose [1] B-nut from the strut fitting.

NOTE: Use one wrench to hold the main fuel supply hose and use another wrench to turn the B-nut.

SUBTASK 73-11-07-420-001-G00

(7) Put the protection covers on the main fuel supply hose [1], main fuel pump inlet port, and strut fitting.

——— END OF TASK ———

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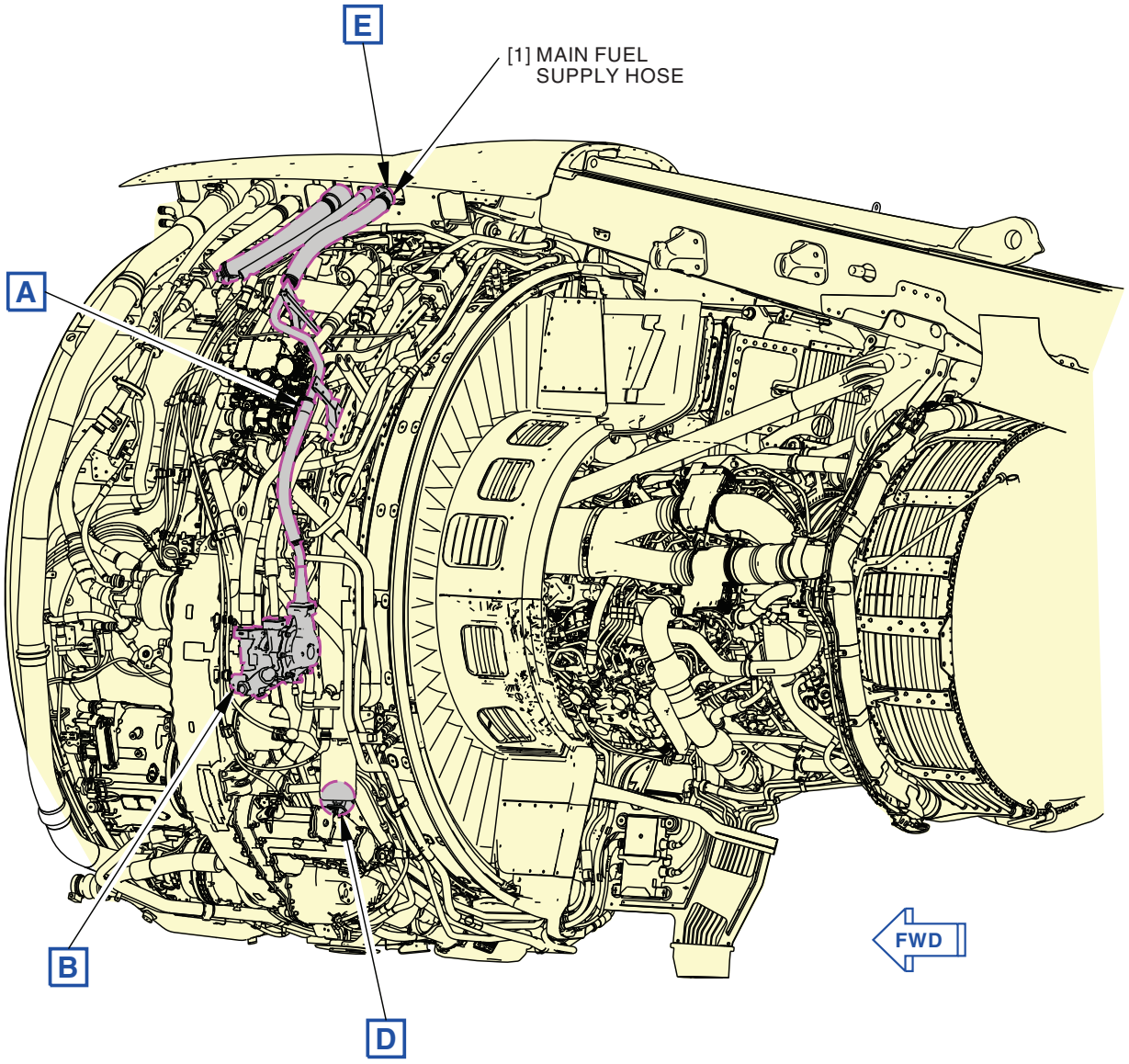
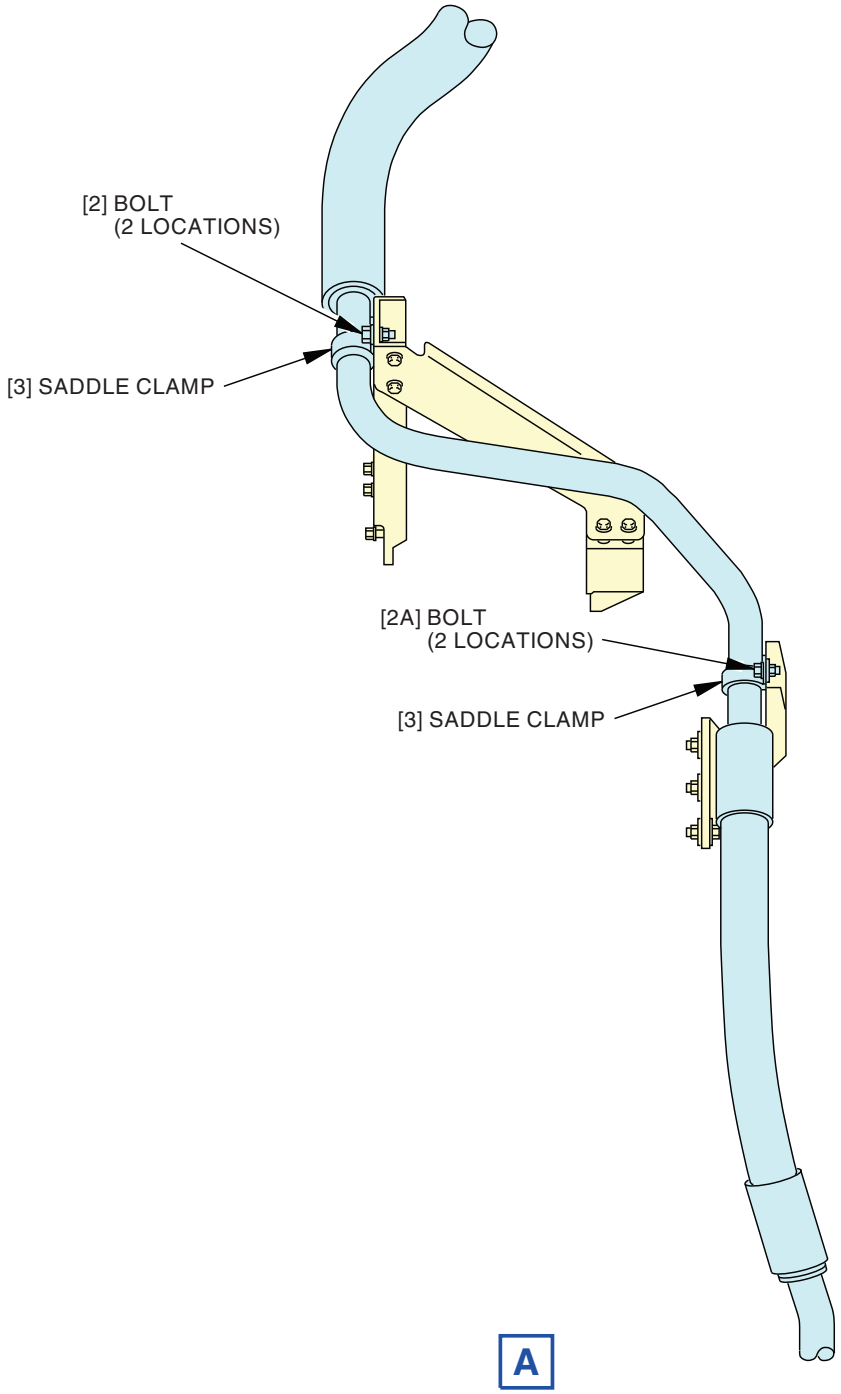


2354777 S00061515918_V1

Main Fuel Supply Hose Installation
Figure 401/73-11-07-990-802-G00 (Sheet 1 of 5)

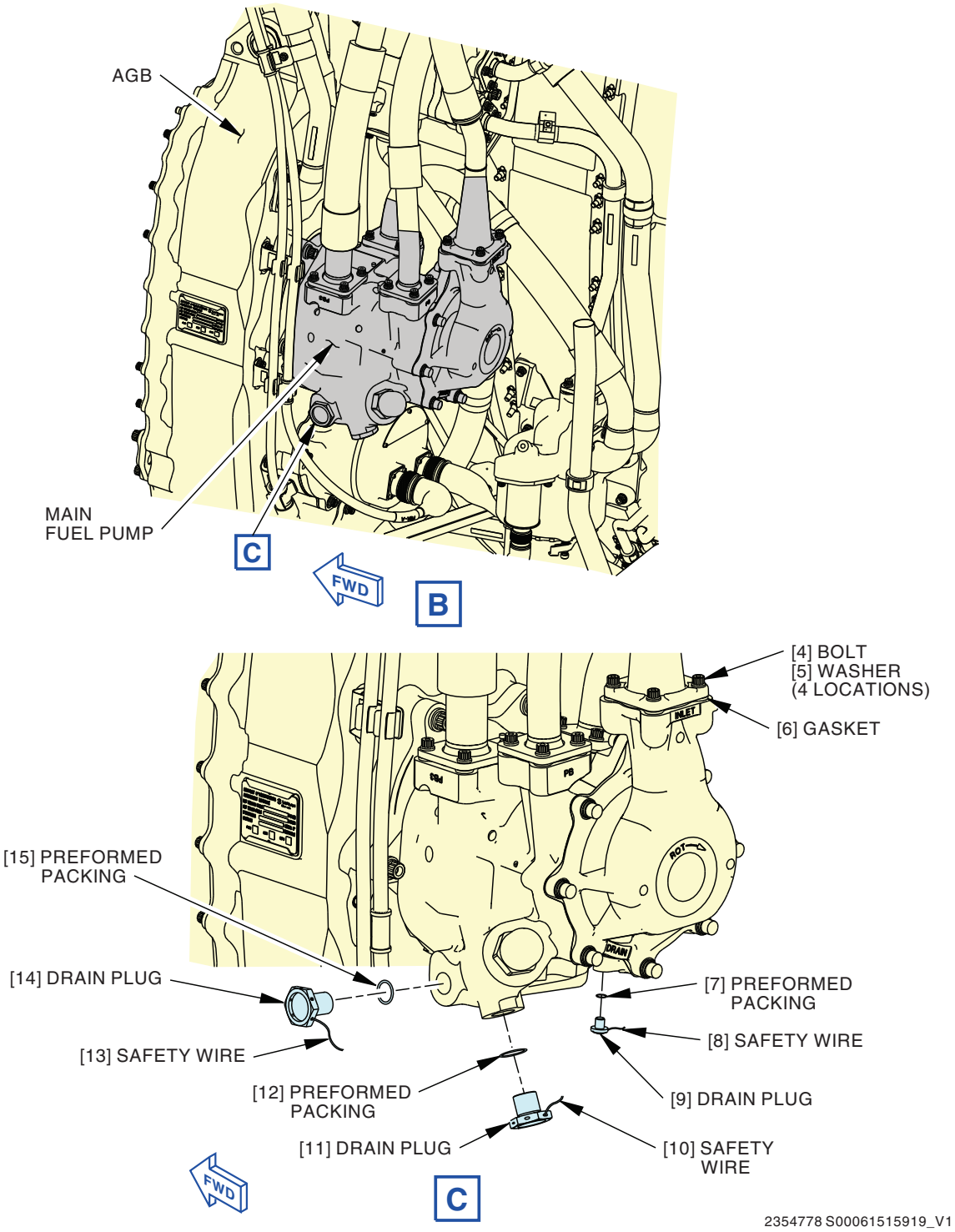
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2454227 S0000570630_V3

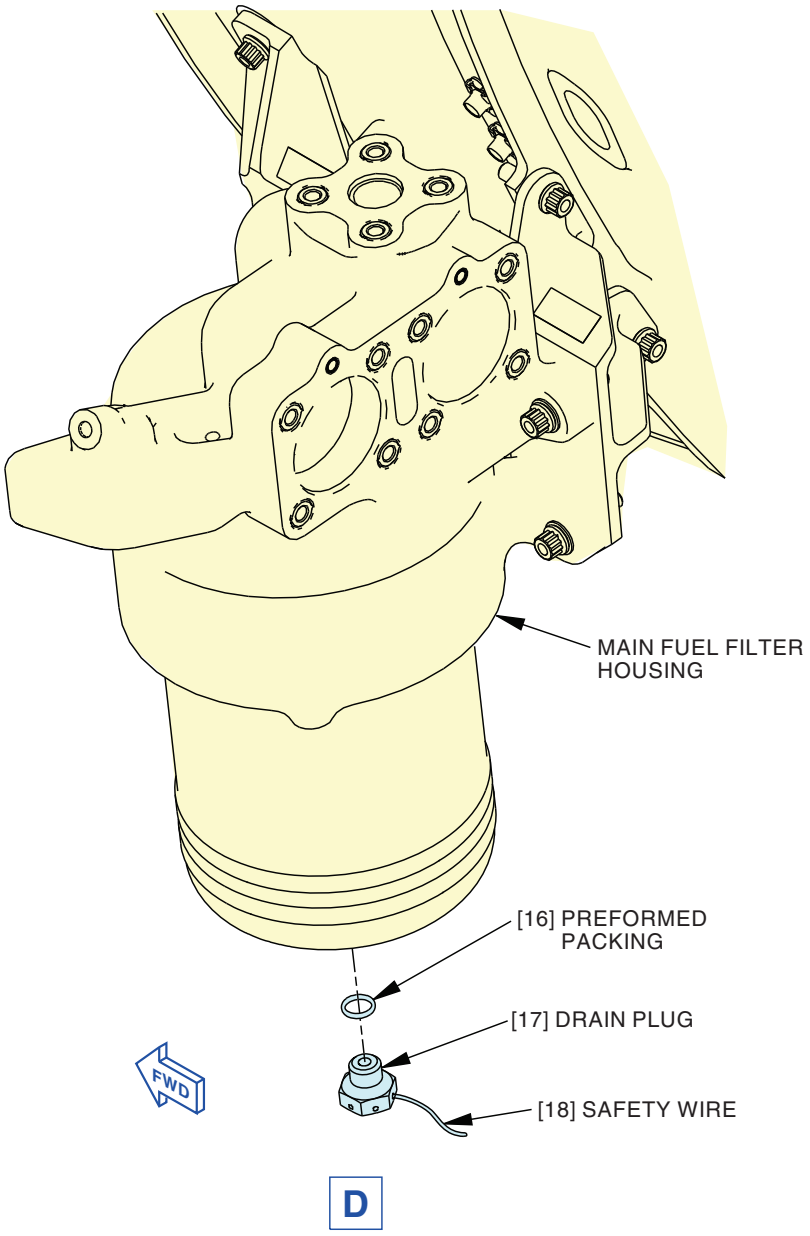
Main Fuel Supply Hose Installation
Figure 401/73-11-07-990-802-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel Supply Hose Installation
Figure 401/73-11-07-990-802-G00 (Sheet 3 of 5)

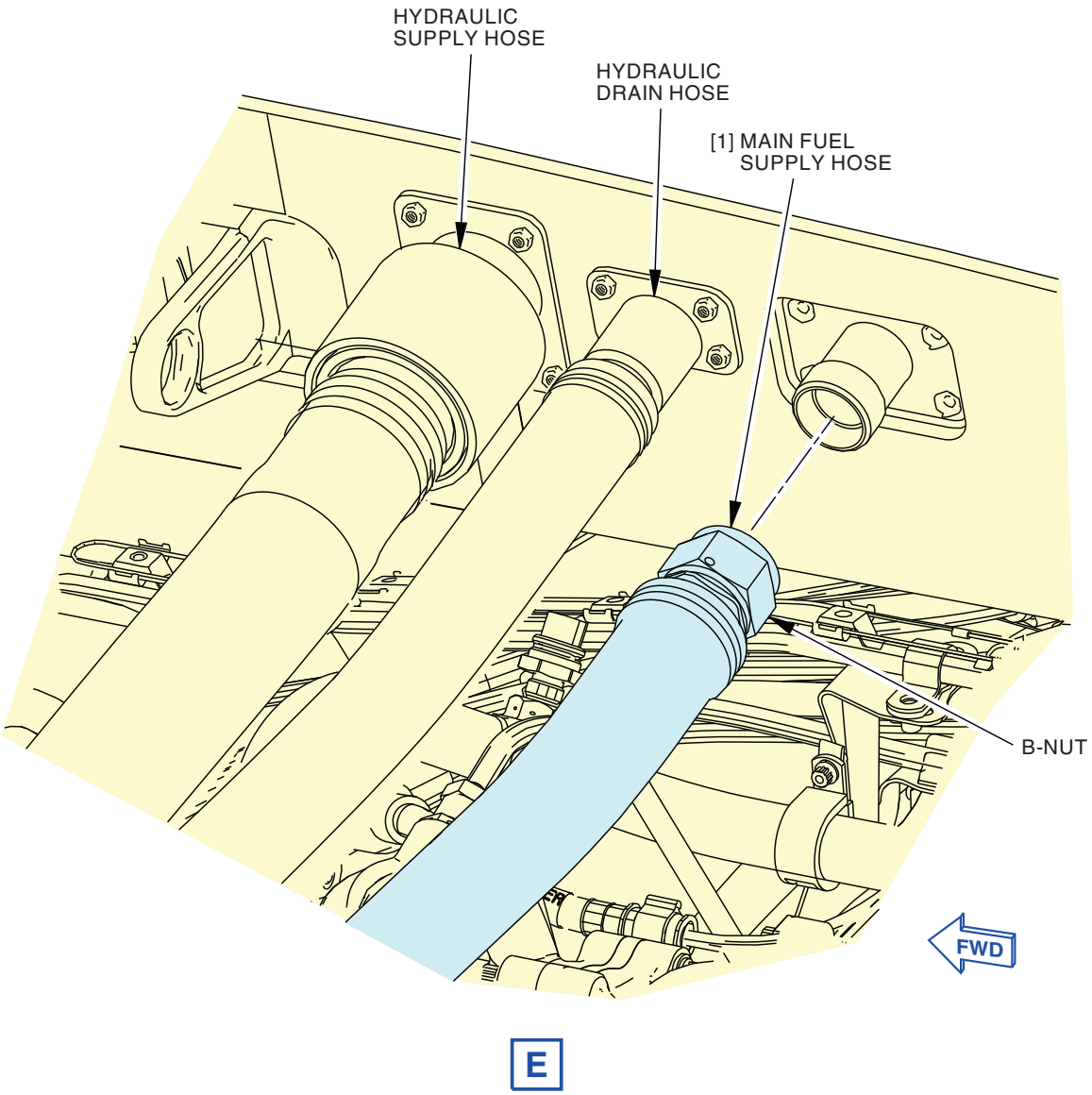
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354779 S00061515920_V1

Main Fuel Supply Hose Installation
Figure 401/73-11-07-990-802-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2354780 S00061515921_V1

Main Fuel Supply Hose Installation
Figure 401/73-11-07-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-07-400-801-G00

3. Main Fuel Supply Hose Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the main fuel supply hose.

(2) Do the tests that are listed in the power plant test reference table after the main fuel supply hose installation.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D50011	Grease - Perfluoropolyether - Christo-lube MCG111	
D50271 [C02-033]	Lubricant - Petrolatum, Soft	VV-P-236
G00834	Cloth - Lint-free Cotton	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Main fuel supply hose	73-11-51-01-030	PTF ALL
6	Gasket seal	73-11-51-01-025	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Installation

SUBTASK 73-11-07-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-07-860-018-G00

(2) For the applicable engine, make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 73-11-07-860-017-G00

(3) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED light, on the P5 forward overhead panel, is on dim.

(c) Make sure that the applicable ENG VALVE CLOSED light, on the P5 forward overhead panel, is on dim.

SUBTASK 73-11-07-860-019-G00

(4) On the applicable engine, make sure that these access panels are open:

(TASK 71-11-04-010-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Main Fuel Supply Hose Installation

SUBTASK 73-11-07-430-001-G00

(1) Connect the main fuel supply hose [1] to the strut fitting as follows:

(a) Remove the protection covers from the main fuel supply hose [1] and strut fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Lubricate the main fuel supply hose [1] B-nut with Christo-lube MCG111 grease, D50011.

(c) Connect the main fuel supply hose [1] B-nut to the strut fitting.

(d) Use two wrenches to tighten the main fuel supply hose [1] B-nut to 1140 in-lb (129 N·m) - 1260 in-lb (142.4 N·m).

NOTE: Use one wrench to hold the main fuel supply hose and use another wrench to turn the B-nut.

(e) Tighten the B-nut again to 1140 in-lb (129 N·m) - 1260 in-lb (142.4 N·m).

SUBTASK 73-11-07-430-002-G00

(2) Connect the main fuel supply hose [1] to the main fuel pump inlet port as follows:

(a) Remove the protection cover on the main fuel supply hose [1] and main fuel pump inlet port.

(b) Lubricate the gasket seal [6] with soft petrolatum, D50271 [C02-033].

(c) Lubricate the threads of the four bolts [4] with Pure Nickel Special compound, D00006.

(d) Put the gasket seal [6] in its position on the main fuel pump inlet port.

(e) Install the main fuel supply hose [1] in its position on the main fuel pump inlet port.

1) Loosely install the four bolts [4] and four washers [5] that attach the main fuel supply hose [1] to the main fuel pump inlet port.

(f) Attach the main fuel supply hose [1] to the fan case brackets.

1) Install the two bolts [2] and saddle clamp [3] that attach the main fuel supply hose [1] to the upper fan case bracket.

2) Install the two bolts [2A] and saddle clamp [3] that attach the main fuel supply hose [1] to the lower fan case bracket.

3) Adjust the main fuel supply hose [1] to distribute the hose equally.

4) Tighten the two bolts [2] and two bolts [2A] to 78 in-lb (8.8 N·m) - 82 in-lb (9.3 N·m).

(g) Tighten the bolts [4] to 50 in-lb (5.6 N·m) - 56 in-lb (6.3 N·m).

SUBTASK 73-11-07-100-001-G00

(3) Use a clean lint-free cloth, G00834, to remove the fuel from the surface of the main fuel supply hose [1].

J. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-07-860-005-G00

(1) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-07-860-006-G00

(2) For the applicable engine, remove the safety tags and close these applicable circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	1	C04016	ENGINE 1 IGNITION PWR CONT
E	2	C04017	ENGINE 2 IGNITION PWR CONT

SUBTASK 73-11-07-860-007-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-07-860-014-G00

(4) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

K. Main Fuel Supply Hose Installation Test

SUBTASK 73-11-07-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FMU FUEL STRAINER DIFFERENTIAL PRESSURE TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) FMU fuel strainer differential pressure transducer removal

(2) FMU fuel strainer differential pressure transducer installation.

TASK 73-11-09-000-801-G00

2. FMU Fuel Strainer Differential Pressure Transducer Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the FMU fuel strainer differential pressure transducer.

(2) The FMU fuel strainer differential pressure transducer is referred to as the fuel strainer DP transducer.

(3) The fuel strainer DP transducer is installed at the 10:00 o'clock position on the left engine case.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-11-09-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-09-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-09-010-001-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fuel Strainer DP Transducer Removal

SUBTASK 73-11-09-020-001-G00

(1) Remove the fuel strainer DP transducer [1] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the fuel strainer DP transducer [1].

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(b) Disconnect the electrical connector [3] as follows:

1) Use a teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] from the fuel strainer DP transducer [1] (TASK 70-00-01-910-803-G00).

2) Install the protection covers on the electrical receptacle of the fuel strainer DP transducer [1] and the electrical connector [3].

(c) Remove the fuel strainer DP transducer [1] from the FMU [2] as follows:

1) Remove the two bolts [4] that attach the fuel strainer DP transducer [1] to the FMU [2].

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

2) Remove the fuel strainer DP transducer [1] from the FMU [2].

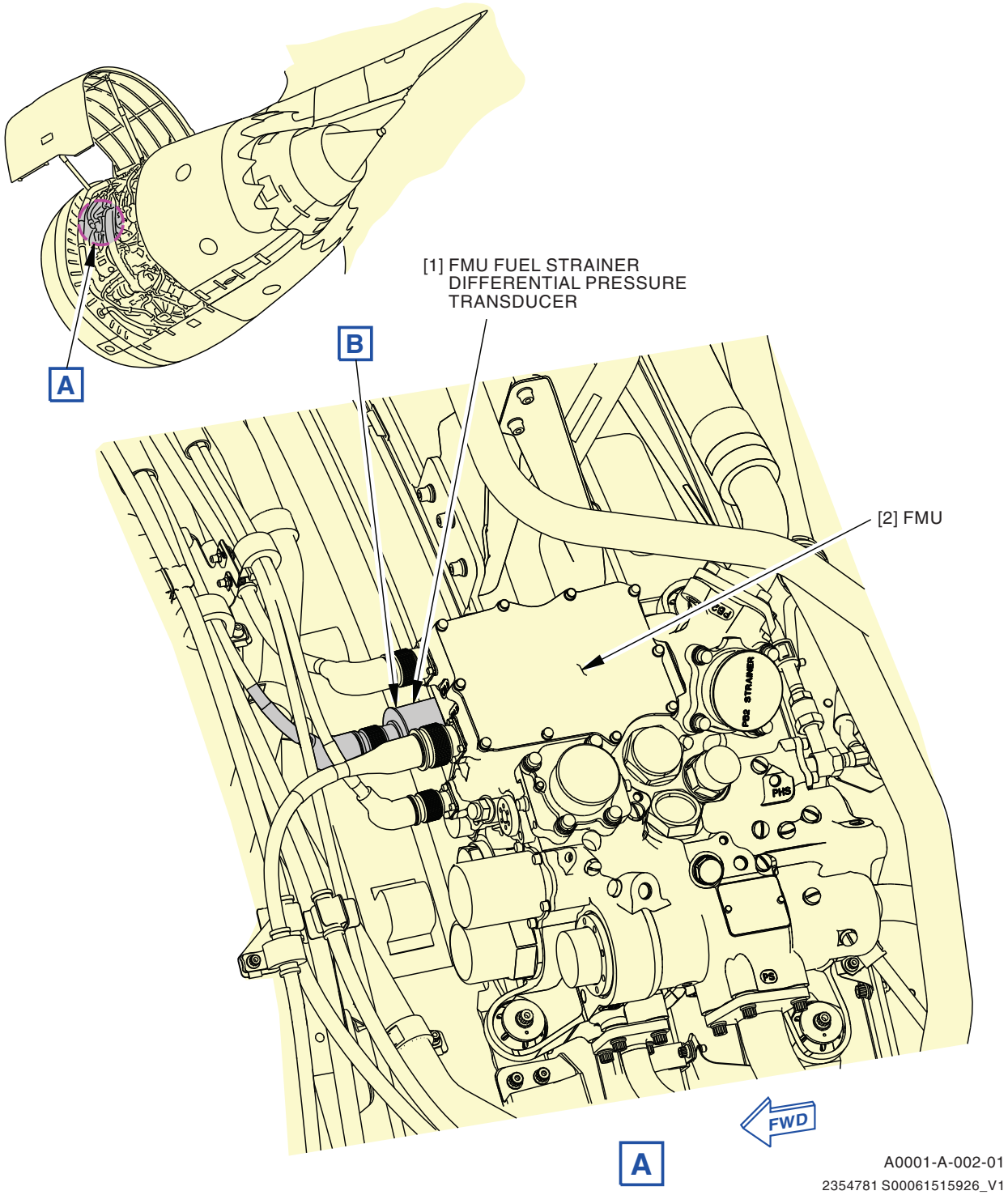
ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Remove and discard the preformed packing [5], the two back-up rings [7] and the preformed packing [6] from the fuel strainer DP transducer [1].

4) Install the protection covers on the fuel strainer DP transducer [1] and the opening port.

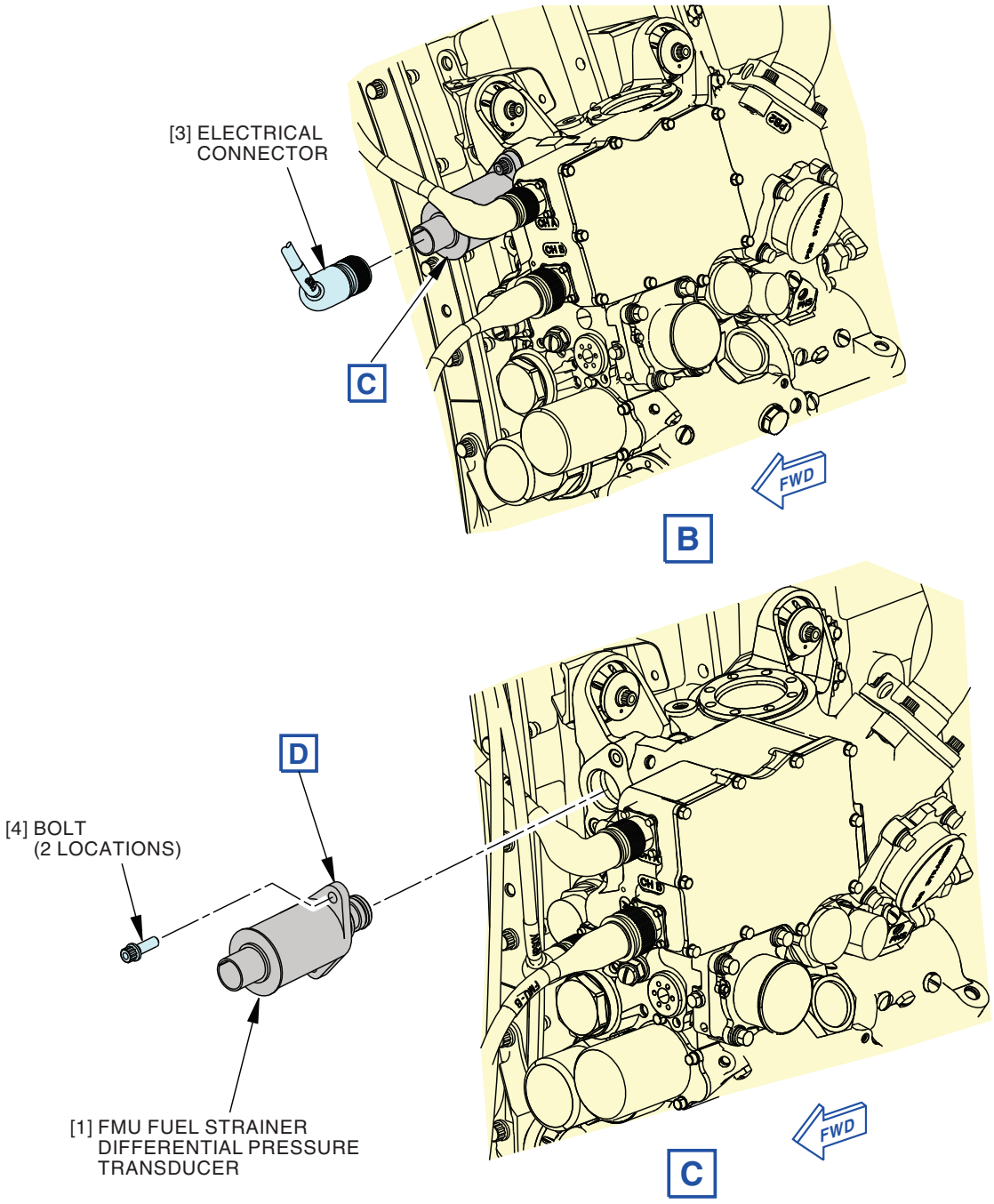
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Strainer DP Transducer Installation
Figure 401/73-11-09-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



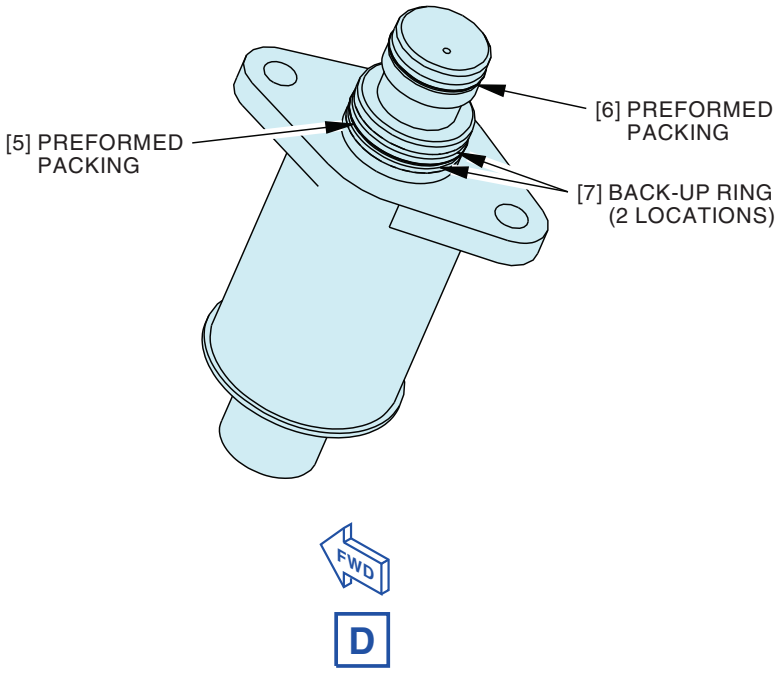
NOTE:

SOME COMPONENTS ARE NOT SHOWN

A0002-A-002-01
2398636 S0000552651_V2

Fuel Strainer DP Transducer Installation
Figure 401/73-11-09-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fuel Strainer DP Transducer Installation
Figure 401/73-11-09-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-09-400-801-G00

3. FMU Fuel Strainer Differential Pressure Transducer Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the FMU fuel strainer differential pressure transducer.

(2) The FMU fuel strainer differential pressure transducer is referred to as the fuel strainer DP transducer.

(3) You must do the tests that are listed in the power plant test reference table after you install the fuel strainer DP transducer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel strainer DP transducer	73-11-09-01-010	PTF ALL
5	Preformed packing	73-11-09-01-050	PTF ALL
6	Preformed packing	73-11-09-01-030	PTF ALL
7	Back-up ring	73-11-09-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Fuel Strainer DP Transducer Installation

SUBTASK 73-11-09-420-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Install the fuel strainer DP transducer [1] as follows:

(a) Remove the protection covers from the fuel strainer DP transducer [1] and the opening port.

(b) Make sure that the mating faces are clean and clear of the unwanted materials.

(c) Lubricate the threads of the two bolts [4], the new new preformed packing [5], the two new back-up rings [7] and the preformed packing [6] with engine oil, D00599 [CP2442].

(d) Install the new preformed packing [5], the two new back-up rings [7] and the preformed packing [6] onto the grooves of the fuel strainer DP transducer [1].

(e) Install the fuel strainer DP transducer [1] on the FMU [2].

(f) Install the two bolts [4] that attach the fuel strainer DP transducer [1] to the FMU [2].

(g) Tighten the bolts [4] to 109.25 in-lb (12.3 N·m) - 120.75 in-lb (13.6 N·m).

SUBTASK 73-11-09-410-002-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(2) Connect the electrical connector [3] as follows:

(a) Remove the protective covers from the electrical receptacle of the fuel strainer DP transducer [1] and the electrical connector [3].

(b) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] to the fuel strainer DP transducer [1] (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-09-100-001-G00

(3) Clean the work area and clear the tools used in the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-09-410-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-09-860-003-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-09-860-004-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

J. Fuel Strainer DP Transducer Installation Test

SUBTASK 73-11-09-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FMU FUEL STRAINER DIFFERENTIAL PRESSURE TRANSDUCER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) FMU Fuel Strainer Differential Pressure Transducer inspection.

TASK 73-11-09-210-801-G00

2. FMU Fuel Strainer Differential Pressure Transducer Inspection

Figure 601

A. General

(1) This task provides the instructions to inspect the FMU fuel strainer differential pressure transducer.

(2) The FMU fuel strainer DP transducer is installed on the fuel metering unit on the left side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-09-000-801-G00	FMU Fuel Strainer Differential Pressure Transducer Removal (P/B 401)
73-11-09-400-801-G00	FMU Fuel Strainer Differential Pressure Transducer Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-11-09-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-09-860-006-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-09-010-003-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. FMU Fuel Strainer Differential Pressure Transducer Inspection

SUBTASK 73-11-09-210-001-G00

(1) Inspect the FMU fuel strainer Differential Pressure (DP) transducer:

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, do the steps in these tasks to replace the transducer preformed packings (TASK 73-11-09-000-801-G00 and TASK 73-11-09-400-801-G00).

(b) Make sure that the FMU fuel strainer DP transducer is not loose.

1) If the transducer is loose, tighten the bolts that attach the transducer to the FMU (TASK 73-11-09-400-801-G00).

(c) Make sure that the electrical connector is not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-09-410-004-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-09-860-007-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-09-860-008-G00

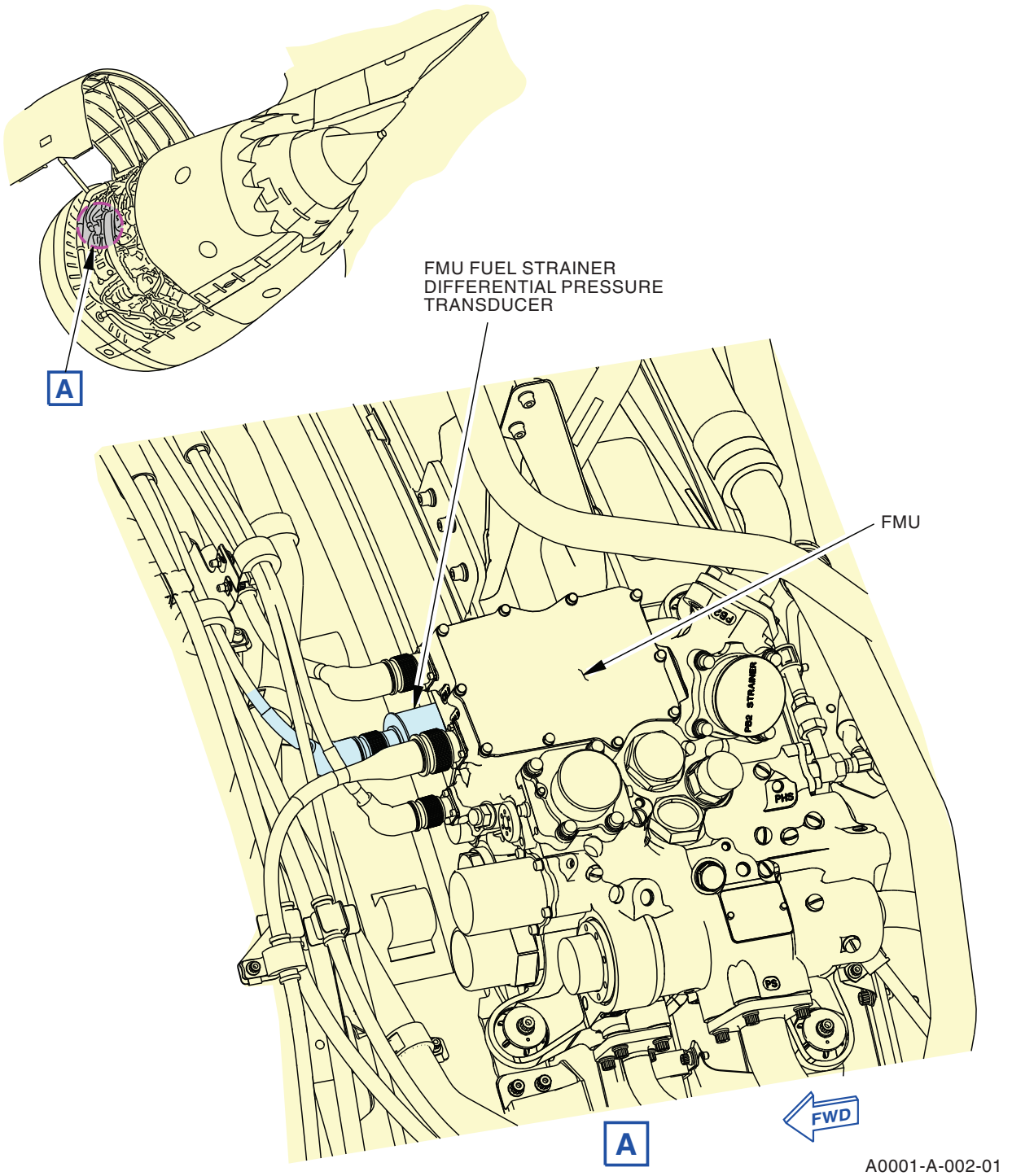
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FMU Fuel Strainer Differential Pressure Transducer Inspection
Figure 601/73-11-09-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL FILTER DIFFERENTIAL PRESSURE TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel filter differential pressure transducer removal

(2) Fuel filter differential pressure transducer installation.

TASK 73-11-10-000-801-G00

2. Fuel Filter Differential Pressure Transducer Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the fuel filter differential pressure transducer.

(2) The fuel filter differential pressure transducer is referred to as the fuel filter DP transducer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-11-10-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-10-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-10-860-003-G00

(3) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-10-860-004-G00

(4) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-10-010-001-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fuel Filter Differential Pressure Transducer Removal

SUBTASK 73-11-10-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] from the fuel filter DP transducer [1] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [3] from the receptacle of the fuel filter DP transducer [1].

(b) Install the protective covers on the electrical connector [3] and the fuel filter DP transducer [1].

SUBTASK 73-11-10-020-001-G00

(2) Remove the fuel filter DP transducer [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Put a fuel resistant container (5 gal), STD-201 below the fuel filter DP transducer [1].

(b) Remove the two bolts [4] that attach the fuel filter DP transducer [1] to the main fuel filter housing [2].

(c) Remove the fuel filter DP transducer [1].

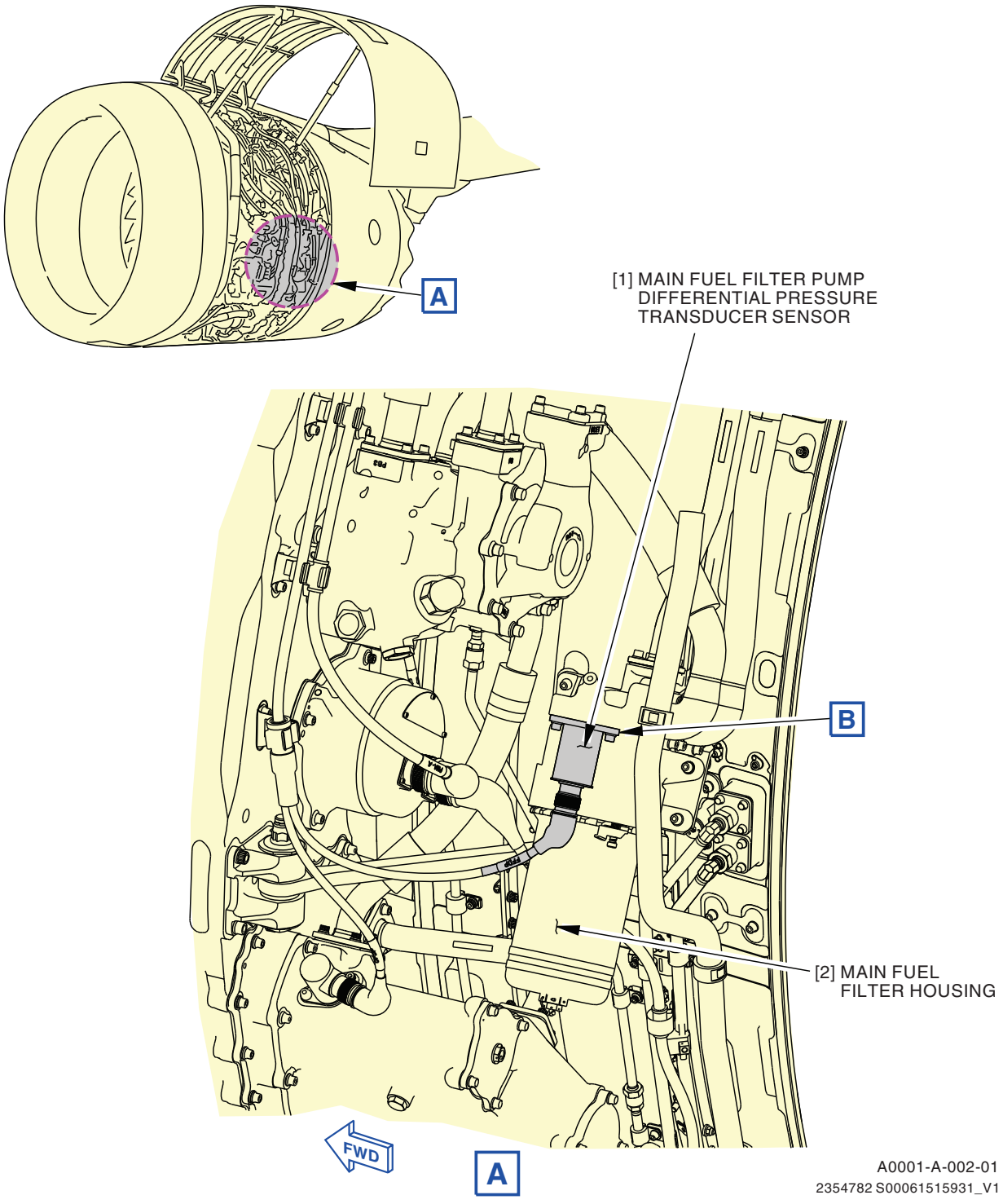
1) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove and discard the preformed packing [5], the two back-up rings [7], and the preformed packing [6] from the fuel filter DP transducer [1].

(e) Install the protective covers on the fuel filter DP transducer [1] and the main fuel filter housing [2].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Filter Differential Pressure Transducer Installation
Figure 401/73-11-10-990-801-G00 (Sheet 1 of 3)

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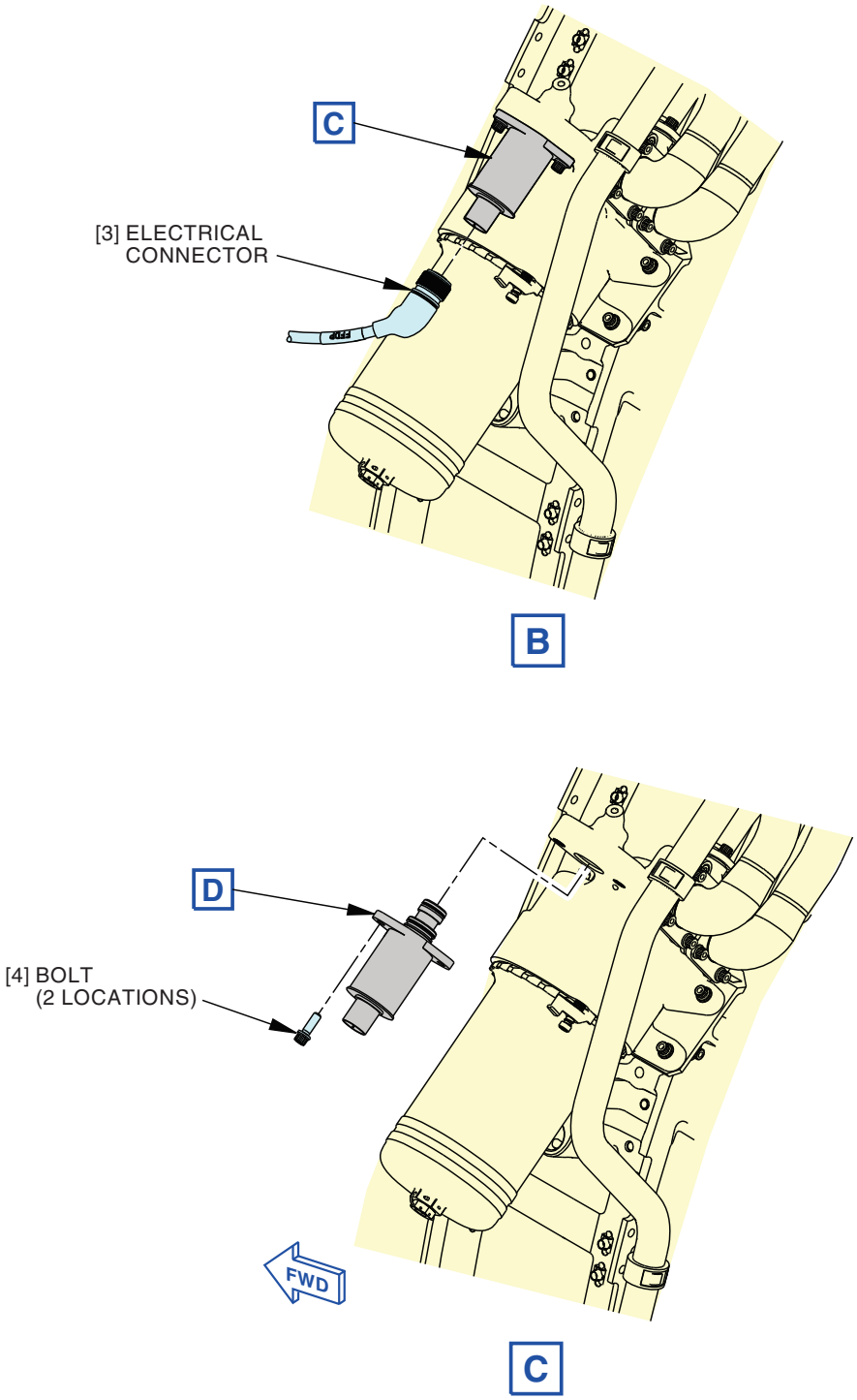

Fuel Filter Differential Pressure Transducer Installation
Figure 401/73-11-10-990-801-G00 (Sheet 2 of 3)

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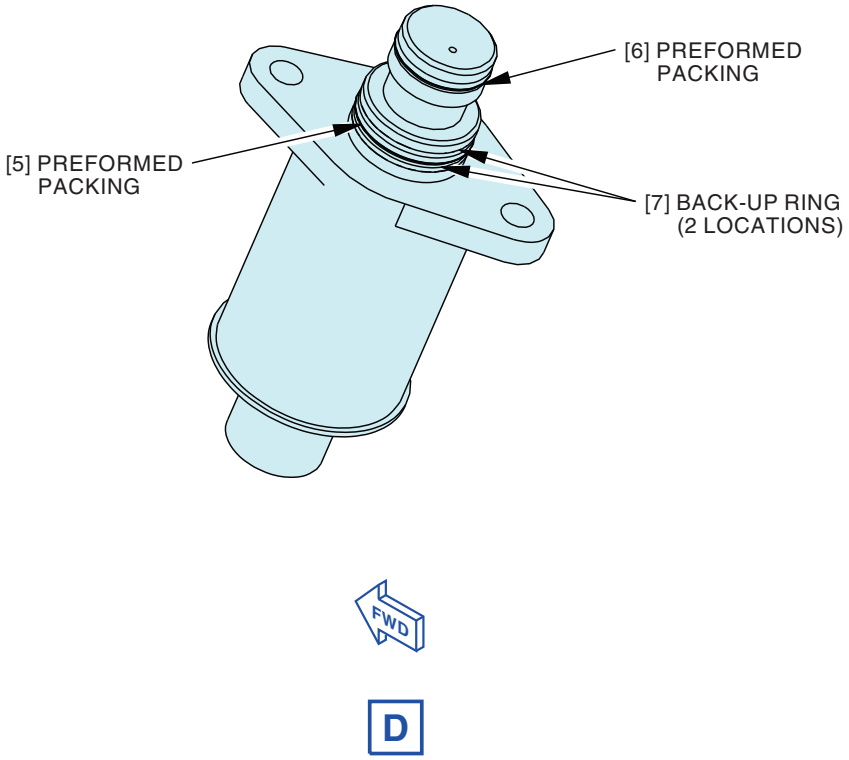

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Fuel Filter Differential Pressure Transducer Installation
Figure 401/73-11-10-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-10-400-801-G00

3. Fuel Filter Differential Pressure Transducer Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the fuel filter differential pressure transducer.

(2) The fuel filter differential pressure transducer is referred to as the fuel filter DP transducer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel filter DP transducer	73-11-10-01-010	PTF ALL
5	Preformed packing	73-11-10-01-050	PTF ALL
6	Preformed packing	73-11-10-01-030	PTF ALL
7	Back-up ring	73-11-10-01-040	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Fuel Filter Differential Pressure Transducer Installation

SUBTASK 73-11-10-540-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Install the fuel filter DP transducer [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the protective covers from the fuel filter DP transducer [1] and the main fuel filter housing [2].

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads of the two bolts [4], the new preformed packing [5], the two new back-up rings [7], and the new preformed packing [6] with the oil, D00599 [CP2442].

(d) Install the new preformed packing [5], the two new back-up rings [7], and the new preformed packing [6] onto the grooves of the fuel filter DP transducer [1].

(e) Install the fuel filter DP transducer [1] on the main fuel filter housing [2].

(f) Install the two bolts [4] that attach the fuel filter DP transducer [1] to the main fuel filter housing [2].

(g) Tighten the bolts [4] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

SUBTASK 73-11-10-430-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connector [3] and the fuel filter DP transducer [1].

(b) Connect the electrical connector [3] to the fuel filter DP transducer [1].

(c) Tighten the electrical connector [3].

SUBTASK 73-11-10-860-005-G00

(3) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-10-860-006-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-10-860-007-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-10-860-008-G00

(3) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-10-860-009-G00

(4) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-10-410-001-G00

(5) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Fuel Filter Differential Pressure Transducer Installation Test

SUBTASK 73-11-10-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL FILTER DIFFERENTIAL PRESSURE TRANSDUCER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fuel Filter Differential Pressure Transducer inspection.

TASK 73-11-10-210-801-G00

2. Fuel Filter Differential Pressure Transducer Inspection

(Figure 601)

A. General

(1) This task provides the instructions to inspect the fuel filter differential pressure transducer.

(2) The fuel filter differential pressure transducer is located on the main fuel filter housing on the left side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-10-000-801-G00	Fuel Filter Differential Pressure Transducer Removal (P/B 401)
73-11-10-400-801-G00	Fuel Filter Differential Pressure Transducer Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-11-10-860-010-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-10-860-011-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-10-010-002-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fuel Filter Differential Pressure Transducer Inspection

SUBTASK 73-11-10-210-001-G00

(1) Inspect the fuel filter Differential Pressure (DP) transducer:

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, do the steps in these tasks to replace the transducer preformed packings (TASK 73-11-10-000-801-G00 and TASK 73-11-10-400-801-G00).

(b) Make sure that the fuel filter DP transducer is not loose.

1) If the transducer is loose, tighten the bolts that attach the transducer to the main fuel filter housing (TASK 73-11-10-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-10-410-002-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-11-10-860-012-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-10-860-013-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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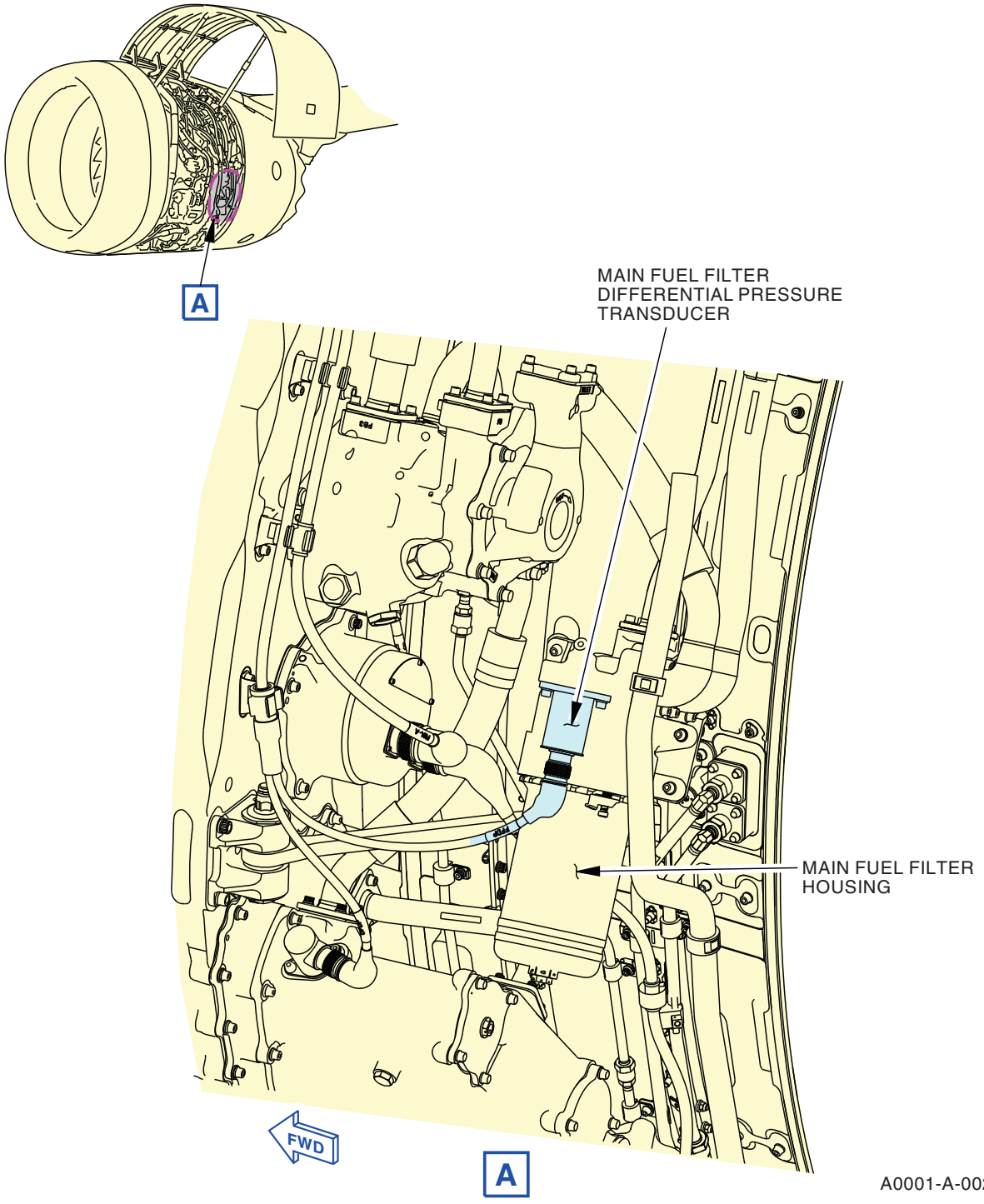

Fuel Filter Differential Pressure Transducer Inspection
Figure 601/73-11-10-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SERVO FUEL HEATER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Servo fuel heater removal

(2) Servo fuel heater installation.

TASK 73-11-11-000-801-G00

2. Servo Fuel Heater Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the servo fuel heater.

B. References

Reference	Title
12-13-11-100-801	Engine Oil System Flushing (P/B 301)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-11-11-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 73-11-11-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-11-860-003-G00

(3) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-11-860-004-G00

(4) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-11-010-001-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-11-11-680-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.

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(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(6) Drain the oil from the oil supply line (TASK 12-13-11-100-801).

NOTE: The oil line supply drain [39] is above the lubrication unit on the oil supply.

SUBTASK 73-11-11-680-002-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

(7) Drain the fuel from the main fuel filter (TASK 73-11-02-000-801-G00).

NOTE: The main fuel filter cartridge drain [38] is on the main fuel filter bowl.

G. Servo Fuel Heater Removal

SUBTASK 73-11-11-860-005-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Put a fuel resistant container (5 gal), STD-201 below the servo fuel heater.

SUBTASK 73-11-11-020-006-G00

(2) Disconnect the fuel tube [18] from the servo fuel heater [1] as follows:

(a) Remove the four bolts [19] from the fuel tube [18].

(b) Disconnect the fuel tube [18] from the servo fuel heater [1].

(c) Remove and discard the gasket seal [20].

(d) Install the protective covers on the openings.

SUBTASK 73-11-11-020-010-G00

(3) Disconnect the fuel tube [18] from the main fuel filter housing as follows:

(a) Remove the four bolts [34] from the fuel tube [18].

(b) Disconnect the fuel tube [18] from the main fuel filter housing.

(c) Remove and discard the gasket seal [35].

(d) Remove the bolt [37] and clamp [36].

(e) Install the protective covers on the openings.

SUBTASK 73-11-11-020-007-G00

(4) Disconnect the fuel tube [21] from the servo fuel heater [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the four bolts [22] from the fuel tube [21].

(b) Disconnect the fuel tube [21] from the servo fuel heater [1].

(c) Remove and discard the gasket seal [23].

(d) Install the protective covers on the openings.

SUBTASK 73-11-11-020-008-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(5) Disconnect the oil tube [24] from the servo fuel heater [1] as follows:

(a) Remove the four bolts [25] from the oil tube [24].

(b) Disconnect the oil tube [24] from the servo fuel heater [1].

(c) Remove and discard the gasket seal [26].

(d) Install the protective covers on the openings.

SUBTASK 73-11-11-020-009-G00

(6) Disconnect the oil tube [27] from the servo fuel heater [1] as follows:

(a) Remove the four bolts [28] from the oil tube [27].

(b) Disconnect the oil tube [27] from the servo fuel heater [1].

(c) Remove and discard the gasket seal [29].

(d) Install the protective covers on the openings.

SUBTASK 73-11-11-010-002-G00

(7) Remove the servo fuel heater [1] from the mounting brackets as follows:

(a) Remove the three bolts [30] with integrated washers from the servo fuel heater [1].

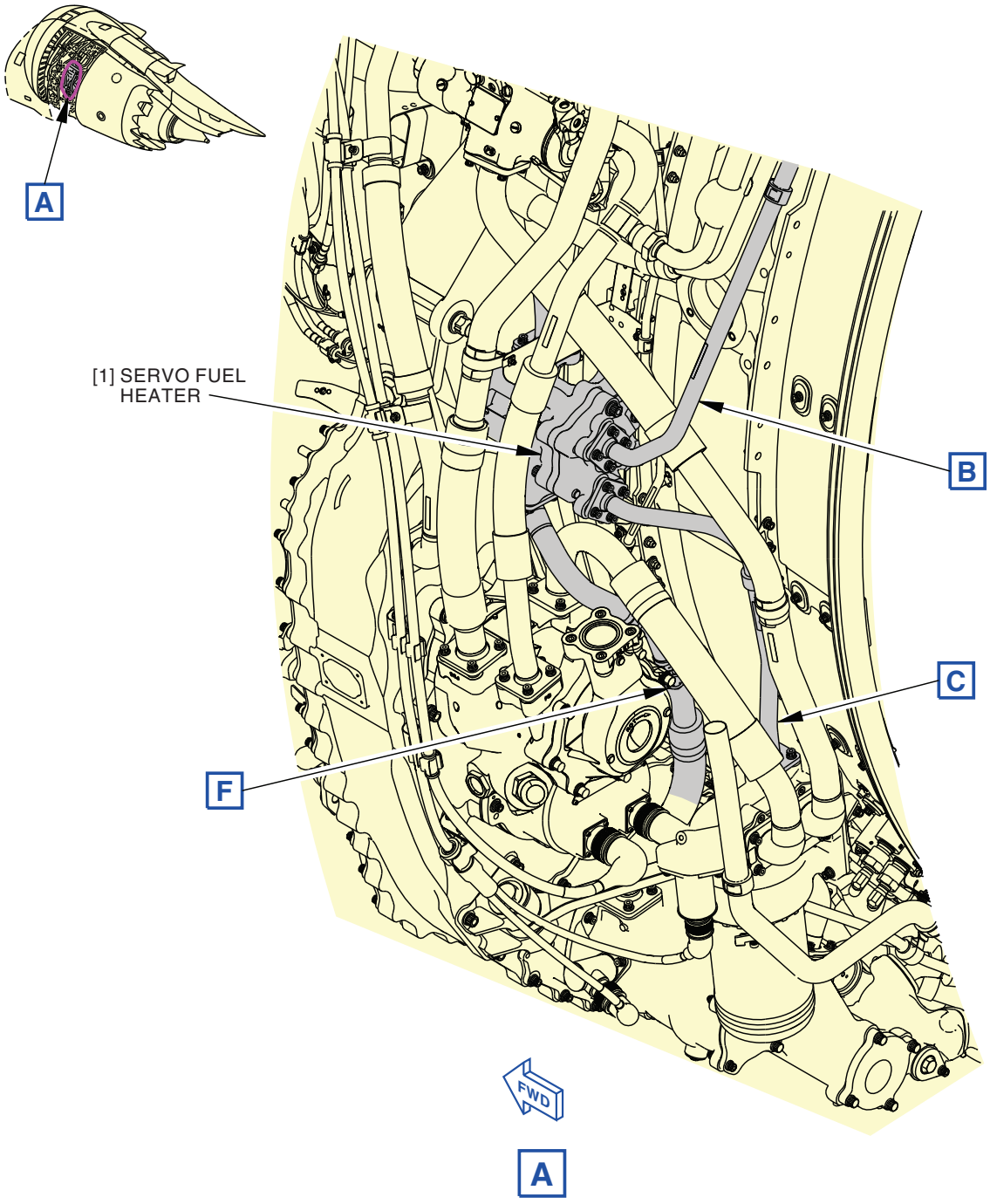
(b) Remove the servo fuel heater [1] from the mounting bracket [31], the mounting bracket [32] and the mounting bracket [33].

SUBTASK 73-11-11-480-001-G00

(8) Install the protective covers on the openings for the fuel tubes, the main fuel pump [2], the main fuel/oil heat exchanger [3], the FMU, and the servo fuel heater [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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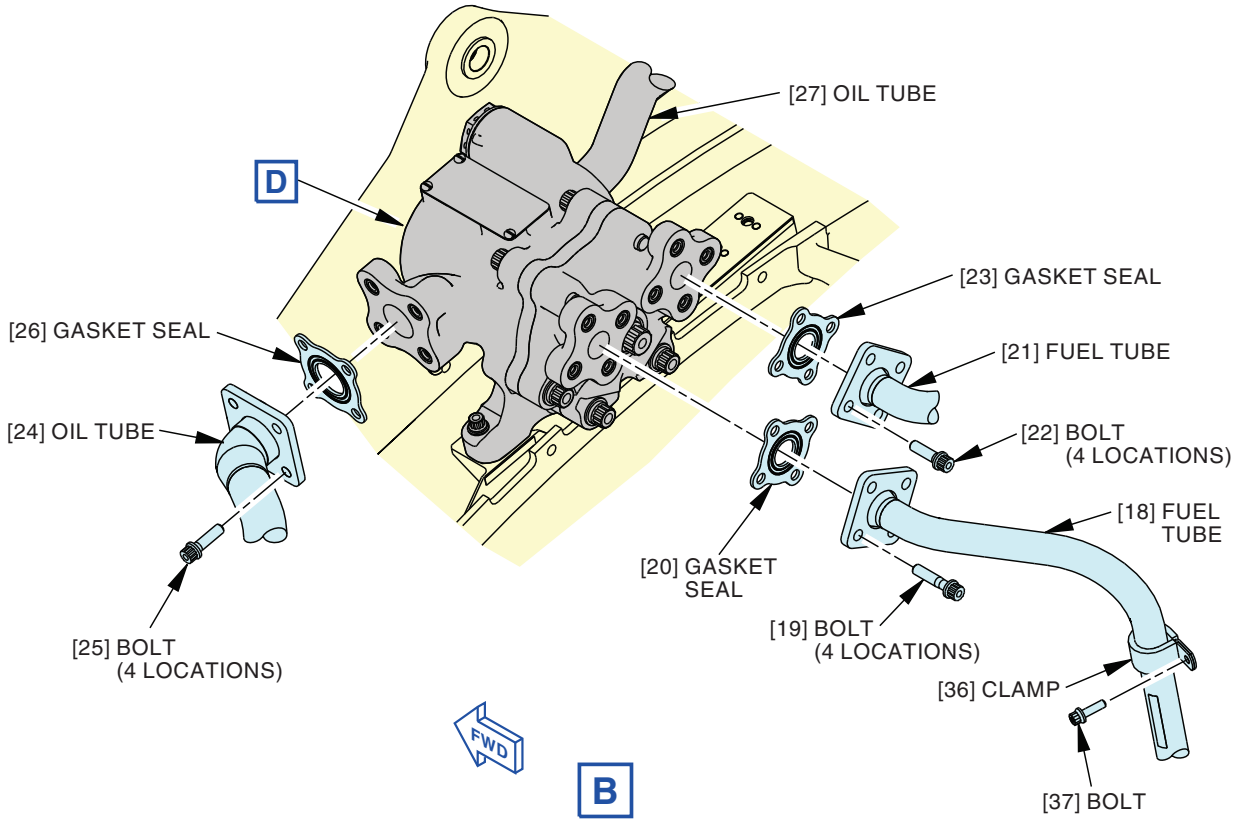
2402758 S0000555298_V4

Servo Fuel Heater Installation
Figure 401/73-11-11-990-801-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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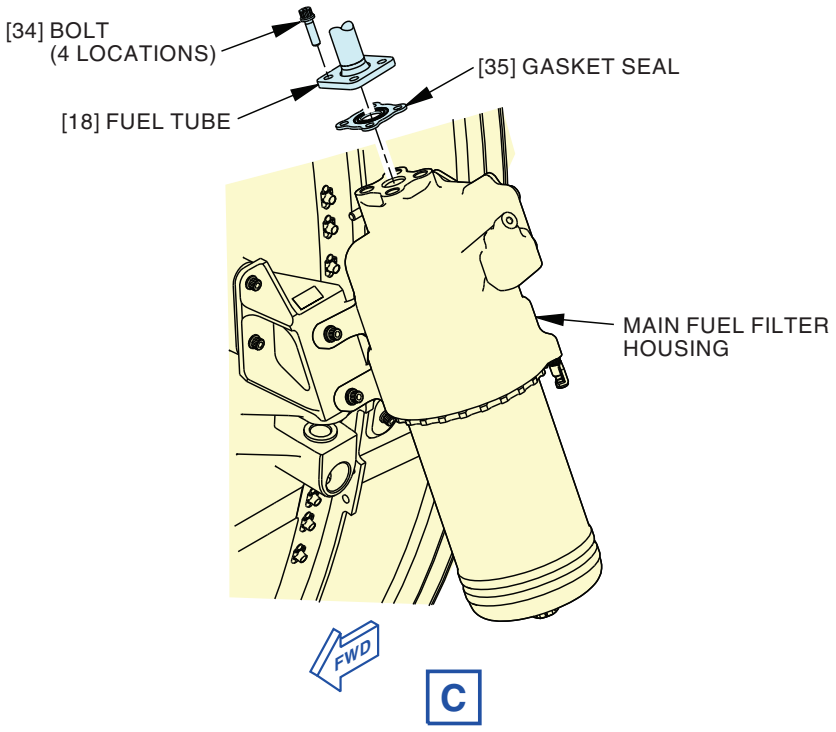
Servo Fuel Heater Installation
Figure 401/73-11-11-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Servo Fuel Heater Installation
Figure 401/73-11-11-990-801-G00 (Sheet 3 of 5)

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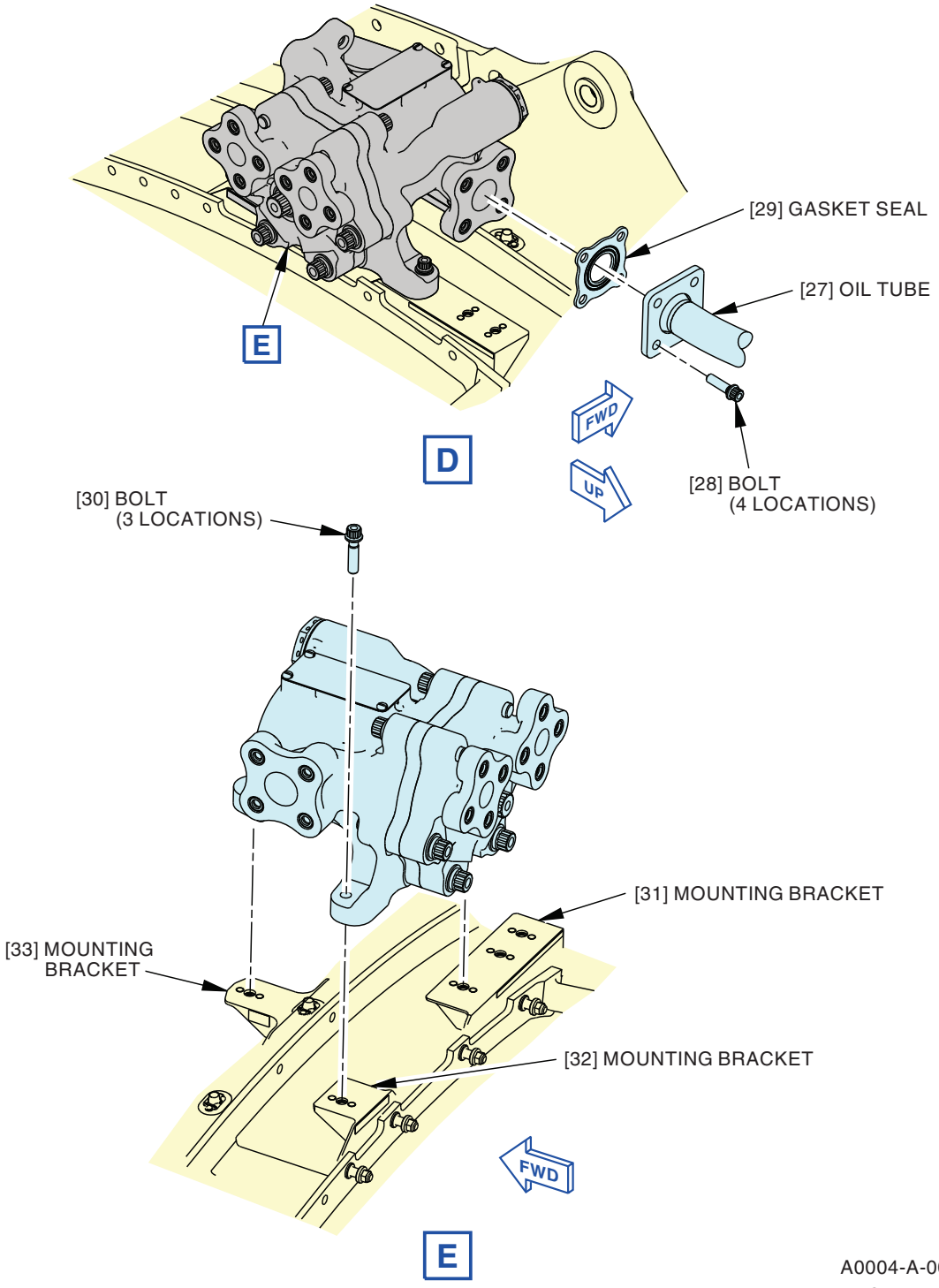

Servo Fuel Heater Installation
Figure 401/73-11-11-990-801-G00 (Sheet 4 of 5)

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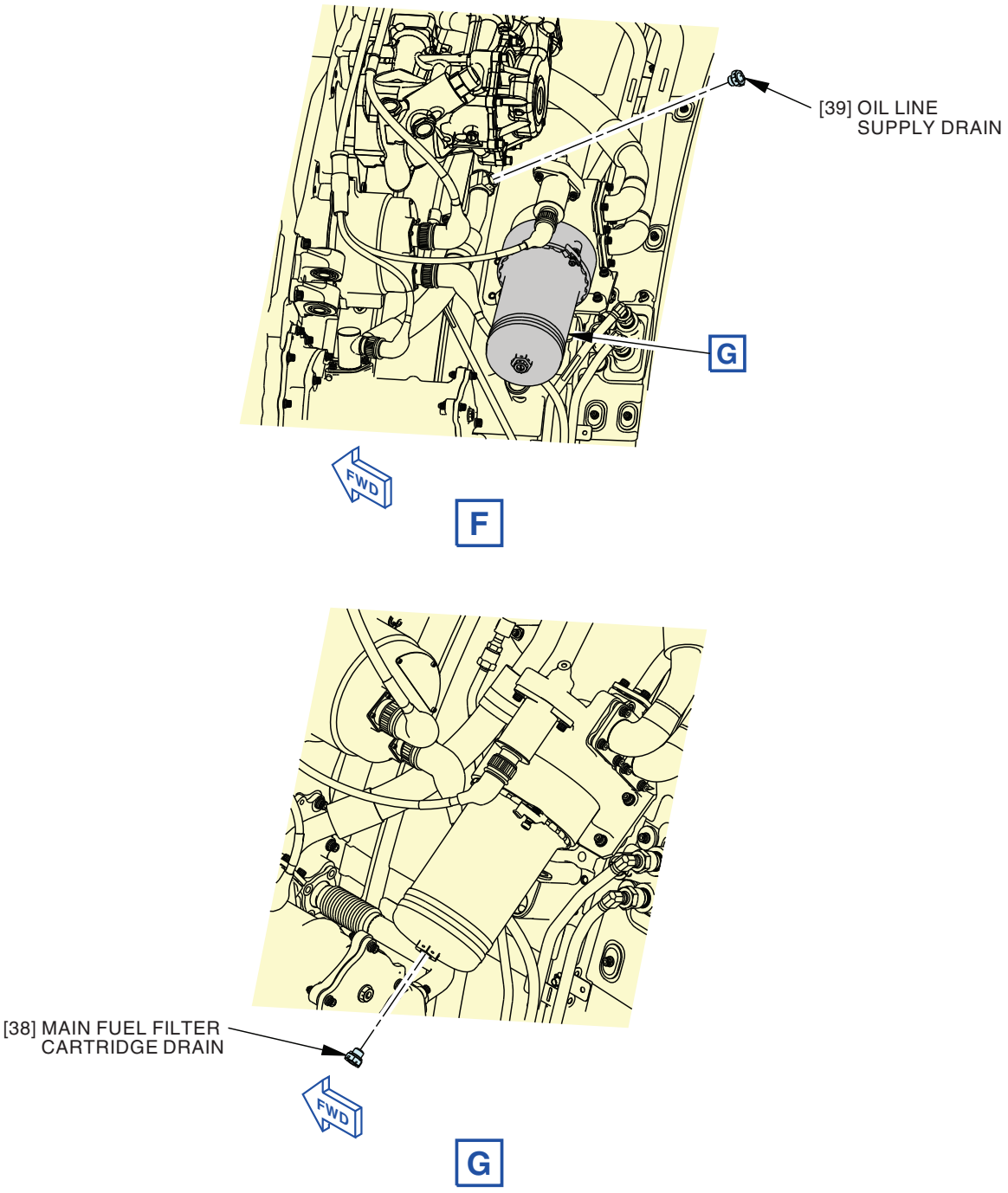

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Servo Fuel Heater Installation
Figure 401/73-11-11-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-11-400-801-G00

3. Servo Fuel Heater Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the servo fuel heater.

(2) You must do the tests that are listed in the power plant test reference table after you install the servo fuel heater.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51328 [CP2674]	Cloth - Lint-Free	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Servo fuel heater	73-11-11-01-100	PTF ALL
20	Gasket seal	73-11-11-01-020	PTF ALL
23	Gasket seal	73-11-11-01-040	PTF ALL
26	Gasket seal	73-11-11-01-060	PTF ALL
29	Gasket seal	73-11-11-01-080	PTF ALL
35	Gasket seal	73-11-03-01-130	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Servo Fuel Heater Installation

SUBTASK 73-11-11-420-001-G00

(1) Install the servo fuel heater [1]:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Lubricate the threads of the three bolts [30] with integrated washers with engine oil, D00599 [CP2442].

(d) Install the three bolts [30] to attach the servo fuel heater [1] to the mounting bracket [31], the mounting bracket [32], and the mounting bracket [33].

(e) Tighten the bolts [30] to 67.5 in-lb (7.6 N·m) - 74.3 in-lb (8.4 N·m).

SUBTASK 73-11-11-420-002-G00

(2) Connect the oil tube [27] to the servo fuel heater [1]:

(a) Lubricate the threads and pressure faces of the four bolts [28] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate each side of the elastomer part of a new gasket seal [29] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(c) Install the new gasket seal [29] and the oil tube [27] in on the servo fuel heater [1].

(d) Install the four bolts [28] on the oil tube [27].

(e) Tighten the bolts [28] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(f) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51328 [CP2674].

SUBTASK 73-11-11-420-003-G00

(3) Connect the oil tube [24] to the servo fuel heater [1]:

(a) Lubricate the threads and pressure faces of the four bolts [25] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate each side of the elastomer part of a new gasket seal [26] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(c) Install the new gasket seal [26] and the oil tube [24] on the servo fuel heater [1].

(d) Install the four bolts [25] on the oil tube [24].

(e) Tighten the bolts [25] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(f) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51328 [CP2674].

SUBTASK 73-11-11-420-004-G00

(4) Connect the fuel tube [21] to the servo fuel heater [1]:

(a) Lubricate the threads and pressure faces of the four bolts [22] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate each side of the elastomer part of a new gasket seal [23] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(c) Install the new gasket seal [23] and the fuel tube [21] on the servo fuel heater [1].

(d) Install the four bolts [22] on the fuel tube [21].

(e) Tighten the bolts [22] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(f) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51328 [CP2674].

SUBTASK 73-11-11-420-005-G00

(5) Connect the fuel tube [18] to the servo fuel heater [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Lubricate the threads and pressure faces of the four bolts [19] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate each side of the elastomer part of a new gasket seal [20] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(c) Install the new gasket seal [20] and the fuel tube [18] in a correct position on the servo fuel heater [1].

(d) Install the four bolts [19] on the fuel tube [18]

(e) Tighten the bolts [19] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(f) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51328 [CP2674].

SUBTASK 73-11-11-420-006-G00

(6) Connect the fuel tube [18] to the main fuel filter housing:

(a) Lubricate the threads and pressure faces of the four bolts [34] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate each side of the elastomer part of a new gasket seal [35] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(c) Install the new gasket seal [35] and the fuel tube [18] on the main fuel filter housing.

(d) Install the four bolts [34] on the fuel tube [18].

(e) Tighten the bolts [34] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(f) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51328 [CP2674].

(g) Install the bolt [37] on the clamp [36] that hold the fuel tube [18].

1) Tighten the bolt [37] to 42.5 in-lb (4.8 N·m) - 46.2 in-lb (5.2 N·m).

SUBTASK 73-11-11-160-001-G00

(7) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-11-610-001-G00

(1) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 73-11-11-860-006-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-11-860-007-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-11-860-008-G00

(4) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-11-860-009-G00

(5) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-11-410-002-G00

(6) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Servo Fuel Heater Installation Test

SUBTASK 73-11-11-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN FUEL/OIL HEAT EXCHANGER DIFFERENTIAL PRESSURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Main fuel/oil heat exchanger differential pressure sensor removal

(2) Main fuel/oil heat exchanger differential pressure sensor installation.

TASK 73-11-12-000-801-G00

2. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the differential pressure sensor for the main fuel/oil heat exchanger.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Preformed packing	73-11-01-01-070	PTF ALL
12	Preformed packing	73-11-01-01-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 73-11-12-210-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-12-860-001-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-12-010-001-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-11-12-480-001-G00

(4) Put a fuel resistant container (5 gal), STD-201 below the main fuel pump [7].

I. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Removal

SUBTASK 73-11-12-020-001-G00

(1) Remove the drain plug [8] from the main fuel pump [7]:

(a) Cut and discard safety cable, G50065 [CP8006], from the drain plug [8], drain plug [11], and drain plug [14].

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD VENTILATION. THE OIL IS POISONOUS AND CAN BE ABSORBED THROUGH YOUR SKIN. THE OIL FUMES CAN IRRITATE YOUR RESPIRATORY TRACT.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Remove the drain plug [8] from the main fuel pump [7].

(c) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(d) Remove and discard the preformed packing [9] from the drain plug [8].

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SUBTASK 73-11-12-420-001-G00

(2) Install the drain plug [8] on the main fuel pump [7] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the drain plug [8] and new preformed packing [9] with vaseline, D00672 [CP5070].

(c) Install the preformed packing [9] on the drain plug [8].

(d) Install the drain plug [8] on the main fuel pump [7].

(e) Tighten the drain plug [8] to 190.0 in-lb (21.5 N·m) - 210.0 in-lb (23.7 N·m).

SUBTASK 73-11-12-020-002-G00

(3) Remove the drain plug [11] from the main fuel pump [7] as follows:

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Remove the drain plug [11] from the main fuel pump [7].

(b) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(c) Remove and discard the preformed packing [12] from the drain plug [11].

SUBTASK 73-11-12-420-002-G00

(4) Install the drain plug [11] on the main fuel pump [7] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the drain plug [11] and new preformed packing [12] with vaseline, D00672 [CP5070].

(c) Install the preformed packing [12] on the drain plug [11].

(d) Install the drain plug [11] on the main fuel pump [7].

(e) Tighten the drain plug [11] to 190.0 in-lb (21.5 N·m) - 210.0 in-lb (23.7 N·m).

(f) Install safety cable, G50065 [CP8006], through the drain plug [8], drain plug [11], and drain plug [14].

SUBTASK 73-11-12-020-004-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(5) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connector [3] from the differential pressure sensor [1] (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-12-020-005-G00

(6) Remove the differential pressure sensor [1] as follows:

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(a) Remove the two bolts [4] that attach the main fuel/oil heat exchanger differential pressure sensor [1] to the main fuel/oil heat exchanger [2].

(b) Remove the differential pressure sensor [1] from the engine.

(c) Remove and discard the preformed packing [5], preformed packing [6], and backup rings [17] from the grooves of the differential pressure sensor [1].

(d) Install protective cover on the opening for the differential pressure sensor [1].

——— END OF TASK ———

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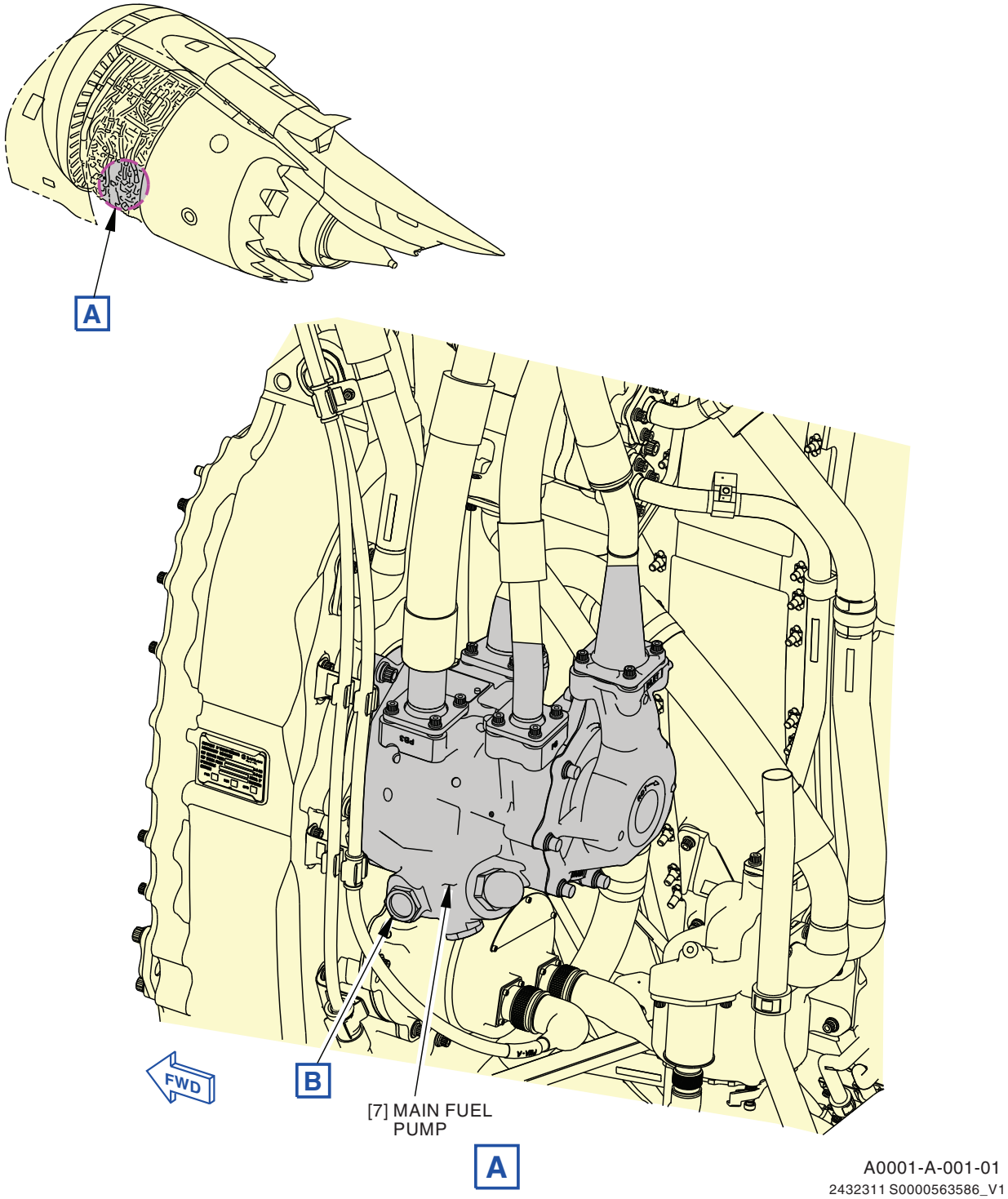

Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation
Figure 401/73-11-12-990-801-G00 (Sheet 1 of 5)

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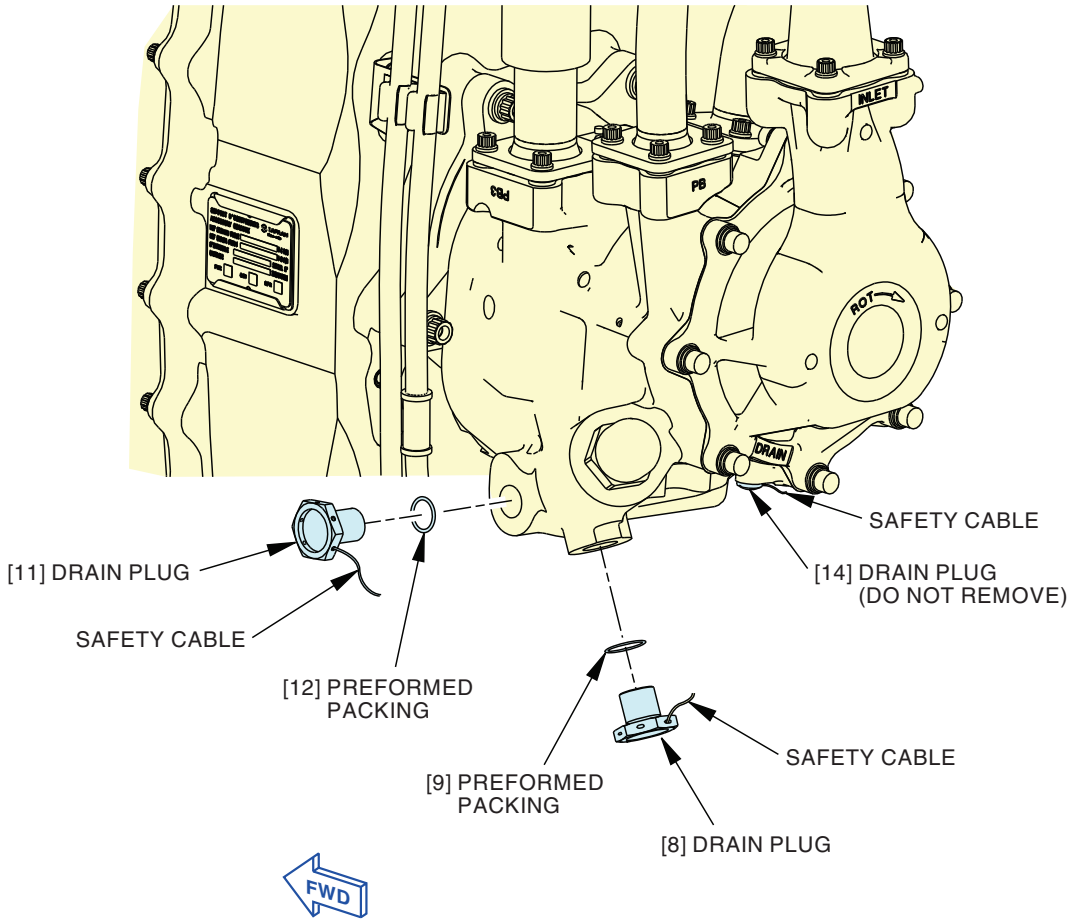

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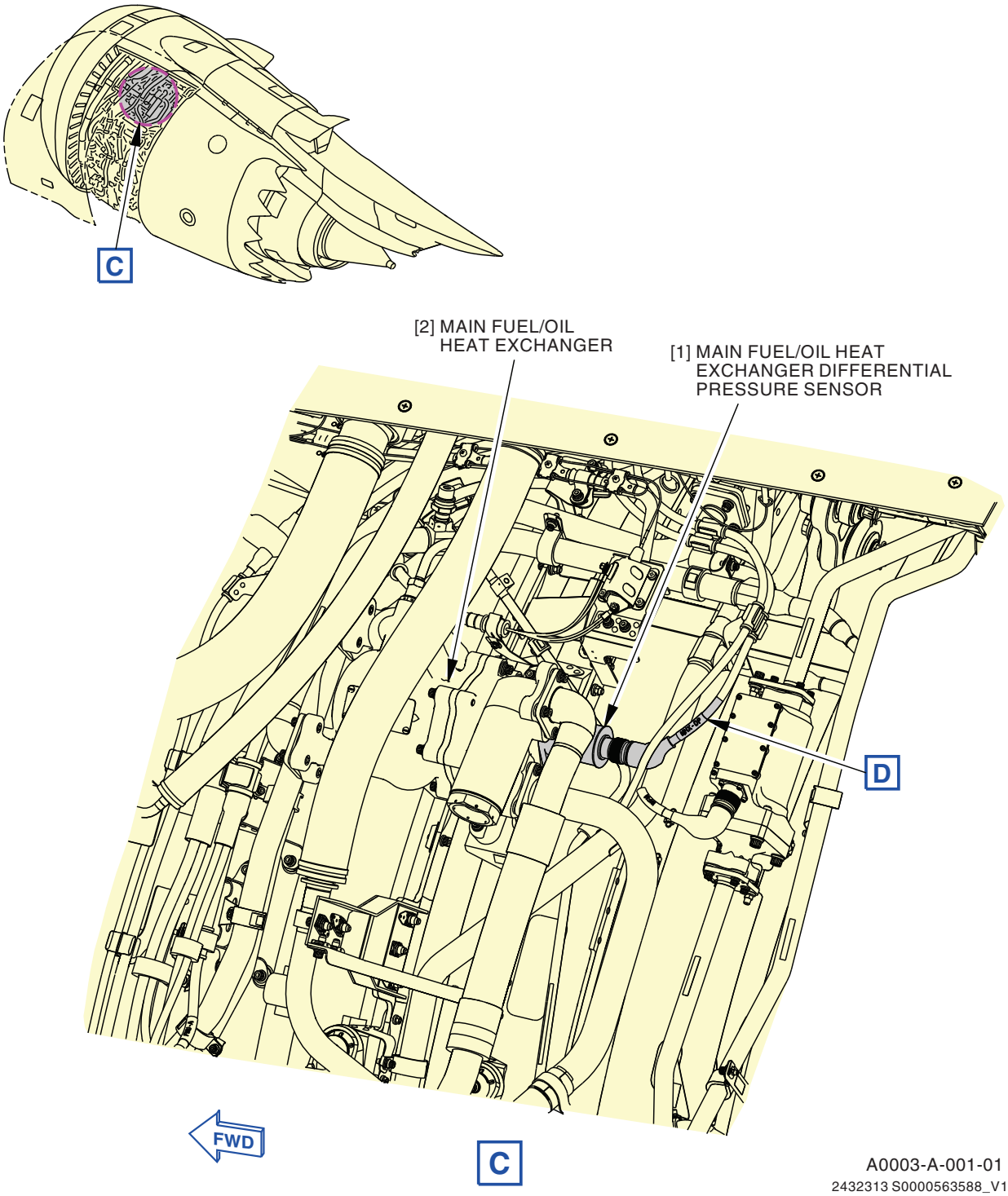
2432312 S0000563587_V2

Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation
Figure 401/73-11-12-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation
Figure 401/73-11-12-990-801-G00 (Sheet 3 of 5)

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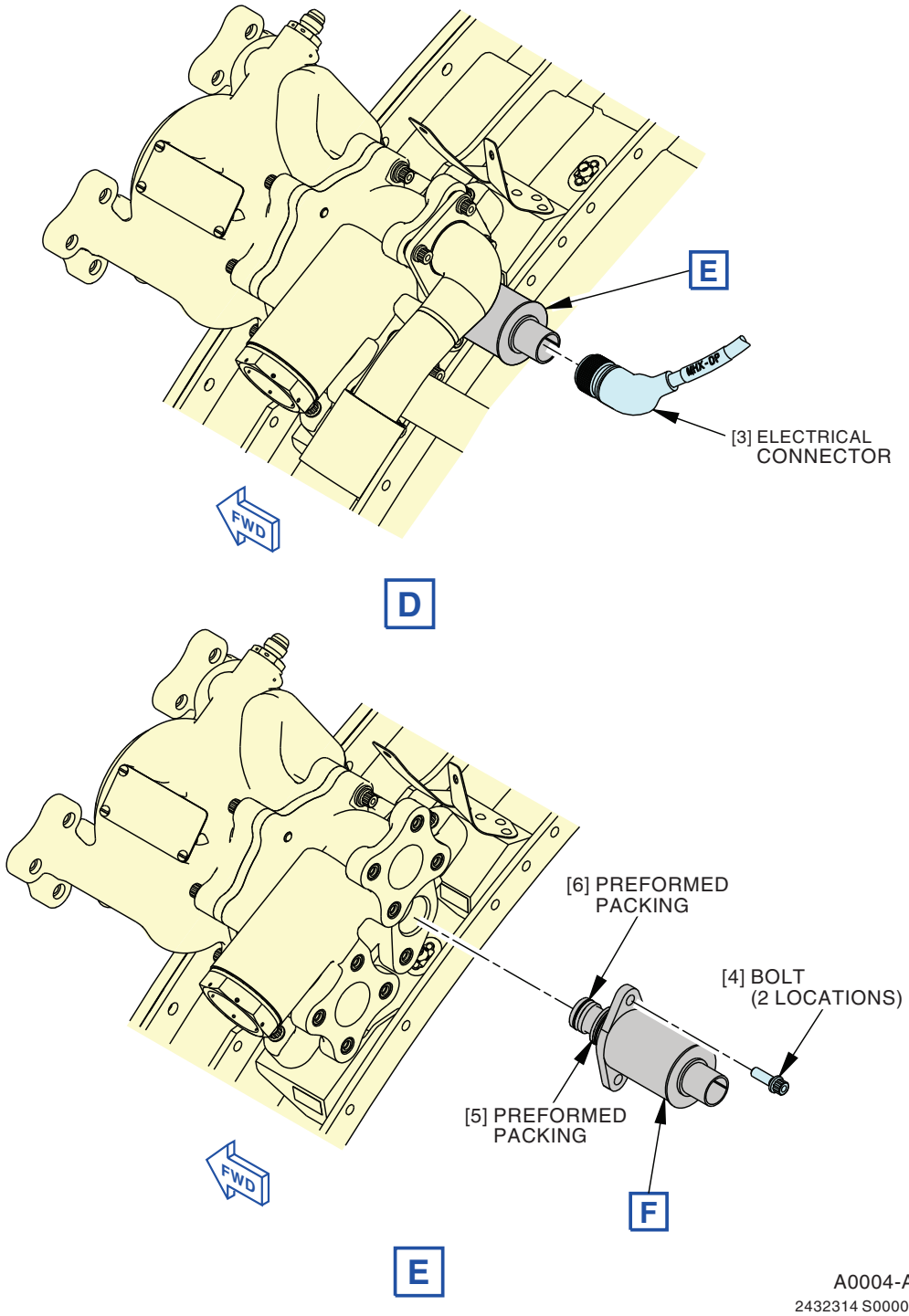

Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation
Figure 401/73-11-12-990-801-G00 (Sheet 4 of 5)

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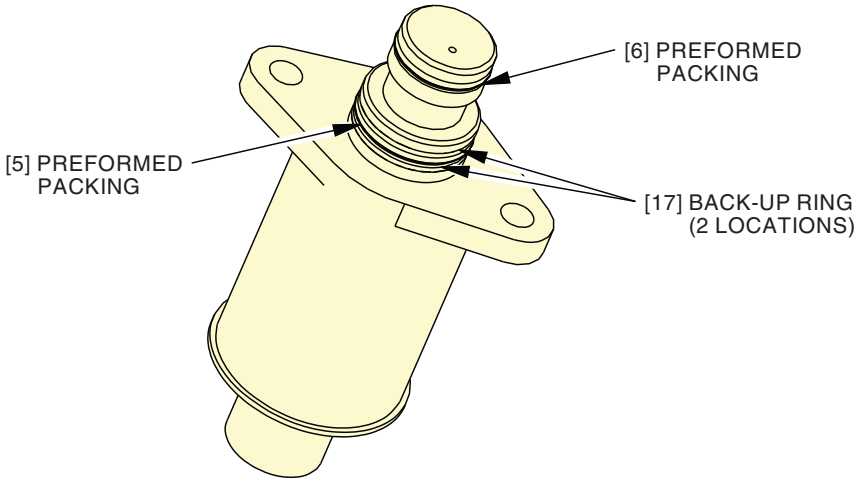

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2489319 S0000586564_V2

Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation
Figure 401/73-11-12-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-12-400-801-G00

3. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the differential pressure sensor for the main fuel/oil heat exchanger.

(2) You must do the tests that are listed in the power plant test reference table after you install the differential pressure sensor for the main fuel/oil heat exchanger.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Differential pressure sensor	73-11-12-01-020	PTF ALL
5	Preformed packing	73-11-12-01-050	PTF ALL
6	Preformed packing	73-11-12-01-030	PTF ALL
17	Backup ring	73-11-12-01-040	PTF ALL
		73-11-12-01-060	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation

SUBTASK 73-11-12-420-004-G00

(1) Install the differential pressure sensor [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective cover from the engine opening for the main fuel/oil heat exchanger differential pressure sensor [1].

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(c) Lubricate the threads of the bolts [4], new preformed packing [5], preformed packing [6] and backup rings [17] with engine oil, D00599 [CP2442].

(d) Install the preformed packing [6], preformed packing [5] and backup rings [17] in the grooves of the main fuel/oil heat exchanger differential pressure sensor [1].

(e) Install the differential pressure sensor [1] on the main fuel/oil heat exchanger [2].

(f) Install the bolts [4] that attach the differential pressure sensor [1] to the main fuel/oil heat exchanger [2].

(g) Tighten the bolts [4] to 109.31 in-lb (12.35 N·m) to 120.81 in-lb (13.65 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use a teflon-jawed pliers, STD-664, to connect the electrical connector [3] to the differential pressure sensor [1] (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-12-160-001-G00

(2) Clean the area and remove all tools from where you did the work.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-12-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-11-12-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-12-860-003-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

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J. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation Test

SUBTASK 73-11-12-700-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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MAIN FUEL/OIL HEAT EXCHANGER DIFFERENTIAL PRESSURE SENSOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Main Fuel/Oil Heat Exchanger Differential Pressure Sensor inspection.

TASK 73-11-12-210-801-G00

2. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection

(Figure 601)

A. General

(1) This task provides the instructions to inspect the differential pressure sensor for the main fuel/oil heat exchanger.

(a) The differential pressure sensor is located on the main fuel/oil heat exchanger on the left side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-12-000-801-G00	Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Removal (P/B 401)
73-11-12-400-801-G00	Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-11-12-860-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 73-11-12-860-005-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-12-010-002-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection

SUBTASK 73-11-12-210-002-G00

(1) Inspect the main fuel/oil heat exchanger Differential Pressure (DP) sensor:

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, do the steps in these tasks to replace the sensor preformed packings (TASK 73-11-12-000-801-G00 and TASK 73-11-12-400-801-G00).

(b) Make sure that the main fuel/oil heat exchanger DP sensor is not loose.

1) If the sensor is loose, tighten the bolts that attach the sensor to the heat exchanger (TASK 73-11-12-400-801-G00).

(c) Make sure that the electrical connector is not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If the electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-12-410-002-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

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SUBTASK 73-11-12-860-006-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-12-860-007-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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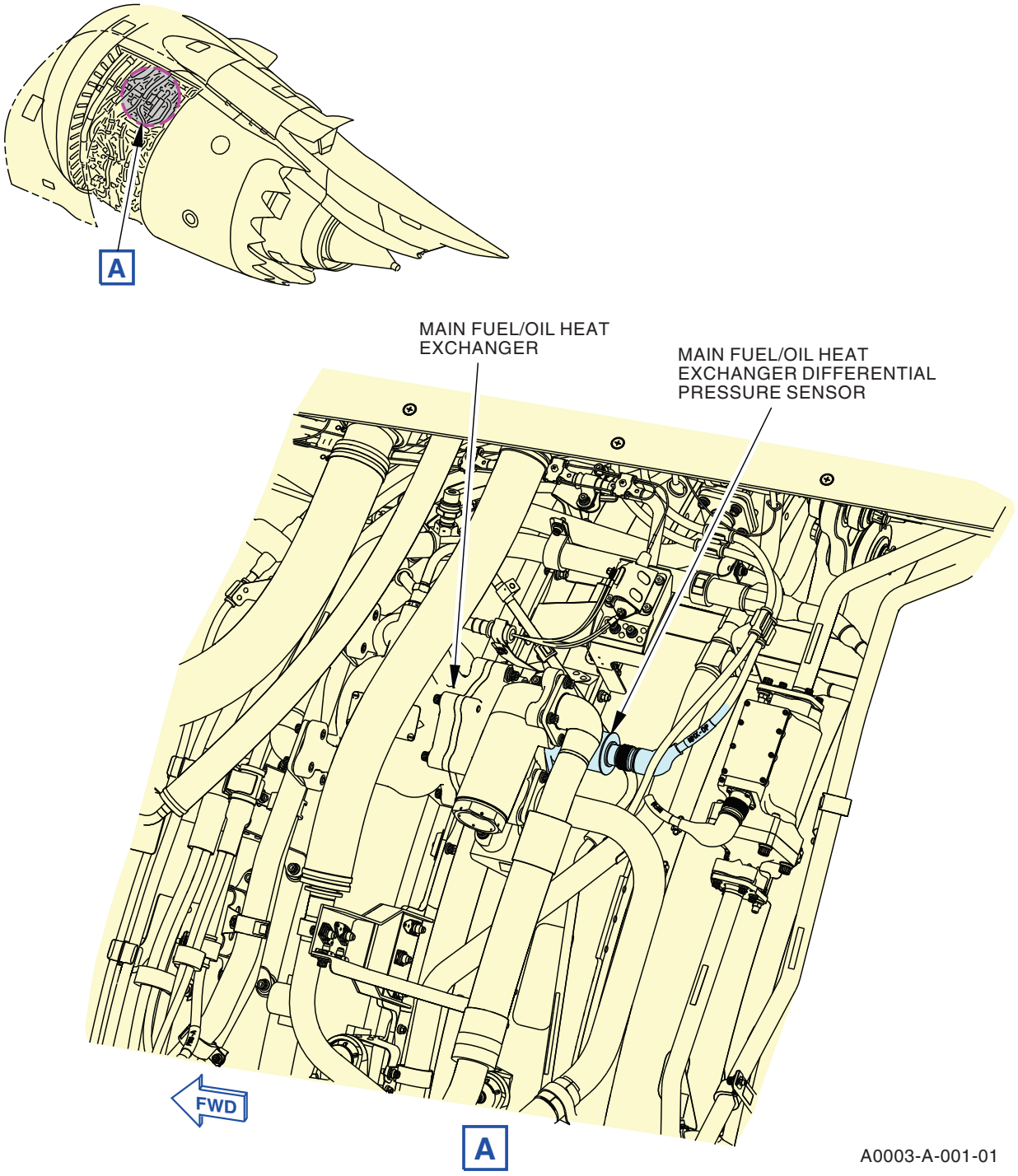

Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection
Figure 601/73-11-12-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) EEC System Configuration Data Functions Check

(2) EEC Shop Faults Function Check

(3) EEC Onboard Software Installation

(4) EEC Onboard Software Installation Using a PMAT

(5) EEC Onboard Software Verification Installation

(6) EEC Maintenance Power Selection.

B. For general information for software installation times and data loaders, do this task: On-Airplane Software Installation, TASK 20-15-11-470-801.

TASK 73-21-00-470-801-G00

2. EEC System Configuration Data Functions Check

A. General

(1) This task gives the instructions to do the Electronic Engine Control (EEC) system configuration data functions check.

(2) This task verifies the system configuration in the Onboard Network System (ONS)'s Line Replaceable Unit (LRU).

B. References

Reference	Title
46-13-00-710-802	System Configuration Check (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Data Functions Check (with MDS/ML)

SUBTASK 73-21-00-860-122-G00

(1) Access the applicable EEC system configuration page (TASK 46-13-00-710-802).

E. EEC System Configuration

SUBTASK 73-21-00-860-119-G00

(1) Verify the software, thrust management, and various engine configuration setting used by the EEC.

NOTE: EEC System Configuration Page gives information on the software, thrust management, and various engine configuration settings.

NOTE: This EEC information can be compared with the engine data plate, the cross-channel EEC, the opposite engine, the aircraft type, and engine Service Bulletin records to verify the engine system configuration.

(a) Table 201 gives the format for the EEC System Configuration Page.

Table 201/73-21-00-993-807-G00 EEC System Configuration

Hardware Part Number	AAAAAAAAAA *[1]

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Table 201/73-21-00-993-807-G00 EEC System Configuration (Continued)

Serial Number	BBBBBBBB *[2]
Software Location Description:	EEC–XZ *[3], *[4]
73 EEC OPS Part Number	CCCCCCCCCCCCCC *[5]
77 EEC AVM Part Number	DDDDDDDDDDDDDD *[6]
77 EEC EHM Part Number	EEEEEEEEEEEEEE *[7]
77 EEC AML Part Number	PPPPPPPPPPPPPP *[8]
Options	
Engine Serial Number	FFF-FFF *[9]
PSS Part Number	GGGGGGGGGG *[10]
Engine Configuration	HHHH *[11]
Engine Rating	1BJJJJJJJJ *[12]
Bump	K *[13]
N1 Trim	L *[14]
Menu Mode Hardware Code	RR *[15]
Auto-detected Hardware Configuration Code	TTTTT *[16]
HPT Deterioration	S *[17]
73 EEC OPS Version	MMMM *[18]
77 EEC AVM Version	NNNN *[19]
77 EEC EHM Version	OOOO *[20]
77 EEC AML Version	QQQQ*[21]

*[1] EEC Hardware Part Number
*[2] EEC Serial Number
*[3] "X" - Left or Right Engine (1 = Left Engine; 2 = Right Engine)
*[4] "Z" - EEC Channel (A or B)
*[5] OPS Software Part Number
*[6] AVM Software Part Number
*[7] EHM Software Part Number
*[8] AML Software Part Number
*[9] Engine Serial Number (ESN)
*[10] Pressure Sub-System (PSS) Part Number
*[11] Engine Configuration
*[12] Engine Rating
*[13] Bump available (Y = Yes; N = No)
*[14] N1 Trim designation
*[15] Menu Mode Hardware Code
*[16] Auto-detected Hardware Configuration Code
*[17] HPT Deterioration indication (Y = Yes; N = No)
*[18] OPS Software Version
*[19] AVM Software Version
*[20] EHM Software Version

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